Chapter 8
Two-Level Fractional Factorial Designs
Solutions

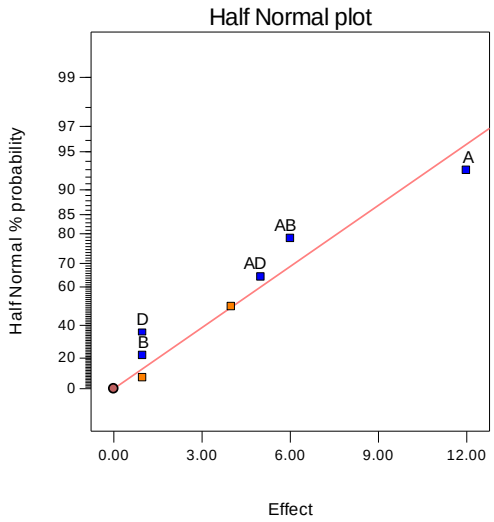
8.1. Suppose that in the chemical process development experiment in Problem 6.11, it was only possible to run a one-half fraction of the 2^4 design. Construct the design and perform the statistical analysis, using the data from replicate 1.

The required design is a 2^{4-1} with $I=ABCD$.

A	*B*	*C*	*D=ABC*		
–	–	–	–	(1)	90
+	–	–	+	*ad*	72
–	+	–	+	*bd*	87
+	+	–	–	*ab*	83
–	–	+	+	*cd*	99
+	–	+	–	*ac*	81
–	+	+	–	*bc*	88
+	+	+	+	*abcd*	80

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	-12	288	64.2857
Model	B	-1	2	0.446429
Model	C	4	32	7.14286
Model	D	-1	2	0.446429
Model	AB	6	72	16.0714
Model	AC	-1	2	0.446429
Model	AD	-5	50	11.1607
Error	BC	Aliased		
Error	BD	Aliased		
Error	CD	Aliased		
Error	ABC	Aliased		
Error	ABD	Aliased		
Error	ACD	Aliased		
Error	BCD	Aliased		
Error	ABCD	Aliased		
	Lenth's ME		22.5856	
	Lenth's SME	54.0516		



The largest effect is *A*. The next largest effects are the *AB* and *AD* interactions. A plausible tentative model would be *A*, *AB* and *AD*, along with *B* and *D* to preserve hierarchy.

Design Expert Output

Response: yield

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	414.00	5	82.80	4.87	0.1791	not significant
A	*288.00*	*1*	*288.00*	*16.94*	*0.0543*	
B	*2.00*	*1*	*2.00*	*0.12*	*0.7643*	
D	*2.00*	*1*	*2.00*	*0.12*	*0.7643*	
AB	*72.00*	*1*	*72.00*	*4.24*	*0.1758*	
AD	*50.00*	*1*	*50.00*	*2.94*	*0.2285*	
Residual	34.00	2	17.00			
Cor Total	448.00	7				

The "Model F-value" of 4.87 implies the model is not significant relative to the noise. There is a 17.91 % chance that a "Model F-value" this large could occur due to noise.

Std. Dev.	4.12	R-Squared	0.9241
Mean	85.00	Adj R-Squared	0.7344
C.V.	4.85	Pred R-Squared	-0.2143
PRESS	544.00	Adeq Precision	6.441

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	85.00	1	1.46	78.73	91.27	
A-A	-6.00	1	1.46	-12.27	0.27	1.00
B-B	-0.50	1	1.46	-6.77	5.77	1.00
D-D	-0.50	1	1.46	-6.77	5.77	1.00
AB	3.00	1	1.46	-3.27	9.27	1.00
AD	-2.50	1	1.46	-8.77	3.77	1.00

Final Equation in Terms of Coded Factors:

yield	=
+85.00	
-6.00	* A
-0.50	* B
-0.50	* D
+3.00	* A * B
-2.50	* A * D

Final Equation in Terms of Actual Factors:

yield	=
+85.00000	
-6.00000	* A
-0.50000	* B
-0.50000	* D
+3.00000	* A * B
-2.50000	* A * D

The Design-Expert output indicates that we really only need the main effect of factor *A*. The updated analysis is shown below:

Design Expert Output

Response: yield

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	288.00	1	288.00	10.80	0.0167	significant
A	*288.00*	*1*	*288.00*	10.80	0.0167	
Residual	160.00	6	26.67			
Cor Total	448.00	7				

The Model F-value of 10.80 implies the model is significant. There is only a 1.67% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	5.16	R-Squared	0.6429
Mean	85.00	Adj R-Squared	0.5833
C.V.	6.08	Pred R-Squared	0.3651
PRESS	284.44	Adeq Precision	4.648

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	85.00	1	1.83	80.53	89.47	
A-A	-6.00	1	1.83	-10.47	-1.53	1.00

Final Equation in Terms of Coded Factors:

yield	=
+85.00	
-6.00	* A

Final Equation in Terms of Actual Factors:

yield	=
+85.00000	
-6.00000	* A

8.2S. Consider the plasma etch experiment described in Example 6.1. Suppose that only a one-half fraction of the design could be run. Set up the design and analyze the data.

Because Example 6.1 is a replicated 2^3 factorial experiment, a half fraction of this design is a 2^{3-1} with four runs. The experiment is replicated to assure an adequate estimate of the MS_E.

A	*B*	*C=AB*	Etch Rate (A/min)		Factor Low (-)	Levels High (+)
–	–	+	1037	*A* (Gap, cm)	0.80	1.20
–	–	+	1052	*B* (C_2F_6 flow, SCCM)	125	200
+	–	–	669	*C* (Power, W)	275	325
+	–	–	650			
–	+	–	633			
–	+	–	601			
+	+	+	729			
+	+	+	860			

The analysis shown below identifies all three main effects as significant. Because this is a resolution III design, the main effects are aliased with two factor interactions. The original analysis from Example 6.1 identifies factors *A*, *C*, and the *AC* interaction as significant. In our replicated half fraction experiment, factor *B* is aliased with the *AC* interaction. This problem points out the concerns of running small resolution III designs.

Design Expert Output

Response: Etch Rate

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	2.225E+005	3	74169.79	31.61	0.0030	significant
A	*21528.13*	*1*	*21528.13*	*9.18*	*0.0388*	
B	*42778.13*	*1*	*42778.13*	*18.23*	*0.0130*	
C	*1.582E+005*	*1*	*1.582E+005*	*67.42*	*0.0012*	
Pure Error	9385.50	4	2346.37			
Cor Total	2.319E+005	7				

The Model F-value of 31.61 implies the model is significant. There is only a 0.30% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	48.44	R-Squared	0.9595
Mean	778.88	Adj R-Squared	0.9292
C.V.	6.22	Pred R-Squared	0.8381
PRESS	37542.00	Adeq Precision	12.481

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	778.88	1	17.13	731.33	826.42	
A-Gap	-51.88	1	17.13	-99.42	-4.33	1.00
B-C2F6 Flow	-73.13	1	17.13	-120.67	-25.58	1.00
C-Power	140.63	1	17.13	93.08	188.17	1.00

Final Equation in Terms of Coded Factors:

Etch Rate =
+778.88
-51.88 * A
-73.13 * B
+140.63 * C

Final Equation in Terms of Actual Factors:

Etch Rate =
-332.37500
-259.37500 * Gap
-1.95000 * C2F6 Flow
+5.62500 * Power

8.3S. Problem 6.24 describes a process improvement study in the manufacturing process of an integrated circuit. Suppose that only eight runs could be made in this process. Set up an appropriate 2^{5-2} design and find the alias structure. Use the appropriate observations from Problem 6.24 as the observations in this design and estimate the factor effects. What conclusions can you draw?

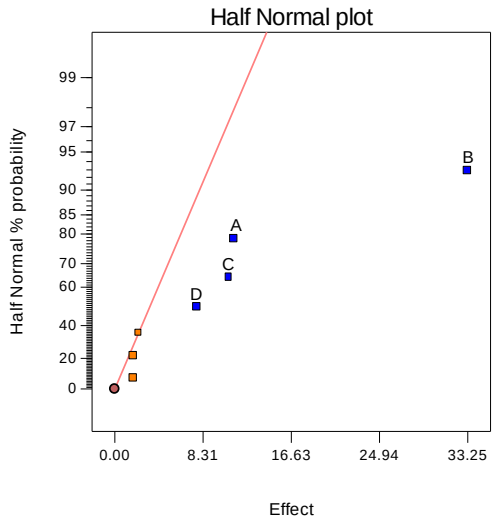
$$I = ABD = ACE = BCDE$$

A	*(ABD)*	*=BD*	*A*	*(ACE)*	*=CE*	*A*	*(BCDE)*	*=ABCDE*	*A=BD=CE=ABCDE*
B	*(ABD)*	*=AD*	*B*	*(ACE)*	*=ABCE*	*B*	*(BCDE)*	*=CDE*	*B=AD=ABCE=CDE*
C	*(ABD)*	*=ABCD*	*C*	*(ACE)*	*=AE*	*C*	*(BCDE)*	*=BDE*	*C=ABCD=AE=BDE*
D	*(ABD)*	*=AB*	*D*	*(ACE)*	*=ACDE*	*D*	*(BCDE)*	*=BCE*	*D=AB=ACDE=BCE*
E	*(ABD)*	*=ABDE*	*E*	*(ACE)*	*=AC*	*E*	*(BCDE)*	*=BCD*	*E=ABDE=AC=BCD*
BC	*(ABD)*	*=ACD*	*BC*	*(ACE)*	*=ABE*	*BC*	*(BCDE)*	*=DE*	*BC=ACD=ABE=DE*
BE	*(ABD)*	*=ADE*	*BE*	*(ACE)*	*=ABC*	*BE*	*(BCDE)*	*=CD*	*BE=ADE=ABC=CD*

A	*B*	*C*	*D=AB*	*E=AC*		
–	–	–	+	+	*de*	6
+	–	–	–	–	*a*	9
–	+	–	–	+	*be*	35
+	+	–	+	–	*abd*	50
–	–	+	+	–	*cd*	18
+	–	+	–	+	*ace*	22
–	+	+	–	–	*bc*	40
+	+	+	+	+	*abcde*	63

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	11.25	253.125	8.91953
Model	B	33.25	2211.13	77.9148
Model	C	10.75	231.125	8.1443
Model	D	7.75	120.125	4.23292
Error	E	2.25	10.125	0.356781
Error	BC	-1.75	6.125	0.215831
Error	BE	1.75	6.125	0.215831
	Lenth's ME	28.232		
	Lenth's SME	67.5646		



The main *A*, *B*, *C*, and *D* are large. However, recall that we are really estimating *A+BD+CE*, *B+AD*, *C+DE* and *D+AD*. There are other possible interpretations of the experiment because of the aliasing.

Design Expert Output

Response: Yield

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Model	2815.50	4	703.88	94.37	0.0017	significant
A	*253.13*	*1*	*253.13*	*33.94*	*0.0101*	
B	*2211.12*	*1*	*2211.12*	*296.46*	*0.0004*	
C	*231.13*	*1*	*231.13*	*30.99*	*0.0114*	
D	*120.13*	*1*	*120.13*	*16.11*	*0.0278*	
Residual	22.38	3	7.46			
Cor Total	2837.88	7				

The Model F-value of 94.37 implies the model is significant. There is only a 0.17% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	2.73	R-Squared	0.9921
Mean	30.38	Adj R-Squared	0.9816
C.V.	8.99	Pred R-Squared	0.9439
PRESS	159.11	Adeq Precision	25.590

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	30.38	1	0.97	27.30	33.45	
A-Aperture	5.63	1	0.97	2.55	8.70	1.00
B-Exposure Time	16.63	1	0.97	13.55	19.70	1.00
C-Develop Time	5.37	1	0.97	2.30	8.45	1.00
D-Mask Dimension	3.87	1	0.97	0.80	6.95	1.00

Final Equation in Terms of Coded Factors:

Yield =
+30.38
+5.63 * A
+16.63 * B
+5.37 * C
+3.87 * D

Final Equation in Terms of Actual Factors:

Aperture small
Mask Dimension Small
Yield =
-6.00000
+0.83125 * Exposure Time
+0.71667 * Develop Time

Aperture large
Mask Dimension Small
Yield =
+5.25000
+0.83125 * Exposure Time
+0.71667 * Develop Time

Aperture small
Mask Dimension Large
Yield =
+1.75000
+0.83125 * Exposure Time
+0.71667 * Develop Time

Aperture large
Mask Dimension Large
Yield =
+13.00000
+0.83125 * Exposure Time
+0.71667 * Develop Time

8.4S. Continuation of Problem 8.3S. Suppose you have made the eight runs in the 2^{5-2} design in Problem 8.3S. What additional runs would be required to identify the factor effects that are of interest? What are the alias relationships in the combined design?

We could fold over the original design by changing the signs on the generators $D = AB$ and $E = AC$ to produce the following new experiment.

A	*B*	*C*	*D=-AB*	*E=-AC*		
–	–	–	–	–	(1)	7
+	–	–	+	+	*ade*	10
–	+	–	+	–	*bd*	32
+	+	–	–	+	*abe*	52
–	–	+	–	+	*ce*	15
+	–	+	+	–	*acd*	21
–	+	+	+	+	*bcde*	41
+	+	+	–	–	*abc*	60

A	(–ABD)	=–BD	A	(–ACE)	=–CE	A	(BCDE)	=ABCDE	A=–BD=–CE=ABCDE
B	(–ABD)	=–AD	B	(–ACE)	=–ABCE	B	(BCDE)	=CDE	B=–AD=–ABCE=CDE
C	(–ABD)	=–ABCD	C	(–ACE)	=–AE	C	(BCDE)	=BDE	C=–ABCD=–AE=BDE
D	(–ABD)	=–AB	D	(–ACE)	=–ACDE	D	(BCDE)	=BCE	D=–AB=–ACDE=BCE
E	(–ABD)	=–ABDE	E	(–ACE)	=–AC	E	(BCDE)	=BCD	E=–ABDE=–AC=BCD
BC	(–ABD)	=–ACD	BC	(–ACE)	=–ABE	BC	(BCDE)	=DE	BC=–ACD=–ABE=DE
BE	(–ABD)	=–ADE	BE	(–ACE)	=–ABC	BE	(BCDE)	=CD	BE=–ADE=–ABC=CD

Assuming all three factor and higher interactions to be negligible, all main effects can be separated from their two-factor interaction aliases in the combined design.

8.5. In the Example 6.6, a 2^4 factorial design was used to improve the response rate to a credit card marketing offer. Suppose that the researchers had used the 2^{4-1} fraction factorial design with *I=ABCD* instead. Set up the design and select the responses for the runs from the full factorial data in Example 6.6. Analyze the data and draw conclusions. Compare your findings with those from the full factorial in Example 6.6.

Based on the Pseudo *p*-Value, the effects appear to be much less significant than those found in Example 6.6. The estimates for the Long-term Interest Rate, Account Opening Fee, and Annual Fee * Long-term Interest Rate are similar to the estimates found in Example 6.6; however, the other estimates are not as similar.

JMP Output

Response Response rate

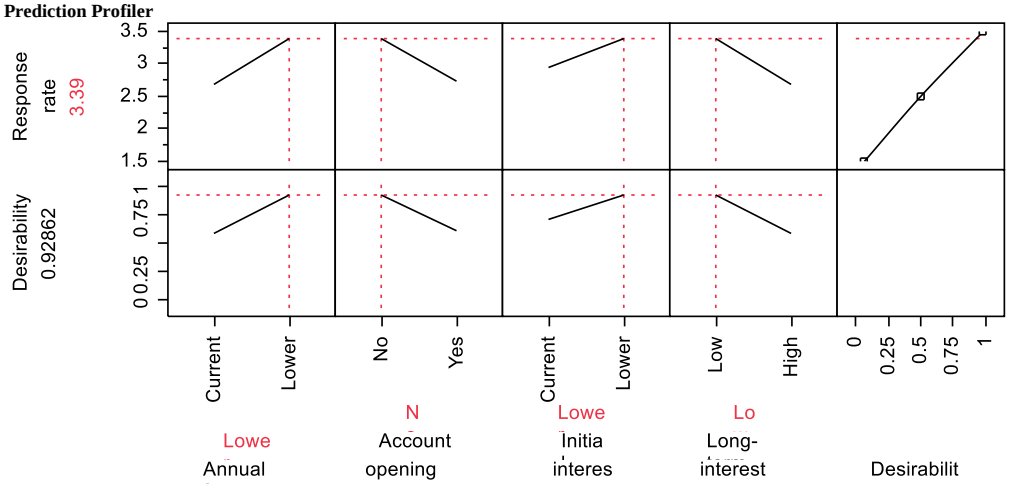
Summary of Fit

RSquare	1
RSquare Adj	.
Root Mean Square Error	.
Mean of Response	2.3375
Observations (or Sum Wgts)	8

Sorted Parameter Estimates

Term	Estimate	Relative Std Error	Pseudo t-Ratio	Pseudo t-Ratio	Pseudo p-Value
Long-term interest rate[Low]	0.275	0.353553	1.22		0.3307
Account-opening fee[No]	0.255	0.353553	1.13		0.3600
Initial interest[Current]	-0.17	0.353553	-0.76		0.5188
Annual fee[Current]	-0.15	0.353553	-0.67		0.5649
Annual fee[Current]*Long-term interest rate[Low]	-0.0775	0.353553	-0.34		0.7591
Annual fee[Current]*Account-opening fee[No]	-0.0725	0.353553	-0.32		0.7739
Annual fee[Current]*Initial interest[Current]	0.0525	0.353553	0.23		0.8344

No error degrees of freedom, so ordinary tests uncomputable. Relative Std Error corresponds to residual standard error of 1. Pseudo t-Ratio and p-Value calculated using Lenth PSE = 0.225 and DFE=2.3333



8.6. Continuation of Problem 8.5. In Example 6.6, we found that all four main effects and the two-factor *AB* interaction were significant. Show that if the alternate fraction (*I=-ABCD*) is added to the 2^{4-1} design in Problem 8.5 that the analysis from the combined design produced results identical to those found in Example 6.6

The estimates for the effects from the alternate fraction are shown below. By combining the estimates from Problem 8.5 with the estimates below, the same estimates as those shown in Example 6.6 are found. For example:

$$\frac{1}{2}\left([A]+[A']\right)=\frac{1}{2}\left(-0.15+\left(-0.2575\right)\right)=-0.20375\rightarrow A$$
$$\frac{1}{2}\left([AB]+[AB']\right)=\frac{1}{2}\left(-0.0725+\left(-0.23\right)\right)=-0.15125\rightarrow AB$$
$$\frac{1}{2}\left([AB]-[AB']\right)=\frac{1}{2}\left(-0.0725-\left(-0.23\right)\right)=0.07875\rightarrow CD$$

The analysis can also be performed as full factorial in two blocks with the block effect confounded with the *ABCD* interaction. This JMP analysis for the full factorial in two blocks is also shown below. The effect estimates are the same as those shown in Example 6.6.

JMP Output

Response Response rate

Summary of Fit

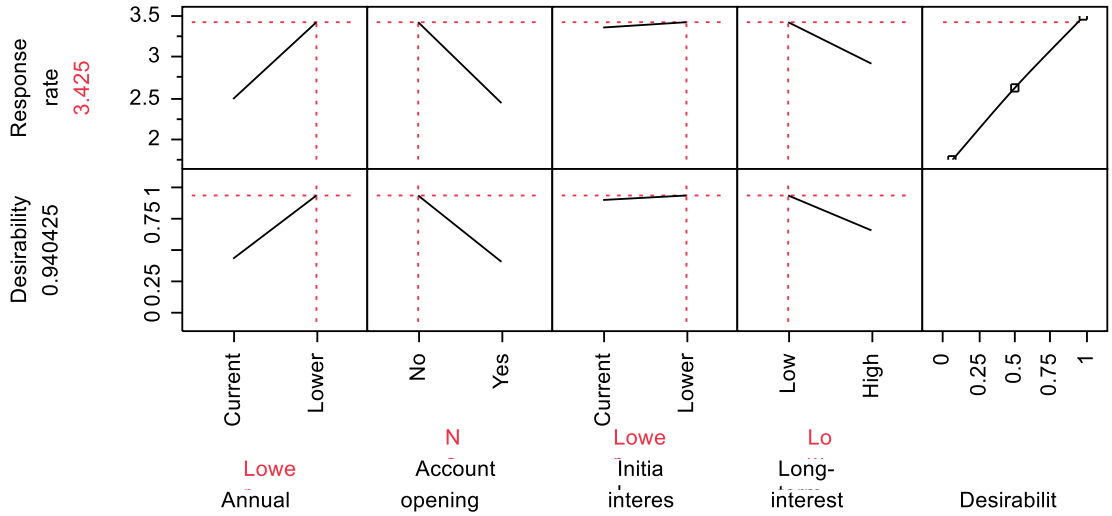
RSquare	1
RSquare Adj	.
Root Mean Square Error	.
Mean of Response	2.39
Observations (or Sum Wgts)	8

Sorted Parameter Estimates

Term	Estimate	Relative Std Error	Pseudo t-Ratio	Pseudo t-Ratio	Pseudo p-Value
Account-opening fee[No]	0.2625	0.353553	0.79		0.5035
Annual fee[Current]	-0.2575	0.353553	-0.77		0.5109
Annual fee[Current]*Account-opening fee[No]	-0.23	0.353553	-0.69		0.5529
Long-term interest rate[Low]	0.2225	0.353553	0.67		0.5649
Initial interest[Current]	-0.0825	0.353553	-0.25		0.8248
Annual fee[Current]*Initial interest[Current]	-0.05	0.353553	-0.15		0.8929
Annual fee[Current]*Long-term interest rate[Low]	-0.03	0.353553	-0.09		0.9355

No error degrees of freedom, so ordinary tests uncomputable. Relative Std Error corresponds to residual standard error of 1. Pseudo t-Ratio and p-Value calculated using Lenth PSE = 0.33375 and DFE=2.3333

Prediction Profiler



JMP Output

Response Response rate

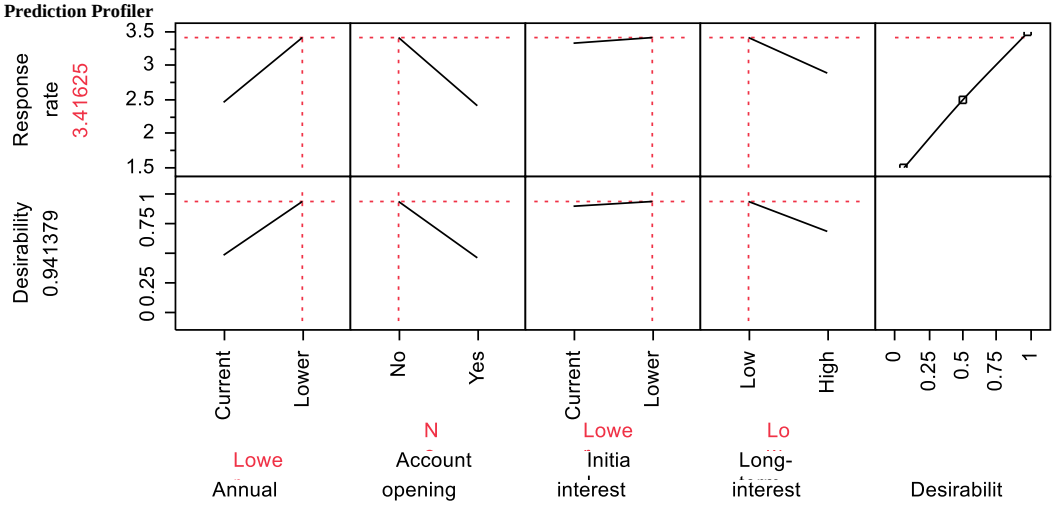
Summary of Fit

RSquare	1
RSquare Adj	.
Root Mean Square Error	.
Mean of Response	2.36375
Observations (or Sum Wgts)	16

Sorted Parameter Estimates

Term	Estimate	Relative Std Error	Pseudo t-Ratio	Pseudo t-Ratio	Pseudo p-Value
Account-opening fee[No]	0.25875	0.25	3.63		0.0150*
Long-term interest rate[Low]	0.24875	0.25	3.49		0.0174*
Annual fee[Current]	-0.20375	0.25	-2.86		0.0354*
Annual fee[Current]*Account-opening fee[No]	-0.15125	0.25	-2.12		0.0872
Initial interest rate[Current]	-0.12625	0.25	-1.77		0.1366
Initial interest rate[Current]*Long-term interest rate[Low]	0.07875	0.25	1.11		0.3194
Annual fee[Current]*Long-term interest rate[Low]	-0.05375	0.25	-0.75		0.4846
Account-opening fee[No]*Initial interest rate[Current]*Long-term interest rate[Low]	0.05375	0.25	0.75		0.4846
Account-opening fee[No]*Long-term interest rate[Low]	0.05125	0.25	0.72		0.5042
Annual fee[Current]*Account-opening fee[No]*Long-term interest rate[Low]	-0.04375	0.25	-0.61		0.5661
Block[1]	-0.02625	0.25	-0.37		0.7276
Annual fee[Current]*Account-opening fee[No]*Initial interest rate[Current]	0.02625	0.25	0.37		0.7276
Account-opening fee[No]*Initial interest rate[Current]	-0.02375	0.25	-0.33		0.7524
Annual fee[Current]*Initial interest rate[Current]*Long-term interest rate[Low]	-0.00375	0.25	-0.05		0.9601
Annual fee[Current]*Initial interest rate[Current]	0.00125	0.25	0.02		0.9867

No error degrees of freedom, so ordinary tests uncomputable. Relative Std Error corresponds to residual standard error of 1. Pseudo t-Ratio and p-Value calculated using Lenth PSE = 0.07125 and DFE=5



8.7. Continuation of Problem 8.5. Reconsider the 2^{4-1} fractional factorial design with $I=ABCD$ from Problem 8.6. Set a partial fold-over of this fraction to isolate the AB interaction. Select the appropriate set of responses from the full factorial data in Example 6.6 and analyze the resulting data.

The fold-over could be based on either factor A or B to isolate the AB interaction. The partial fold-over shown below in block 2 was based on factor B, with the + level chosen (Yes).

Annual Fee	Account opening fee	Initial interest Rate	Long-term interest rate	Block	Response rate
Current	No	Current	Low	1	2.45
Lower	Yes	Current	Low	1	2.29
Lower	No	Lower	Low	1	3.39
Current	Yes	Lower	Low	1	2.32
Lower	No	Current	High	1	2.24
Current	Yes	Current	High	1	1.69
Current	No	Lower	High	1	2.29
Lower	Yes	Lower	High	1	2.03
Current	Yes	Current	Low	2	2.16
Lower	Yes	Lower	Low	2	2.44
Lower	Yes	Current	High	2	1.87
Current	Yes	Lower	High	2	2.04

The analysis is shown in the JMP Output below.

JMP Output

Response Response rate

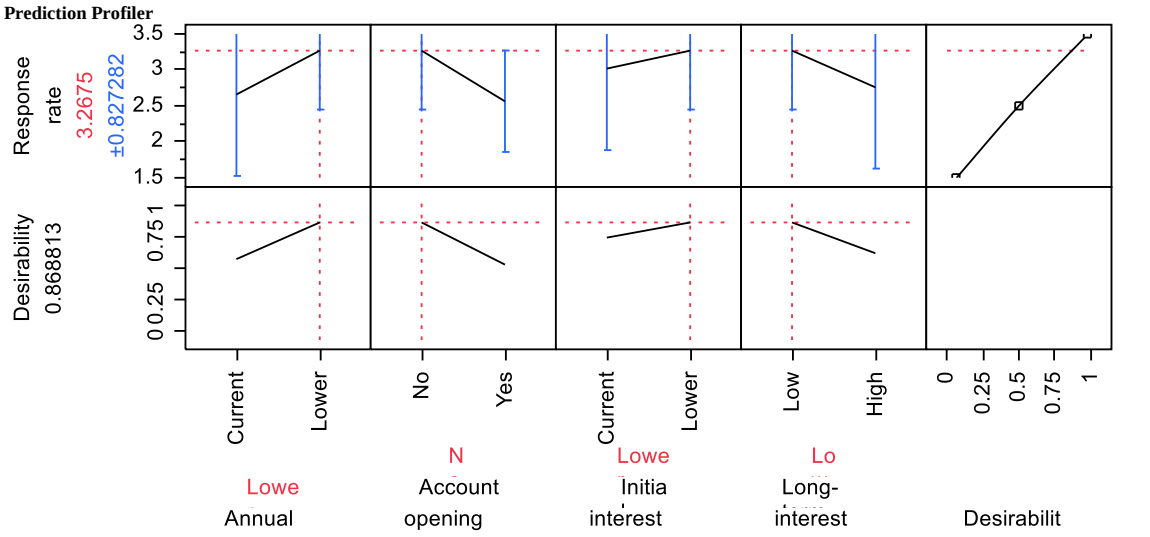
Summary of Fit

RSquare	0.956417
RSquare Adj	0.760293
Root Mean Square Error	0.205548
Mean of Response	2.2675
Observations (or Sum Wgts)	12

Sorted Parameter Estimates

Term	Estimate	Std Error	t Ratio	t Ratio	Prob>|t|
Long-term interest rate[Low]	0.23625	0.062936	3.75		0.0642
Account-opening fee[No]	0.255	0.072672	3.51		0.0725
Initial interest rate[Current]	-0.13625	0.062936	-2.16		0.1628
Annual fee[Current]	-0.15	0.072672	-2.06		0.1751
Annual fee[Current]*Account-opening fee[No]	-0.0975	0.072672	-1.34		0.3118
Annual fee[Current]*Long-term interest rate[Low]	-0.04375	0.062936	-0.70		0.5589
Initial interest rate[Current]*Long-term interest rate[Low]	0.025	0.072672	0.34		0.7636
Block[1]	-0.0225	0.072672	-0.31		0.7861

Term	Estimate	Std Error	t Ratio	t Ratio	Prob>|t|
Annual fee[Current]*Initial interest rate[Current]	0.01375	0.062936	0.22		0.8473



8.8. An article by J.J. Pignatiello, Jr. And J.S. Ramberg in the *Journal of Quality Technology*, (Vol. 17, 1985, pp. 198-206) describes the use of a replicated fractional factorial to investigate the effects of five factors on the free height of leaf springs used in an automotive application. The factors are A = furnace temperature, B = heating time, C = transfer time, D = hold down time, and E = quench oil temperature. The data are shown in Table P81.

Table P8.1

A	*B*	*C*	*D*	*E*	Free Height		
–	–	–	–	–	7.78	7.78	7.81
+	–	–	+	–	8.15	8.18	7.88
–	+	–	+	–	7.50	7.56	7.50
+	+	–	–	–	7.59	7.56	7.75
–	–	+	+	–	7.54	8.00	7.88
+	–	+	–	–	7.69	8.09	8.06
–	+	+	–	–	7.56	7.52	7.44
+	+	+	+	–	7.56	7.81	7.69
–	–	–	–	+	7.50	7.25	7.12
+	–	–	+	+	7.88	7.88	7.44
–	+	–	+	+	7.50	7.56	7.50
+	+	–	–	+	7.63	7.75	7.56
–	–	+	+	+	7.32	7.44	7.44
+	–	+	–	+	7.56	7.69	7.62
–	+	+	–	+	7.18	7.18	7.25
+	+	+	+	+	7.81	7.50	7.59

(a) Write out the alias structure for this design. What is the resolution of this design?

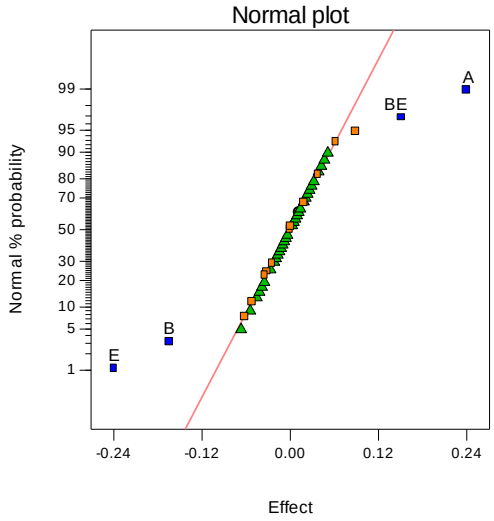
I=ABCD, Resolution IV

A	*(ABCD)=*	*BCD*
B	*(ABCD)=*	*ACD*
C	*(ABCD)=*	*ABD*
D	*(ABCD)=*	*ABC*
E	*(ABCD)=*	*ABCDE*
AB	*(ABCD)=*	*CD*
AC	*(ABCD)=*	*BD*
AD	*(ABCD)=*	*BC*
AE	*(ABCD)=*	*BCDE*
BE	*(ABCD)=*	*ACDE*
CE	*(ABCD)=*	*ABDE*
DE	*(ABCD)=*	*ABCE*

(b) Analyze the data. What factors influence the mean free height?

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	0.242083	0.703252	24.3274
Model	B	-0.16375	0.321769	11.1309
Model	C	-0.0495833	0.0295021	1.02056
Model	D	0.09125	0.0999188	3.45646
Model	E	-0.23875	0.684019	23.6621
Model	AB	-0.0295833	0.0105021	0.363296
Model	AC	0.00125	1.875E-005	0.000648614
Model	AD	-0.0229167	0.00630208	0.218006
Model	AE	0.06375	0.0487687	1.68704
Error	BC	Aliased		
Error	BD	Aliased		
Model	BE	0.152917	0.280602	9.70679
Error	CD	Aliased		
Model	CE	-0.0329167	0.0130021	0.449777
Model	DE	0.0395833	0.0188021	0.650415
Error	Pure Error		0.627067	21.6919
	Lenth's ME	0.088057		
	Lenth's SME	0.135984		



Design Expert Output

Response:Free Height

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	1.99	4	0.50	23.74	< 0.0001	significant
A	*0.70*	*1*	*0.70*	*33.56*	*< 0.0001*	
B	*0.32*	*1*	*0.32*	*15.35*	*0.0003*	
E	*0.68*	*1*	*0.68*	*32.64*	*< 0.0001*	
BE	*0.28*	*1*	*0.28*	*13.39*	*0.0007*	
Residual	0.90	43	0.021			
Lack of Fit	*0.27*	*11*	*0.025*	*1.27*	*0.2844 not significant*	
Pure Error	*0.63*	*32*	*0.020*			
Cor Total	2.89	47				

The Model F-value of 23.74 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.14	R-Squared	0.6883
Mean	7.63	Adj R-Squared	0.6593
C.V.	1.90	Pred R-Squared	0.6116
PRESS	1.12	Adeq Precision	13.796

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	7.63	1	0.021	7.58	7.67	
A-Furnace Temp	0.12	1	0.021	0.079	0.16	1.00
B-Heating Time	-0.082	1	0.021	-0.12	-0.040	1.00
E-Quench Temp	-0.12	1	0.021	-0.16	-0.077	1.00
BE	0.076	1	0.021	0.034	0.12	1.00

Final Equation in Terms of Coded Factors:

```
Free Height    =
     +7.63
     +0.12    * A
    -0.082    * B
     -0.12    * E
    +0.076    * B * E
```

Final Equation in Terms of Actual Factors:

```
Free Height    =
    +7.62562
    +0.12104  * Furnace Temp
   -0.081875  * Heating Time
    -0.11937  * Quench Temp
   +0.076458  * Heating Time * Quench Temp
```

(c) Calculate the range and standard deviation of the free height for each run. Is there any indication that any of these factors affects variability in the free height?

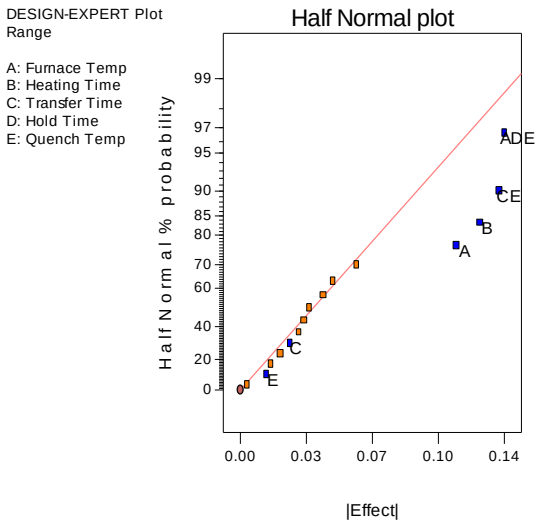
Design Expert Output (Range)

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	0.11375	0.0517563	16.2198
Model	B	-0.12625	0.0637563	19.9804
Model	C	0.02625	0.00275625	0.863774
Error	D	0.06125	0.0150063	4.70277
Model	E	-0.01375	0.00075625	0.236999
Error	AB	0.04375	0.00765625	2.39937
Error	AC	-0.03375	0.00455625	1.42787
Error	AD	0.03625	0.00525625	1.64724
Error	AE	-0.00375	5.625E-005	0.017628
Model	BC	Aliased		
Error	BD	Aliased		
Model	BE	0.01625	0.00105625	0.331016

Error	CD	Aliased		
Model	CE	-0.13625	0.0742562	23.271
Error	DE	-0.02125	0.00180625	0.566056
Error	ABC	Aliased		
Error	ABD	Aliased		
Error	ABE	0.03125	0.00390625	1.22417
Error	ACD	Aliased		
Error	ACE	0.04875	0.00950625	2.97914
Error	ADE	0.13875	0.0770062	24.1328
Error	BCD	Aliased		
Model	BCE	Aliased		
Error	BDE	Aliased		
Error	CDE	Aliased		

Lenth's ME 0.130136
Lenth's SME 0.264194

Interaction *ADE* is aliased with *BCE*. Although the plot below identifies *ADE*, *BCE* was included in the analysis.



Design Expert Output (Range)

Response: Range

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F
Model	0.28	8	0.035	5.70	0.0167 significant
A	0.052	1	0.052	8.53	0.0223
B	0.064	1	0.064	10.50	0.0142
C	2.756E-003	1	2.756E-003	0.45	0.5220
E	7.562E-004	1	7.562E-004	0.12	0.7345
BC	5.256E-003	1	5.256E-003	0.87	0.3831
BE	1.056E-003	1	1.056E-003	0.17	0.6891
CE	0.074	1	0.074	12.23	0.0100
BCE	0.077	1	0.077	12.69	0.0092
Residual	0.042	7	6.071E-003		
Cor Total	0.32	15			

The Model F-value of 5.70 implies the model is significant. There is only a 1.67% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.078	R-Squared	0.8668
Mean	0.22	Adj R-Squared	0.7146
C.V.	35.52	Pred R-Squared	0.3043
PRESS	0.22	Adeq Precision	7.166

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	0.22	1	0.019	0.17	0.27	
A-Furn Temp	0.057	1	0.019	0.011	0.10	1.00
B-Heat Time	-0.063	1	0.019	-0.11	-0.017	1.00
C-Transfer Time	0.013	1	0.019	-0.033	0.059	1.00
E-Qnch Temp	-6.875E-003	1	0.019	-0.053	0.039	1.00
BC	0.018	1	0.019	-0.028	0.064	1.00
BE	8.125E-003	1	0.019	-0.038	0.054	1.00
CE	-0.068	1	0.019	-0.11	-0.022	1.00
BCE	0.069	1	0.019	0.023	0.12	1.00

Final Equation in Terms of Coded Factors:

```
Range    =
  +0.22
 +0.057  * A
 -0.063  * B
 +0.013  * C
-6.875E-003  * E
 +0.018  * B * C
+8.125E-003  * B * E
 -0.068  * C * E
 +0.069  * B * C * E
```

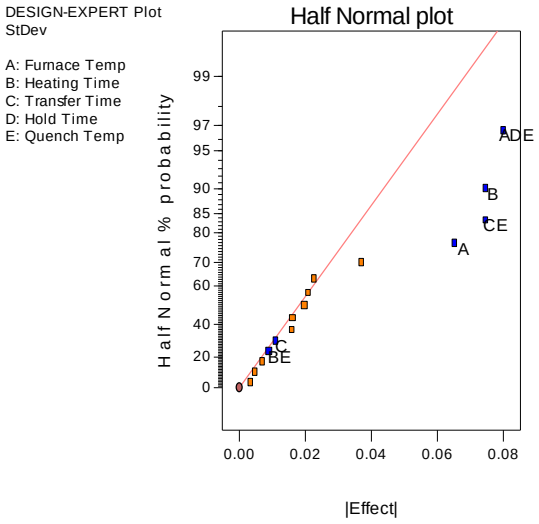
Final Equation in Terms of Actual Factors:

```
Range    =
 +0.21937
+0.056875  * Furnace Temp
-0.063125  * Heating Time
+0.013125  * Transfer Time
-6.87500E-003  * Quench Temp
+0.018125  * Heating Time * Transfer Time
+8.12500E-003  * Heating Time * Quench Temp
-0.068125  * Transfer Time * Quench Temp
+0.069375  * Heating Time * Transfer Time * Quench Temp
```

Design Expert Output (StDev)

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	0.0625896	0.0156698	16.873
Model	B	-0.0714887	0.0204425	22.0121
Model	C	0.010567	0.000446646	0.48094
Error	D	0.0353616	0.00500176	5.3858
Model	E	-0.00684034	0.000187161	0.201532
Error	AB	0.0153974	0.000948317	1.02113
Error	AC	-0.0218505	0.00190978	2.05641
Error	AD	0.0190608	0.00145326	1.56484
Error	AE	-0.00329035	4.33057E-005	0.0466308
Model	BC	Aliased		
Error	BD	Aliased		
Model	BE	0.0087666	0.000307413	0.331017
Error	CD	Aliased		
Model	CE	-0.0714816	0.0204385	22.0078
Error	DE	-0.00467792	8.75317E-005	0.0942525
Error	ABC	Aliased		
Error	ABD	Aliased		
Error	ABE	0.0155599	0.000968437	1.0428
Error	ACD	Aliased		
Error	ACE	0.0199742	0.00159587	1.7184
Error	ADE	Aliased		
Error	BCD	Aliased		
Model	BCE	0.0764346	0.023369	25.1633
Error	BDE	Aliased		
Error	CDE	Aliased		
	Lenth's ME	0.0596836		
	Lenth's SME	0.121166		

Interaction *ADE* is aliased with *BCE*. Although the plot below identifies *ADE*, *BCE* was included in the analysis.



Design Expert Output (StDev)

Response: StDev

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	0.082	8	0.010	6.82	0.0101	significant
A	*0.016*	*1*	*0.016*	*10.39*	*0.0146*	
B	*0.020*	*1*	*0.020*	*13.56*	*0.0078*	
C	*4.466E-004*	*1*	*4.466E-004*	*0.30*	*0.6032*	
E	*1.872E-004*	*1*	*1.872E-004*	*0.12*	*0.7350*	
BC	*1.453E-003*	*1*	*1.453E-003*	*0.96*	*0.3589*	
BE	*3.074E-004*	*1*	*3.074E-004*	*0.20*	*0.6653*	
CE	*0.020*	*1*	*0.020*	*13.55*	*0.0078*	
BCE	*0.023*	*1*	*0.023*	*15.50*	*0.0056*	
Residual	0.011	7	1.508E-003			
Cor Total	0.093	15				

The Model F-value of 6.82 implies the model is significant. There is only a 1.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.039	R-Squared	0.8863
Mean	0.12	Adj R-Squared	0.7565
C.V.	33.07	Pred R-Squared	0.4062
PRESS	0.055	Adeq Precision	7.826

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	0.12	1	9.708E-003	0.094	0.14	
A-Furnace Temp	0.031	1	9.708E-003	8.340E-003	0.054	1.00
B-Heating Time	-0.036	1	9.708E-003	-0.059	-0.013	1.00
C-Transfer Time	5.283E-003	1	9.708E-003	-0.018	0.028	1.00
E-Quench Temp	-3.420E-003	1	9.708E-003	-0.026	0.020	1.00
BC	9.530E-003	1	9.708E-003	-0.013	0.032	1.00
BE	4.383E-003	1	9.708E-003	-0.019	0.027	1.00
CE	-0.036	1	9.708E-003	-0.059	-0.013	1.00
BCE	0.038	1	9.708E-003	0.015	0.061	1.00

Final Equation in Terms of Coded Factors:

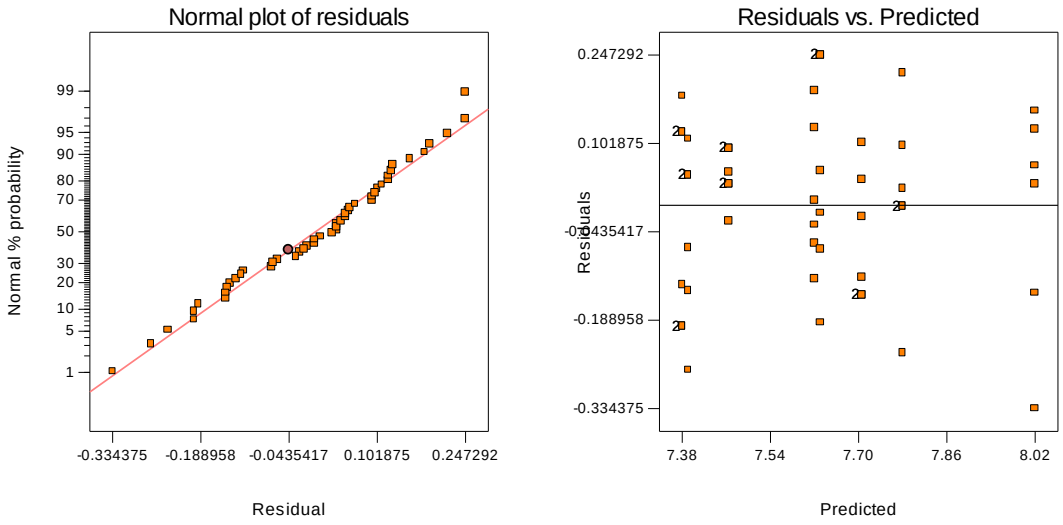
StDev	=
+0.12	
+0.031	* A
-0.036	* B
+5.283E-003	* C
-3.420E-003	* E
+9.530E-003	* B * C
+4.383E-003	* B * E
-0.036	* C * E
+0.038	* B * C * E

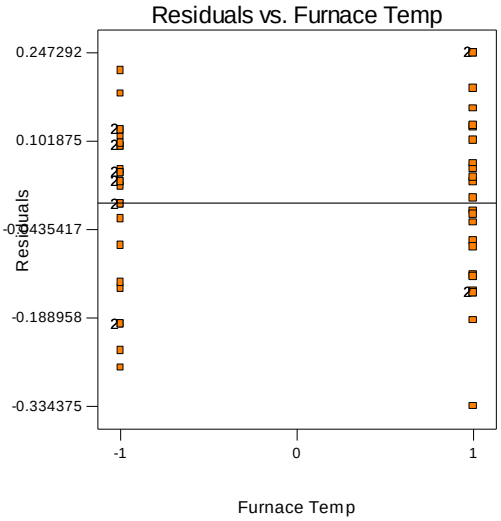
Final Equation in Terms of Actual Factors:

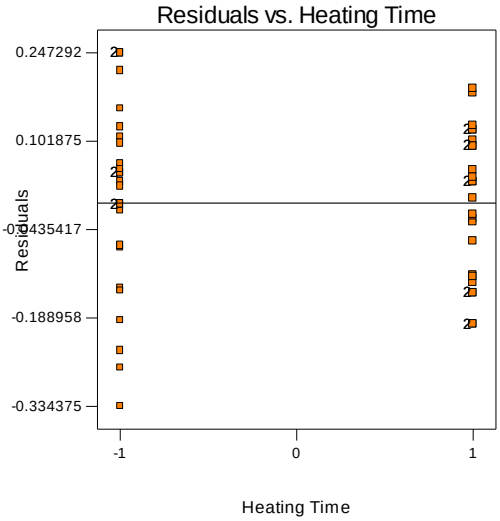
StDev	=
+0.11744	
+0.031295	* Furnace Temp
-0.035744	* Heating Time
+5.28350E-003	* Transfer Time
-3.42017E-003	* Quench Temp
+9.53040E-003	* Heating Time * Transfer Time
+4.38330E-003	* Heating Time * Quench Temp
-0.035741	* Transfer Time * Quench Temp
+0.038217	* Heating Time * Transfer Time * Quench Temp

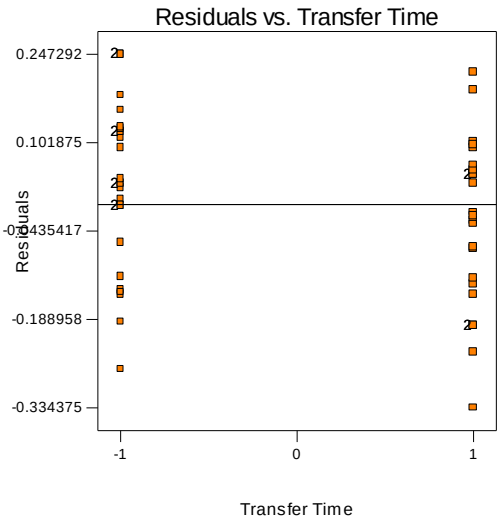
(d) Analyze the residuals from this experiment, and comment on your findings.

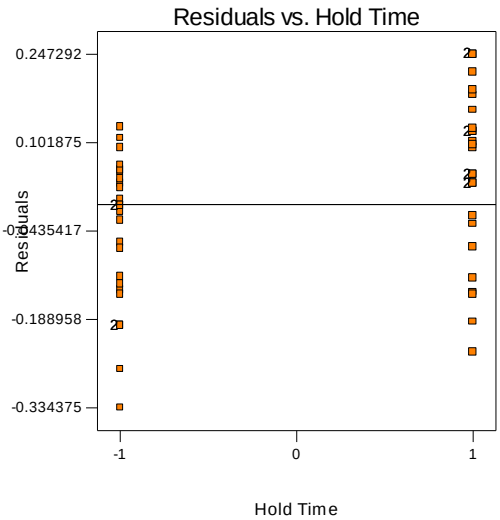
The residual plot follows. All plots are satisfactory.

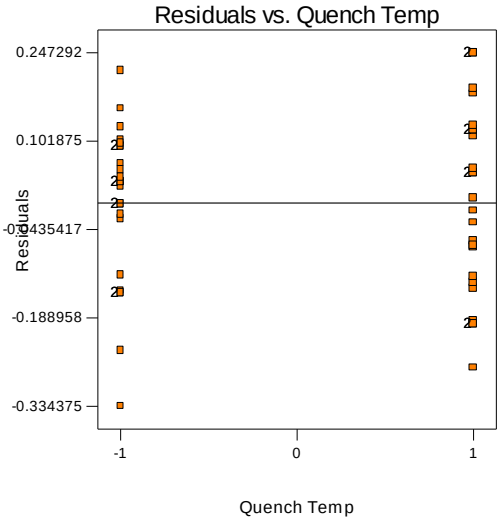



Residuals vs. Furnace Temp
0.247292
0.101875
-0.0435417
-0.188958
-0.334375
Residuals
-1
0
1
Furnace Temp


Residuals vs. Heating Time
0.247292
0.101875
-0.0435417
-0.188958
-0.334375
Residuals
-1
0
1
Heating Time


Residuals vs. Transfer Time
0.247292
0.101875
-0.0435417
-0.188958
-0.334375
Residuals
-1
0
1
Transfer Time

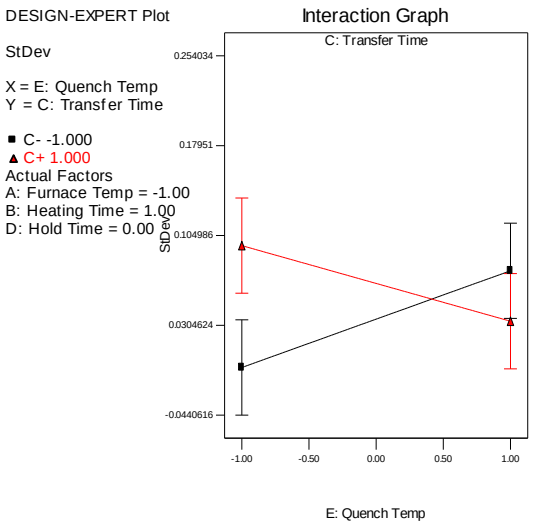

Residuals vs. Hold Time
0.247292
0.101875
-0.0435417
-0.188958
-0.334375
Residuals
-1
0
1
Hold Time


Residuals vs. Quench Temp
0.247292
0.101875
-0.0435417
-0.188958
-0.334375
Residuals
-1
0
1
Quench Temp

(e) Is this the best possible design for five factors in 16 runs? Specifically, can you find a fractional design for five factors in 16 runs with a higher resolution than this one?

This was not the best design. A resolution V design is possible by setting the generator equal to the highest order interaction, *ABCDE*.

8.9. Consider the leaf spring experiment in Problem 8.8. Suppose that factor *E* (quench oil temperature) is very difficult to control during manufacturing. Where would you set factors *A, B, C* and *D* to reduce variability in the free height as much as possible regardless of the quench oil temperature used?



Run the process with *A* at the low level, *B* at the high level, *C* at the low level and *D* at either level (the low level of *D* may give a faster process).

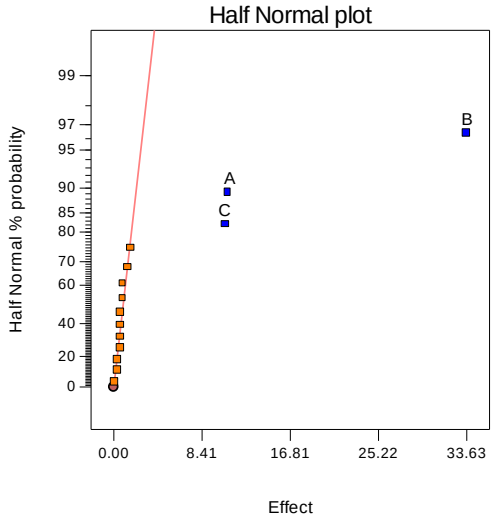
8.10. Consider the 2^5 design in Problem 6.24. Suppose that only a one-half fraction could be run. Furthermore, two days were required to take the 16 observations, and it was necessary to confound the 2^{5-1} design in two blocks. Construct the design and analyze the data.

A	*B*	*C*	*D*	*E=ABCD*		Data	Blocks = *AB*	Block
–	–	–	–	+	*e*	8	+	1
+	–	–	–	–	*a*	9	–	2
–	+	–	–	–	*b*	34	–	2
+	+	–	–	+	*abe*	52	+	1
–	–	+	–	–	*c*	16	+	1
+	–	+	–	+	*ace*	22	–	2
–	+	+	–	+	*bce*	45	–	2
+	+	+	–	–	*abc*	60	+	1
–	–	–	+	–	*d*	8	+	1
+	–	–	+	+	*ade*	10	–	2
–	+	–	+	+	*bde*	30	–	2
+	+	–	+	–	*abd*	50	+	1
–	–	+	+	+	*cde*	15	+	1
+	–	+	+	–	*acd*	21	–	2
–	+	+	+	–	*bcd*	44	–	2
+	+	+	+	+	*abcde*	63	+	1

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	10.875	473.063	8.6343
Model	B	33.625	4522.56	82.5455
Model	C	10.625	451.562	8.24188
Error	D	-0.625	1.5625	0.0285186
Error	E	0.375	0.5625	0.0102667
Error	AB	Aliased		
Error	AC	0.625	1.5625	0.0285186
Error	AD	0.875	3.0625	0.0558965
Error	AE	1.375	7.5625	0.13803
Error	BC	0.875	3.0625	0.0558965
Error	BD	-0.375	0.5625	0.0102667
Error	BE	0.125	0.0625	0.00114075
Error	CD	0.625	1.5625	0.0285186
Error	CE	0.625	1.5625	0.0285186
Error	DE	-1.625	10.5625	0.192786
	Lenth's ME	2.46263		
	Lenth's SME	5.0517		

The *AB* interaction in the above table is aliased with the three-factor interaction *BCD*, and is also confounded with blocks.



Design Expert Output

Response: Yield

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Block	203.06	1	203.06			
Model	5447.19	3	1815.73	630.31	< 0.0001	significant
A	*473.06*	*1*	*473.06*	*164.22*	*< 0.0001*	
B	*4522.56*	*1*	*4522.56*	*1569.96*	*< 0.0001*	
C	*451.56*	*1*	*451.56*	*156.76*	*< 0.0001*	
Residual	31.69	11	2.88			
Cor Total	5681.94	15				

The Model F-value of 630.31 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	1.70	R-Squared	0.9942
Mean	30.44	Adj R-Squared	0.9926
C.V.	5.58	Pred R-Squared	0.9878
PRESS	67.04	Adeq Precision	58.100

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	30.44	1	0.42	29.50	31.37	
Block 1	3.56	1				
Block 2	-3.56					
A-Aperture	5.44	1	0.42	4.50	6.37	1.00
B-Exposure Time	16.81	1	0.42	15.88	17.75	1.00
C-Develop Time	5.31	1	0.42	4.38	6.25	1.00

Final Equation in Terms of Coded Factors:

Yield =
+30.44
+5.44 * A
+16.81 * B
+5.31 * C

Final Equation in Terms of Actual Factors:

Aperture small
Yield =
-1.56250
+0.84063 * Exposure Time
+0.70833 * Develop Time

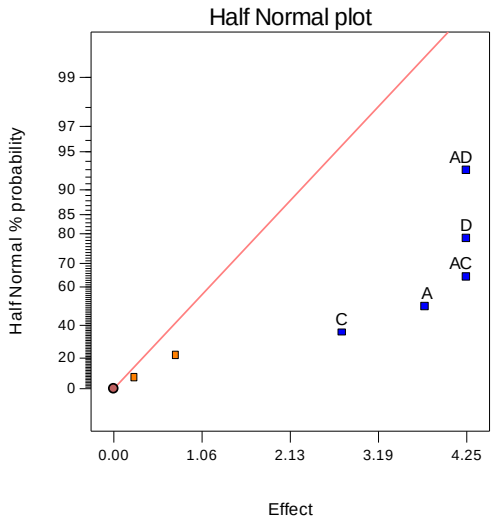
Aperture large
Yield =
+9.31250
+0.84063 * Exposure Time
+0.70833 * Develop Time

8.11. Analyze the data in Problem 6.26 as if it came from a 2^{4-1}_{IV} design with $I=ABCD$. Project the design into a full factorial in the subset of the original four factors that appear to be significant.

Run Number	*A*	*B*	*C*	*D=ABC*		Yield (lbs)		Factor Low (-)	Levels High (+)
1	–	–	–	–	(1)	12	*A* (h)	2.5	3.0
2	+	–	–	+	*ad*	25	*B* (%)	14	18
3	–	+	–	+	*bd*	13	*C* (psi)	60	80
4	+	+	–	–	*ab*	16	*D* (°C)	225	250
5	–	–	+	+	*cd*	19			
6	+	–	+	–	*ac*	15			
7	–	+	+	–	*bc*	20			
8	+	+	+	+	*abcd*	23			

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	3.75	28.125	18.3974
Error	B	0.25	0.125	0.0817661
Model	C	2.75	15.125	9.8937
Model	D	4.25	36.125	23.6304
Error	AB	-0.75	1.125	0.735895
Model	AC	-4.25	36.125	23.6304
Model	AD	4.25	36.125	23.6304
	Lenth's ME	21.174		
	Lenth's SME	50.6734		



Design Expert Output

Response: Yield in lbs

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	151.63	5	30.32	48.52	0.0203	significant
A	*28.13*	*1*	*28.13*	*45.00*	*0.0215*	
C	*15.13*	*1*	*15.13*	*24.20*	*0.0389*	
D	*36.12*	*1*	*36.12*	*57.80*	*0.0169*	
AC	*36.12*	*1*	*36.12*	*57.80*	*0.0169*	
AD	*36.13*	*1*	*36.13*	*57.80*	*0.0169*	
Residual	1.25	2	0.62			
Cor Total	152.88	7				

The Model F-value of 48.52 implies the model is significant. There is only a 2.03% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.79	R-Squared	0.9918
Mean	17.88	Adj R-Squared	0.9714
C.V.	4.42	Pred R-Squared	0.8692
PRESS	20.00	Adeq Precision	17.892

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	17.88	1	0.28	16.67	19.08	
A-Time	1.87	1	0.28	0.67	3.08	1.00
C-Pressure	1.37	1	0.28	0.17	2.58	1.00
D-Temperature	2.13	1	0.28	0.92	3.33	1.00
AC	-2.13	1	0.28	-3.33	-0.92	1.00
AD	2.13	1	0.28	0.92	3.33	1.00

Final Equation in Terms of Coded Factors:

Yield =
+17.88
+1.87 * A
+1.37 * C
+2.13 * D
-2.13 * A * C
+2.13 * A * D

Final Equation in Terms of Actual Factors:

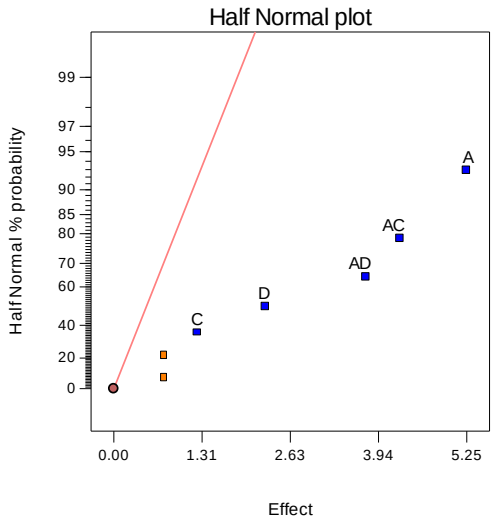
Yield	=	
+227.75000		
-94.50000	*	Time
+2.47500	*	Pressure
-1.70000	*	Temperature
-0.85000	*	Time * Pressure
+0.68000	*	Time * Temperature

8.12. Repeat Problem 8.11 using $I=-ABCD$. Does use of the alternate fraction change your interpretation of the data?

Run Number	*A*	*B*	*C*	*D=ABC*		Yield (lbs)		Factor Low (-)	Levels High (+)
1	–	–	–	+	*d*	10	*A* (h)	2.5	3.0
2	+	–	–	–	*a*	18	*B* (%)	14	18
3	–	+	–	–	*b*	13	*C* (psi)	60	80
4	+	+	–	+	*abd*	24	*D* (°C)	225	250
5	–	–	+	–	*c*	17			
6	+	–	+	+	*acd*	21			
7	–	+	+	+	*bcd*	17			
8	+	+	+	–	*abc*	15			

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	5.25	55.125	40.8712
Error	B	0.75	1.125	0.834106
Model	C	1.25	3.125	2.31696
Model	D	2.25	10.125	7.50695
Error	AB	-0.75	1.125	0.834106
Model	AC	-4.25	36.125	26.7841
Model	AD	3.75	28.125	20.8526
	Lenth's ME	12.7044		
	Lenth's SME	30.404		



Design Expert Output

Response: Yield in lbs

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	132.63	5	26.52	23.58	0.0412	significant
A	*55.13*	*1*	*55.13*	*49.00*	*0.0198*	
C	*3.13*	*1*	*3.13*	*2.78*	*0.2375*	
D	*10.13*	*1*	*10.13*	*9.00*	*0.0955*	
AC	*36.13*	*1*	*36.13*	*32.11*	*0.0298*	
AD	*28.13*	*1*	*28.13*	*25.00*	*0.0377*	
Residual	2.25	2	1.12			
Cor Total	134.88	7				

The Model F-value of 23.58 implies the model is significant. There is only a 4.12% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	1.06	R-Squared	0.9833
Mean	16.88	Adj R-Squared	0.9416
C.V.	6.29	Pred R-Squared	0.7331
PRESS	36.00	Adeq Precision	14.425

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	16.88	1	0.37	15.26	18.49	
A-Time	2.63	1	0.37	1.01	4.24	1.00
C-Pressure	0.63	1	0.37	-0.99	2.24	1.00
D-Temperature	1.13	1	0.37	-0.49	2.74	1.00
AC	-2.13	1	0.37	-3.74	-0.51	1.00
AD	1.88	1	0.37	0.26	3.49	1.00

Final Equation in Terms of Coded Factors:

```
Yield    =
+16.88
 +2.63   * A
 +0.63   * C
 +1.13   * D
 -2.13   * A * C
 +1.88   * A * D
```

Final Equation in Terms of Actual Factors:

```
Yield      =
+190.50000
 -72.50000 * Time
  +2.40000 * Pressure
  -1.56000 * Temperature
  -0.85000 * Time * Pressure
  +0.60000 * Time * Temperature
```

8.13. Project the 2^{4-1}_{IV} design in Example 8.1 into two replicates of a 2^2 design in the factors *A* and *B*. Analyze the data and draw conclusions.

The *Design Expert* output below does not identify a significant effect.

Design Expert Output

Response: Filtration Rate

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	728.50	3	242.83	0.41	0.7523	not significant
A	*722.00*	*1*	*722.00*	*1.23*	*0.3291*	
B	*4.50*	*1*	*4.50*	*7.682E-003*	*0.9344*	
AB	*2.00*	*1*	*2.00*	*3.414E-003*	*0.9562*	
Residual	2343.00	4	585.75			
Lack of Fit	*0.000*	*0*				
Pure Error	*2343.00*	*4*	*585.75*			
Cor Total	3071.50	7				

The "Model F-value" of 0.41 implies the model is not significant relative to the noise. There is a 75.23 % chance that a "Model F-value" this large could occur due to noise.

Std. Dev.	24.20	R-Squared	0.2372
Mean	70.75	Adj R-Squared	-0.3349
C.V.	34.21	Pred R-Squared	-2.0513
PRESS	9372.00	Adeq Precision	1.198

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	70.75	1	8.56	46.99	94.51	
A-Temperature	9.50	1	8.56	-14.26	33.26	1.00
B-Pressure	0.75	1	8.56	-23.01	24.51	1.00
AB	-0.50	1	8.56	-24.26	23.26	1.00

Final Equation in Terms of Coded Factors:

Filtration Rate =
+70.75
+9.50 * A
+0.75 * B
-0.50 * A * B

Final Equation in Terms of Actual Factors:

Filtration Rate =
+70.75000
+9.50000 * Temperature
+0.75000 * Pressure
-0.50000 * Temperature * Pressure

8.14. Construct a 2^{5-2}_{III} design. Determine the effects that may be estimated if a full fold over of this design is performed.

The 2^{5-2}_{III} design is shown below.

A	*B*	*C*	*D=AB*	*E=AC*	
–	–	–	+	+	*de*
+	–	–	–	–	*a*
–	+	–	–	+	*be*
+	+	–	+	–	*abd*
–	–	+	+	–	*cd*
+	–	+	–	+	*ace*
–	+	+	–	–	*bc*
+	+	+	+	+	*abcde*

The design with the fold over included is as follows.

Block	*A*	*B*	*C*	*D=AB*	*E=AC*	
1	–	–	–	+	+	*de*
1	+	–	–	–	–	*a*
1	–	+	–	–	+	*be*
1	+	+	–	+	–	*abd*
1	–	–	+	+	–	*cd*
1	+	–	+	–	+	*ace*
1	–	+	+	–	–	*bc*
1	+	+	+	+	+	*abcde*
2	+	+	+	–	–	*abc*
2	–	+	+	+	+	*bcde*
2	+	–	+	+	–	*acd*
2	–	–	+	–	+	*ce*
2	+	+	–	–	+	*abe*
2	–	+	–	+	–	*bd*
2	+	–	–	+	+	*ade*
2	–	–	–	–	–	(1)

The effects are shown in the table below.

Principal Fraction	Second Fraction
$\ell_A = A+BD+CE$	$\ell^*_A = A-BD-CE$
$\ell_B = B + AD$	$\ell^*_B = B - AD$
$\ell_C = C + AE$	$\ell^*_C = C - AE$
$\ell_D = D + AB$	$\ell^*_D = D - AB$
$\ell_E = E + AC$	$\ell^*_E = E - AC$
$\ell_{BC} = BC + DE$	$\ell^*_{BC} = BC + DE$
$\ell_{BE} = BE + CD$	$\ell^*_{BE} = BE + CD$

By combining the two fractions we can estimate the following:

$(\ell_i + \ell^*_i)/2$	$(\ell_i - \ell^*_i)/2$
A	*BD+CE*
B	*AD*
C	*AE*
D	*AB*
E	*AC*
BC+DE	
BE+CD	

These estimates are confirmed with the Design Expert output shown below.

Design Expert Output

Design Matrix Evaluation for Factorial Reduced 3FI Model

Factorial Effects Aliases

[Est. Terms] Aliased Terms

[Intercept] = Intercept
[Block 1] = Block 1 + ABD + ACE
[Block 2] = Block 2 - ABD - ACE
[A] = A
[B] = B + CDE
[C] = C + BDE
[D] = D + BCE
[E] = E + BCD
[AB] = AB
[AC] = AC
[AD] = AD

[AE] = AE
[BC] = BC + DE
[BD] = BD + CE
[BE] = BE + CD
[ABC] = ABC + ADE
[ABE] = ABE + ACD

8.15. Fold over the 2^{7-4}_{III} design in Table 8.19 to produce an eight-factor design. Verify that the resulting design is a 2^{8-4}_{IV} design. Is this a minimal design?

	H	*A*	*B*	*C*	*D=AB*	*E=AC*	*F=BC*	*G=ABC*
	+	–	–	–	+	+	+	–
Original	+	+	–	–	–	–	+	+
Design	+	–	+	–	–	+	–	+
	+	+	+	–	+	–	–	–
	+	–	–	+	+	–	–	+
	+	+	–	+	–	+	–	–
	+	–	+	+	–	–	+	–
	+	+	+	+	+	+	+	+
	–	+	+	+	–	–	–	+
Second	–	–	+	+	+	+	–	–
Set of	–	+	–	+	+	–	+	–
Runs w/	–	–	–	+	–	+	+	+
all Signs	–	+	+	–	–	+	+	–
Switched	–	–	+	–	+	–	+	+
	–	+	–	–	+	+	–	+
	–	–	–	–	–	–	–	–

After folding the original design over, we add a new factor *H*, and we have a design with generators *D=ABH*, *E=ACH*, *F=BCH*, and *G=ABC*. This is a 2^{8-4}_{IV} design. It is a minimal design, since it contains $2k=2(8)=16$ runs.

8.16. Fold over a 2^{5-2}_{III} design to produce a six-factor design. Verify that the resulting design is a 2^{6-2}_{IV} design. Compare this 2^{6-2}_{IV} design to the in Table 8.10.

	F	*A*	*B*	*C*	*D=AB*	*E=BC*
	+	–	–	–	+	+
Original	+	+	–	–	–	+
Design	+	–	+	–	–	–
	+	+	+	–	+	–
	+	–	–	+	+	–
	+	+	–	+	–	–
	+	–	+	+	–	+
	+	+	+	+	+	+
	–	+	+	+	–	–
Second	–	–	+	+	+	–
Set of	–	+	–	+	+	+
Runs w/	–	–	–	+	–	+
all Signs	–	+	+	–	–	+
Switched	–	–	+	–	+	+
	–	+	–	–	+	–
	–	–	–	–	–	–

If we relabel the factors from left to right as *A, B, C, D, E, F*, then this design becomes 2^{6-2}_{IV} with generators *I=ABDF* and *I=BCEF*. It is not a minimal design, since $2k = 2(6) = 12$ runs, and the design contains 16 runs.

8.17. An industrial engineer is conducting an experiment using a Monte Carlo simulation model of an inventory system. The independent variables in her model are the order quantity (*A*), the reorder point (B), the setup cost (*C*), the backorder cost (*D*), and the carrying cost rate (*E*). The response variable is average annual cost. To conserve computer time, she decides to investigate these factors using a 2^{5-2}_{III} design with *I=ABD* and *I=BCE*. The results she obtains are *de* = 95, *ae* = 134, *b* = 158, *abd* = 190, *cd* = 92, *ac* = 187, *bce* = 155, and *abcde* = 185.

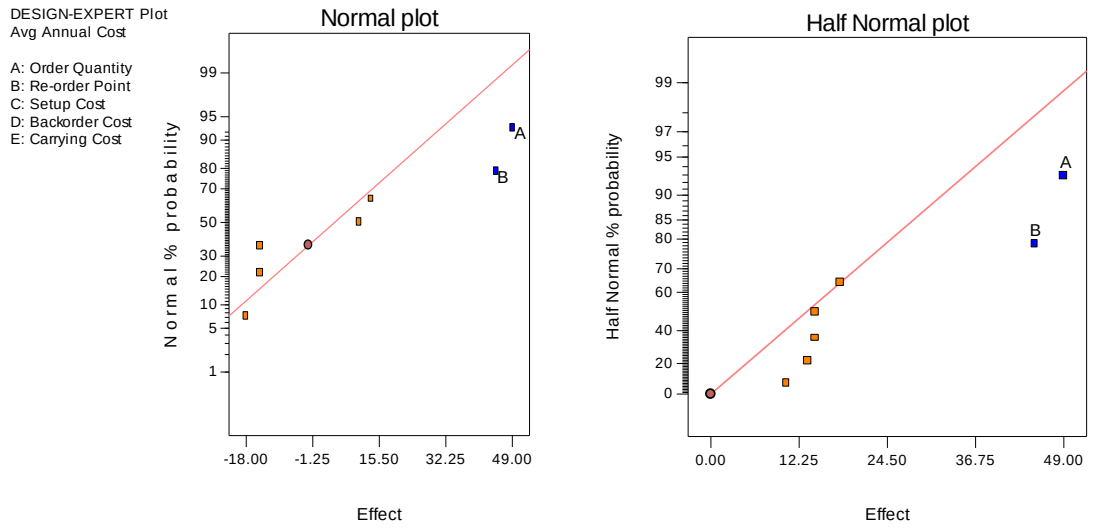
(a) Verify that the treatment combinations given are correct. Estimate the effects, assuming three-factor and higher interactions are negligible.

The treatment combinations given are correct. The effects are shown in the *Design Expert* output and the normal and half normal probability plot of effects identify factors *A* and *B* as important.

A	*B*	*C*	*D=AB*	*E=BC*	
–	–	–	+	+	*de*
+	–	–	–	+	*ae*
–	+	–	–	–	*b*
+	+	–	+	–	*abd*
–	–	+	+	–	*cd*
+	–	+	–	–	*ac*
–	+	+	–	+	*bce*
+	+	+	+	+	*abcde*

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	49	4802	43.9502
Model	B	45	4050	37.0675
Error	C	10.5	220.5	2.01812
Error	D	-18	648	5.93081
Error	E	-14.5	420.5	3.84862
Error	AC	13.5	364.5	3.33608
Error	AE	-14.5	420.5	3.84862
	Lenth's ME	81.8727		
	Lenth's SME	195.937		



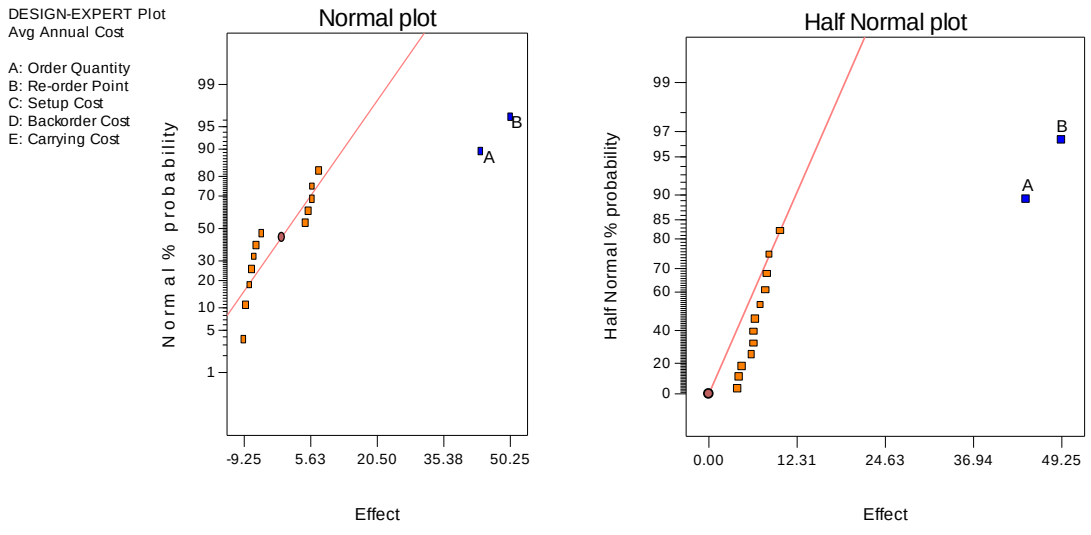
(b) Suppose that a second fraction is added to the first, for example *ade* = 136, *e* = 93, *ab* = 187, *bd* = 153, *acd* = 139, *c* = 99, *abce* = 191, and *bcde* = 150. How was this second fraction obtained? Add this data to the original fraction, and estimate the effects.

This second fraction is formed by reversing the signs of factor *A*.

A	*B*	*C*	*D=AB*	*E=BC*	
+	–	–	+	+	*ade*
–	–	–	–	+	*e*
+	+	–	–	–	*ab*
–	+	–	+	–	*bd*
+	–	+	+	–	*acd*
–	–	+	–	–	*c*
+	+	+	–	+	*abce*
–	+	+	+	+	*bcde*

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	44.25	7832.25	39.5289
Model	B	49.25	9702.25	48.9666
Error	C	6.5	169	0.852932
Error	D	-8	256	1.29202
Error	E	-8.25	272.25	1.37403
Error	AB	-10	400	2.01877
Error	AC	7.25	210.25	1.06112
Error	AD	-4.25	72.25	0.364641
Error	AE	-6	144	0.726759
Error	BD	4.75	90.25	0.455486
Error	CD	-8.5	289	1.45856
Error	DE	6.25	156.25	0.788584
Error	ACD	-6.25	156.25	0.788584
Error	ADE	4	64	0.323004
	Lenth's ME	25.1188		
	Lenth's SME	51.5273		



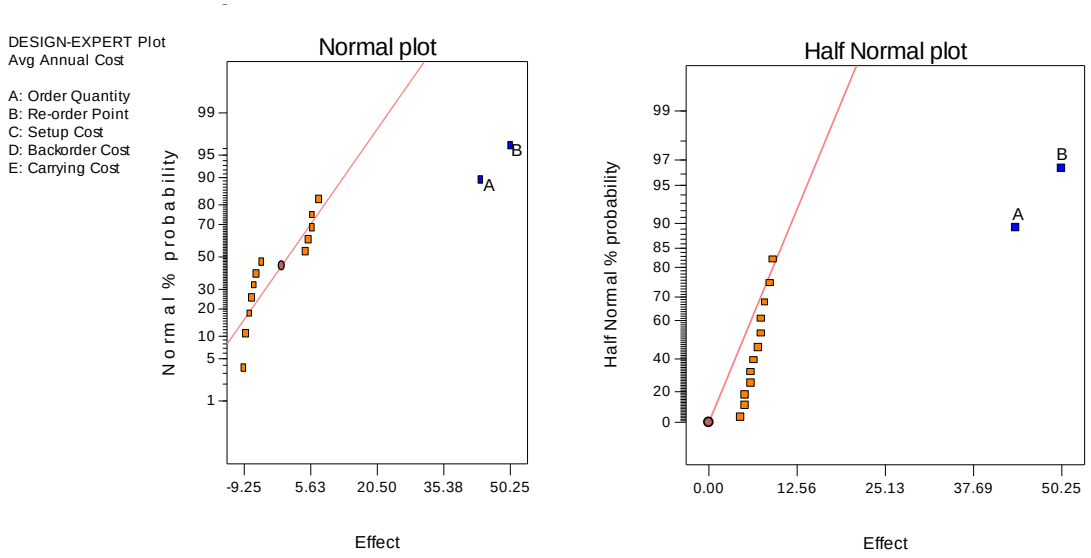
(c) Suppose that the fraction *abc* = 189, *ce* = 96, *bcd* = 154, *acde* = 135, *abe* = 193, *bde* = 152, *ad* = 137, and (1) = 98 was run. How was this fraction obtained? Add this data to the original fraction and estimate the effects.

This second fraction is formed by reversing the signs of all factors.

A	*B*	*C*	*D=AB*	*E=BC*	
+	+	+	-	-	*abc*
-	+	+	+	-	*bcd*
+	-	+	+	+	*acde*
-	-	+	-	+	*ce*
+	+	-	-	+	*abe*
-	+	-	+	+	*bde*
+	-	-	+	-	*ad*
-	-	-	-	-	(1)

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	43.75	7656.25	38.1563
Model	B	50.25	10100.3	50.3364
Error	C	4.5	81	0.403678
Error	D	-8.75	306.25	1.52625
Error	E	-7.5	225	1.12133
Error	AB	-9.25	342.25	1.70566
Error	AC	6	144	0.71765
Error	AD	-5.25	110.25	0.549451
Error	AE	-6.5	169	0.842242
Error	BC	-7	196	0.976801
Error	BD	5.25	110.25	0.549451
Error	BE	6	144	0.71765
Error	ABC	-8	256	1.27582
Error	ABE	7.5	225	1.12133
	Lenth's ME	26.5964		
	Lenth's SME	54.5583		



8.18. Construct a 2^{5-1} design. Show how the design may be run in two blocks of eight observations each. Are any main effects or two-factor interactions confounded with blocks?

A	*B*	*C*	*D*	*E=ABCD*		Blocks = *AB*	Block
–	–	–	–	+	*e*	+	1

+	–	–	–	–	*a*	–	2
–	+	–	–	–	*b*	–	2
+	+	–	–	+	*abe*	+	1
–	–	+	–	–	*c*	+	1
+	–	+	–	+	*ace*	–	2
–	+	+	–	+	*bce*	–	2
+	+	+	–	–	*abc*	+	1
–	–	–	+	–	*d*	+	1
+	–	–	+	+	*ade*	–	2
–	+	–	+	+	*bde*	–	2
+	+	–	+	–	*abd*	+	1
–	–	+	+	+	*cde*	+	1
+	–	+	+	–	*acd*	–	2
–	+	+	+	–	*bcd*	–	2
+	+	+	+	+	*abcde*	+	1

Blocks are confounded with *AB* and *CDE*.

8.19. Construct a 2^{7-2} design. Show how the design may be run in four blocks of eight observations each. Are any main effects or two-factor interactions confounded with blocks?

	A	*B*	*C*	*D*	*E*	*F=CDE*	*G=ABC*		Block=ACE	Block=BFG	Block assignment
1	–	–	–	–	–	–	–	(1)	–	–	1
2	+	–	–	–	–	–	+	*ag*	+	+	4
3	–	+	–	–	–	–	+	*bg*	–	–	1
4	+	+	–	–	–	–	–	*ab*	+	+	4
5	–	–	+	–	–	+	+	*cfg*	+	–	3
6	+	–	+	–	–	+	–	*acf*	–	+	2
7	–	+	+	–	–	+	–	*bcf*	+	–	3
8	+	+	+	–	–	+	+	*abcfg*	–	+	2
9	–	–	–	+	–	+	–	*df*	–	+	2
10	+	–	–	+	–	+	+	*adfg*	+	–	3
11	–	+	–	+	–	+	+	*bdfg*	–	+	2
12	+	+	–	+	–	+	–	*abdf*	+	–	3
13	–	–	+	+	–	–	+	*cdg*	+	+	4
14	+	–	+	+	–	–	–	*acd*	–	–	1
15	–	+	+	+	–	–	–	*bcd*	+	+	4
16	+	+	+	+	–	–	+	*abcdg*	–	–	1
17	–	–	–	–	+	+	–	*ef*	+	+	4
18	+	–	–	–	+	+	+	*aefg*	–	–	1
19	–	+	–	–	+	+	+	*befg*	+	+	4
20	+	+	–	–	+	+	–	*abef*	–	–	1
21	–	–	+	–	+	–	+	*ceg*	–	+	2
22	+	–	+	–	+	–	–	*ace*	+	–	3
23	–	+	+	–	+	–	–	*bce*	–	+	2
24	+	+	+	–	+	–	+	*abceg*	+	–	3
25	–	–	–	+	+	–	–	*de*	+	–	3
26	+	–	–	+	+	–	+	*adeg*	–	+	2
27	–	+	–	+	+	–	+	*bdeg*	+	–	3
28	+	+	–	+	+	–	–	*abde*	–	+	2
29	–	–	+	+	+	+	+	*cdefg*	–	–	1
30	+	–	+	+	+	+	–	*acdef*	+	+	4
31	–	+	+	+	+	+	–	*bcdef*	–	–	1
32	+	+	+	+	+	+	+	*abcdefg*	+	+	4

Blocks are confounded with *ACE*, *BFG*, and *ABCEFG*.

8.20. ***Nonregular fractions of the 2^k*** **[John (1971)].** Consider a 2^4 design. We must estimate the four main effects and the six two-factor interactions, but the full 2^4 factorial cannot be run. The largest possible block contains 12 runs. These 12 runs can be obtained from the four one-quarter fractions defined by $I = \pm AB = \pm ACD = \pm BCD$ by omitting the principal fraction. Show how the remaining three 2^{4-2} fractions can be combined to estimate the required effects, assuming that three-factor and higher interactions are negligible. This design could be thought of as a three-quarter fraction.

The four 2^{4-2} fractions are as follows:

(1) $I = +AB = +ACD = +BCD$
Runs: *c,d,ab,abcd*

(2) $I = +AB = -ACD = -BCD$
Runs: (1), *cd*, *abc*, *abd*

(3) $I = -AB = +ACD = -BCD$
Runs: *a*, *bc*, *bd*, *acd*

(4) $I = -AB = -ACD = +BCD$
Runs: *b*, *ac*, *ad*, *bcd*

If we do not run the principal fraction (1), then we can combine the remaining 3 fractions to 3 one-half fractions of the 2^4 as follows:

Fraction 1: (2) + (3) implies $I = -BCD$. This fraction estimates: *A*, *AB*, *AC*, and *AD*
Fraction 2: (2) + (4) implies $I = -ACD$. This fraction estimates: *B*, *BC*, *BD*, and *AB*
Fraction 3: (3) + (4) implies $I = -AB$. This fraction estimates: *C*, *D*, and *CD*

In estimating these effects we assume that all three-factor and higher interactions are negligible. Note that *AB* is estimated in two of the one-half fractions: 1 and 2. We would average these quantities and obtain a single estimate of *AB*. John (1971, pp. 161-163) discusses this design and shows that the estimates obtained above are also the least squares estimates. John also derives the variances and covariances of these estimators.

8.21S. Carbon anodes used in a smelting process are baked in a ring furnace. An experiment is run in the furnace to determine which factors influence the weight of packing material that is stuck to the anodes after baking. Six variables are of interest, each at two levels: *A* = pitch/fines ratio (0.45, 0.55); *B* = packing material type (1, 2); *C* = packing material temperature (ambient, 325 C); *D* = flue location (inside, outside); *E* = pit temperature (ambient, 195 C); and *F* = delay time before packing (zero, 24 hours). A 2^{6-3} design is run, and three replicates are obtained at each of the design points. The weight of packing material stuck to the anodes is measured in grams. The data in run order are as follows: *abd* = (984, 826, 936); *abcdef* = (1275, 976, 1457); *be* = (1217, 1201, 890); *af* = (1474, 1164, 1541); *def* = (1320, 1156, 913); *cd* = (765, 705, 821); *ace* = (1338, 1254, 1294); and *bcf* = (1325, 1299, 1253). We wish to minimize the amount stuck packing material.

(a) Verify that the eight runs correspond to a 2^{6-3}_{III} design. What is the alias structure?

A	*B*	*C*	*D=AB*	*E=AC*	*F=BC*	
–	–	–	+	+	+	*def*
+	–	–	–	–	+	*af*
–	+	–	–	+	–	*be*
+	+	–	+	–	–	*abd*
–	–	+	+	–	–	*cd*
+	–	+	–	+	–	*ace*
–	+	+	–	–	+	*bcf*

+	+	+	+	+	+	*abcdef*

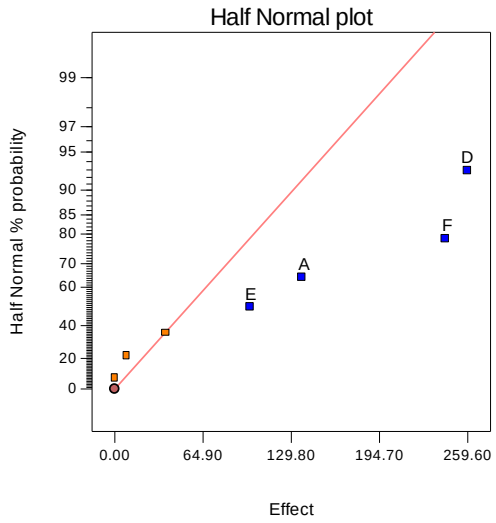
I=ABD=ACE=BCF=BCDE=ACDF=ABEF=DEF, Resolution III

A=BD=CE=CDF=BEF
B=AD=CF=CDE=AEF
C=AE=BF=BDE=ADF
D=AB=EF=BCE=ACF
E=AC=DF=BCD=ABF
F=BC=DE=ACD=ABE
CD=BE=AF=ABC=ADE=BDF=CEF

(b) Use the average weight as a response. What factors appear to be influential?

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	A	137.833	37996.1	12.0947
Error	B	-8.83333	156.056	0.049675
Error	C	11.6667	272.222	0.0866527
Model	D	-259.667	134854	42.926
Model	E	99.8333	19933.4	6.34511
Model	F	243.5	118585	37.7473
Error	AF	-34.3333	2357.56	0.750447
	Lenth's ME	563.698		
	Lenth's SME	1349.04		



Factors *A, D, E* and *F* (and their aliases) are apparently important.

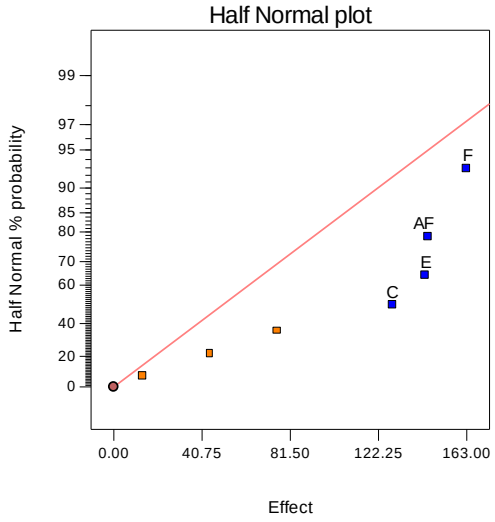
(c) Use the range of the weights as a response. What factors appear to be influential?

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Error	A	44.5	3960.5	2.13311
Error	B	13.5	364.5	0.196319
Model	C	-129	33282	17.9256
Error	D	75.5	11400.5	6.14028
Model	E	144	41472	22.3367
Model	F	163	53138	28.62
Model	AF	145	42050	22.648
	Lenth's ME	728.384		

Lenth's SME 1743.17

Factors *C, E, F* and the *AF* interaction (and their aliases) appear to be large.



(d) What recommendations would you make to the process engineers?

It is not known exactly what to do here, since *A, D, E* and *F* are large effects, and because the design is resolution III, the main effects are aliased with two-factor interactions. Note, for example, that *D* is aliased with *EF* and the main effect could really be an *EF* interaction. If the main effects are really important, then setting all factors at the low level would minimize the amount of material stuck to the anodes. It would be necessary to run additional experiments to confirm these findings.

8.22S. A 16-run experiment was performed in a semiconductor manufacturing plant to study the effects of six factors on the curvature or camber of the substrate devices produced. The six variables and their levels are shown in Table P8.2.

Table P8.2

Run	Lamination Temperature (c)	Lamination Time (s)	Lamination Pressure (tn)	Firing Temperature (c)	Firing Cycle Time (h)	Firing Dew Point (c)
1	55	10	5	1580	17.5	20
2	75	10	5	1580	29	26
3	55	25	5	1580	29	20
4	75	25	5	1580	17.5	26
5	55	10	10	1580	29	26
6	75	10	10	1580	17.5	20
7	55	25	10	1580	17.5	26
8	75	25	10	1580	29	20
9	55	10	5	1620	17.5	26
10	75	10	5	1620	29	20
11	55	25	5	1620	29	26
12	75	25	5	1620	17.5	20
13	55	10	10	1620	29	20
14	75	10	10	1620	17.5	26
15	55	25	10	1620	17.5	20
16	75	25	10	1620	29	26

Each run was replicated four times, and a camber measurement was taken on the substrate. The data are shown in Table P8.3.

Table P8.3

	Camber	for	Replicate	(in/in)	Total	Mean	Standard
Run	1	2	3	4	(10^{-4} in/in)	(10^{-4} in/in)	Deviation
1	0.0167	0.0128	0.0149	0.0185	629	157.25	24.418
2	0.0062	0.0066	0.0044	0.0020	192	48.00	20.976
3	0.0041	0.0043	0.0042	0.0050	176	44.00	4.083
4	0.0073	0.0081	0.0039	0.0030	223	55.75	25.025
5	0.0047	0.0047	0.0040	0.0089	223	55.75	22.410
6	0.0219	0.0258	0.0147	0.0296	920	230.00	63.639
7	0.0121	0.0090	0.0092	0.0086	389	97.25	16.029
8	0.0255	0.0250	0.0226	0.0169	900	225.00	39.420
9	0.0032	0.0023	0.0077	0.0069	201	50.25	26.725
10	0.0078	0.0158	0.0060	0.0045	341	85.25	50.341
11	0.0043	0.0027	0.0028	0.0028	126	31.50	7.681
12	0.0186	0.0137	0.0158	0.0159	640	160.00	20.083
13	0.0110	0.0086	0.0101	0.0158	455	113.75	31.120
14	0.0065	0.0109	0.0126	0.0071	371	92.75	29.510
15	0.0155	0.0158	0.0145	0.0145	603	150.75	6.750
16	0.0093	0.0124	0.0110	0.0133	460	115.00	17.450

(a) What type of design did the experimenters use?

The 2_{IV}^{6-2}, a 16-run design.

(b) What are the alias relationships in this design?

The defining relation is *I=ABCE=ACDF=BDEF*. The aliases are shown below.

A(ABCE)=	*BCE*	*A(ACDF)=*	*CDF*	*A(BDEF)=*	*ABDEF*	*A=BCE=CDF=ABDEF*
B(ABCE)=	*ACE*	*B(ACDF)=*	*ABCDF*	*B(BDEF)=*	*DEF*	*B=ACE=ABCDF=DEF*
C(ABCE)=	*ABE*	*C(ACDF)=*	*ADF*	*C(BDEF)=*	*BCDEF*	*C=ABE=ADF=BCDEF*
D(ABCE)=	*ABCDE*	*D(ACDF)=*	*ACF*	*D(BDEF)=*	*BEF*	*D=ABCDE=ACF=BEF*
E(ABCE)=	*ABC*	*E(ACDF)=*	*ACDEF*	*E(BDEF)=*	*BDF*	*E=ABC=ACDEF=BDF*
F(ABCE)=	*ABCEF*	*F(ACDF)=*	*ACD*	*F(BDEF)=*	*BDE*	*F=ABCEF=ACD=BDE*
AB(ABCE)=	*CE*	*AB(ACDF)=*	*BCDF*	*AB(BDEF)=*	*ADEF*	*AB=CE=BCDF=ADEF*
AC(ABCE)=	*BE*	*AC(ACDF)=*	*DF*	*AC(BDEF)=*	*ABCDEF*	*AC=BE=DF=ABCDEF*
AD(ABCE)=	*BCDE*	*AD(ACDF)=*	*CF*	*AD(BDEF)=*	*ABEF*	*AD=BCDE=CF=ABEF*
AE(ABCE)=	*BC*	*AE(ACDF)=*	*CDEF*	*AE(BDEF)=*	*ABDF*	*AE=BC=CDEF=ABDF*
AF(ABCE)=	*BCEF*	*AF(ACDF)=*	*CD*	*AF(BDEF)=*	*ABDE*	*AF=BCEF=CD=ABDE*
BD(ABCE)=	*ACDE*	*BD(ACDF)=*	*ABCF*	*BD(BDEF)=*	*EF*	*BD=ACDE=ABCF=EF*
BF(ABCE)=	*ACEF*	*BF(ACDF)=*	*ABCD*	*BF(BDEF)=*	*DE*	*BF=ACEF=ABCD=DE*

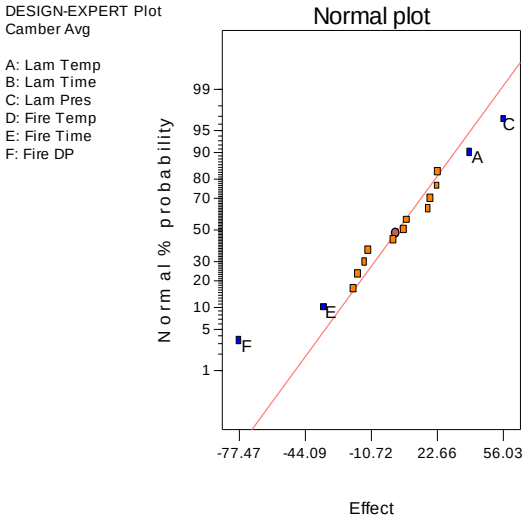
(c) Do any of the process variables affect average camber?

Yes, per the analysis below, variables *A*, *C*, *E*, and *F* affect average camber.

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	38.9063	6054.79	10.2962
Error	B	5.78125	133.691	0.227344
Model	C	56.0313	12558	21.355
Error	D	-14.2188	808.691	1.37519
Model	E	-34.4687	4752.38	8.08148
Model	F	-77.4688	24005.6	40.8219
Error	AB	19.1563	1467.85	2.49609
Error	AC	22.4063	2008.16	3.4149

Error	AD	-12.2188	597.191	1.01553
Error	AE	18.1563	1318.6	2.24229
Error	AF	-19.7187	1555.32	2.64483
Error	BC	Aliased		
Error	BD	23.0313	2121.75	3.60807
Error	BE	Aliased		
Error	BF	7.40625	219.41	0.37311
Error	CD	Aliased		
Error	CE	Aliased		
Error	CF	Aliased		
Error	DE	Aliased		
Error	DF	Aliased		
Error	EF	Aliased		
Error	ABC	Aliased		
Error	ABD	0.53125	1.12891	0.00191972
Error	ABE	Aliased		
Error	ABF	-17.3438	1203.22	2.04609
	Lenth's ME	71.9361		
	Lenth's SME	146.041		



Design Expert Output

Response: Camber Avg in in/in

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	47370.80	4	11842.70	11.39	0.0007	significant
A	*6054.79*	*1*	*6054.79*	*5.82*	*0.0344*	
C	*12558.00*	*1*	*12558.00*	*12.08*	*0.0052*	
E	*4752.38*	*1*	*4752.38*	*4.57*	*0.0558*	
F	*24005.63*	*1*	*24005.63*	*23.09*	*0.0005*	
Residual	11435.01	11	1039.55			
Cor Total	58805.81	15				

The Model F-value of 11.39 implies the model is significant. There is only a 0.07% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	32.24	R-Squared	0.8055
Mean	107.02	Adj R-Squared	0.7348
C.V.	30.13	Pred R-Squared	0.5886
PRESS	24193.08	Adeq Precision	11.478

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	107.02	1	8.06	89.27	124.76	

A-Lam Temp	19.45	1	8.06	1.71	37.19	1.00
C-Lam Pres	28.02	1	8.06	10.27	45.76	1.00
E-Fire Time	-17.23	1	8.06	-34.98	0.51	1.00
F-Fire DP	-38.73	1	8.06	-56.48	-20.99	1.00

Final Equation in Terms of Coded Factors:

Camber Avg =
+107.02
+19.45 * A
+28.02 * C
-17.23 * E
-38.73 * F

Final Equation in Terms of Actual Factors:

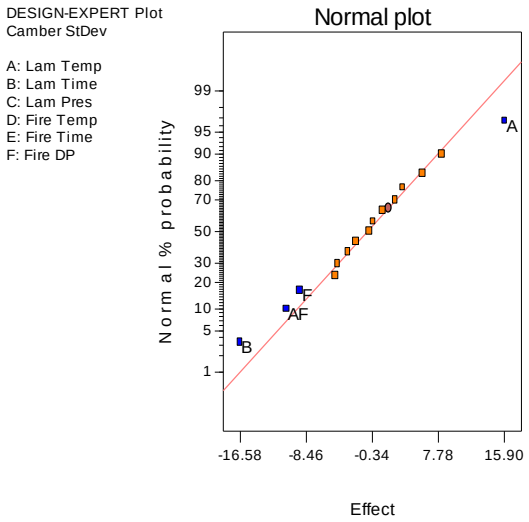
Camber Avg =
+263.17380
+1.94531 * Lam Temp
+11.20625 * Lam Pres
-2.99728 * Fire Time
-12.91146 * Fire DP

(d) Do any of the process variables affect the variability in camber measurements?

Yes, *A*, *B*, *F*, and *AF* interaction affect the variability in camber measurements.

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	15.9035	1011.69	27.6623
Model	B	-16.5773	1099.22	30.0558
Error	C	5.8745	138.039	3.77437
Error	D	-3.2925	43.3622	1.18564
Error	E	-2.33725	21.851	0.597466
Model	F	-9.256	342.694	9.37021
Error	AB	0.95525	3.65001	0.0998014
Error	AC	2.524	25.4823	0.696757
Error	AD	-4.6265	85.618	2.34103
Error	AE	-0.18025	0.12996	0.00355347
Model	AF	-10.8745	473.019	12.9337
Error	BC	Aliased		
Error	BD	-4.85575	94.3132	2.57879
Error	BE	Aliased		
Error	BF	8.21825	270.159	7.38689
Error	CD	Aliased		
Error	CE	Aliased		
Error	CF	Aliased		
Error	DE	Aliased		
Error	DF	Aliased		
Error	EF	Aliased		
Error	ABC	Aliased		
Error	ABD	-0.68125	1.85641	0.0507593
Error	ABE	Aliased		
Error	ABF	3.39825	46.1924	1.26303
	Lenth's ME	17.8392		
	Lenth's SME	36.2162		



Response: Camber StDev

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	2926.62	4	731.65	11.02	0.0008	significant
A	*1011.69*	*1*	*1011.69*	*15.23*	*0.0025*	
B	*1099.22*	*1*	*1099.22*	*16.55*	*0.0019*	
F	*342.69*	*1*	*342.69*	*5.16*	*0.0442*	
AF	*473.02*	*1*	*473.02*	*7.12*	*0.0218*	
Residual	730.65	11	66.42			
Cor Total	3657.27	15				

The Model F-value of 11.02 implies the model is significant. There is only a 0.08% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	8.15	R-Squared	0.8002
Mean	25.35	Adj R-Squared	0.7276
C.V.	32.15	Pred R-Squared	0.5773
PRESS	1545.84	Adeq Precision	9.516

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	25.35	1	2.04	20.87	29.84	
A-Lam Temp	7.95	1	2.04	3.47	12.44	1.00
B-Lam Time	-8.29	1	2.04	-12.77	-3.80	1.00
F-Fire DP	-4.63	1	2.04	-9.11	-0.14	1.00
AF	-5.44	1	2.04	-9.92	-0.95	1.00

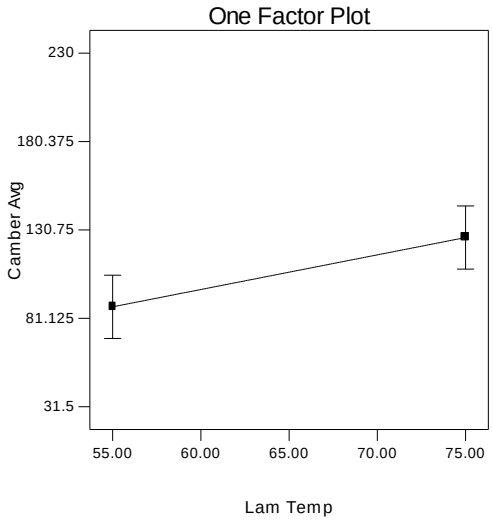
Final Equation in Terms of Coded Factors:

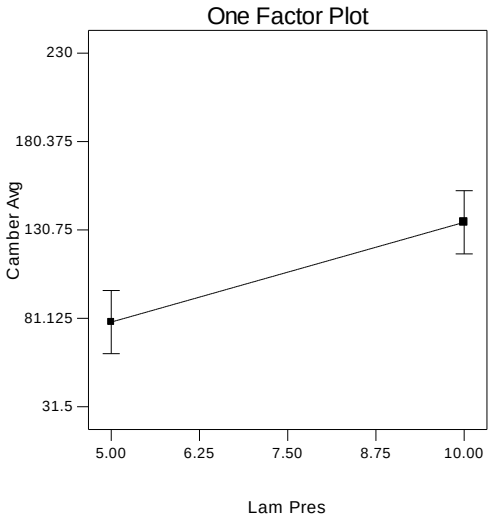
Camber StDev =
+25.35
+7.95 * A
-8.29 * B
-4.63 * F
-5.44 * A * F

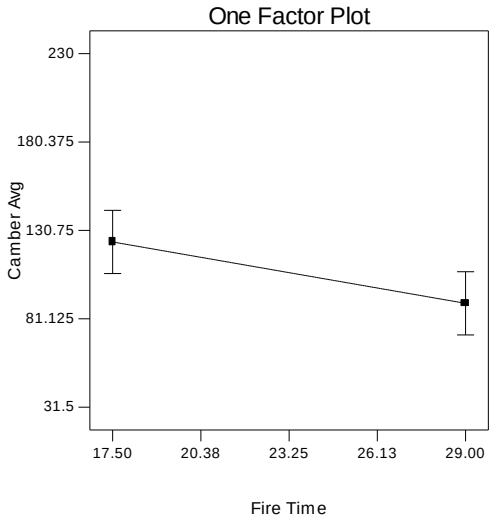
Final Equation in Terms of Actual Factors:

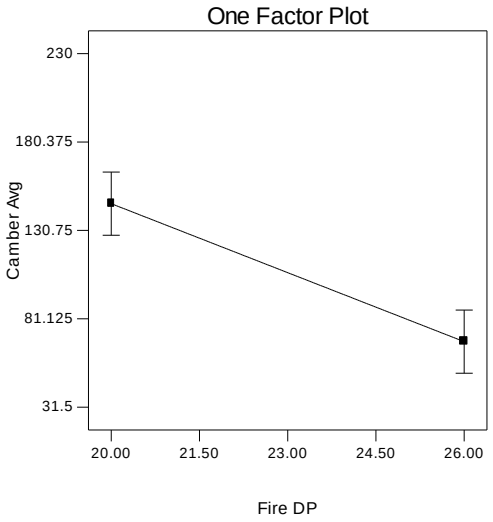
Camber StDev =
-242.46746
+4.96373 * Lam Temp
-1.10515 * Lam Time
+10.23804 * Fire DP
-0.18124 * Lam Temp * Fire DP

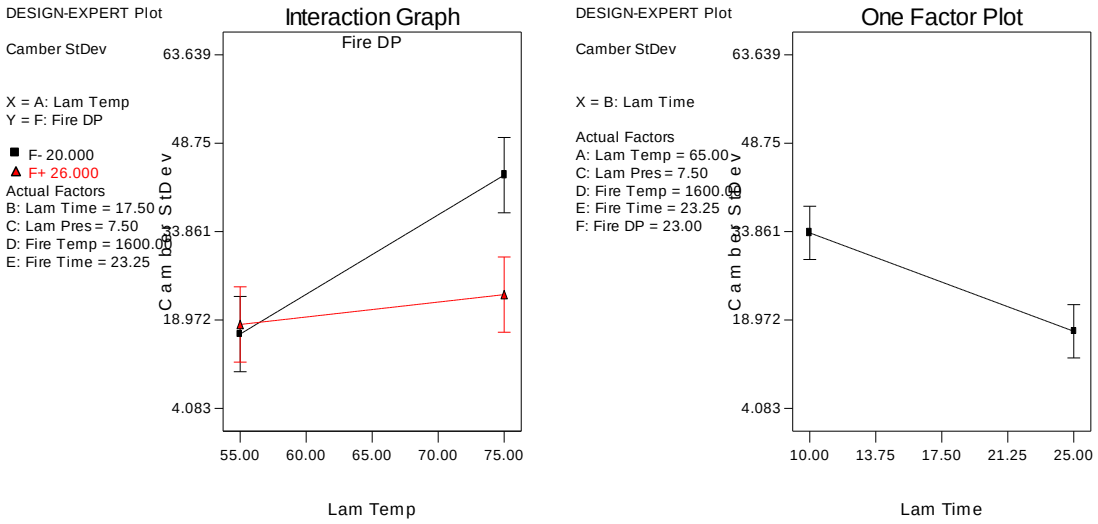
(e) If it is important to reduce camber as much as possible, what recommendations would you make?











Run *A* and *C* at the low level and *E* and *F* at the high level. *B* at the low level enables a lower variation without affecting the average camber.

8.23. A spin coater is used to apply photoresist to a bare silicon wafer. This operation usually occurs early in the semiconductor manufacturing process, and the average coating thickness and the variability in the coating thickness has an important impact on downstream manufacturing steps. Six variables are used in the experiment. The variables and their high and low levels are as follows:

Factor	Low Level	High Level
Final Spin Speed	7350 rpm	6650 rpm
Acceleration Rate	5	20
Volume of Resist Applied	3 cc	5 cc
Time of Spin	14 s	6 s
Resist Batch Variation	Batch 1	Batch 2
Exhaust Pressure	Cover Off	Cover On

The experimenter decides to use a 2^{6-1} design and to make three readings on resist thickness on each test wafer. The data are shown in table P8.4.

Table P8.4

	A	*B*	*C*	*D*	*E*	*F*	Resist Thickness				
Run	Volume	Batch	Time	Speed	Acc.	Cover	Left	Center	Right	Avg.	Range
1	5	2	14	7350	5	Off	4531	4531	4515	4525.7	16
2	5	1	6	7350	5	Off	4446	4464	4428	4446	36
3	3	1	6	6650	5	Off	4452	4490	4452	4464.7	38
4	3	2	14	7350	20	Off	4316	4328	4308	4317.3	20
5	3	1	14	7350	5	Off	4307	4295	4289	4297	18
6	5	1	6	6650	20	Off	4470	4492	4495	4485.7	25
7	3	1	6	7350	5	On	4496	4502	4482	4493.3	20
8	5	2	14	6650	20	Off	4542	4547	4538	4542.3	9
9	5	1	14	6650	5	Off	4621	4643	4613	4625.7	30
10	3	1	14	6650	5	On	4653	4670	4645	4656	25
11	3	2	14	6650	20	On	4480	4486	4470	4478.7	16
12	3	1	6	7350	20	Off	4221	4233	4217	4223.7	16
13	5	1	6	6650	5	On	4620	4641	4619	4626.7	22
14	3	1	6	6650	20	On	4455	4480	4466	4467	25
15	5	2	14	7350	20	On	4255	4288	4243	4262	45
16	5	2	6	7350	5	On	4490	4534	4523	4515.7	44

17	3	2	14	7350	5	On	4514	4551	4540	4535	37
18	3	1	14	6650	20	Off	4494	4503	4496	4497.7	9
19	5	2	6	7350	20	Off	4293	4306	4302	4300.3	13
20	3	2	6	7350	5	Off	4534	4545	4512	4530.3	33
21	5	1	14	6650	20	On	4460	4457	4436	4451	24
22	3	2	6	6650	5	On	4650	4688	4656	4664.7	38
23	5	1	14	7350	20	Off	4231	4244	4230	4235	14
24	3	2	6	7350	20	On	4225	4228	4208	4220.3	20
25	5	1	14	7350	5	On	4381	4391	4376	4382.7	15
26	3	2	6	6650	20	Off	4533	4521	4511	4521.7	22
27	3	1	14	7350	20	On	4194	4230	4172	4198.7	58
28	5	2	6	6650	5	Off	4666	4695	4672	4677.7	29
29	5	1	6	7350	20	On	4180	4213	4197	4196.7	33
30	5	2	6	6650	20	On	4465	4496	4463	4474.7	33
31	5	2	14	6650	5	On	4653	4685	4665	4667.7	32
32	3	2	14	6650	5	Off	4683	4712	4677	4690.7	35

(a) Verify that this is a 2^{6-1} design. Discuss the alias relationships in this design.

I=ABCDEF. This is a resolution VI design where main effects are aliased with five-factor interactions and two-factor interactions are aliased with four-factor interactions.

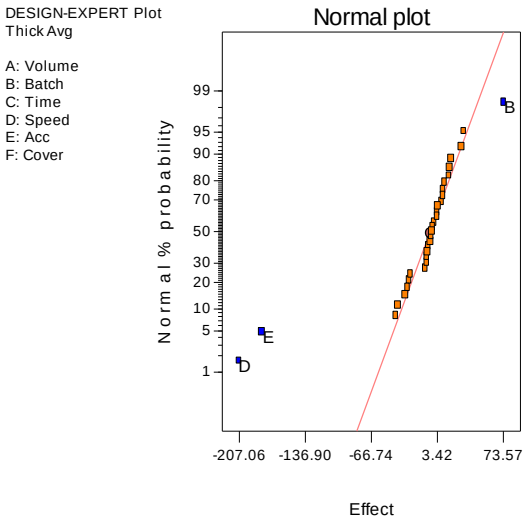
(b) What factors appear to affect average resist thickness?

Factors *B*, *D*, and *E* appear to affect the average resist thickness.

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Error	A	9.925	788.045	0.107795
Model	B	73.575	43306.2	5.92378
Error	C	3.375	91.125	0.0124648
Model	D	-207.062	342999	46.9182
Model	E	-182.925	267692	36.6172
Error	F	-5.6625	256.511	0.0350877
Error	AB	-9	648	0.0886387
Error	AC	-7.3	426.32	0.0583155
Error	AD	-3.8625	119.351	0.0163258
Error	AE	-7.1	403.28	0.0551639
Error	AF	-26.9875	5826.6	0.79701
Error	BC	10.875	946.125	0.129419
Error	BD	18.1125	2624.5	0.359001
Error	BE	-28.35	6429.78	0.879518
Error	BF	-30.2375	7314.45	1.00053
Error	CD	-24.9875	4995	0.683257
Error	CE	8.2	537.92	0.0735811
Error	CF	-6.7875	368.561	0.0504148
Error	DE	-38.5375	11881.1	1.6252
Error	DF	-3.2	81.92	0.0112057
Error	EF	-41.1625	13554.8	1.85414
Error	ABC	0.375	1.125	0.000153887
Error	ABD	Aliased		
Error	ABE	16.5	2178	0.297925
Error	ABF	31.4125	7893.96	1.0798
Error	ACD	15.5875	1943.76	0.265883
Error	ACE	Aliased		
Error	ACF	Aliased		
Error	ADE	9.5375	727.711	0.0995423
Error	ADF	Aliased		
Error	AEF	Aliased		
Error	BCD	29.0875	6768.66	0.925873
Error	BCE	-1.625	21.125	0.00288965
Error	BCF	Aliased		
Error	BDE	-1.8875	28.5013	0.00389863
Error	BDF	3.95	124.82	0.0170739
Error	BEF	Aliased		
Error	CDE	Aliased		

Error	CDF	Aliased		
Error	CEF	3.1375	78.7512	0.0107722
Error	DEF	Aliased		
	Lenth's ME	28.6178		
	Lenth's SME	54.4118		



Design Expert Output

Response: Thick Avg

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	6.540E+005	3	2.180E+005	79.21	< 0.0001	significant
B	*43306.24*	*1*	*43306.24*	*15.74*	*0.0005*	
D	*3.430E+005*	*1*	*3.430E+005*	*124.63*	*< 0.0001*	
E	*2.677E+005*	*1*	*2.677E+005*	*97.27*	*< 0.0001*	
Residual	77059.83	28	2752.14			
Cor Total	7.311E+005	31				

The Model F-value of 79.21 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	52.46	R-Squared	0.8946
Mean	4458.51	Adj R-Squared	0.8833
C.V.	1.18	Pred R-Squared	0.8623
PRESS	1.006E+005	Adeq Precision	24.993

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	4458.51	1	9.27	4439.52	4477.51	
B-Batch	36.79	1	9.27	17.79	55.78	1.00
D-Speed	-103.53	1	9.27	-122.53	-84.53	1.00
E-Acc	-91.46	1	9.27	-110.46	-72.47	1.00

Final Equation in Terms of Coded Factors:

Thick Avg =
+4458.51
+36.79 * B
-103.53 * D
-91.46 * E

Final Equation in Terms of Actual Factors:

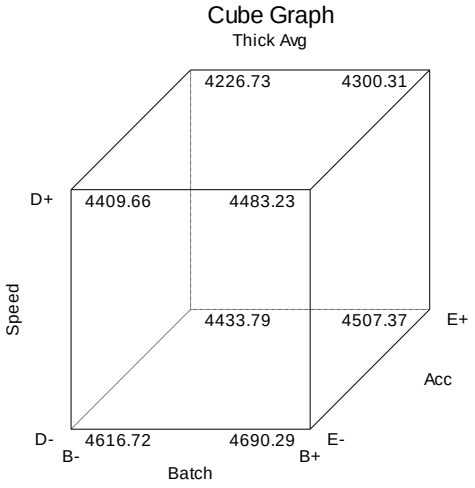
```
Batch       Batch 1
Thick Avg   =
+6644.78750
-0.29580    * Speed
-12.19500   * Acc

Batch       Batch 2
Thick Avg   =
+6718.36250
-0.29580    * Speed
-12.19500   * Acc
```

(c) Since the volume of resist applied has little effect on average thickness, does this have any important practical implications for the process engineers?

Yes, less material could be used.

(d) Project this design into a smaller design involving only the significant factors. Graphically display the results. Does this aid in interpretation?



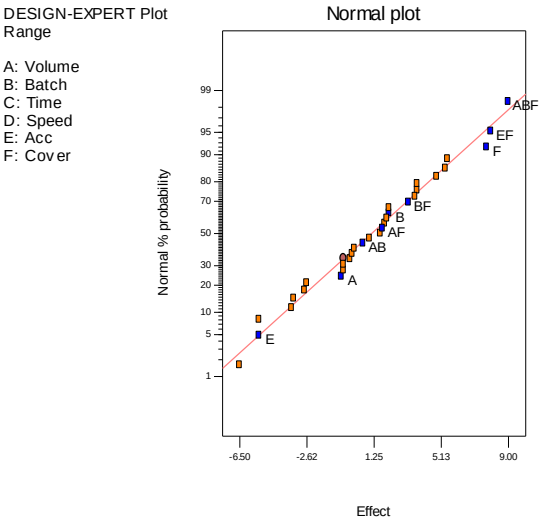
The cube plot usually assists the experimenter in drawing conclusions.

(e) Use the range of resist thickness as a response variable. Is there any indication that any of these factors affect the variability in resist thickness?

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	-0.625	3.125	0.0777387
Model	B	2.125	36.125	0.89866
Error	C	-2.75	60.5	1.50502
Error	D	1.625	21.125	0.525514
Model	E	-5.375	231.125	5.74956
Model	F	7.75	480.5	11.9531
Model	AB	0.625	3.125	0.0777387
Error	AC	-3.5	98	2.43789
Error	AD	-0.125	0.125	0.00310955
Error	AE	1.875	28.125	0.699649
Model	AF	1.75	24.5	0.609472
Error	BC	0	0	0
Error	BD	0.125	0.125	0.00310955

Error	BE	-5.375	231.125	5.74956
Model	BF	3.25	84.5	2.10206
Error	CD	3.75	112.5	2.79859
Error	CE	3.75	112.5	2.79859
Error	CF	4.875	190.125	4.72962
Error	DE	5.375	231.125	5.74956
Error	DF	5.5	242	6.02009
Model	EF	8	512	12.7367
Error	ABC	Aliased		
Error	ABD	Aliased		
Error	ABE	3.625	105.125	2.61513
Model	ABF	9	648	16.1199
Error	ACD	-6.5	338	8.40822
Error	ACE	Aliased		
Error	ACF	Aliased		
Error	ADE	-3.375	91.125	2.26686
Error	ADF	-0.5	2	0.0497528
Error	AEF	1	8	0.199011
Error	BCD	Aliased		
Error	BCE	Aliased		
Error	BCF	Aliased		
Error	BDE	-2.625	55.125	1.37131
Error	BDF	-0.5	2	0.0497528
Error	BEF	Aliased		
Error	CDE	Aliased		
Error	CDF	Aliased		
Error	CEF	2.125	36.125	0.89866
Error	DEF	2	32	0.796045
	Lenth's ME	9.15104		
	Lenth's SME	17.3991		



Design Expert Output

Response: Thick Range

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	2023.00	9	224.78	2.48	0.0400	significant
A	*3.13*	*1*	*3.13*	*0.034*	*0.8545*	
B	*36.13*	*1*	*36.13*	*0.40*	*0.5346*	
E	*231.12*	*1*	*231.12*	*2.55*	*0.1248*	
F	*480.50*	*1*	*480.50*	*5.29*	*0.0313*	
AB	*3.12*	*1*	*3.12*	*0.034*	*0.8545*	
AF	*24.50*	*1*	*24.50*	*0.27*	*0.6086*	
BF	*84.50*	*1*	*84.50*	*0.93*	*0.3451*	

EF	*512.00*	*1*	*512.00*	*5.64*	*0.0267*
ABF	*648.00*	*1*	*648.00*	*7.14*	*0.0139*
Residual	1996.88	22	90.77		
Cor Total	4019.88	31			

The Model F-value of 2.48 implies the model is significant. There is only a 4.00% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	9.53	R-Squared	0.5032
Mean	26.56	Adj R-Squared	0.3000
C.V.	35.87	Pred R-Squared	-0.0510
PRESS	4224.79	Adeq Precision	5.586

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	26.56	1	1.68	23.07	30.06	
A-Volume	-0.31	1	1.68	-3.81	3.18	1.00
B-Batch	1.06	1	1.68	-2.43	4.56	1.00
E-Acc	-2.69	1	1.68	-6.18	0.81	1.00
F-Cover	3.88	1	1.68	0.38	7.37	1.00
AB	0.31	1	1.68	-3.18	3.81	1.00
AF	0.88	1	1.68	-2.62	4.37	1.00
BF	1.63	1	1.68	-1.87	5.12	1.00
EF	4.00	1	1.68	0.51	7.49	1.00
ABF	4.50	1	1.68	1.01	7.99	1.00

Final Equation in Terms of Coded Factors:

```
Thick Range    =
    +26.56
      -0.31    * A
      +1.06    * B
      -2.69    * E
      +3.88    * F
      +0.31    * A * B
      +0.88    * A * F
      +1.63    * B * F
      +4.00    * E * F
      +4.50    * A * B * F
```

Final Equation in Terms of Actual Factors:

```
     Batch    Batch 1
     Cover    Off
Thick Range   =
  +22.39583
   +3.00000    * Volume
   -0.89167    * Acc

     Batch    Batch 2
     Cover    Off
Thick Range   =
  +54.77083
   -5.37500    * Volume
   -0.89167    * Acc

     Batch    Batch 1
     Cover    On
Thick Range   =
  +42.56250
   -4.25000    * Volume
   +0.17500    * Acc

     Batch    Batch 2
     Cover    On
Thick Range   =
   +9.43750
   +5.37500    * Volume
   +0.17500    * Acc
```

The model for thickness range is not very strong. Notice the small value of R^2, and in particular, the adjusted R^2. Often we find that obtaining a good model for a response that expresses variability is not as easy as finding a satisfactory model for a response that measures the mean.

(f) Where would you recommend that the process engineers run the process?

Considering only the average thickness results, the engineers could use factors *B, D* and *E* to put the process mean at target. Then the engineer could consider the other factors on the range model to try to set the factors to reduce the variation in thickness at that mean.

8.24S. Harry Peterson-Nedry (a friend of the author) owns a vineyard and winery in Newberg, Oregon. They grow several varieties of grapes and manufacture wine. Harry and Judy have used factorial designs for process and product development in the winemaking segment of their business. This problem describes the experiment conducted for their 1985 Pinot Noir. Eight variables, shown in Table P8.5, were originally studied in this experiment:

Table P8.5

	Variable	Low Level	High Level
A	Pinot Noir Clone	Pommard	Wadenswil
B	Oak Type	Allier	Troncais
C	Age of Barrel	Old	New
D	Yeast/Skin Contact	Champagne	Montrachet
E	Stems	None	All
F	Barrel Toast	Light	Medium
G	Whole Cluster	None	10%
H	Fermentation Temperature	Low (75 F Max)	High (92 F Max)

Harry decided to use a 2^{8-4}_{IV} design with 16 runs. The wine was taste-tested by a panel of experts on 8 March 1986. Each expert ranked the 16 samples of wine tasted, with rank 1 being the best. The design and taste-test panel results are shown in Table P8.6.

Table P8.6

Run	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	HPN	JPN	CAL	DCM	RGB	y_{bar}	*s*
1	-	-	-	-	-	-	-	-	12	6	13	10	7	9.6	3.05
2	+	-	-	-	-	+	+	+	10	7	14	14	9	10.8	3.11
3	-	+	-	-	+	-	+	+	14	13	10	11	15	12.6	2.07
4	+	+	-	-	+	+	-	-	9	9	7	9	12	9.2	1.79
5	-	-	+	-	+	+	+	-	8	8	11	8	10	9.0	1.41
6	+	-	+	-	+	-	-	+	16	12	15	16	16	15.0	1.73
7	-	+	+	-	-	+	-	+	6	5	6	5	3	5.0	1.22
8	+	+	+	-	-	-	+	-	15	16	16	15	14	15.2	0.84
9	-	-	-	+	+	+	-	+	1	2	3	3	2	2.2	0.84
10	+	-	-	+	+	-	+	-	7	11	4	7	6	7.0	2.55
11	-	+	-	+	-	+	+	-	13	3	8	12	8	8.8	3.96
12	+	+	-	+	-	-	-	+	3	1	5	1	4	2.8	1.79
13	-	-	+	+	-	-	+	+	2	10	2	4	5	4.6	3.29
14	+	-	+	+	-	+	-	-	4	4	1	2	1	2.4	1.52
15	-	+	+	+	+	-	-	-	5	15	9	6	11	9.2	4.02
16	+	+	+	+	+	+	+	+	11	14	12	13	13	12.6	1.14

(a) What are the alias relationships in the design selected by Harry?

$$E = BCD,\ F = ACD,\ G = ABC,\ H = ABD$$

Defining Contrast : $I = BCDE = ACDF = ABEF = ABCG = ADEG = BDFG = CEFG = ABDH = ACEH = BCFH = DEFH = CDGH = BEGH = AFGH = ABCDEFGH$

Aliases:

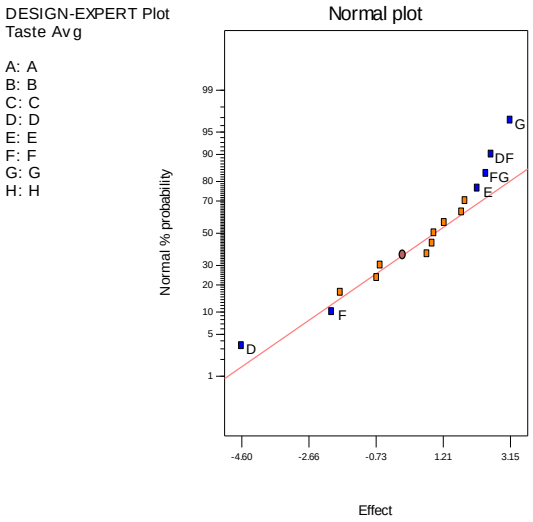
$$A = BCG = BDH = BEF = CDF = CEH = DEG = FGH$$
$$B = ACG = ADH = AEF = CDE = CFH = DFG = EGH$$
$$C = ABG = ADF = AEH = BDE = BFH = DGH = EFG$$
$$D = ABH = ACF = AEG = BCE = BFG = CGH = EFH$$
$$E = ABF = ACH = ADG = BCD = BGH = CFG = DFH$$
$$F = ABE = ACD = AGH = BCH = BDG = CEG = DEH$$
$$G = ABC = ADE = AFH = BDF = BEH = CDH = CEF$$
$$H = ABD = ACE = AFG = BCF = BEG = CDG = DEF$$
$$AB = CG = DH = EF$$
$$AC = BG = DF = EH$$
$$AD = BH = CF = EG$$
$$AE = BF = CH = DG$$
$$AF = BE = CD = GH$$
$$AG = BC = DE = FH$$
$$AH = BD = CE = FG$$

(b) Use the average ranks ($\bar{y}$) as a response variable. Analyze the data and draw conclusions. You will find it helpful to examine a normal probability plot of effect estimates.

The effects list and normal probability plot of effects are shown below. Factors *D*, *E*, *F*, and *G* appear to be significant. Also note that the *DF* and *FG* interactions were chosen instead of *AC* and *AH* based on the alias structure shown above.

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Require	Intercept			
Error	A	1.75	12.25	4.57636
Error	B	1.85	13.69	5.11432
Error	C	1.25	6.25	2.33488
Model	D	-4.6	84.64	31.6198
Model	E	2.2	19.36	7.23252
Model	F	-2	16	5.97729
Model	G	3.15	39.69	14.8274
Error	H	-0.6	1.44	0.537956
Error	AB	-0.7	1.96	0.732218
Ignore	AC	Aliased		
Error	AD	-1.75	12.25	4.57636
Error	AE	0.95	3.61	1.34863
Error	AF	0.75	2.25	0.840556
Error	AG	0.9	3.24	1.2104
Ignore	AH	Aliased		
Model	DF	2.6	27.04	10.1016
Model	FG	2.45	24.01	8.96967
	Lenth's ME	6.74778		
	Lenth's SME	13.699		



Design Expert Output

Response: Taste Avg

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	210.74	6	35.12	5.55	0.0115	significant
D	*84.64*	*1*	*84.64*	*13.38*	*0.0053*	
E	*19.36*	*1*	*19.36*	*3.06*	*0.1142*	
F	*16.00*	*1*	*16.00*	*2.53*	*0.1462*	
G	*39.69*	*1*	*39.69*	*6.27*	*0.0336*	
DF	*27.04*	*1*	*27.04*	*4.27*	*0.0687*	
FG	*24.01*	*1*	*24.01*	*3.80*	*0.0832*	
Residual	56.94	9	6.33			
Cor Total	267.68	15				

The Model F-value of 5.55 implies the model is significant. There is only a 1.15% chance that a "Model F-Value" this large could occur due to noise.

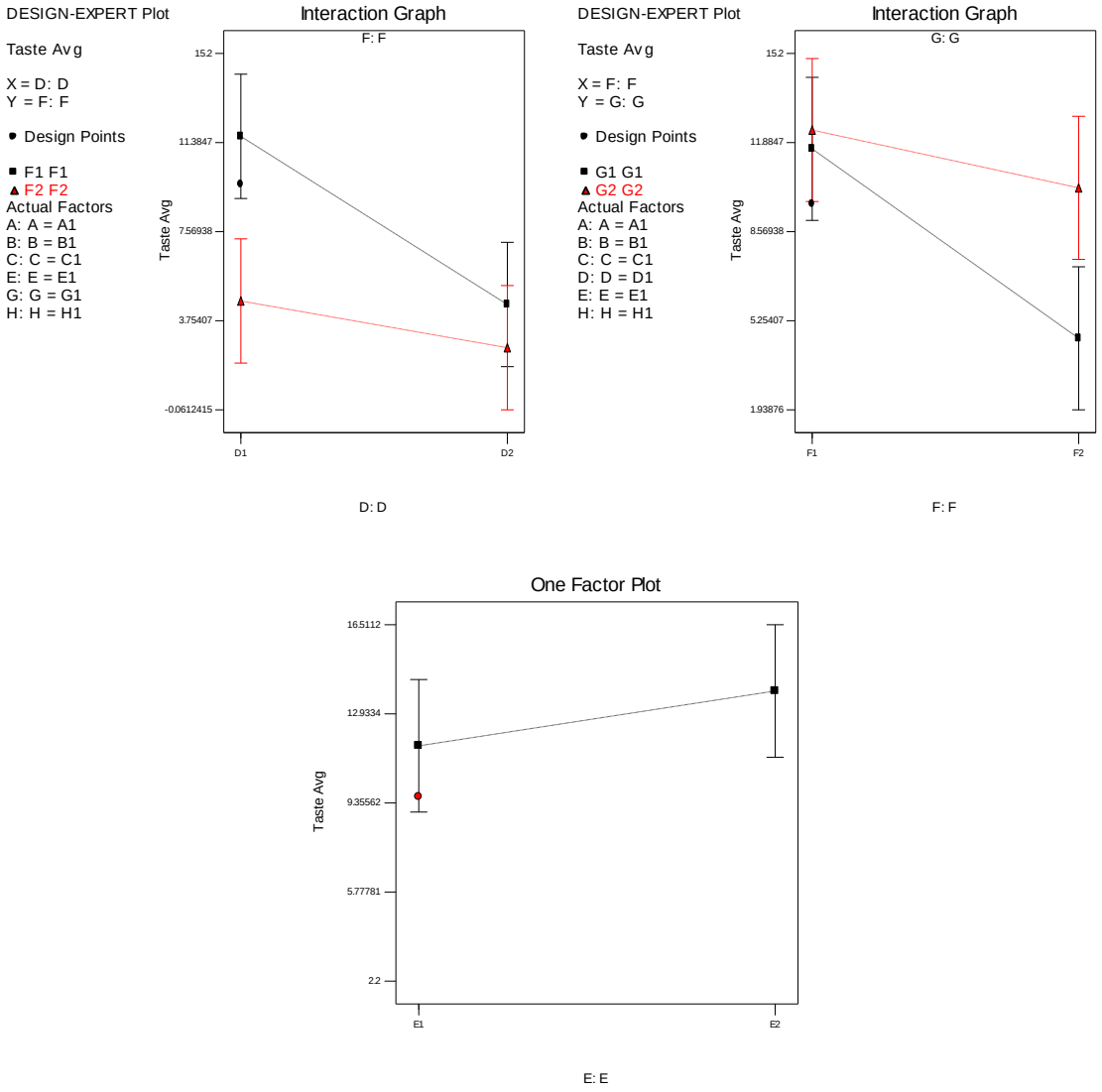
Std. Dev.	2.52	R-Squared	0.7873
Mean	8.50	Adj R-Squared	0.6455
C.V.	29.59	Pred R-Squared	0.3277
PRESS	179.96	Adeq Precision	7.183

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	8.50	1	0.63	7.08	9.92	
D-D	-2.30	1	0.63	-3.72	-0.88	1.00
E-E	1.10	1	0.63	-0.32	2.52	1.00
F-F	-1.00	1	0.63	-2.42	0.42	1.00
G-G	1.57	1	0.63	0.15	3.00	1.00
DF	1.30	1	0.63	-0.12	2.72	1.00
FG	1.23	1	0.63	-0.20	2.65	1.00

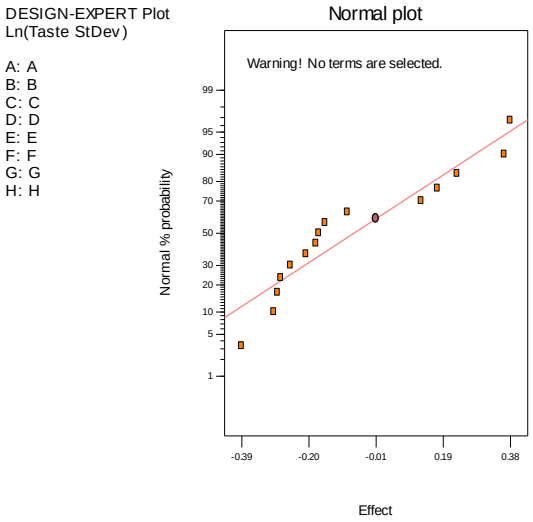
Final Equation in Terms of Coded Factors:

Taste Avg =
+8.50
-2.30 * D
+1.10 * E
-1.00 * F
+1.57 * G
+1.30 * D * F
+1.23 * F * G

Factors *D* and *G* and are important. Factor *E* and the *DF* and *FG* interactions are moderately important and were included in the model because the PRESS statistic showed improvement with their inclusion. Factor *F* is added to the model to preserve hierarchy. As stated earlier, the interactions are aliased with other two-factor interactions that could also be important. So the interpretation of the two-factor interaction is somewhat uncertain. Normally, we would add runs to the design to isolate the significant interactions, but that will not work very well here because each experiment requires a full growing season. In other words, it would require a very long time to add runs to de-alias the alias chains of interest.



(c) Use the standard deviation of the ranks (or some appropriate transformation such as log *s*) as a response variable. What conclusions can you draw about the effects of the eight variables on variability in wine quality?

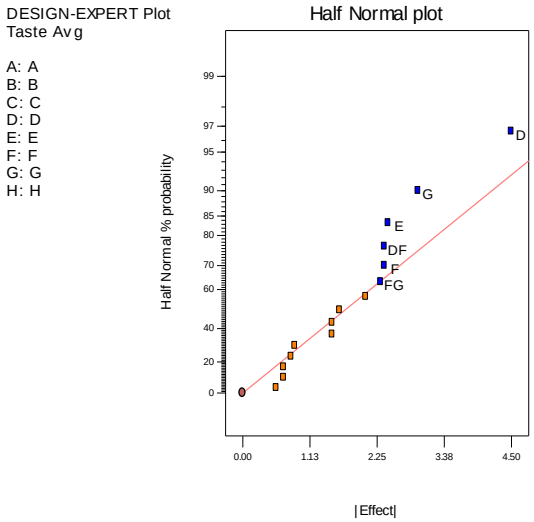


There do not appear to be any significant factors.

(d) After looking at the results, Harry decides that one of the panel members (DCM) knows more about beer than he does about wine, so they decide to delete his ranking. What affect would this have on the results and on conclusions from parts (b) and (c)?

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Require	Intercept			
Error	A	1.625	10.5625	4.02957
Error	B	2.0625	17.0156	6.49142
Error	C	1.5	9	3.43348
Model	D	-4.5	81	30.9013
Model	E	2.4375	23.7656	9.06652
Model	F	-2.375	22.5625	8.60753
Model	G	2.9375	34.5156	13.1676
Error	H	-0.6875	1.89063	0.721268
Error	AB	-0.5625	1.26563	0.482833
Ignore	AC	Aliased		
Error	AD	-1.5	9	3.43348
Error	AE	0.6875	1.89063	0.721268
Error	AF	0.875	3.0625	1.16834
Error	AG	0.8125	2.64062	1.00739
Ignore	AH	Aliased		
Model	DF	2.375	22.5625	8.60753
Model	FG	2.3125	21.3906	8.16047
	Lenth's ME	6.26579		
	Lenth's SME	12.7205		



Design Expert Output

Response: Taste Avg

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	205.80	6	34.30	5.48	0.0120	significant
D	*81.00*	*1*	*81.00*	*12.94*	*0.0058*	
E	*23.77*	*1*	*23.77*	*3.80*	*0.0831*	
F	*22.56*	*1*	*22.56*	*3.60*	*0.0901*	
G	*34.52*	*1*	*34.52*	*5.51*	*0.0434*	
DF	*22.56*	*1*	*22.56*	*3.60*	*0.0901*	
FG	*21.39*	*1*	*21.39*	*3.42*	*0.0975*	
Residual	56.33	9	6.26			
Cor Total	262.13	15				

The Model F-value of 5.48 implies the model is significant. There is only a 1.20% chance that a "Model F-Value" this large could occur due to noise.

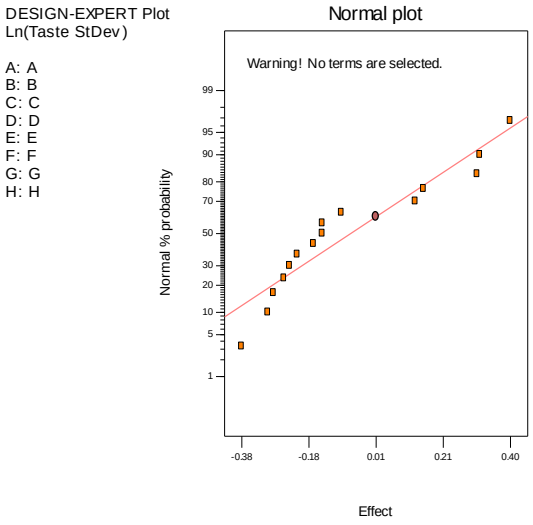
Std. Dev.	2.50	R-Squared	0.7851
Mean	8.50	Adj R-Squared	0.6418
C.V.	29.43	Pred R-Squared	0.3208
PRESS	178.02	Adeq Precision	7.403

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	8.50	1	0.63	7.09	9.91	
D-D	-2.25	1	0.63	-3.66	-0.84	1.00
E-E	1.22	1	0.63	-0.20	2.63	1.00
F-F	-1.19	1	0.63	-2.60	0.23	1.00
G-G	1.47	1	0.63	0.054	2.88	1.00
DF	1.19	1	0.63	-0.23	2.60	1.00
FG	1.16	1	0.63	-0.26	2.57	1.00

Final Equation in Terms of Coded Factors:

Taste Avg =
+8.50
-2.25 * D
+1.22 * E
-1.19 * F
+1.47 * G
+1.19 * D * F
+1.16 * F * G

The results are very similar for average taste without DCM as they were with DCM.



The standard deviation response is much the same with or without DCM's responses. Again, there are no significant factors.

(e) Suppose that just before the start of the experiment, Harry discovered that the eight new barrels they ordered from France for use in the experiment would not arrive in time, and all 16 runs would have to be made with old barrels. If Harry just drops column C from their design, what does this do to the alias relationships? Do they need to start over and construct a new design?

The resulting design is a 2_{IV}^{7-3}, with defining relations: $I = ABEF = ADEG = BDFG = ABDH = DEFH = BEGH = AFGH$.

(f) Harry knows from experience that some treatment combinations are unlikely to produce good results. For example, the run with all eight variables at the high level generally results in a poorly rated wine. This was confirmed in the 8 March 1986 taste test. He wants to set up a new design for their 1986 Pinot Noir using these same eight variables, but they do not want to make the run with all eight factors at the high level. What design would you suggest?

By changing the sign of any of the design generators, a design that does not include the principal fraction will be generated. This will give a design without an experimental run combination with all of the variables at the high level.

8.25. Consider the 2^5 factorial in Problem 6.35. Suppose that the experimenters could only afford 16 runs. Set up the 2^{5-1} fractional factorial design with $I=ABCDE$ and select the responses from the full factorial data in Problem 6.35.

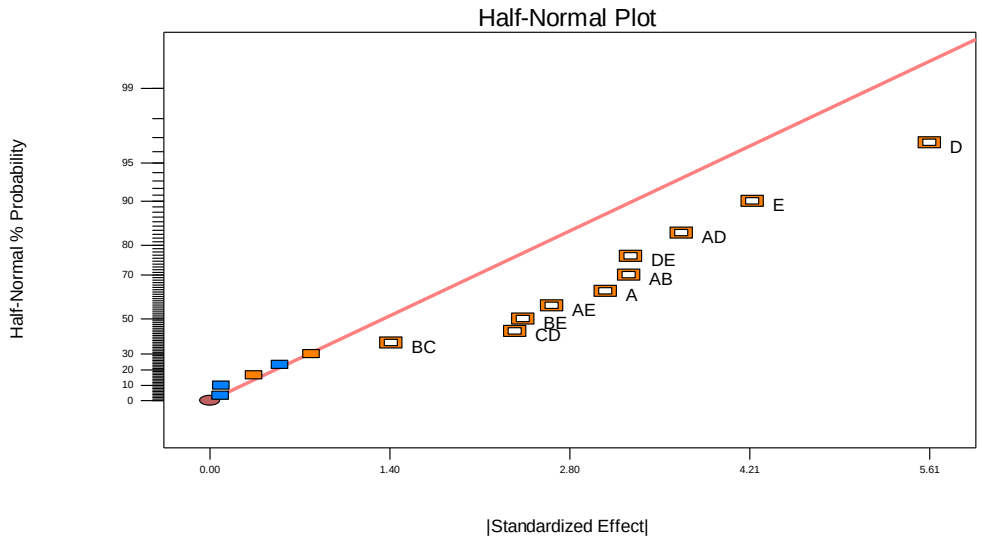
The runs for the 2^{5-1} fractional factorial design with $I=ABCDE$ are shown below.

A	B	C	D	E	y
-1	-1	-1	-1	1	7.93
1	-1	-1	-1	-1	5.56
-1	1	-1	-1	-1	5.77
1	1	-1	-1	1	12

-1	-1	1	-1	-1	9.17
1	-1	1	-1	1	3.65
-1	1	1	-1	1	6.4
1	1	1	-1	-1	5.69
-1	-1	-1	1	-1	8.82
1	-1	-1	1	1	17.55
-1	1	-1	1	1	8.87
1	1	-1	1	-1	8.94
-1	-1	1	1	1	13.06
1	-1	1	1	-1	11.49
-1	1	1	1	-1	6.25
1	1	1	1	1	26.05

(a) Analyze the data and draw conclusions.

The analysis which identifies the significant effects is shown below.



Design Expert Output

Response 1 y

ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	460.75	12	38.40	92.45	0.0016	significant
A-A	*38.01*	*1*	*38.01*	*91.51*	*0.0024*	
B-B	*0.47*	*1*	*0.47*	*1.13*	*0.3658*	
C-C	*2.50*	*1*	*2.50*	*6.01*	*0.0915*	
D-D	*125.78*	*1*	*125.78*	*302.84*	*0.0004*	
E-E	*71.49*	*1*	*71.49*	*172.13*	*0.0010*	
AB	*42.64*	*1*	*42.64*	*102.67*	*0.0020*	
AD	*54.02*	*1*	*54.02*	*130.08*	*0.0014*	
AE	*28.41*	*1*	*28.41*	*68.40*	*0.0037*	
BC	*7.98*	*1*	*7.98*	*19.22*	*0.0220*	
BE	*23.81*	*1*	*23.81*	*57.34*	*0.0048*	
CD	*22.61*	*1*	*22.61*	*54.44*	*0.0051*	
DE	*43.03*	*1*	*43.03*	*103.62*	*0.0020*	
Residual	1.25	3	0.42			
Cor Total	461.99	15				

(b) Compare your findings with those from the full factorial in Problem 6.35.

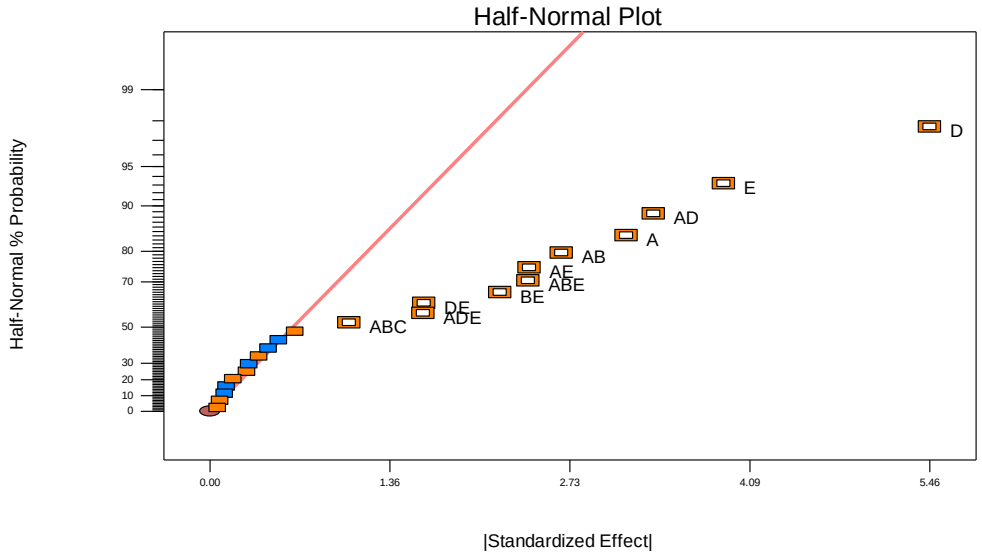
The results are similar to those found in Problem 6.35 with the exception that the *BC* and *CD* interactions are now significant, and factor *C* remains in the model because of these interactions.

(c) Are there any potential interactions that need further study? What additional runs do you recommend? Select these runs from the full factorial design in Problem 6.35 and analyze the new design. Discuss your conclusions.

The *BC* interaction is aliased with the *ADE* interaction, and the *CD* interaction is aliased with the *ABE* interaction. The *ADE* and *ABE* interactions in Problem 6.43 were both found to be significant. Including these effects would also allow us to remove the insignificant factor *C* from the model.

The additional runs are shown below as block 2. This is a partial fold-over where factor A was chosen for the fold-over, and factor *C* at the -1 level was chosen for the runs. This allowed both the *ABE* and *ADE* interactions to be de-aliased.

Block	*A*	*B*	*C*	*D*	*E*	*y*
1	-1	-1	-1	-1	1	7.93
1	1	-1	-1	-1	-1	5.56
1	-1	1	-1	-1	-1	5.77
1	1	1	-1	-1	1	12
1	-1	-1	1	-1	-1	9.17
1	1	-1	1	-1	1	3.65
1	-1	1	1	-1	1	6.4
1	1	1	1	-1	-1	5.69
1	-1	-1	-1	1	-1	8.82
1	1	-1	-1	1	1	17.55
1	-1	1	-1	1	1	8.87
1	1	1	-1	1	-1	8.94
1	-1	-1	1	1	1	13.06
1	1	-1	1	1	-1	11.49
1	-1	1	1	1	-1	6.25
1	1	1	1	1	1	26.05
2	1	-1	-1	-1	1	5
2	-1	-1	-1	-1	-1	8.11
2	1	1	-1	-1	-1	5.82
2	-1	1	-1	-1	1	7.47
2	1	-1	-1	1	-1	14.23
2	-1	-1	-1	1	1	12.43
2	1	1	-1	1	1	25.38
2	-1	1	-1	1	-1	9.2



Design Expert Output

Response 1 y

ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Block	6.81	1	6.81			
Model	764.14	15	50.94	89.46	< 0.0001	significant
A-A	*51.83*	*1*	*51.83*	*91.03*	*< 0.0001*	
B-B	*0.47*	*1*	*0.47*	*0.82*	*0.3942*	
C-C	*2.50*	*1*	*2.50*	*4.38*	*0.0746*	
D-D	*178.72*	*1*	*178.72*	*313.85*	*< 0.0001*	
E-E	*91.03*	*1*	*91.03*	*159.85*	*< 0.0001*	
AB	*42.64*	*1*	*42.64*	*74.88*	*< 0.0001*	
AC	*0.074*	*1*	*0.074*	*0.13*	*0.7298*	
AD	*67.88*	*1*	*67.88*	*119.20*	*< 0.0001*	
AE	*35.19*	*1*	*35.19*	*61.79*	*0.0001*	
BC	*0.073*	*1*	*0.073*	*0.13*	*0.7310*	
BE	*29.02*	*1*	*29.02*	*50.96*	*0.0002*	
DE	*15.80*	*1*	*15.80*	*27.75*	*0.0012*	
ABC	*6.68*	*1*	*6.68*	*11.73*	*0.0111*	
ABE	*34.92*	*1*	*34.92*	*61.32*	*0.0001*	
ADE	*9.58*	*1*	*9.58*	*16.82*	*0.0046*	
Residual	3.99	7	0.57			
Cor Total	774.93	23				

8.26. Consider the 2^4 factorial experiment for surfactin production in Problem 6.36. Suppose that the experimenters could only afford eight runs. Set up the 2^{4-1} fractional factorial with I = *ABCD* and select the responses for the runs from the full factorial data in Problem 6.36. Analyze and draw conclusions. Compare your findings with those from the full factorial in Problem 6.36.

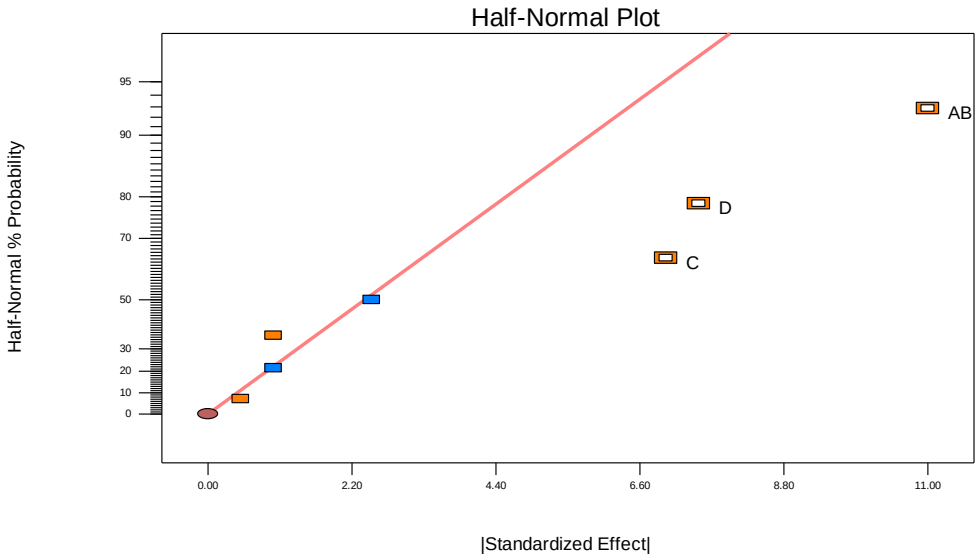
The runs for the 2^{4-1} fractional factorial with I = *ABCD* are shown below.

Run	Glucose (g dm^{-3})	NH_4NO_3 (g dm^{-3})	$FeSO_4$ (g dm^{-3} x 10^{-4})	$MnSO_4$ (g dm^{-3} x 10^{-2})	y (CMC)$^{-1}$
1	20.00	2.00	6.00	4.00	23
2	60.00	2.00	6.00	20.00	16

3	20.00	6.00	6.00	20.00	18
4	60.00	6.00	6.00	4.00	18
5	20.00	2.00	30.00	20.00	36
6	60.00	2.00	30.00	4.00	16
7	20.00	6.00	30.00	4.00	17
8	60.00	6.00	30.00	20.00	34

(a) Analyze the data and draw conclusions.

The analysis below identifies the factors *C* and *D*, and the *AB* interaction as being significant.



Design Expert Output

Response 1 y
ANOVA for selected factorial model
Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	467.00	5	93.40	74.72	0.0133	significant
A-Glucose	*12.50*	*1*	*12.50*	*10.00*	*0.0871*	
B-NH4NO3	*2.00*	*1*	*2.00*	*1.60*	*0.3333*	
C-FeSO4	*98.00*	*1*	*98.00*	*78.40*	*0.0125*	
D-MnSO4	*112.50*	*1*	*112.50*	*90.00*	*0.0109*	
AB	*242.00*	*1*	*242.00*	*193.60*	*0.0051*	
Residual	2.50	2	1.25			
Cor Total	469.50	7				

(b) Compare your findings with those from the full factorial in Problem 6.36.

In this 2^{4-1} fractional factorial with $I = ABCD$, the *AB* interaction is aliased with the *CD* interaction. The full factorial experiment in Problem 6.44 found the *CD* interaction to be significant. Additional runs, such as a fold-over, should be run to de-alias the *CD* from the *AB* interaction.

8.27. Consider the 2^4 factorial experiment in Problem 6.38. Suppose that the experimenters could only afford eight runs. Set up the 2^{4-1} fractional factorial design with $I = ABCD$ and select the responses from the full factorial data in Problem 6.38.

Problem 6.38 was originally analyzed as a replicated 2^3 full factorial experiment. The purpose for this experiment was to compare the fishbone apatite with the hydroxyapatite. However, as shown in the solutions for this problem, the experimenters might have observed better results had they analyzed the experiment as a replicated 2^4 full factorial experiment. This analysis is shown in the solutions for part (i) of Problem 6.38.

The solution presented here is for the replicated 2^{4-1} 2 fractional factorial experiment with $I = ABCD$ as shown below.

Apatite	pH	Pb	Type	Pb,mM	pH
+	+	+	Hydroxyapatite	0.11	3.49
+	+	+	Hydroxyapatite	0.12	3.46
+	+	–	Fishbone	0.01	6.84
+	+	–	Fishbone	0	6.61
+	–	+	Fishbone	1.11	3.35
+	–	+	Fishbone	1.04	3.34
+	–	–	Hydroxyapatite	0.03	3.36
+	–	–	Hydroxyapatite	0.05	3.24
–	+	+	Fishbone	2.11	5.29
–	+	+	Fishbone	2.18	5.06
–	+	–	Hydroxyapatite	0	5.53
–	+	–	Hydroxyapatite	0	5.43
–	–	+	Hydroxyapatite	1.34	2.82
–	–	+	Hydroxyapatite	1.26	2.79
–	–	–	Fishbone	0.05	4.5
–	–	–	Fishbone	0.05	4.74

(a) Analyze the data for all of the responses and draw conclusions.

Due to non-constant variance observed in the residuals plots, the same transformations that were used in the solution of Problem 6.38 part (i) were also used here.

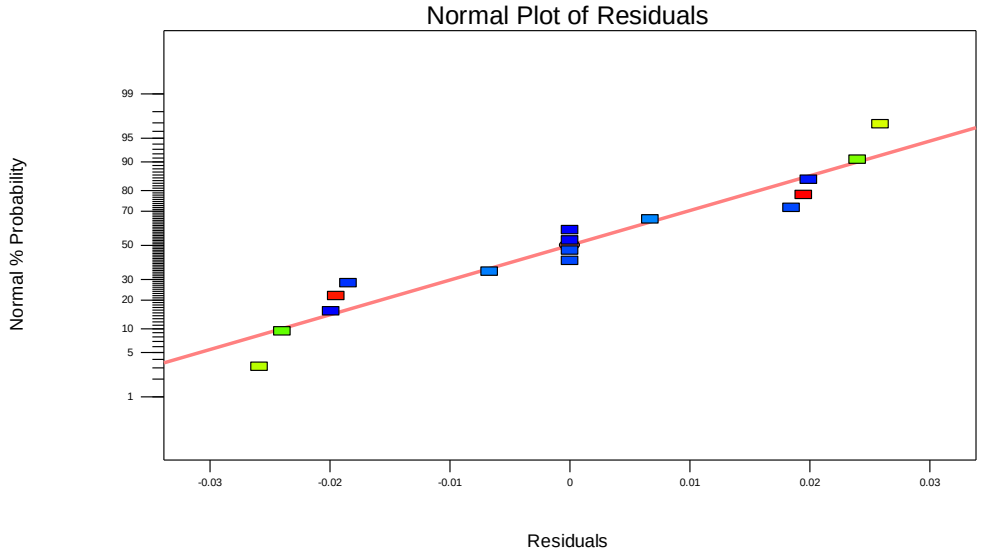
The analysis for the Pb Response identifies all for main effects and the *AB*, *AC*, and *AD* interactions as being significant. However, the *AB* interaction is aliased with *CD*, *AC* is aliased with *BD*, and *AD* is aliased with *BC.*

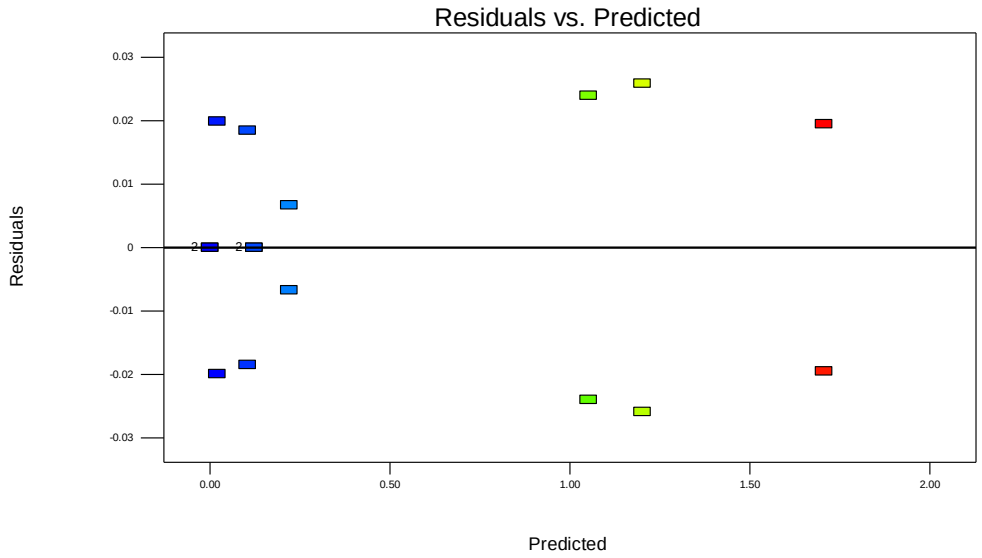
Design Expert Output

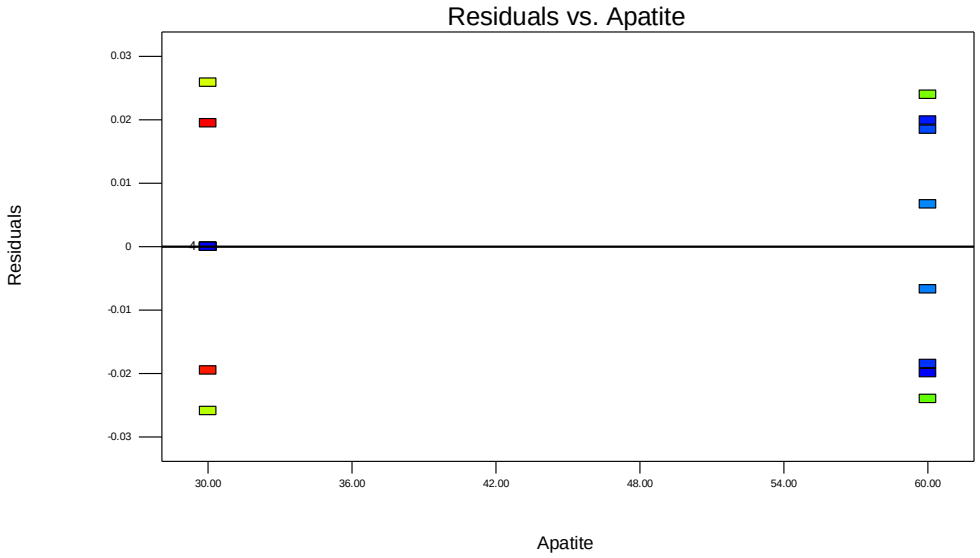
Response 1 Pb Response
Transform: Power Lambda: 0.7 Constant: 0
ANOVA for selected factorial model
Analysis of variance table [Partial sum of squares - Type III]

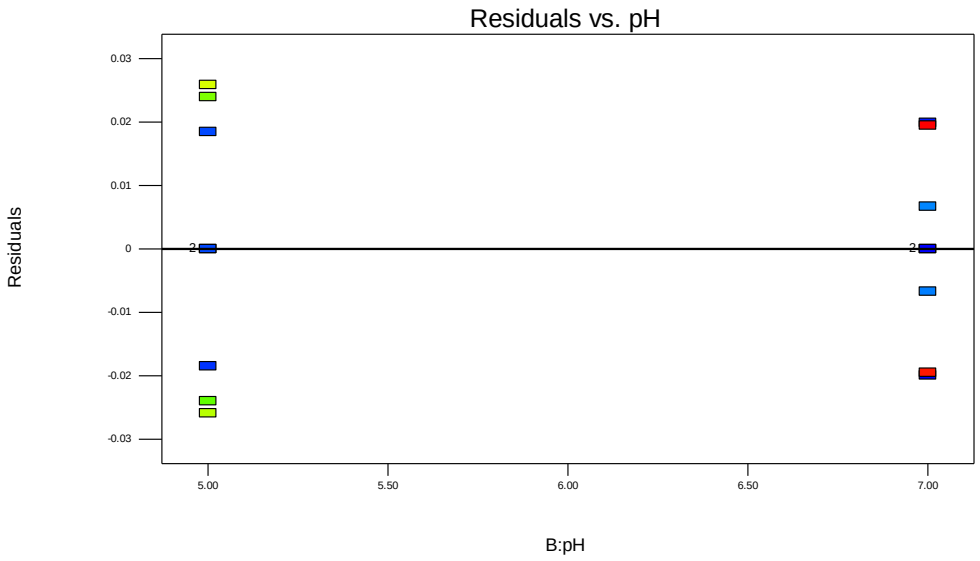
Source	**Sum of Squares**	**df**	**Mean Square**	**F Value**	**p-value Prob > F**	
Model	6.17	7	0.88	1465.63	< 0.0001	significant
A-Apatite	*0.67*	*1*	*0.67*	*1109.89*	*< 0.0001*	
B-pH	*0.071*	*1*	*0.071*	*118.74*	*< 0.0001*	
C-Pb	*3.87*	*1*	*3.87*	*6425.51*	*< 0.0001*	
D-Type	*0.47*	*1*	*0.47*	*785.36*	*< 0.0001*	
AB	*0.42*	*1*	*0.42*	*700.14*	*< 0.0001*	
AC	*0.67*	*1*	*0.67*	*1113.81*	*< 0.0001*	
AD	*3.600E-003*	*1*	*3.600E-003*	*5.98*	*0.0402*	
Pure Error	4.813E-003	8	6.016E-004			
Cor Total	6.18	15				

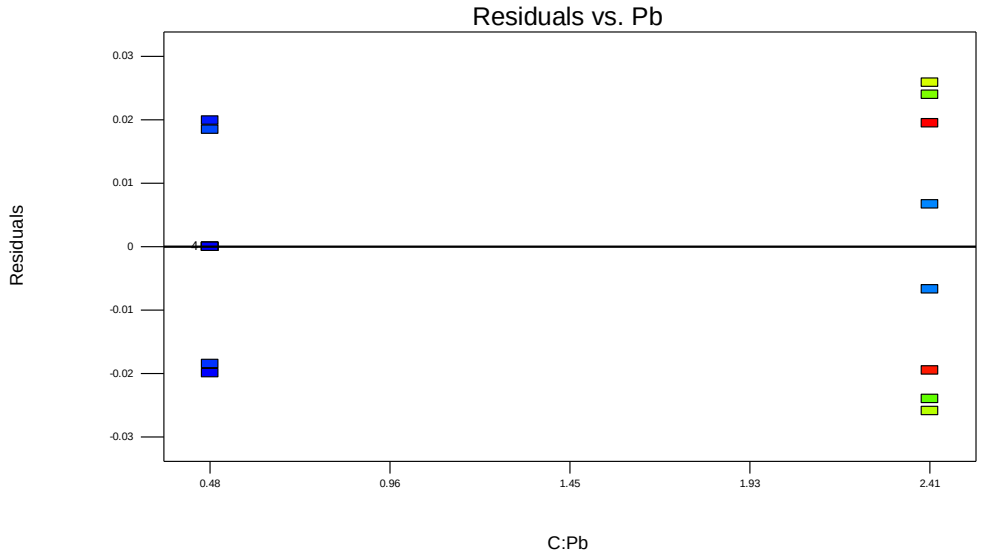
There are no concerns with the residual plots shown below.

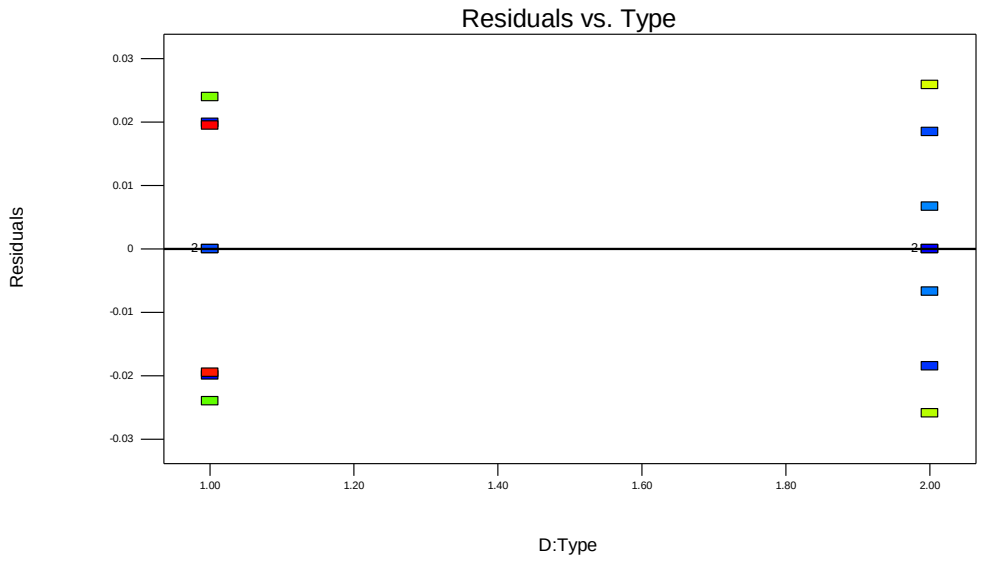





Residuals vs. Apatite
Residuals
0.03
0.02
0.01
0
-0.01
-0.02
-0.03
30.00
36.00
42.00
48.00
54.00
60.00
Apatite


Residuals vs. pH
Residuals
0.03
0.02
0.01
0
-0.01
-0.02
-0.03
5.00
5.50
6.00
6.50
7.00
B:pH





The analysis for the pH Response identifies all for main effects and the *AD* interaction as being significant. However, the *AD* interaction is aliased with *BC*.

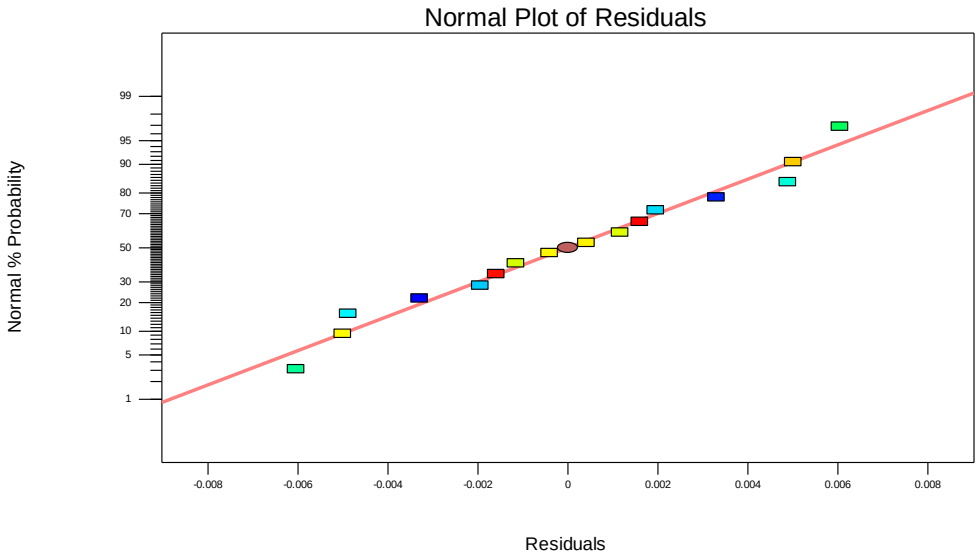
Design Expert Output

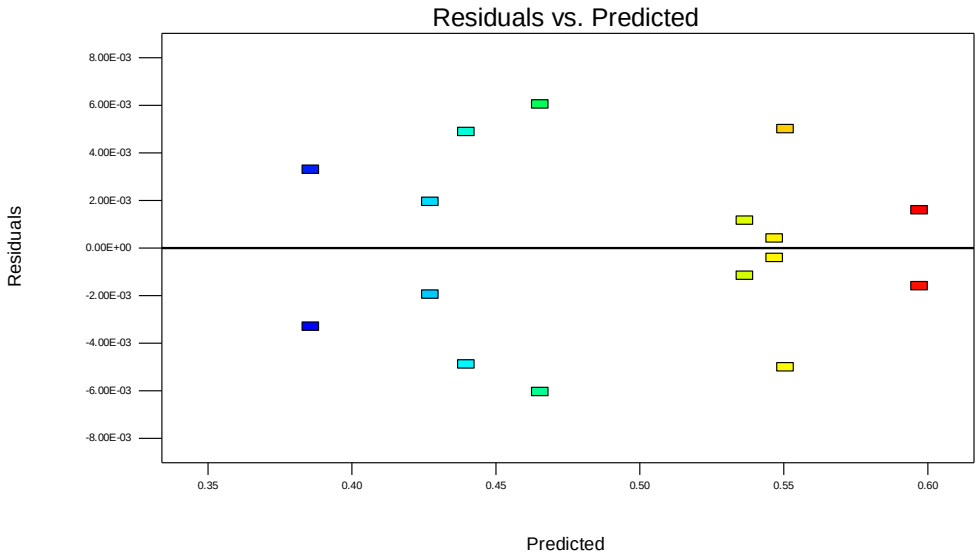
Response 2 pH Response
Transform: Inverse Sqrt **Constant: 0**
ANOVA for selected factorial model
Analysis of variance table [Partial sum of squares - Type III]

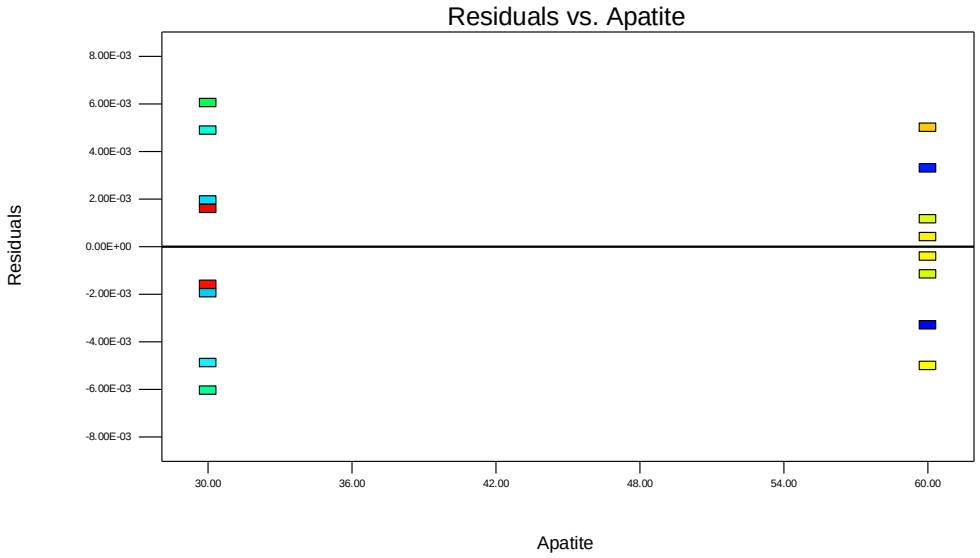
Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	0.077	7	0.011	421.02	< 0.0001	significant
A-Apatite	*2.030E-003*	*1*	*2.030E-003*	*77.93*	*< 0.0001*	
B-pH	*0.034*	*1*	*0.034*	*1319.61*	*< 0.0001*	
C-Pb	*0.021*	*1*	*0.021*	*813.91*	*< 0.0001*	
D-Type	*0.019*	*1*	*0.019*	*719.62*	*< 0.0001*	
AB	*1.038E-004*	*1*	*1.038E-004*	*3.98*	*0.0810*	
AC	*1.961E-006*	*1*	*1.961E-006*	*0.075*	*0.7907*	

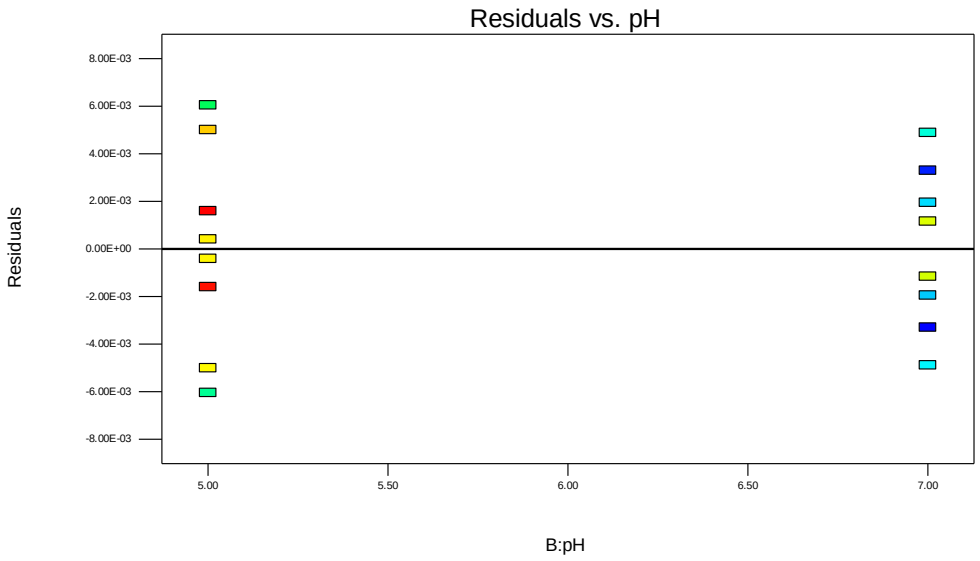
AD	*3.119E-004*	*1*	*3.119E-004*	*11.97*	*0.0086*
Pure Error	2.084E-004	8	2.605E-005		
Cor Total	0.077	15			

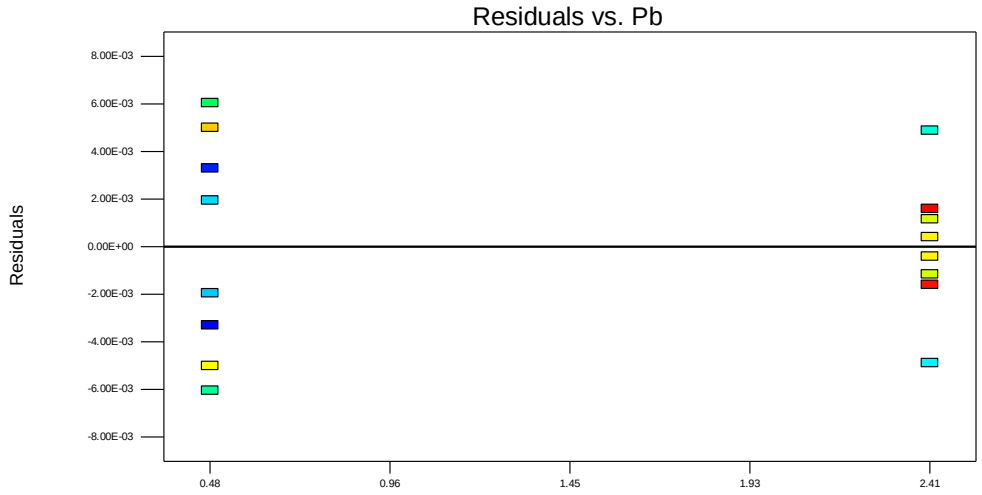
There are no concerns with the residual plots shown below.

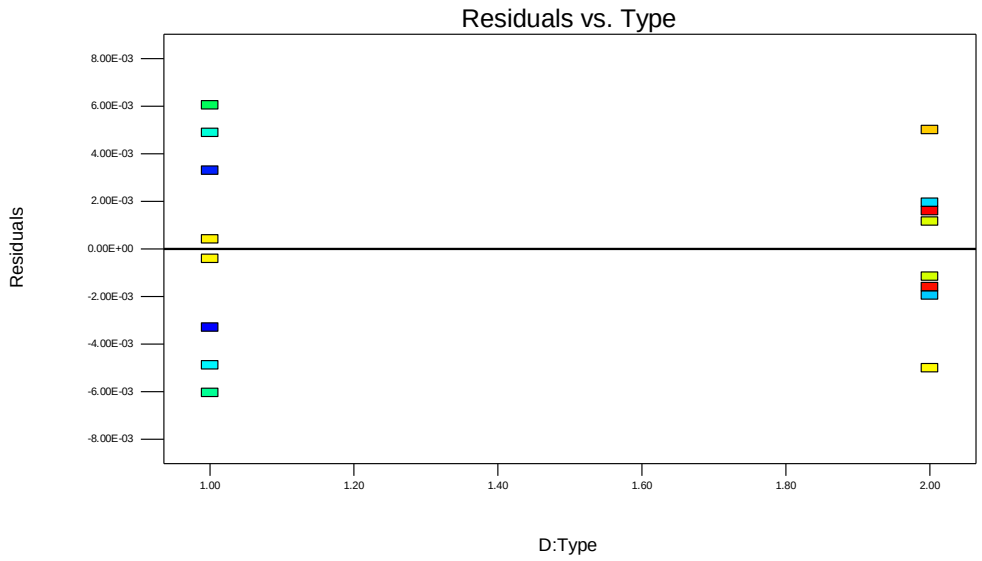





Residuals vs. Apatite
Residuals
8.00E-03
6.00E-03
4.00E-03
2.00E-03
0.00E+00
-2.00E-03
-4.00E-03
-6.00E-03
-8.00E-03
30.00
36.00
42.00
48.00
54.00
60.00
Apatite


Residuals vs. pH
Residuals
8.00E-03
6.00E-03
4.00E-03
2.00E-03
0.00E+00
-2.00E-03
-4.00E-03
-6.00E-03
-8.00E-03
5.00
5.50
6.00
6.50
7.00
B:pH





(b) Compare your findings with those from the full factorial in Problem 6.38.

In Problem 6.38, part (i), the main effects, and several two and three factor interactions were found to be very significant. In the 2^{4-1} fractional factorial experiment in this problem, the main effects are aliased with the three factor interactions, and the two factor interactions are aliased with each other. Based on the results of Problem 6.38, part (i), these aliases are a significant concern.

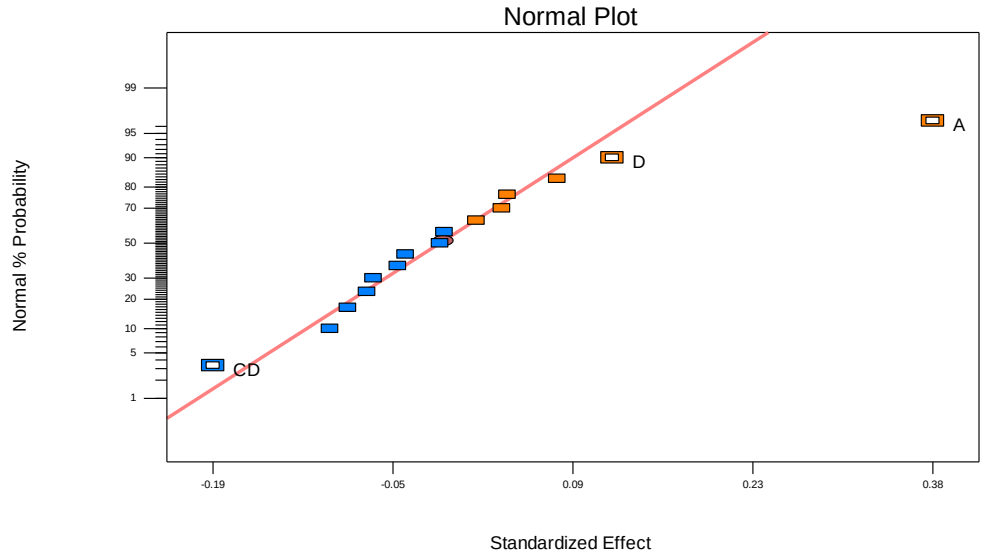
8.28. An article in the *Journal of Chromatography A* ("Simultaneous Supercritical Fluid Derivatization and Extraction of Formaldehyde by the Hantzsch Reaction," 2000, Vol. 896, pp. 51-59) describes an experiment where the Hantzsch reaction is used to produce the chemical derivatization of formaldehyde in a supercritical medium. Pressure, temperature, and other parameters such as static and dynamic extraction time must be optimized to increase the yield of this kinetically controlled reaction. A 2^{5-1} fractional factorial design with one center run was used to study the significant parameters affecting the supercritical

process in terms of resolution and sensitivity. Ultraviolet-visible spectrophotometry was used as the detection technique. The experimental design and the responses are shown in the table below:

Experiment	*P* (MPa)	*T* (°C)	*s* (min)	*d* (min)	*c* (µl)	Resolution	Sensitivity
1	13.8	50	2	2	100	0.00025	0.057
2	55.1	50	2	2	10	0.33333	0.094
3	13.8	120	2	2	10	0.02857	0.017
4	55.1	120	2	2	100	0.20362	1.561
5	13.8	50	15	2	10	0.00027	0.010
6	55.1	50	15	2	100	0.52632	0.673
7	13.8	120	15	2	100	0.00026	0.028
8	55.1	120	15	2	10	0.52632	1.144
9	13.8	50	2	15	10	0.42568	0.142
10	55.1	50	2	15	100	0.60150	0.399
11	13.8	120	2	15	100	0.06098	0.767
12	55.1	120	2	15	10	0.74165	1.086
13	13.8	50	15	15	100	0.08780	0.252
14	55.1	50	15	15	10	0.40000	0.379
15	13.8	120	15	15	10	0.00026	0.028
16	55.1	120	15	15	100	0.28091	3.105
Central	34.5	85	8.5	8.5	55	0.75000	1.836

(a) Analyze the data from this experiment and draw conclusions.

The first analysis shown below is for the Resolution response. Factor *A* and the *CD* interaction are significant with factor *D* being moderately significant. Due to hierarchy, factor *C* is also included in the model. Curvature is also significant.

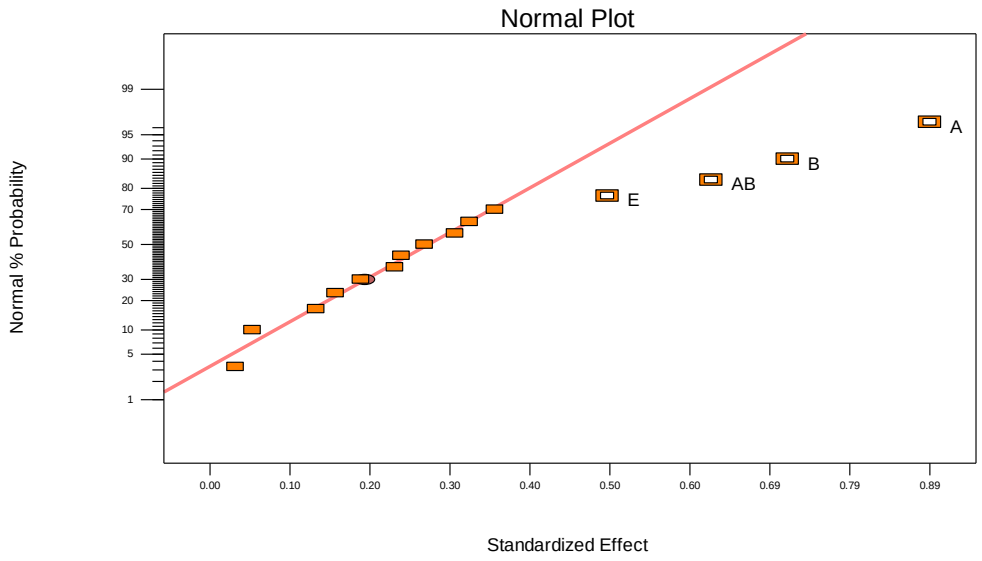


Design Expert Output

Response 1 Resolution
ANOVA for selected factorial model
Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	0.80	4	0.20	15.37	0.0002	significant
A-P	*0.57*	*1*	*0.57*	*43.70*	*< 0.0001*	
C-s	*0.021*	*1*	*0.021*	*1.59*	*0.2339*	
D-d	*0.060*	*1*	*0.060*	*4.63*	*0.0544*	
CD	*0.15*	*1*	*0.15*	*11.56*	*0.0059*	
Curvature	0.22	1	0.22	17.19	0.0016	significant
Residual	0.14	11	0.013			
Cor Total	1.16	16				

The following analysis is for the Sensitivity response. Factors *A*, *B*, and the *AB* interaction are significant, with factor *E* as being moderately significant. Curvature is also significant.



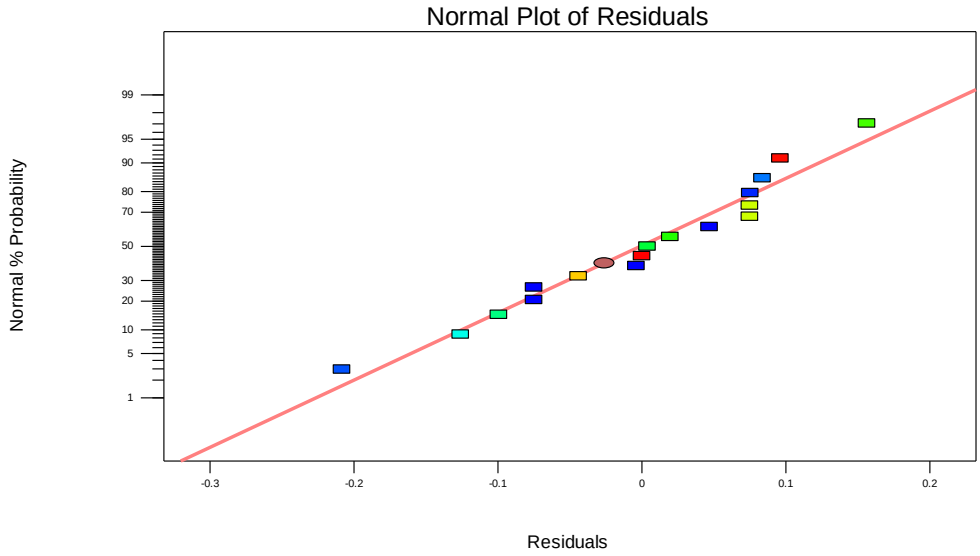
Design Expert Output

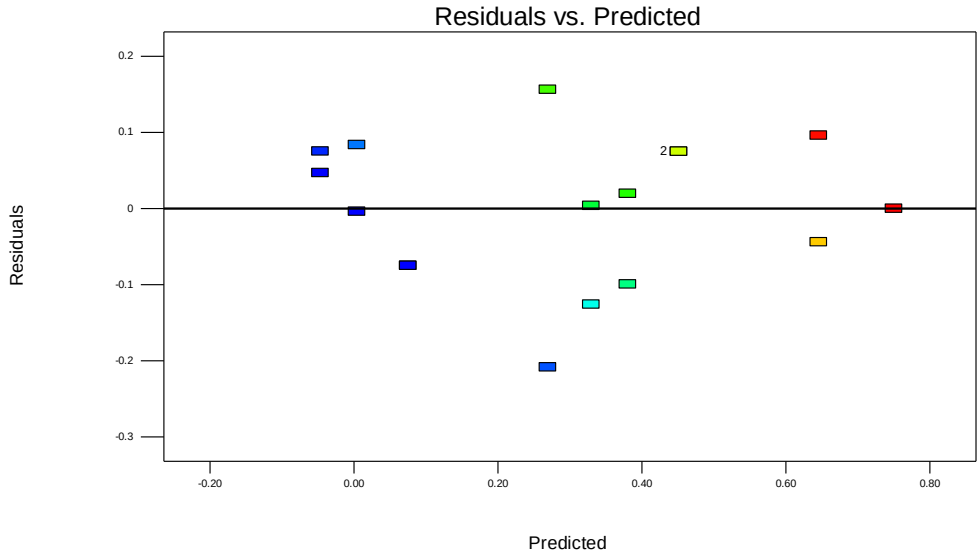
Response 1 Sensitivity
ANOVA for selected factorial model
Analysis of variance table [Partial sum of squares - Type III]

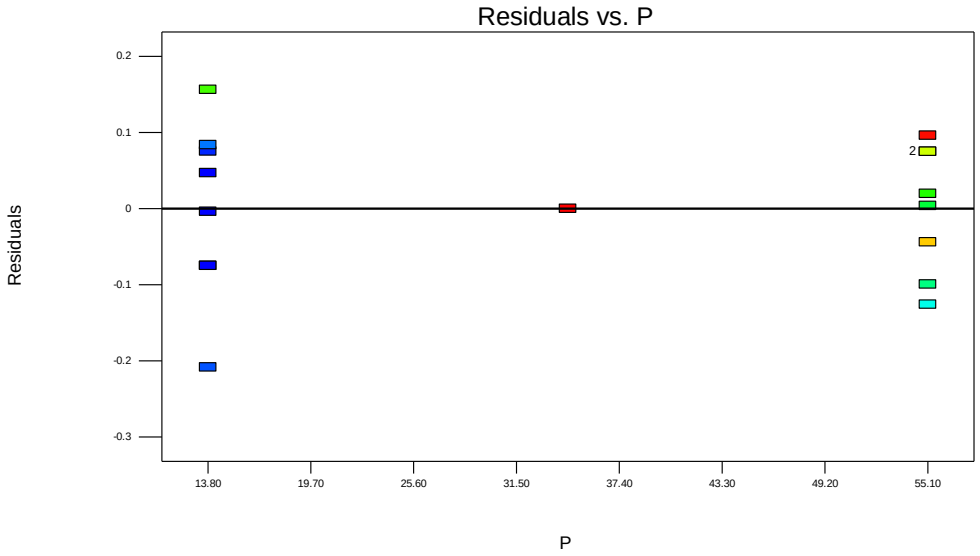
Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	7.75	4	1.94	9.19	0.0016	significant
A-P	*3.19*	*1*	*3.19*	*15.10*	*0.0025*	
B-T	*2.05*	*1*	*2.05*	*9.72*	*0.0098*	
E-c	*0.97*	*1*	*0.97*	*4.60*	*0.0551*	
AB	*1.55*	*1*	*1.55*	*7.32*	*0.0204*	
Curvature	1.42	1	1.42	6.72	0.0251	significant
Residual	2.32	11	0.21			
Cor Total	11.49	16				

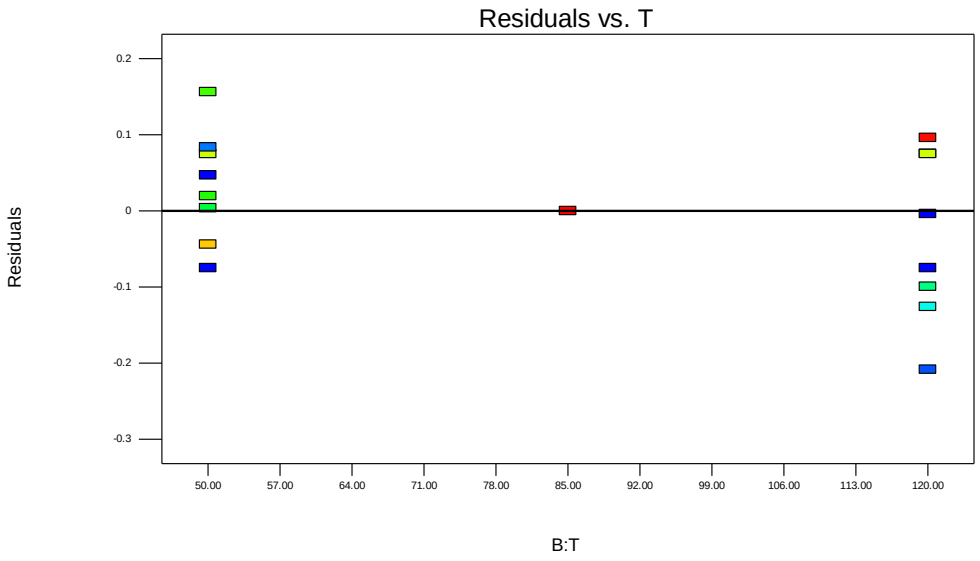
(b) Are there any concerns about model adequacy or violations of the assumptions?

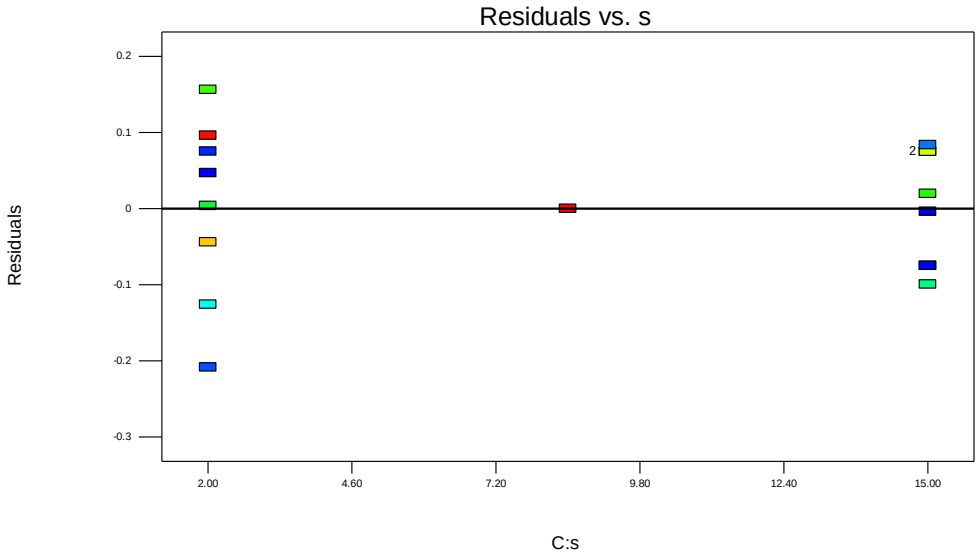
The first set of plots shown below are the residuals for the Resolution response. There does not appear to be any concerns with these residual plots.

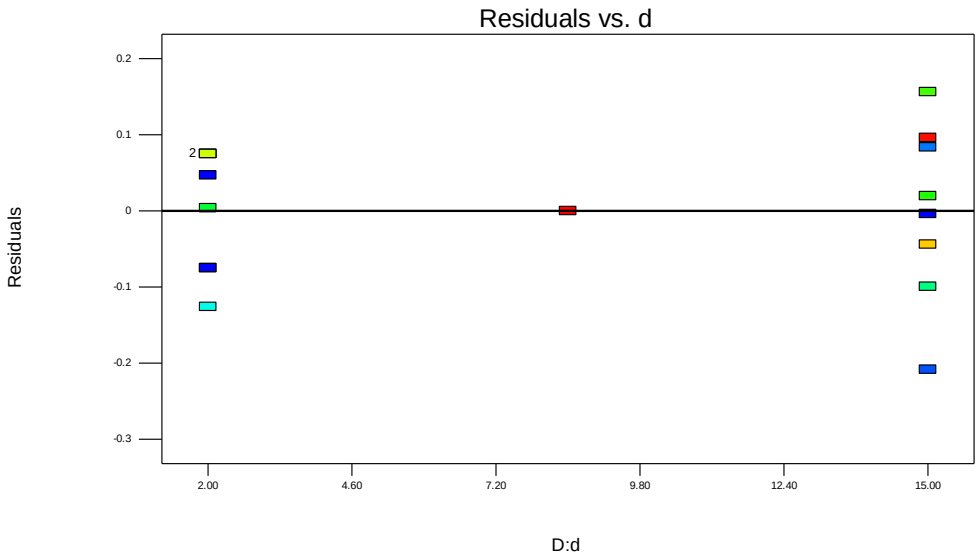


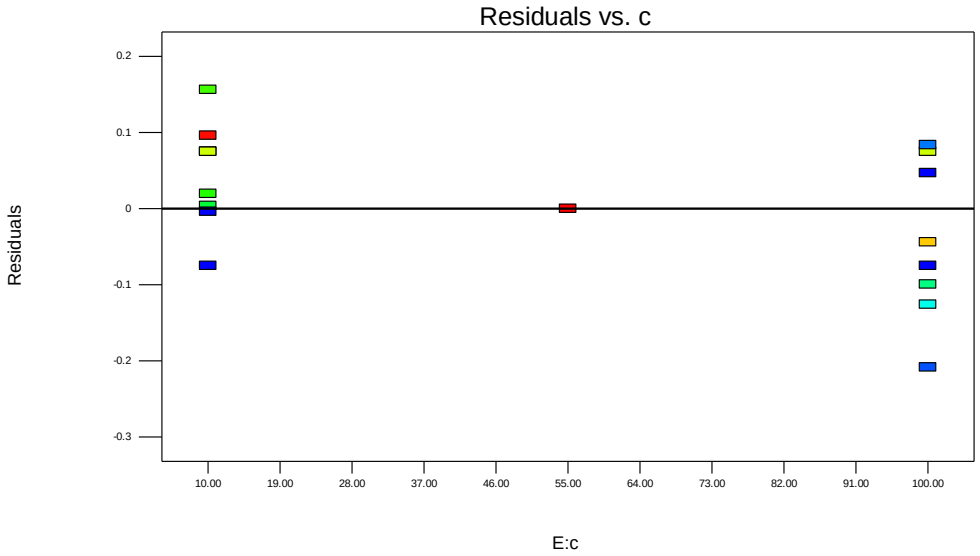



Residuals vs. P
Residuals
0.2
0.1
0
-0.1
-0.2
-0.3
2
13.80
19.70
25.60
31.50
37.40
43.30
49.20
55.10
P

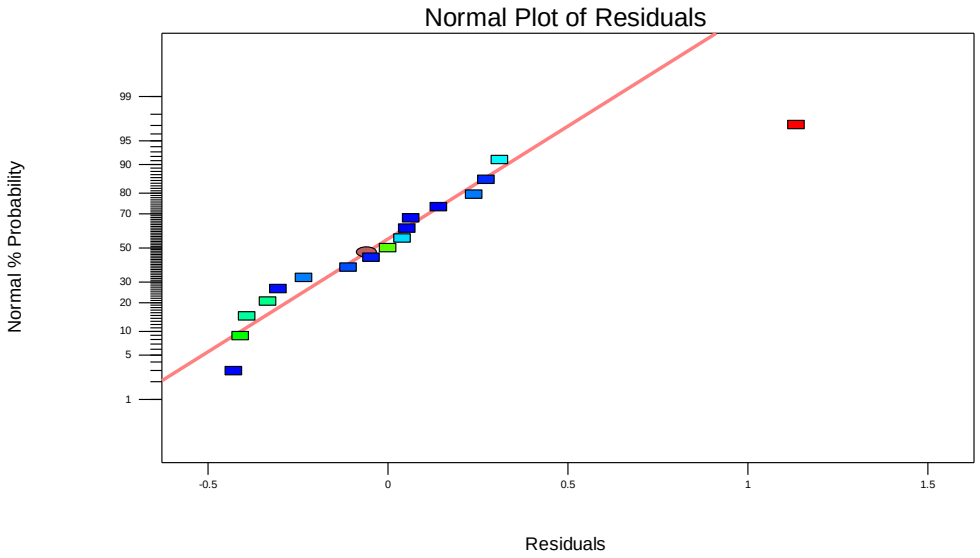

Residuals vs. T
Residuals
0.2
0.1
0
-0.1
-0.2
-0.3
50.00
57.00
64.00
71.00
78.00
85.00
92.00
99.00
106.00
113.00
120.00
B:T

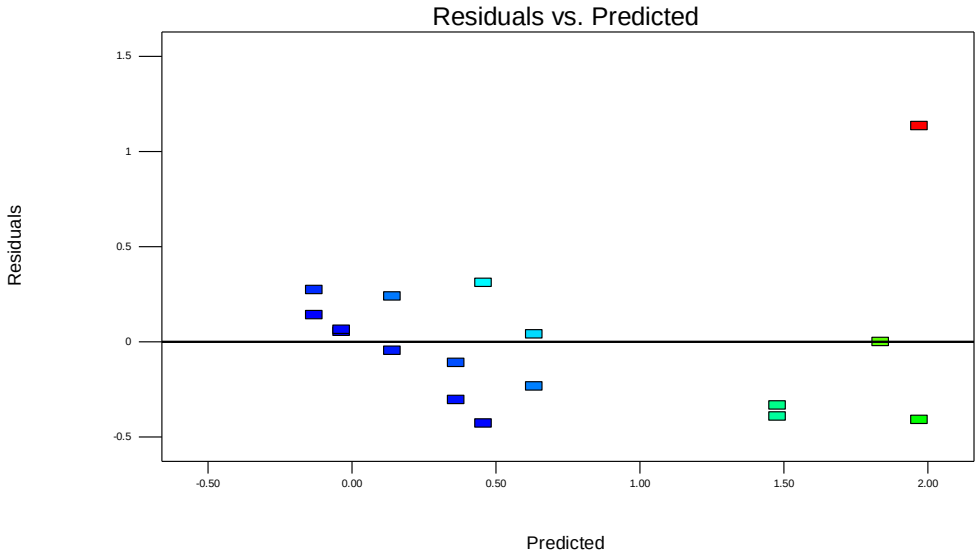

Residuals vs. s
Residuals
0.2
0.1
0
-0.1
-0.2
-0.3
2
2.00
4.60
7.20
9.80
12.40
15.00
C:s

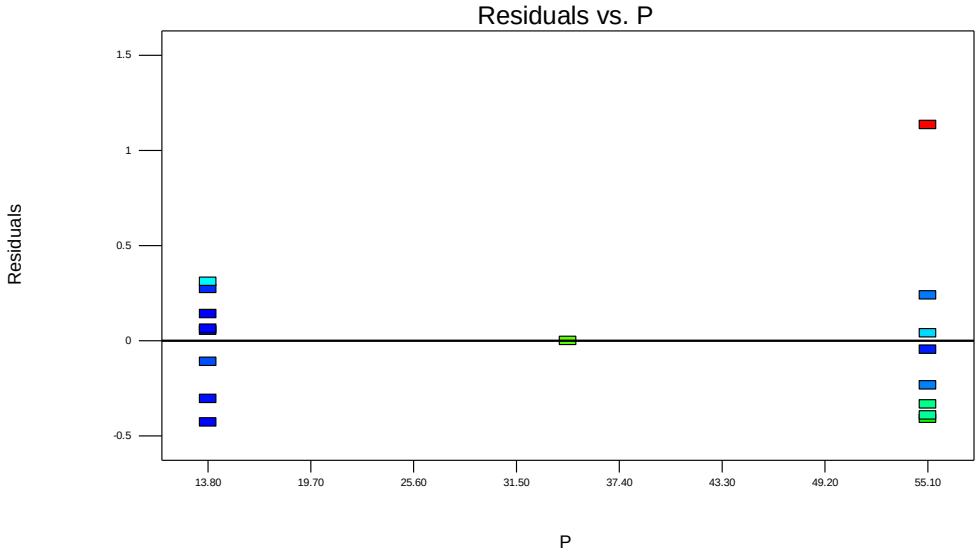

Residuals vs. d
Residuals
0.2
0.1
0
-0.1
-0.2
-0.3
2
2.00
4.60
7.20
9.80
12.40
15.00
D:d

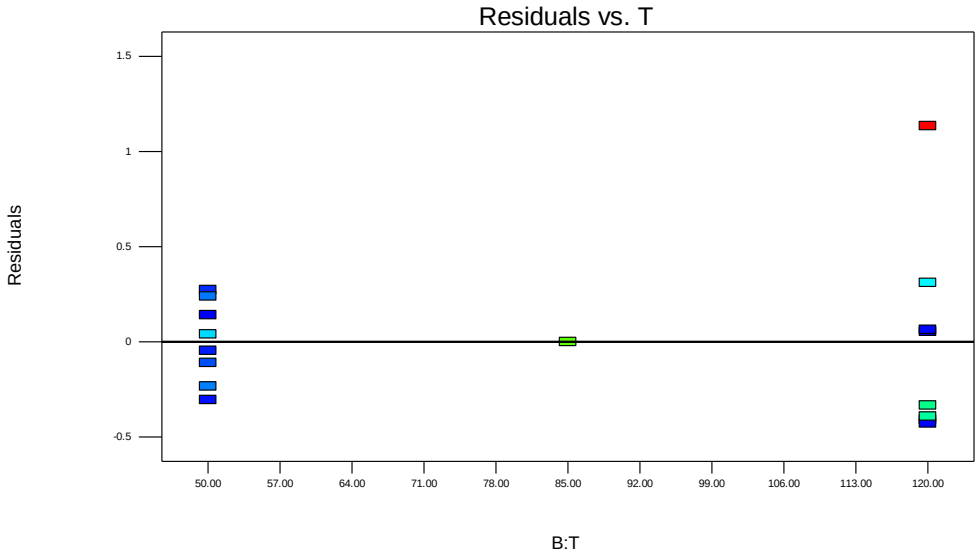


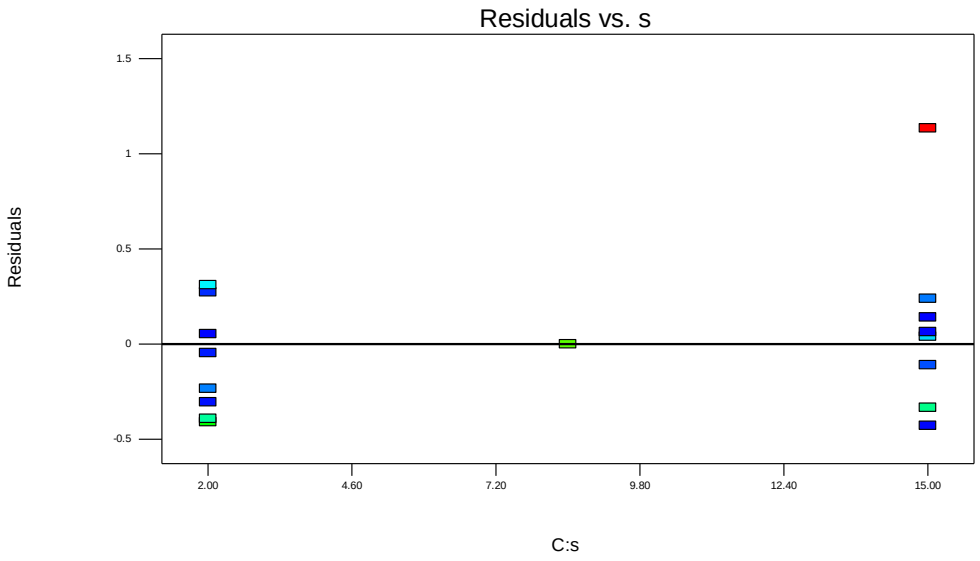
Residual plots for the Sensitivity response are shown below. There appears to be an outlier with the 3.105 Sensitivity value.

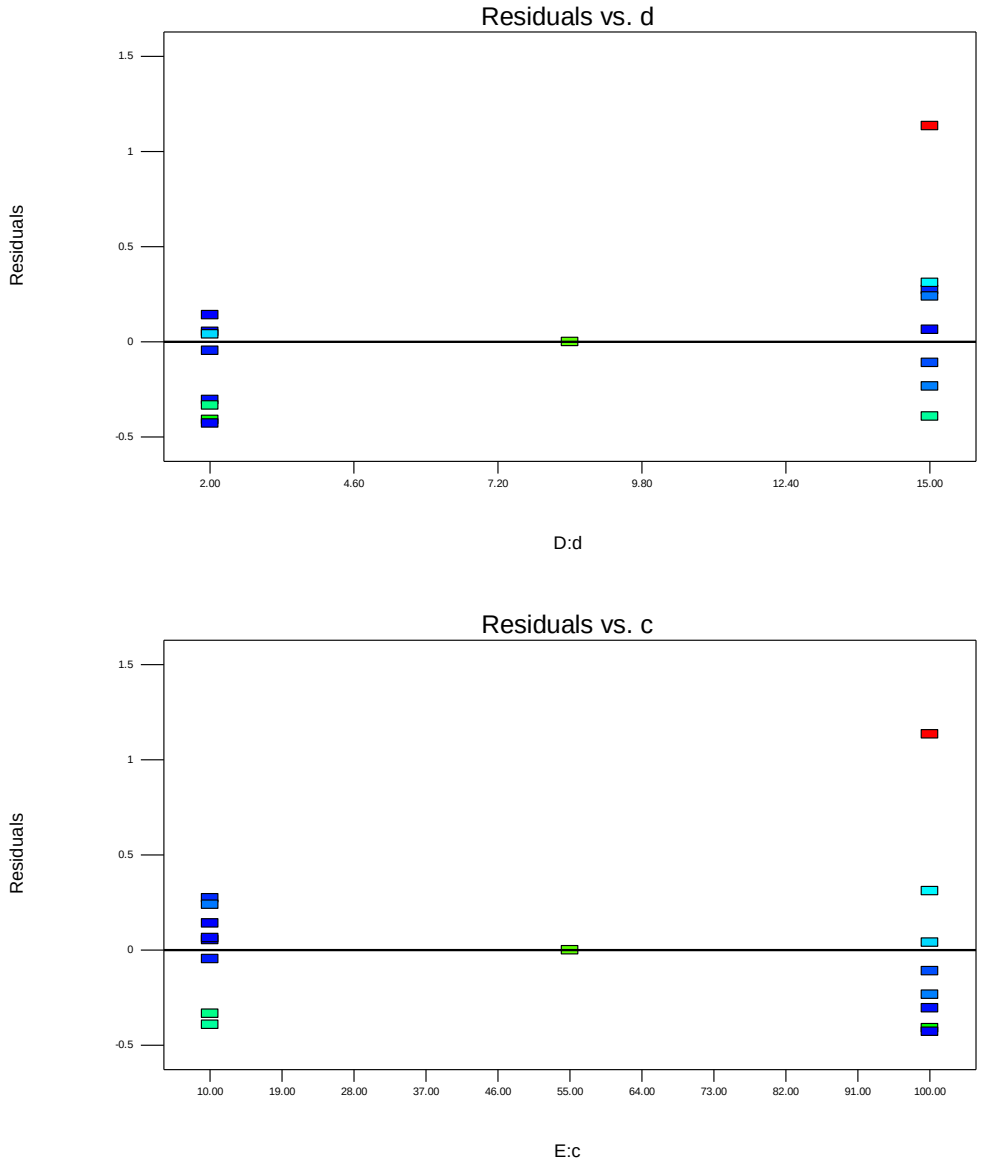



Residuals vs. Predicted
Residuals
1.5
1
0.5
0
-0.5
-0.50
0.00
0.50
1.00
1.50
2.00
Predicted


Residuals vs. P
Residuals
1.5
1
0.5
0
-0.5
13.80
19.70
25.60
31.50
37.40
43.30
49.20
55.10
P


Residuals vs. T
Residuals
1.5
1
0.5
0
-0.5
50.00
57.00
64.00
71.00
78.00
85.00
92.00
99.00
106.00
113.00
120.00
B:T


Residuals vs. s
Residuals
1.5
1
0.5
0
-0.5
2.00
4.60
7.20
9.80
12.40
15.00
C:s



(c) Does the single center point cause any concerns about curvature or the possible need for second-order terms?

Yes, for both Resolution and Sensitivity, there appears to be curvature. Additional runs, such as the axial points for a central composite design, along with additional center points for blocking, could be run to identify the second-order terms.

(d) Do you think that running one center point was a good choice in this design?

It was a good idea to run a center point; however, additional center points would have been an improvement, especially if there was potential need to perform additional runs in a second block to identify the second-order terms.

8.29. An article in *Soldering & Surface Mount Technology* ("Characterization of Solder Paste Printing Process and Its Optimization," 1999, Vol.11, No. 3, pp. 23-26) describes the use of a 2^{8-3} fractional factorial experiment to study the effect of eight factors on two responses: percentage volume matching (PVM) – the ration of the actual printed solder paste volume to the designed volume; and non-conformities per unit (NPU) – the number of solder paste printing defects determined by visual inspection (20' magnification) after printing according to an industry workmanship standard. The factor levels are shown below and the test matrix and response data are shown in Table 8.9.

	Levels	
Parameters	(–)	(+)
A. Squeegee pressure, *MPa*	0.1	0.3
B. Printing speed, *mm/s*	24	32
C. Squeegee angle, *deg*	45	65
D. Temperature, *°C*	20	28
E. Viscosity, *kCps*	1,100-1,150	1,250-1,300
F. Cleaning interval, *stroke*	8	15
G. Separation speed, *mm/s*	0.4	0.8
H. Relative humidity, *%*	30	70

Table P8.9 – The Solder Paste Experiment

Run	Parameters									
Order	A	B	C	D	E	F	G	H	PVM	NPU (%)
4	–	–	–	–	–	–	–	+	1.00	5
13	+	–	–	–	–	+	+	+	1.04	13
6	–	+	–	–	–	+	+	–	1.02	16
3	+	+	–	–	–	–	–	–	0.99	12
19	–	–	+	–	–	+	–	–	1.02	15
25	+	–	+	–	–	–	+	–	1.01	9
21	–	+	+	–	–	–	+	+	1.01	12
14	+	+	+	–	–	+	–	+	1.03	17
10	–	–	–	+	–	–	+	–	1.04	21
22	+	–	–	+	–	+	–	–	1.14	20
1	–	+	–	+	–	+	–	+	1.20	25
2	+	+	–	+	–	–	+	+	1.13	21
30	–	–	+	+	–	+	+	+	1.14	25
8	+	–	+	+	–	–	–	+	1.07	13
9	–	+	+	+	–	–	–	–	1.06	20
20	+	+	+	+	–	+	+	–	1.13	26
17	–	–	–	–	+	–	–	–	1.02	10
18	+	–	–	–	+	+	+	–	1.10	13
5	–	+	–	–	+	+	+	+	1.09	17
26	+	+	–	–	+	–	–	+	0.96	13
31	–	–	+	–	+	+	–	+	1.02	14
11	+	–	+	–	+	–	+	+	1.07	11
29	–	+	+	–	+	–	+	–	0.98	10
23	+	+	+	–	+	+	–	–	0.95	14
32	–	–	–	+	+	–	+	+	1.10	28
7	+	–	–	+	+	+	–	+	1.12	24
15	–	+	–	+	+	+	–	–	1.19	22
27	+	+	–	+	+	–	+	–	1.13	15
12	–	–	+	+	+	+	+	–	1.20	21
28	+	–	+	+	+	–	–	–	1.07	19
24	–	+	+	+	+	–	–	+	1.12	21
16	+	+	+	+	+	+	+	+	1.21	27

(a) Verify that the generators are $I = ABCF$, $I = ABDG$, and $I = BCDEH$ for this design.

This can be done by recognizing that Column *F* = *ABC*, Column *G* = *ABD*, and Column *H* = *BCDE*.

(b) What are the aliases for the main effects and two-factor interactions? You can ignore all interactions order three and higher.

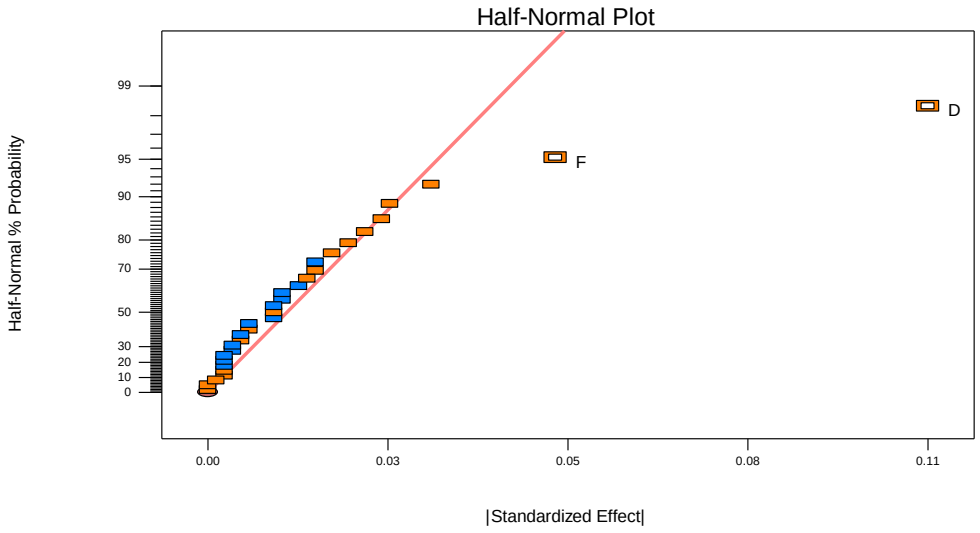
The full defining relationship is: I + ABCF + ABDG + CDFG + ADEFH + ACEGH + BCDEH; and the alias structure for the main effects and two-factor interactions are:

```
A + BCF + BDG + CEGH + DEFH
B + ACF + ADG + CDEH + EFGH
C + ABF + DFG + AEGH + BDEH
D + ABG + CFG + AEFH + BCEH
E + ACGH + ADFH + BCDH + BFGH
F + ABC + CDG + ADEH + BEGH
G + ABD + CDF + ACEH + BEFH
H + ACEG + ADEF + BCDE + BEFG
AB + CF + DG
AC + BF + EGH + ADFG + BCDG
AD + BG + EFH + ACFG + BCDF
AE + CGH + DFH + BCEF + BDEG
AF + BC + DEH + ACDG + BDFG
AG + BD + CEH + ACDF + BCFG
AH + CEG + DEF + BCFH + BDGH
BE + CDH + FGH + ACEF + ADEG
BH + CDE + EFG + ACFH + ADGH
CD + FG + BEH + ABCG + ABDF
CE + AGH + BDH + ABEF + DEFG
CG + DF + AEH + ABCD + ABFG
CH + AEG + BDE + ABFH + DFGH
DE + AFH + BCH + ABEG + CEFG
DH + AEF + BCE + ABGH + CFGH
EF + ADH + BGH + ABCE + CDEG
EG + ACH + BFH + ABDE + CDEF
EH + ACG + ADF + BCD + BFG
FH + ADE + BEG + ABCH + CDGH
GH + ACE + BEF + ABDH + CDFH
```

(c) Analyze both the PVM and NPU responses.

For both the PVM and NPU responses, only factors D and F appear to be significant.

The PVM analysis is shown below.



Design Expert Output

Response 1 PVM

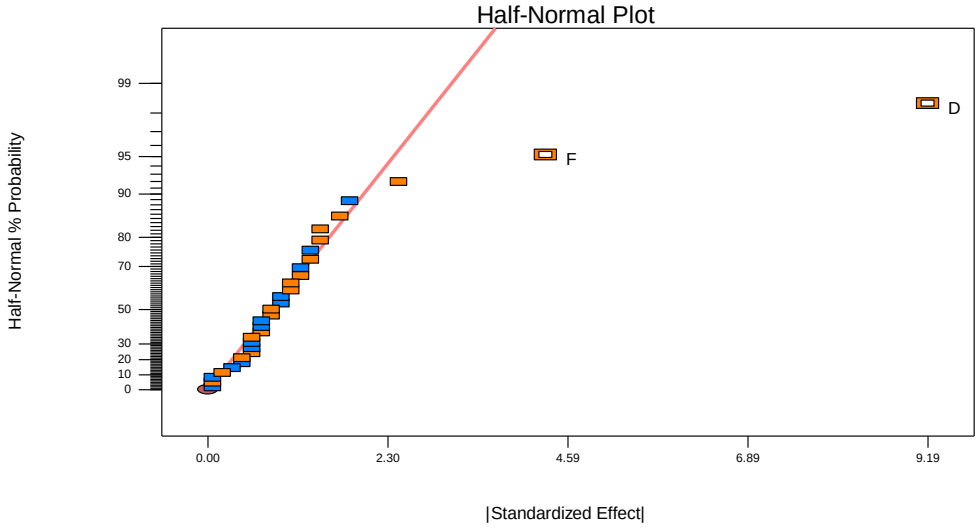
ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	0.12	2	0.058	37.52	< 0.0001	significant
D-Temperature	*0.095*	*1*	*0.095*	*60.85*	*< 0.0001*	
F-Cleaning interval	*0.022*	*1*	*0.022*	*14.18*	*0.0008*	
Residual	0.045	29	1.555E-003			
Cor Total	0.16	31				

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	1.07	1	6.970E-003	1.06	1.09	
D-Temperature	0.054	1	6.970E-003	0.040	0.069	1.00
F-Cleaning interval	0.026	1	6.970E-003	0.012	0.041	1.00

The analysis for the NPU response is shown below.



Design Expert Output

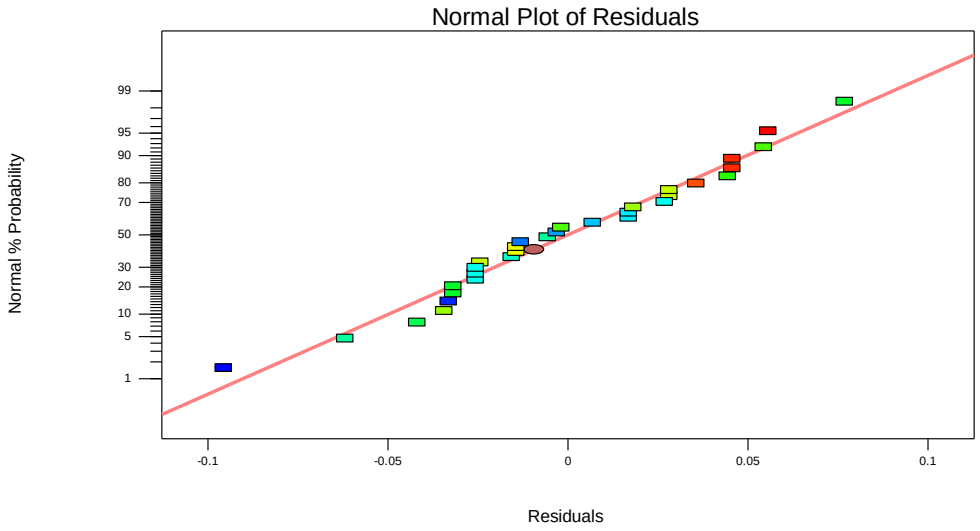
Response 2 NPU
ANOVA for selected factorial model
Analysis of variance table [Partial sum of squares - Type III]

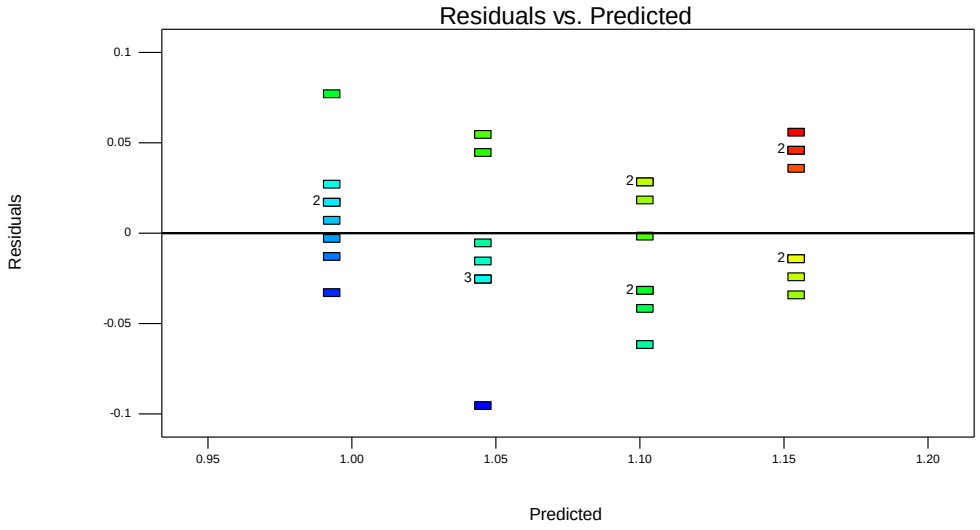
Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	824.06	2	412.03	48.15	< 0.0001	significant
D-Temperature	*675.28*	*1*	*675.28*	*78.91*	*< 0.0001*	
F-Cleaning interval	*148.78*	*1*	*148.78*	*17.39*	*0.0003*	
Residual	248.16	29	8.56			
Cor Total	1072.22	31				

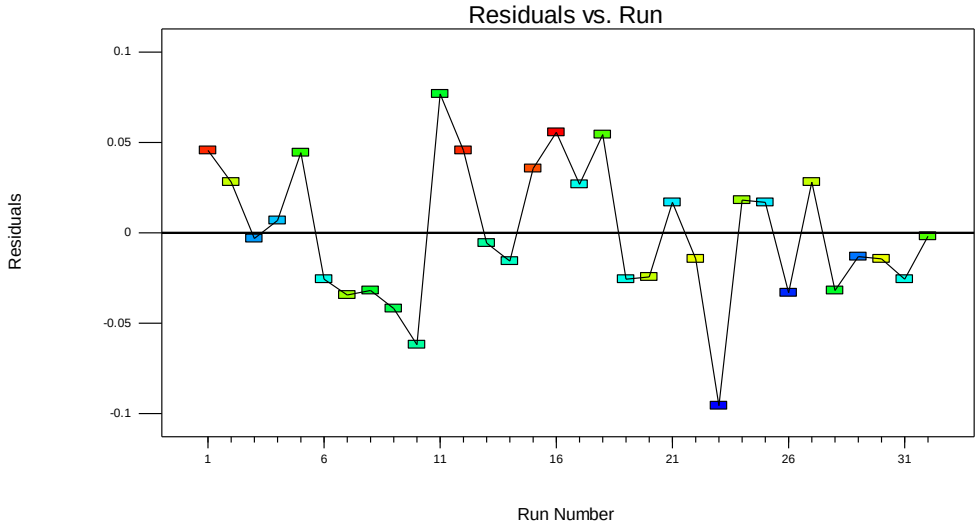
Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	17.16	1	0.52	16.10	18.21	
D-Temperature	4.59	1	0.52	3.54	5.65	1.00
F-Cleaning interval	2.16	1	0.52	1.10	3.21	1.00

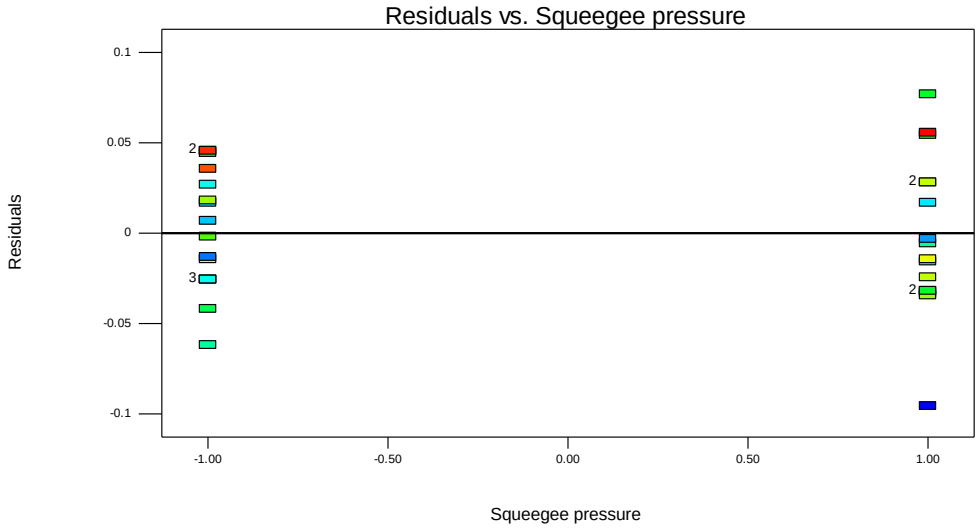
(d) Analyze the residuals for both responses. Are there any problems with model adequacy?

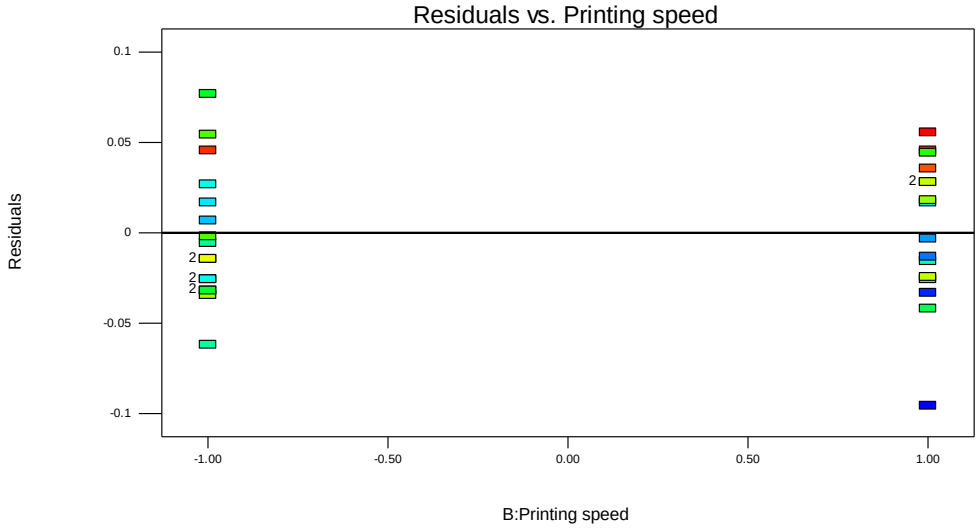
The residual plots for the PVM response shown below do not identify any concerns.

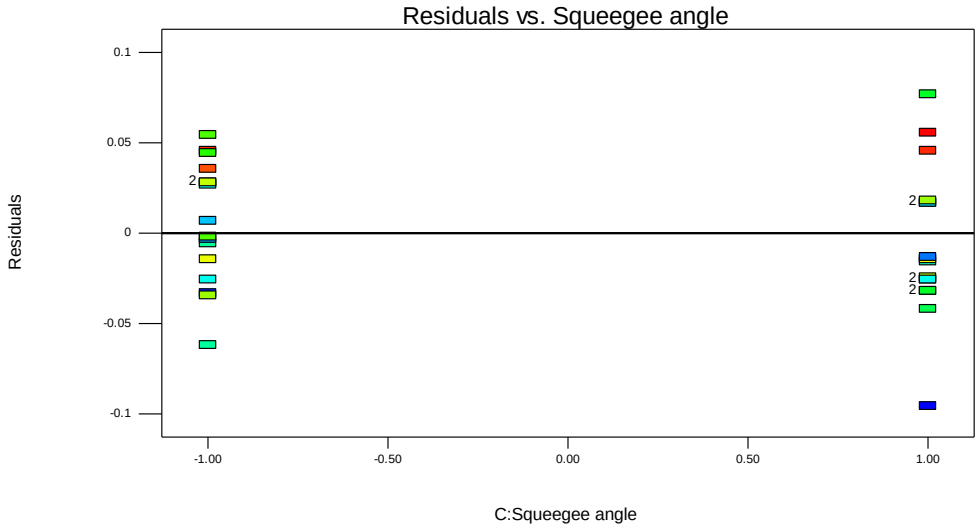

Normal Plot of Residuals
Normal % Probability
99
95
90
80
70
50
30
20
10
5
1
-0.1
-0.05
0
0.05
0.1
Residuals

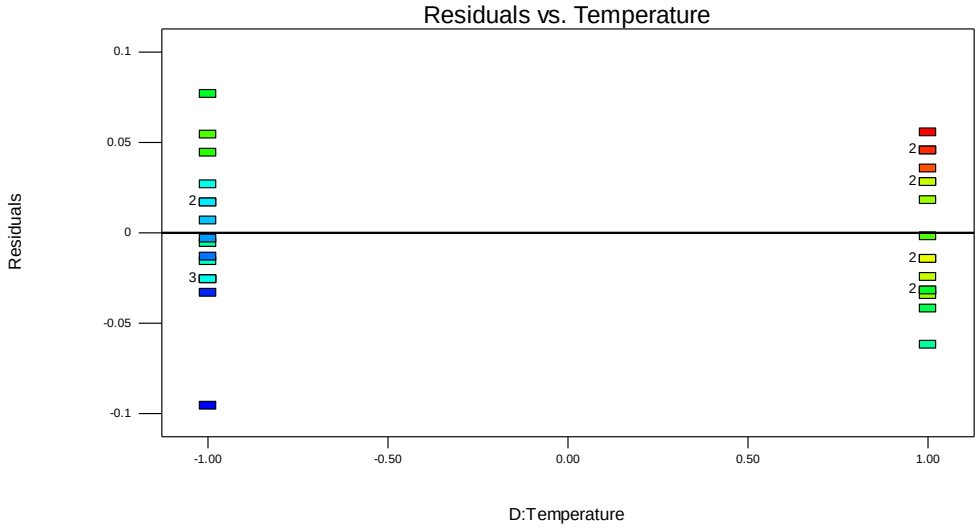

Residuals vs. Predicted
Residuals
0.1
0.05
0
-0.05
-0.1
0.95
1.00
1.05
1.10
1.15
1.20
Predicted

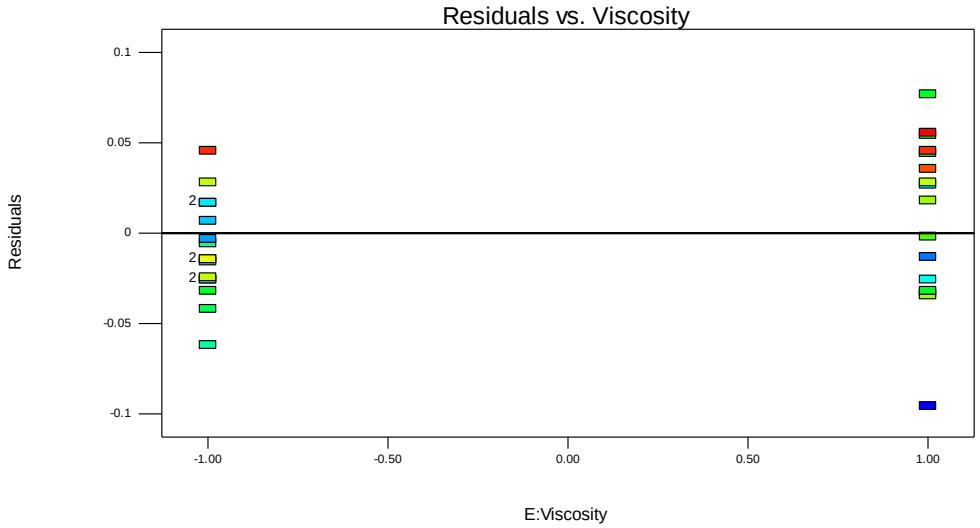

Residuals vs. Run
Residuals
0.1
0.05
0
-0.05
-0.1
1
6
11
16
21
26
31
Run Number

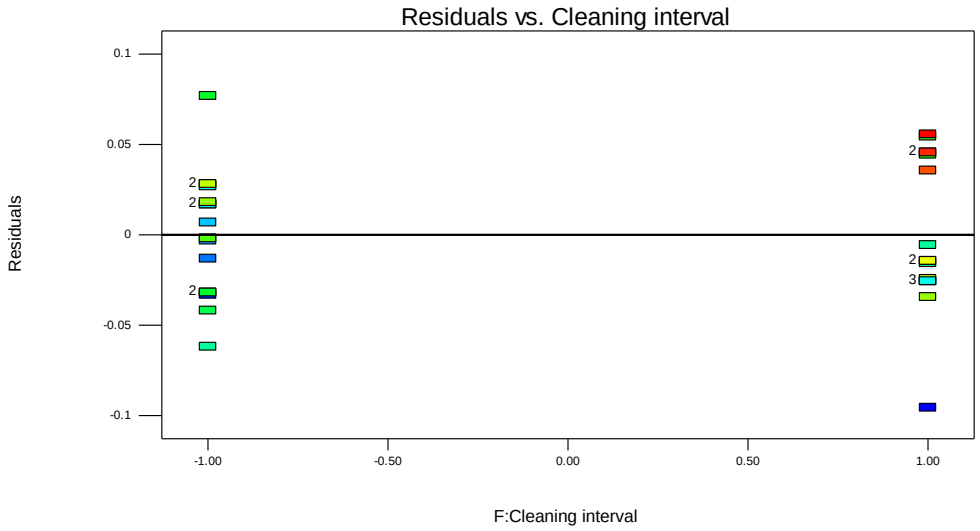

Residuals vs. Squeegee pressure
Residuals
0.1
0.05
0
-0.05
-0.1
2
3
2
2
-1.00
-0.50
0.00
0.50
1.00
Squeegee pressure

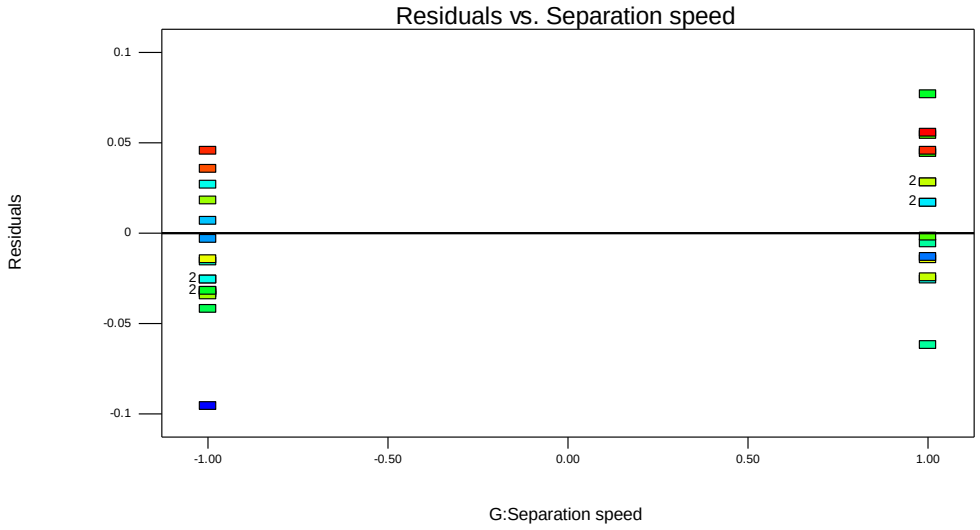

Residuals vs. Printing speed
Residuals
0.1
0.05
0
-0.05
-0.1
-1.00
-0.50
0.00
0.50
1.00
B:Printing speed

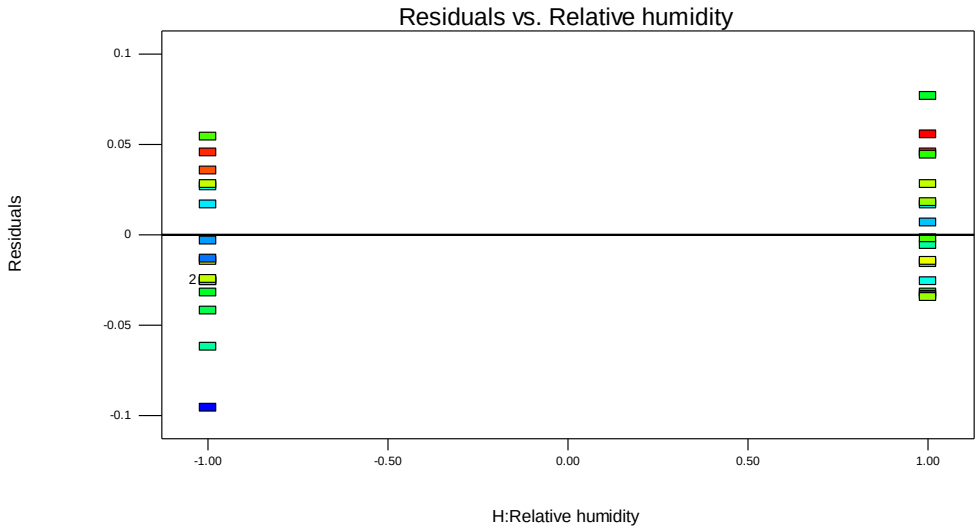

Residuals vs. Squeegee angle
Residuals
0.1
0.05
0
-0.05
-0.1
-1.00
-0.50
0.00
0.50
1.00
C:Squeegee angle


Residuals vs. Temperature
Residuals
0.1
0.05
0
-0.05
-0.1
-1.00
-0.50
0.00
0.50
1.00
D:Temperature

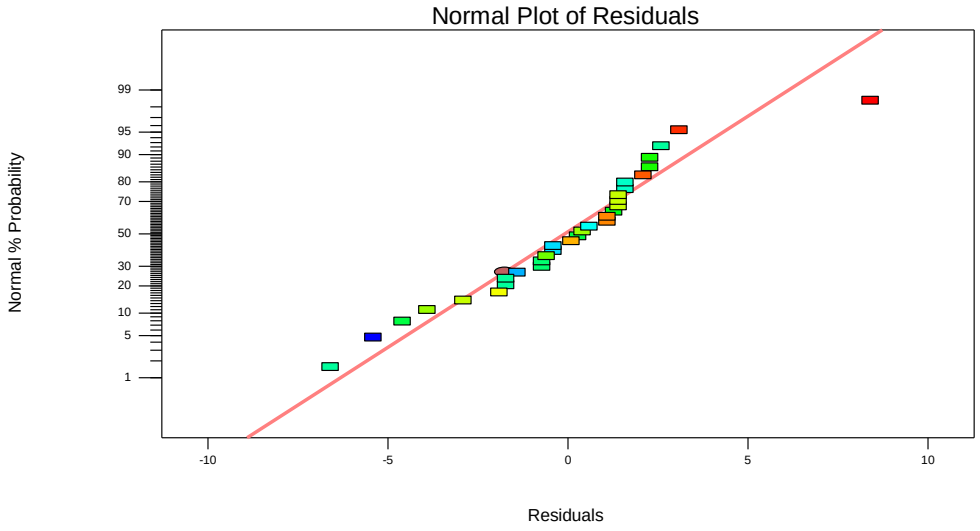

Residuals vs. Viscosity
Residuals
0.1
0.05
0
-0.05
-0.1
-1.00
-0.50
0.00
0.50
1.00
E:Viscosity

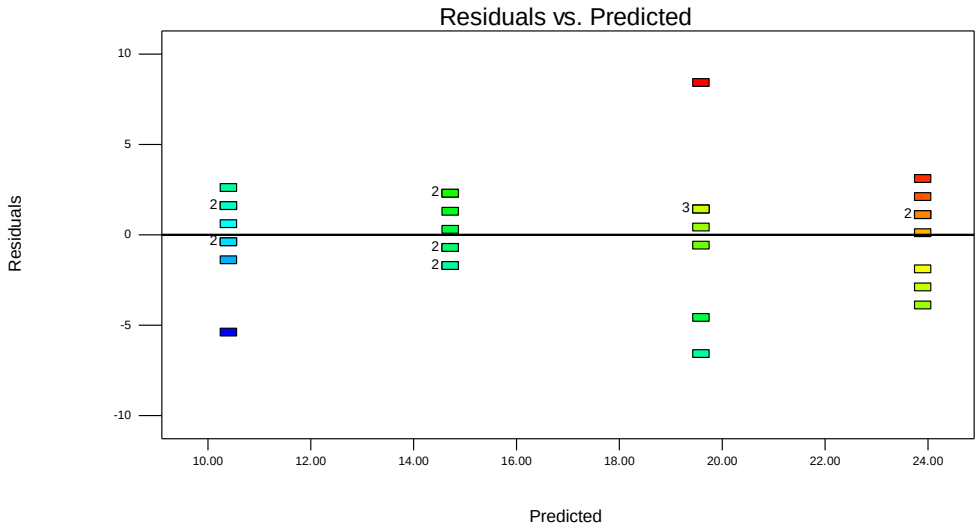

Residuals vs. Cleaning interval
Residuals
0.1
0.05
0
-0.05
-0.1
-1.00
-0.50
0.00
0.50
1.00
F:Cleaning interval

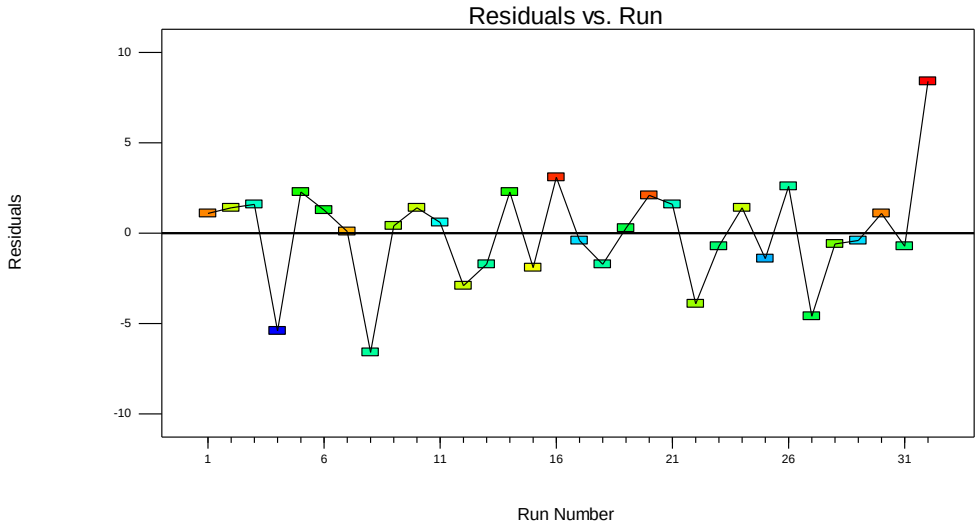

Residuals vs. Separation speed
Residuals
0.1
0.05
0
-0.05
-0.1
-1.00
-0.50
0.00
0.50
1.00
G:Separation speed

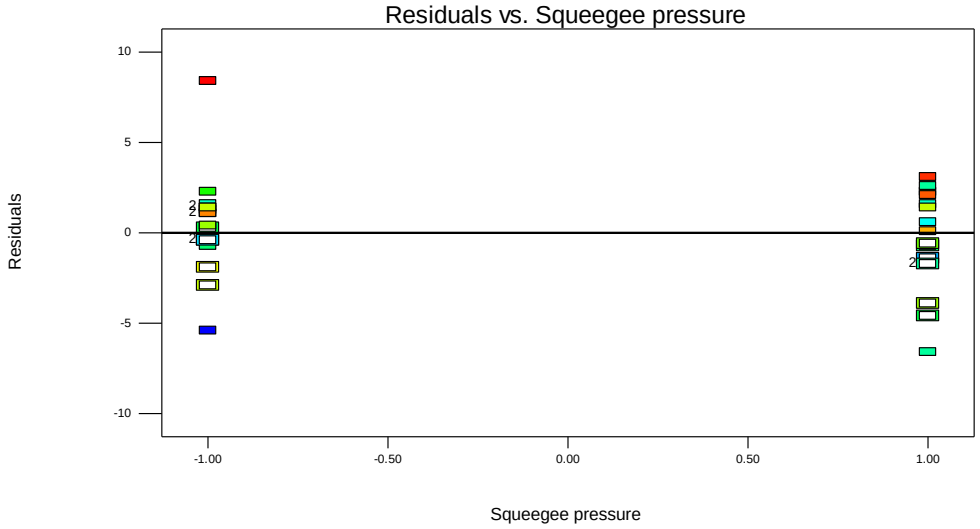


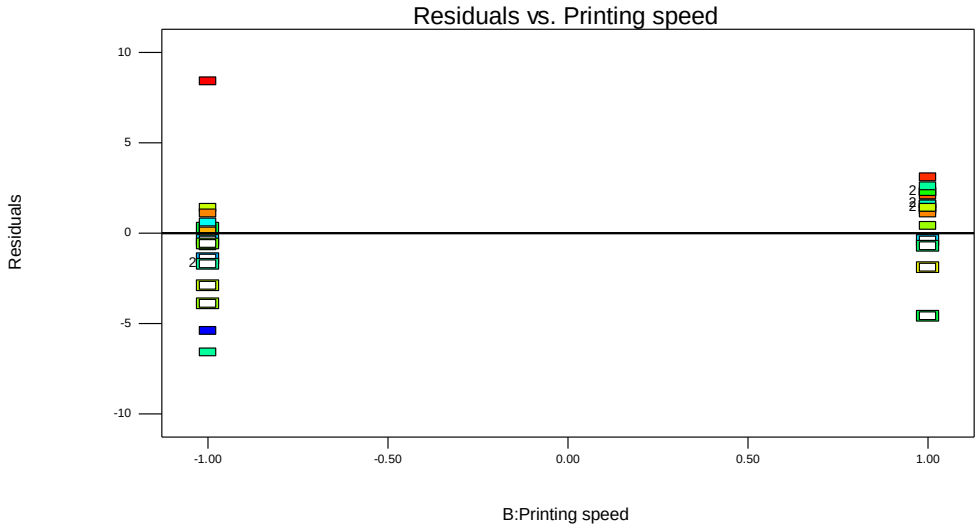
The residual plots for the NPU response shown below identify an outlier with run number 32, NPU value of 28. The experimenters should investigate this run.

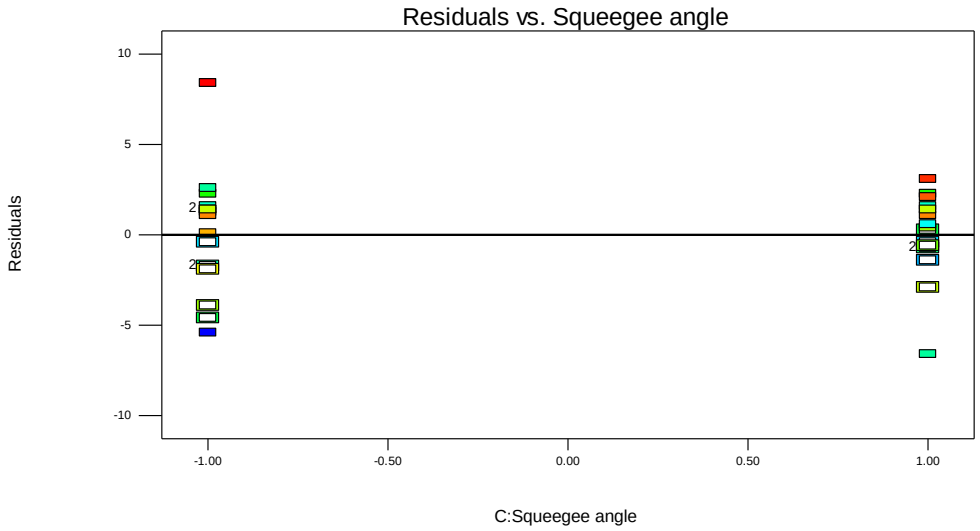


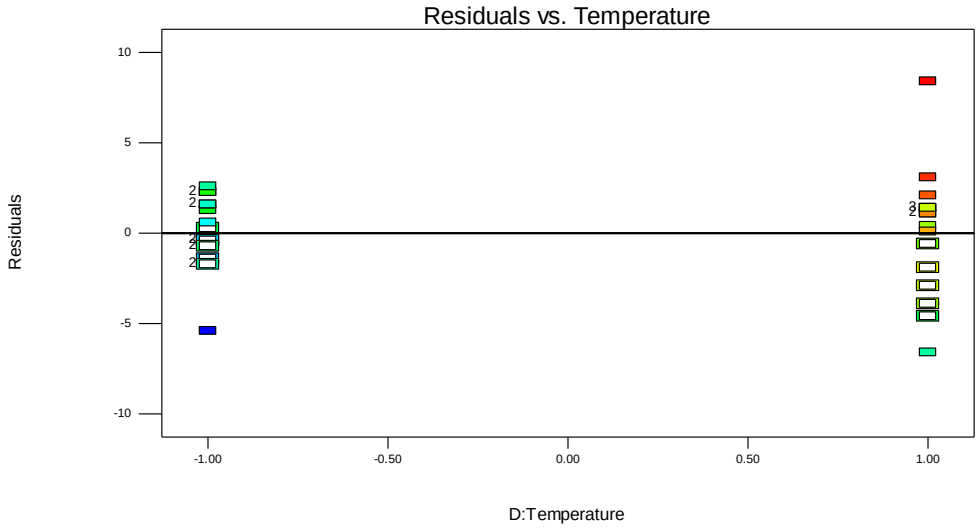

Residuals vs. Predicted
Residuals
Predicted
10
5
0
-5
-10
10.00
12.00
14.00
16.00
18.00
20.00
22.00
24.00

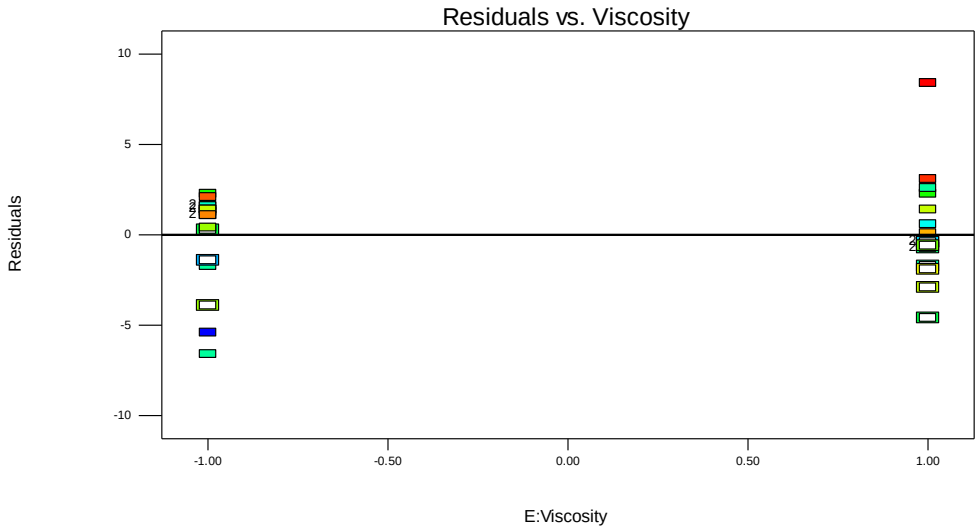

Residuals vs. Run
Residuals
Run Number
10
5
0
-5
-10
1
6
11
16
21
26
31

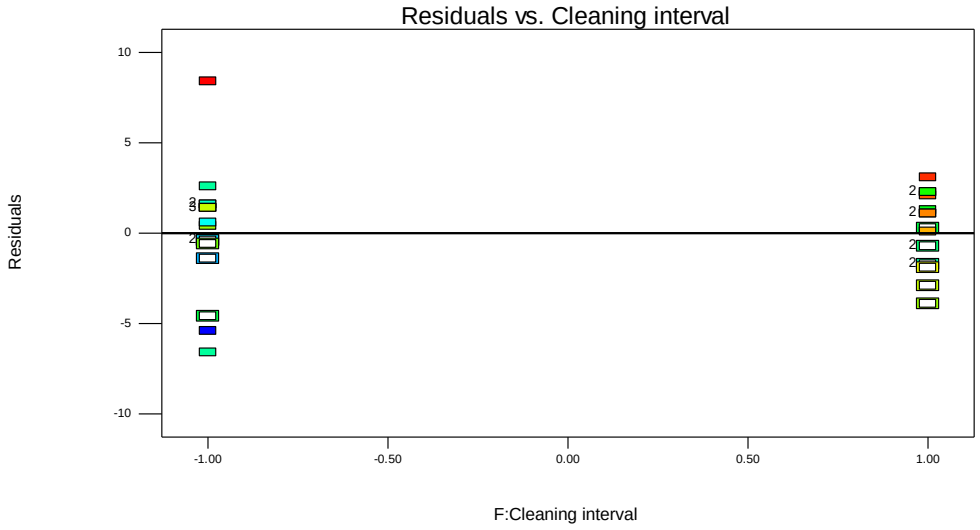

Residuals vs. Squeegee pressure
Residuals
10
5
0
-5
-10
-1.00
-0.50
0.00
0.50
1.00
Squeegee pressure

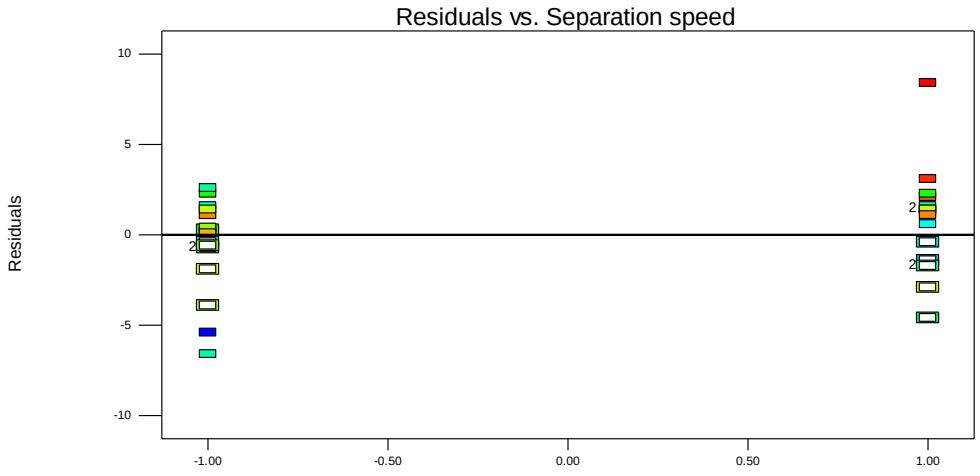

Residuals vs. Printing speed
Residuals
10
5
0
-5
-10
-1.00
-0.50
0.00
0.50
1.00
B:Printing speed

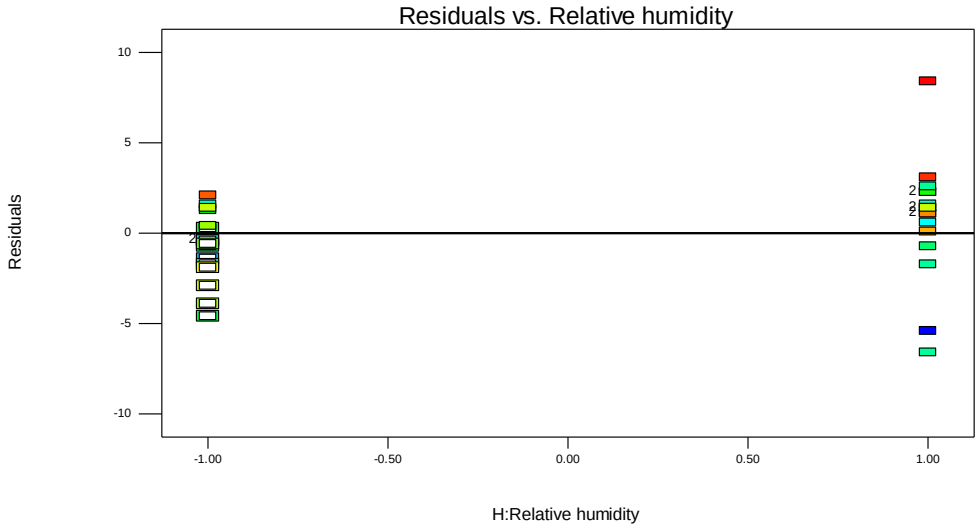

Residuals vs. Squeegee angle
Residuals
10
5
0
-5
-10
-1.00
-0.50
0.00
0.50
1.00
C:Squeegee angle


Residuals vs. Temperature
Residuals
10
5
0
-5
-10
-1.00
-0.50
0.00
0.50
1.00
D:Temperature


Residuals vs. Viscosity
Residuals
10
5
0
-5
-10
-1.00
-0.50
0.00
0.50
1.00
E:Viscosity


Residuals vs. Cleaning interval
Residuals
10
5
0
-5
-10
-1.00
-0.50
0.00
0.50
1.00
F:Cleaning interval





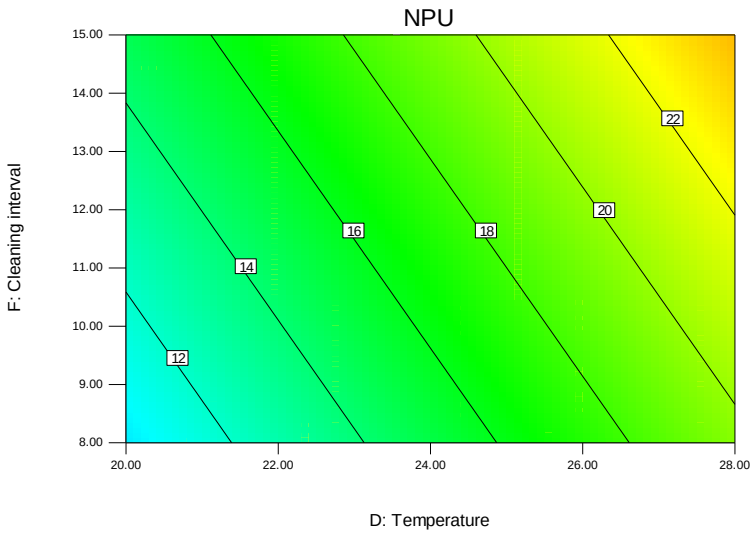
(e) The ideal value of PVM is unity and the NPU response should be as small as possible. Recommend suitable operating conditions for the process based on the experimental results.

The contour plots of both the PVM and NPU responses show that running the Temperature at 20C and the Cleaning Interval at 8 is the best operating conditions. This is confirmed with the desirability plot also shown below. The other factors should be set based on process controls, costs, robustness, or other process performance requirements, as they are not significant to either the PVM or NPU response.

Design-Expert® Software
Factor Coding: Actual
NPU
28
5

X1 = D: Temperature
X2 = F: Cleaning interval

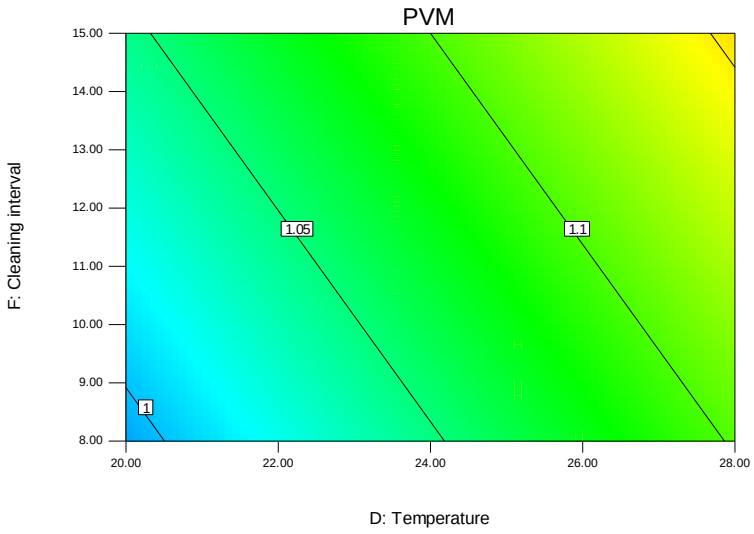
Actual Factors
A: Squeegee pressure = 0.20
B: Printing speed = 28.00
C: Squeegee angle = 55.00
E: Viscosity = 1200.00
G: Separation speed = 0.60
H: Relative humidity = 50.00



Design-Expert® Software
Factor Coding: Actual
PVM
1.21
0.95

X1 = D: Temperature
X2 = F: Cleaning interval

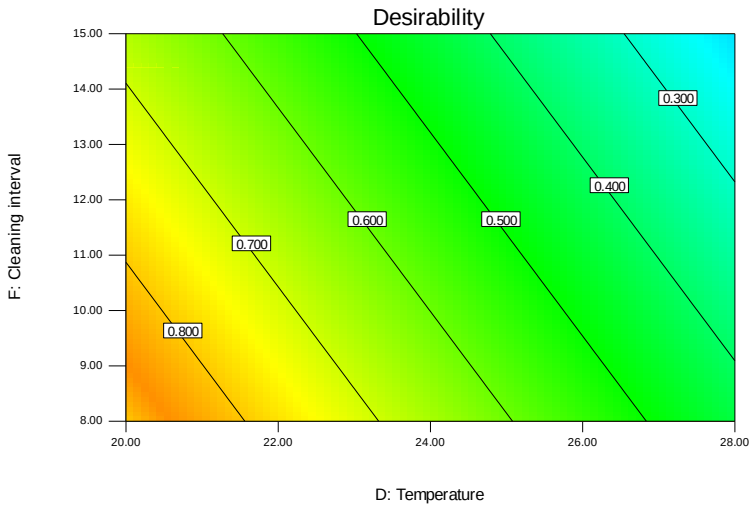
Actual Factors
A: Squeegee pressure = 0.20
B: Printing speed = 28.00
C: Squeegee angle = 55.00
E: Viscosity = 1200.00
G: Separation speed = 0.61
H: Relative humidity = 49.46



Design-Expert® Software
Factor Coding: Actual
Desirability
1.000
0.000

X1 = D: Temperature
X2 = F: Cleaning interval

Actual Factors
A: Squeegee pressure = 0.20
B: Printing speed = 28.00
C: Squeegee angle = 55.00
E: Viscosity = 1200.00
G: Separation speed = 0.60
H: Relative humidity = 50.00



8.30. An article in the *International Journal of Research in Marketing* ("Experimental design on the front lines of marketing: Testing new ideas to increase direct mail sales," 2006. Vol. 23, pp. 309-319) describes the use of a 20-run Plackett-Burman design to improve the effects of 19 factors to improve the response rate to a direct mail campaign to attract new customers to a credit card. The 19 factors are as follows:

Factor	(–) Control	(+) New Idea
A: Envelope teaser	General offer	Product-specific offer
B: Return address	Blind	Add company name
C: "Official" ink-stamp on envelope	Yes	No
D: Postage	Pre-printed	Stamp
E: Additional graphic on envelop	Yes	No
F: Price graphic on letter	Small	Large
G: Sticker	Yes	No
H: Personalize letter copy	No	Yes
I: Copy Message	Targeted	Generic
J: Letter headline	Headline 1	Headline 2
K: List of Benefits	Standard layout	Creative layout
L: Postscript on letter	Control version	New P.S.
M: Signature	Manager	Senior Executive
N: Product selection	Many	Few
O: Value of free gift	High	Low
P: Reply envelope	Control	New Style
Q: Information on buckslip	Product info	Free gift info
R: 2nd buckslip	No	Yes
S: Interest rate	Low	High

The 20-run Plackett-Burman design is shown in the following table. Each test combination in the table was mailed to 5,000 potential customers, and the response rate is the percentage of customers who responded positively to the offer.

Test Cell	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*	*J*	*K*	*L*	*M*	*N*	*O*	*P*	*Q*	*R*	*S*	Orders	Resp Rate
1	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	52	1.04%
2	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+	38	0.76%
3	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+	42	0.84%
4	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	134	2.68%
5	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	104	2.08%
6	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–	60	1.20%
7	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	61	1.22%
8	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	68	1.36%
9	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–	57	1.14%
10	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+	30	0.60%
11	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–	108	2.16%
12	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+	39	0.78%
13	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	40	0.80%
14	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+	49	0.98%
15	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	37	0.74%
16	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–	99	1.98%
17	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	86	1.72%
18	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	43	0.86%
19	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	47	0.94%
20	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	104	2.08%

(a) Verify that in this design each main effect is aliased to all two-factor interactions except those that involve that main effect. That is, in the 19 factor design, the main effect for each factor is aliased

with all two-factor interactions involving the other 18 factors, or 153 two-factor interactions (18!/2!16!).

The Design Expert output shown below identifies the aliases for the main effects and the two factor interactions. Notice that the main effects for each factor is aliased with all two-factor interactions involving the 18 other factors, but do not include the interactions that involve the same main effect.

Design Expert Output

```
[A] = A - 0.2 * BC + 0.2 * BD + 0.2 * BE + 0.2 * BF - 0.2 * BG + 0.2 * BH
  - 0.6 * BI - 0.2 * BJ - 0.2 * BK + 0.2 * BL + 0.2 * BM - 0.2 * BN + 0.2 * BO
  - 0.2 * BP - 0.2 * BQ - 0.2 * BR - 0.2 * BS - 0.2 * CD - 0.2 * CE - 0.6 * CF
  - 0.2 * CG + 0.2 * CH - 0.2 * CI - 0.2 * CJ - 0.2 * CK + 0.2 * CL + 0.2 * CM
  + 0.2 * CN - 0.2 * CO + 0.2 * CP + 0.2 * CQ - 0.2 * CR + 0.2 * CS - 0.2 * DE
  + 0.2 * DF - 0.2 * DG + 0.2 * DH + 0.2 * DI - 0.2 * DJ + 0.2 * DK - 0.2 * DL
  - 0.2 * DM + 0.2 * DN - 0.2 * DO - 0.2 * DP - 0.2 * DQ - 0.6 * DR + 0.2 * DS
  - 0.2 * EF + 0.2 * EG - 0.2 * EH - 0.2 * EI + 0.2 * EJ + 0.2 * EK - 0.2 * EL
  + 0.2 * EM - 0.6 * EN - 0.2 * EO - 0.2 * EP + 0.2 * EQ - 0.2 * ER + 0.2 * ES
  + 0.2 * FG - 0.2 * FH - 0.2 * FI - 0.2 * FJ - 0.2 * FK - 0.2 * FL - 0.2 * FM
  + 0.2 * FN - 0.2 * FO + 0.2 * FP + 0.2 * FQ + 0.2 * FR - 0.2 * FS - 0.2 * GH
  + 0.2 * GI + 0.2 * GJ - 0.6 * GK + 0.2 * GL - 0.2 * GM - 0.2 * GN - 0.2 * GO
  + 0.2 * GP - 0.2 * GQ - 0.2 * GR + 0.2 * GS + 0.2 * HI + 0.2 * HJ - 0.2 * HK
  + 0.2 * HL - 0.2 * HM - 0.2 * HN - 0.2 * HO - 0.2 * HP + 0.2 * HQ - 0.2 * HR
  - 0.6 * HS + 0.2 * IJ + 0.2 * IK - 0.2 * IL - 0.2 * IM - 0.2 * IN + 0.2 * IO
  + 0.2 * IP - 0.2 * IQ - 0.2 * IR - 0.2 * IS - 0.2 * JK - 0.2 * JL + 0.2 * JM
  + 0.2 * JN - 0.2 * JO - 0.6 * JP - 0.2 * JQ + 0.2 * JR - 0.2 * JS - 0.2 * KL
  - 0.2 * KM - 0.2 * KN + 0.2 * KO - 0.2 * KP + 0.2 * KQ + 0.2 * KR + 0.2 * KS
  - 0.6 * LM - 0.2 * LN + 0.2 * LO - 0.2 * LP - 0.2 * LQ + 0.2 * LR + 0.2 * LS
  + 0.2 * MN + 0.2 * MO + 0.2 * MP - 0.2 * MQ - 0.2 * MR - 0.2 * MS - 0.2 * NO
  - 0.2 * NP - 0.2 * NQ + 0.2 * NR + 0.2 * NS + 0.2 * OP - 0.6 * OQ + 0.2 * OR
  - 0.2 * OS + 0.2 * PQ - 0.2 * PR - 0.2 * PS + 0.2 * QR - 0.2 * QS - 0.2 * RS
[B] = B - 0.2 * AC + 0.2 * AD + 0.2 * AE + 0.2 * AF - 0.2 * AG + 0.2 * AH
  - 0.6 * AI - 0.2 * AJ - 0.2 * AK + 0.2 * AL + 0.2 * AM - 0.2 * AN + 0.2 * AO
  - 0.2 * AP - 0.2 * AQ - 0.2 * AR - 0.2 * AS - 0.2 * CD + 0.2 * CE + 0.2 * CF
  + 0.2 * CG - 0.2 * CH + 0.2 * CI - 0.6 * CJ - 0.2 * CK - 0.2 * CL + 0.2 * CM
  + 0.2 * CN - 0.2 * CO + 0.2 * CP - 0.2 * CQ - 0.2 * CR - 0.2 * CS - 0.2 * DE
  - 0.2 * DF - 0.6 * DG - 0.2 * DH + 0.2 * DI - 0.2 * DJ - 0.2 * DK - 0.2 * DL
  + 0.2 * DM + 0.2 * DN + 0.2 * DO - 0.2 * DP + 0.2 * DQ + 0.2 * DR - 0.2 * DS
  - 0.2 * EF + 0.2 * EG - 0.2 * EH + 0.2 * EI + 0.2 * EJ - 0.2 * EK + 0.2 * EL
  - 0.2 * EM - 0.2 * EN + 0.2 * EO - 0.2 * EP - 0.2 * EQ - 0.2 * ER - 0.6 * ES
  - 0.2 * FG + 0.2 * FH - 0.2 * FI - 0.2 * FJ + 0.2 * FK + 0.2 * FL - 0.2 * FM
  + 0.2 * FN - 0.6 * FO - 0.2 * FP - 0.2 * FQ + 0.2 * FR - 0.2 * FS + 0.2 * GH
  - 0.2 * GI - 0.2 * GJ - 0.2 * GK - 0.2 * GL - 0.2 * GM - 0.2 * GN + 0.2 * GO
  - 0.2 * GP + 0.2 * GQ + 0.2 * GR + 0.2 * GS - 0.2 * HI + 0.2 * HJ + 0.2 * HK
  - 0.6 * HL + 0.2 * HM - 0.2 * HN - 0.2 * HO - 0.2 * HP + 0.2 * HQ - 0.2 * HR
  - 0.2 * HS + 0.2 * IJ + 0.2 * IK - 0.2 * IL + 0.2 * IM - 0.2 * IN - 0.2 * IO
  - 0.2 * IP - 0.2 * IQ + 0.2 * IR - 0.2 * IS + 0.2 * JK + 0.2 * JL - 0.2 * JM
  - 0.2 * JN - 0.2 * JO + 0.2 * JP + 0.2 * JQ - 0.2 * JR - 0.2 * JS - 0.2 * KL
  - 0.2 * KM + 0.2 * KN + 0.2 * KO - 0.2 * KP - 0.6 * KQ - 0.2 * KR + 0.2 * KS
  - 0.2 * LM - 0.2 * LN - 0.2 * LO + 0.2 * LP - 0.2 * LQ + 0.2 * LR + 0.2 * LS
  - 0.6 * MN - 0.2 * MO + 0.2 * MP - 0.2 * MQ - 0.2 * MR + 0.2 * MS + 0.2 * NO
  + 0.2 * NP + 0.2 * NQ - 0.2 * NR - 0.2 * NS - 0.2 * OP - 0.2 * OQ - 0.2 * OR
  + 0.2 * OS + 0.2 * PQ - 0.6 * PR + 0.2 * PS + 0.2 * QR - 0.2 * QS + 0.2 * RS
[C] = C - 0.2 * AB - 0.2 * AD - 0.2 * AE - 0.6 * AF - 0.2 * AG + 0.2 * AH
  - 0.2 * AI - 0.2 * AJ - 0.2 * AK + 0.2 * AL + 0.2 * AM + 0.2 * AN - 0.2 * AO
  + 0.2 * AP + 0.2 * AQ - 0.2 * AR + 0.2 * AS - 0.2 * BD + 0.2 * BE + 0.2 * BF
  + 0.2 * BG - 0.2 * BH + 0.2 * BI - 0.6 * BJ - 0.2 * BK - 0.2 * BL + 0.2 * BM
  + 0.2 * BN - 0.2 * BO + 0.2 * BP - 0.2 * BQ - 0.2 * BR - 0.2 * BS - 0.2 * DE
  + 0.2 * DF + 0.2 * DG + 0.2 * DH - 0.2 * DI + 0.2 * DJ - 0.6 * DK - 0.2 * DL
  - 0.2 * DM + 0.2 * DN + 0.2 * DO - 0.2 * DP + 0.2 * DQ - 0.2 * DR - 0.2 * DS
  - 0.2 * EF - 0.2 * EG - 0.6 * EH - 0.2 * EI + 0.2 * EJ - 0.2 * EK - 0.2 * EL
  - 0.2 * EM + 0.2 * EN + 0.2 * EO + 0.2 * EP - 0.2 * EQ + 0.2 * ER + 0.2 * ES
  - 0.2 * FG + 0.2 * FH - 0.2 * FI + 0.2 * FJ + 0.2 * FK - 0.2 * FL + 0.2 * FM
  - 0.2 * FN - 0.2 * FO + 0.2 * FP - 0.2 * FQ - 0.2 * FR - 0.2 * FS - 0.2 * GH
  + 0.2 * GI - 0.2 * GJ - 0.2 * GK + 0.2 * GL + 0.2 * GM - 0.2 * GN + 0.2 * GO
  - 0.6 * GP - 0.2 * GQ - 0.2 * GR + 0.2 * GS + 0.2 * HI - 0.2 * HJ - 0.2 * HK
  - 0.2 * HL - 0.2 * HM - 0.2 * HN - 0.2 * HO + 0.2 * HP - 0.2 * HQ + 0.2 * HR
  + 0.2 * HS - 0.2 * IJ + 0.2 * IK + 0.2 * IL - 0.6 * IM + 0.2 * IN - 0.2 * IO
```

- 0.2 * IP - 0.2 * IQ + 0.2 * IR - 0.2 * IS + 0.2 * JK + 0.2 * JL - 0.2 * JM
+ 0.2 * JN - 0.2 * JO - 0.2 * JP - 0.2 * JQ - 0.2 * JR + 0.2 * JS + 0.2 * KL
+ 0.2 * KM - 0.2 * KN - 0.2 * KO - 0.2 * KP + 0.2 * KQ + 0.2 * KR - 0.2 * KS
- 0.2 * LM - 0.2 * LN + 0.2 * LO + 0.2 * LP - 0.2 * LQ - 0.6 * LR - 0.2 * LS
- 0.2 * MN - 0.2 * MO - 0.2 * MP + 0.2 * MQ - 0.2 * MR + 0.2 * MS - 0.6 * NO
- 0.2 * NP + 0.2 * NQ - 0.2 * NR - 0.2 * NS + 0.2 * OP + 0.2 * OQ + 0.2 * OR
- 0.2 * OS - 0.2 * PQ - 0.2 * PR - 0.2 * PS + 0.2 * QR - 0.6 * QS + 0.2 * RS

[D] = D + 0.2 * AB - 0.2 * AC - 0.2 * AE + 0.2 * AF - 0.2 * AG + 0.2 * AH
+ 0.2 * AI - 0.2 * AJ + 0.2 * AK - 0.2 * AL - 0.2 * AM + 0.2 * AN - 0.2 * AO
- 0.2 * AP - 0.2 * AQ - 0.6 * AR + 0.2 * AS - 0.2 * BC - 0.2 * BE - 0.2 * BF
- 0.6 * BG - 0.2 * BH + 0.2 * BI - 0.2 * BJ - 0.2 * BK - 0.2 * BL + 0.2 * BM
+ 0.2 * BN + 0.2 * BO - 0.2 * BP + 0.2 * BQ + 0.2 * BR - 0.2 * BS - 0.2 * CE
+ 0.2 * CF + 0.2 * CG + 0.2 * CH - 0.2 * CI + 0.2 * CJ - 0.6 * CK - 0.2 * CL
- 0.2 * CM + 0.2 * CN + 0.2 * CO - 0.2 * CP + 0.2 * CQ - 0.2 * CR - 0.2 * CS
- 0.2 * EF + 0.2 * EG + 0.2 * EH + 0.2 * EI - 0.2 * EJ + 0.2 * EK - 0.6 * EL
- 0.2 * EM - 0.2 * EN + 0.2 * EO + 0.2 * EP - 0.2 * EQ + 0.2 * ER - 0.2 * ES
- 0.2 * FG - 0.2 * FH - 0.6 * FI - 0.2 * FJ + 0.2 * FK - 0.2 * FL - 0.2 * FM
- 0.2 * FN + 0.2 * FO + 0.2 * FP + 0.2 * FQ - 0.2 * FR + 0.2 * FS - 0.2 * GH
+ 0.2 * GI - 0.2 * GJ + 0.2 * GK + 0.2 * GL - 0.2 * GM + 0.2 * GN - 0.2 * GO
- 0.2 * GP + 0.2 * GQ - 0.2 * GR - 0.2 * GS - 0.2 * HI + 0.2 * HJ - 0.2 * HK
- 0.2 * HL + 0.2 * HM + 0.2 * HN - 0.2 * HO + 0.2 * HP - 0.6 * HQ - 0.2 * HR
- 0.2 * HS + 0.2 * IJ - 0.2 * IK - 0.2 * IL - 0.2 * IM - 0.2 * IN - 0.2 * IO
- 0.2 * IP + 0.2 * IQ - 0.2 * IR + 0.2 * IS - 0.2 * JK + 0.2 * JL + 0.2 * JM
- 0.6 * JN + 0.2 * JO - 0.2 * JP - 0.2 * JQ - 0.2 * JR + 0.2 * JS + 0.2 * KL
+ 0.2 * KM - 0.2 * KN + 0.2 * KO - 0.2 * KP - 0.2 * KQ - 0.2 * KR - 0.2 * KS
+ 0.2 * LM + 0.2 * LN - 0.2 * LO - 0.2 * LP - 0.2 * LQ + 0.2 * LR + 0.2 * LS
- 0.2 * MN - 0.2 * MO + 0.2 * MP + 0.2 * MQ - 0.2 * MR - 0.6 * MS - 0.2 * NO
- 0.2 * NP - 0.2 * NQ + 0.2 * NR - 0.2 * NS - 0.6 * OP - 0.2 * OQ + 0.2 * OR
- 0.2 * OS + 0.2 * PQ + 0.2 * PR + 0.2 * PS - 0.2 * QR - 0.2 * QS + 0.2 * RS

[E] = E + 0.2 * AB - 0.2 * AC - 0.2 * AD - 0.2 * AF + 0.2 * AG - 0.2 * AH
- 0.2 * AI + 0.2 * AJ + 0.2 * AK - 0.2 * AL + 0.2 * AM - 0.6 * AN - 0.2 * AO
- 0.2 * AP + 0.2 * AQ - 0.2 * AR + 0.2 * AS + 0.2 * BC - 0.2 * BD - 0.2 * BF
+ 0.2 * BG - 0.2 * BH + 0.2 * BI + 0.2 * BJ - 0.2 * BK + 0.2 * BL - 0.2 * BM
- 0.2 * BN + 0.2 * BO - 0.2 * BP - 0.2 * BQ - 0.2 * BR - 0.6 * BS - 0.2 * CD
- 0.2 * CF - 0.2 * CG - 0.6 * CH - 0.2 * CI + 0.2 * CJ - 0.2 * CK - 0.2 * CL
- 0.2 * CM + 0.2 * CN + 0.2 * CO + 0.2 * CP - 0.2 * CQ + 0.2 * CR + 0.2 * CS
- 0.2 * DF + 0.2 * DG + 0.2 * DH + 0.2 * DI - 0.2 * DJ + 0.2 * DK - 0.6 * DL
- 0.2 * DM - 0.2 * DN + 0.2 * DO + 0.2 * DP - 0.2 * DQ + 0.2 * DR - 0.2 * DS
- 0.2 * FG + 0.2 * FH + 0.2 * FI + 0.2 * FJ - 0.2 * FK + 0.2 * FL - 0.6 * FM
- 0.2 * FN - 0.2 * FO + 0.2 * FP + 0.2 * FQ - 0.2 * FR + 0.2 * FS - 0.2 * GH
- 0.2 * GI - 0.6 * GJ - 0.2 * GK + 0.2 * GL - 0.2 * GM - 0.2 * GN - 0.2 * GO
+ 0.2 * GP + 0.2 * GQ + 0.2 * GR - 0.2 * GS - 0.2 * HI + 0.2 * HJ - 0.2 * HK
+ 0.2 * HL + 0.2 * HM - 0.2 * HN + 0.2 * HO - 0.2 * HP - 0.2 * HQ + 0.2 * HR
- 0.2 * HS - 0.2 * IJ + 0.2 * IK - 0.2 * IL - 0.2 * IM + 0.2 * IN + 0.2 * IO
- 0.2 * IP + 0.2 * IQ - 0.6 * IR - 0.2 * IS + 0.2 * JK - 0.2 * JL - 0.2 * JM
- 0.2 * JN - 0.2 * JO - 0.2 * JP - 0.2 * JQ + 0.2 * JR - 0.2 * JS - 0.2 * KL
+ 0.2 * KM + 0.2 * KN - 0.6 * KO + 0.2 * KP - 0.2 * KQ - 0.2 * KR - 0.2 * KS
+ 0.2 * LM + 0.2 * LN - 0.2 * LO + 0.2 * LP - 0.2 * LQ - 0.2 * LR - 0.2 * LS
+ 0.2 * MN + 0.2 * MO - 0.2 * MP - 0.2 * MQ - 0.2 * MR + 0.2 * MS - 0.2 * NO
- 0.2 * NP + 0.2 * NQ + 0.2 * NR - 0.2 * NS - 0.2 * OP - 0.2 * OQ - 0.2 * OR
+ 0.2 * OS - 0.6 * PQ - 0.2 * PR + 0.2 * PS + 0.2 * QR + 0.2 * QS - 0.2 * RS

[F] = F + 0.2 * AB - 0.6 * AC + 0.2 * AD - 0.2 * AE + 0.2 * AG - 0.2 * AH
- 0.2 * AI - 0.2 * AJ - 0.2 * AK - 0.2 * AL - 0.2 * AM + 0.2 * AN - 0.2 * AO
+ 0.2 * AP + 0.2 * AQ + 0.2 * AR - 0.2 * AS + 0.2 * BC - 0.2 * BD - 0.2 * BE
- 0.2 * BG + 0.2 * BH - 0.2 * BI - 0.2 * BJ + 0.2 * BK + 0.2 * BL - 0.2 * BM
+ 0.2 * BN - 0.6 * BO - 0.2 * BP - 0.2 * BQ + 0.2 * BR - 0.2 * BS + 0.2 * CD
- 0.2 * CE - 0.2 * CG + 0.2 * CH - 0.2 * CI + 0.2 * CJ + 0.2 * CK - 0.2 * CL
+ 0.2 * CM - 0.2 * CN - 0.2 * CO + 0.2 * CP - 0.2 * CQ - 0.2 * CR - 0.2 * CS
- 0.2 * DE - 0.2 * DG - 0.2 * DH - 0.6 * DI - 0.2 * DJ + 0.2 * DK - 0.2 * DL
- 0.2 * DM - 0.2 * DN + 0.2 * DO + 0.2 * DP + 0.2 * DQ - 0.2 * DR + 0.2 * DS
- 0.2 * EG + 0.2 * EH + 0.2 * EI + 0.2 * EJ - 0.2 * EK + 0.2 * EL - 0.6 * EM
- 0.2 * EN - 0.2 * EO + 0.2 * EP + 0.2 * EQ - 0.2 * ER + 0.2 * ES - 0.2 * GH
+ 0.2 * GI + 0.2 * GJ + 0.2 * GK - 0.2 * GL + 0.2 * GM - 0.6 * GN - 0.2 * GO
- 0.2 * GP + 0.2 * GQ + 0.2 * GR - 0.2 * GS - 0.2 * HI - 0.2 * HJ - 0.6 * HK
- 0.2 * HL + 0.2 * HM - 0.2 * HN - 0.2 * HO - 0.2 * HP + 0.2 * HQ + 0.2 * HR
+ 0.2 * HS - 0.2 * IJ + 0.2 * IK - 0.2 * IL + 0.2 * IM + 0.2 * IN - 0.2 * IO
+ 0.2 * IP - 0.2 * IQ - 0.2 * IR + 0.2 * IS - 0.2 * JK + 0.2 * JL - 0.2 * JM
- 0.2 * JN + 0.2 * JO + 0.2 * JP - 0.2 * JQ + 0.2 * JR - 0.6 * JS + 0.2 * KL
- 0.2 * KM - 0.2 * KN - 0.2 * KO - 0.2 * KP - 0.2 * KQ - 0.2 * KR + 0.2 * KS
- 0.2 * LM + 0.2 * LN + 0.2 * LO - 0.6 * LP + 0.2 * LQ - 0.2 * LR - 0.2 * LS

```
  + 0.2 * MN + 0.2 * MO - 0.2 * MP + 0.2 * MQ - 0.2 * MR - 0.2 * MS + 0.2 * NO
  + 0.2 * NP - 0.2 * NQ - 0.2 * NR - 0.2 * NS - 0.2 * OP - 0.2 * OQ + 0.2 * OR
  + 0.2 * OS - 0.2 * PQ - 0.2 * PR - 0.2 * PS - 0.6 * QR - 0.2 * QS + 0.2 * RS
[G] = G - 0.2 * AB - 0.2 * AC - 0.2 * AD + 0.2 * AE + 0.2 * AF - 0.2 * AH
  + 0.2 * AI + 0.2 * AJ - 0.6 * AK + 0.2 * AL - 0.2 * AM - 0.2 * AN - 0.2 * AO
  + 0.2 * AP - 0.2 * AQ - 0.2 * AR + 0.2 * AS + 0.2 * BC - 0.6 * BD + 0.2 * BE
  - 0.2 * BF + 0.2 * BH - 0.2 * BI - 0.2 * BJ - 0.2 * BK - 0.2 * BL - 0.2 * BM
  - 0.2 * BN + 0.2 * BO - 0.2 * BP + 0.2 * BQ + 0.2 * BR + 0.2 * BS + 0.2 * CD
  - 0.2 * CE - 0.2 * CF - 0.2 * CH + 0.2 * CI - 0.2 * CJ - 0.2 * CK + 0.2 * CL
  + 0.2 * CM - 0.2 * CN + 0.2 * CO - 0.6 * CP - 0.2 * CQ - 0.2 * CR + 0.2 * CS
  + 0.2 * DE - 0.2 * DF - 0.2 * DH + 0.2 * DI - 0.2 * DJ + 0.2 * DK + 0.2 * DL
  - 0.2 * DM + 0.2 * DN - 0.2 * DO - 0.2 * DP + 0.2 * DQ - 0.2 * DR - 0.2 * DS
  - 0.2 * EF - 0.2 * EH - 0.2 * EI - 0.6 * EJ - 0.2 * EK + 0.2 * EL - 0.2 * EM
  - 0.2 * EN - 0.2 * EO + 0.2 * EP + 0.2 * EQ + 0.2 * ER - 0.2 * ES - 0.2 * FH
  + 0.2 * FI + 0.2 * FJ + 0.2 * FK - 0.2 * FL + 0.2 * FM - 0.6 * FN - 0.2 * FO
  - 0.2 * FP + 0.2 * FQ + 0.2 * FR - 0.2 * FS - 0.2 * HI + 0.2 * HJ + 0.2 * HK
  + 0.2 * HL - 0.2 * HM + 0.2 * HN - 0.6 * HO - 0.2 * HP - 0.2 * HQ + 0.2 * HR
  + 0.2 * HS - 0.2 * IJ - 0.2 * IK - 0.6 * IL - 0.2 * IM + 0.2 * IN - 0.2 * IO
  - 0.2 * IP - 0.2 * IQ + 0.2 * IR + 0.2 * IS - 0.2 * JK + 0.2 * JL - 0.2 * JM
  + 0.2 * JN + 0.2 * JO - 0.2 * JP + 0.2 * JQ - 0.2 * JR - 0.2 * JS - 0.2 * KL
  + 0.2 * KM - 0.2 * KN - 0.2 * KO + 0.2 * KP + 0.2 * KQ - 0.2 * KR + 0.2 * KS
  + 0.2 * LM - 0.2 * LN - 0.2 * LO - 0.2 * LP - 0.2 * LQ - 0.2 * LR - 0.2 * LS
  - 0.2 * MN + 0.2 * MO + 0.2 * MP - 0.6 * MQ + 0.2 * MR - 0.2 * MS + 0.2 * NO
  + 0.2 * NP - 0.2 * NQ + 0.2 * NR - 0.2 * NS + 0.2 * OP + 0.2 * OQ - 0.2 * OR
  - 0.2 * OS - 0.2 * PQ - 0.2 * PR + 0.2 * PS - 0.2 * QR - 0.2 * QS - 0.6 * RS
[H] = H + 0.2 * AB + 0.2 * AC + 0.2 * AD - 0.2 * AE - 0.2 * AF - 0.2 * AG
  + 0.2 * AI + 0.2 * AJ - 0.2 * AK + 0.2 * AL - 0.2 * AM - 0.2 * AN - 0.2 * AO
  - 0.2 * AP + 0.2 * AQ - 0.2 * AR - 0.6 * AS - 0.2 * BC - 0.2 * BD - 0.2 * BE
  + 0.2 * BF + 0.2 * BG - 0.2 * BI + 0.2 * BJ + 0.2 * BK - 0.6 * BL + 0.2 * BM
  - 0.2 * BN - 0.2 * BO - 0.2 * BP + 0.2 * BQ - 0.2 * BR - 0.2 * BS + 0.2 * CD
  - 0.6 * CE + 0.2 * CF - 0.2 * CG + 0.2 * CI - 0.2 * CJ - 0.2 * CK - 0.2 * CL
  - 0.2 * CM - 0.2 * CN - 0.2 * CO + 0.2 * CP - 0.2 * CQ + 0.2 * CR + 0.2 * CS
  + 0.2 * DE - 0.2 * DF - 0.2 * DG - 0.2 * DI + 0.2 * DJ - 0.2 * DK - 0.2 * DL
  + 0.2 * DM + 0.2 * DN - 0.2 * DO + 0.2 * DP - 0.6 * DQ - 0.2 * DR - 0.2 * DS
  + 0.2 * EF - 0.2 * EG - 0.2 * EI + 0.2 * EJ - 0.2 * EK + 0.2 * EL + 0.2 * EM
  - 0.2 * EN + 0.2 * EO - 0.2 * EP - 0.2 * EQ + 0.2 * ER - 0.2 * ES - 0.2 * FG
  - 0.2 * FI - 0.2 * FJ - 0.6 * FK - 0.2 * FL + 0.2 * FM - 0.2 * FN - 0.2 * FO
  - 0.2 * FP + 0.2 * FQ + 0.2 * FR + 0.2 * FS - 0.2 * GI + 0.2 * GJ + 0.2 * GK
  + 0.2 * GL - 0.2 * GM + 0.2 * GN - 0.6 * GO - 0.2 * GP - 0.2 * GQ + 0.2 * GR
  + 0.2 * GS - 0.2 * IJ + 0.2 * IK + 0.2 * IL + 0.2 * IM - 0.2 * IN + 0.2 * IO
  - 0.6 * IP - 0.2 * IQ - 0.2 * IR + 0.2 * IS - 0.2 * JK - 0.2 * JL - 0.6 * JM
  - 0.2 * JN + 0.2 * JO - 0.2 * JP - 0.2 * JQ - 0.2 * JR + 0.2 * JS - 0.2 * KL
  + 0.2 * KM - 0.2 * KN + 0.2 * KO + 0.2 * KP - 0.2 * KQ + 0.2 * KR - 0.2 * KS
  - 0.2 * LM + 0.2 * LN - 0.2 * LO - 0.2 * LP + 0.2 * LQ + 0.2 * LR - 0.2 * LS
  + 0.2 * MN - 0.2 * MO - 0.2 * MP - 0.2 * MQ - 0.2 * MR - 0.2 * MS - 0.2 * NO
  + 0.2 * NP + 0.2 * NQ - 0.6 * NR + 0.2 * NS + 0.2 * OP + 0.2 * OQ - 0.2 * OR
  + 0.2 * OS + 0.2 * PQ + 0.2 * PR - 0.2 * PS - 0.2 * QR - 0.2 * QS - 0.2 * RS
[I] = I - 0.6 * AB - 0.2 * AC + 0.2 * AD - 0.2 * AE - 0.2 * AF + 0.2 * AG
  + 0.2 * AH + 0.2 * AJ + 0.2 * AK - 0.2 * AL - 0.2 * AM - 0.2 * AN + 0.2 * AO
  + 0.2 * AP - 0.2 * AQ - 0.2 * AR - 0.2 * AS + 0.2 * BC + 0.2 * BD + 0.2 * BE
  - 0.2 * BF - 0.2 * BG - 0.2 * BH + 0.2 * BJ + 0.2 * BK - 0.2 * BL + 0.2 * BM
  - 0.2 * BN - 0.2 * BO - 0.2 * BP - 0.2 * BQ + 0.2 * BR - 0.2 * BS - 0.2 * CD
  - 0.2 * CE - 0.2 * CF + 0.2 * CG + 0.2 * CH - 0.2 * CJ + 0.2 * CK + 0.2 * CL
  - 0.6 * CM + 0.2 * CN - 0.2 * CO - 0.2 * CP - 0.2 * CQ + 0.2 * CR - 0.2 * CS
  + 0.2 * DE - 0.6 * DF + 0.2 * DG - 0.2 * DH + 0.2 * DJ - 0.2 * DK - 0.2 * DL
  - 0.2 * DM - 0.2 * DN - 0.2 * DO - 0.2 * DP + 0.2 * DQ - 0.2 * DR + 0.2 * DS
  + 0.2 * EF - 0.2 * EG - 0.2 * EH - 0.2 * EJ + 0.2 * EK - 0.2 * EL - 0.2 * EM
  + 0.2 * EN + 0.2 * EO - 0.2 * EP + 0.2 * EQ - 0.6 * ER - 0.2 * ES + 0.2 * FG
  - 0.2 * FH - 0.2 * FJ + 0.2 * FK - 0.2 * FL + 0.2 * FM + 0.2 * FN - 0.2 * FO
  + 0.2 * FP - 0.2 * FQ - 0.2 * FR + 0.2 * FS - 0.2 * GH - 0.2 * GJ - 0.2 * GK
  - 0.6 * GL - 0.2 * GM + 0.2 * GN - 0.2 * GO - 0.2 * GP - 0.2 * GQ + 0.2 * GR
  + 0.2 * GS - 0.2 * HJ + 0.2 * HK + 0.2 * HL + 0.2 * HM - 0.2 * HN + 0.2 * HO
  - 0.6 * HP - 0.2 * HQ - 0.2 * HR + 0.2 * HS - 0.2 * JK + 0.2 * JL + 0.2 * JM
  + 0.2 * JN - 0.2 * JO + 0.2 * JP - 0.6 * JQ - 0.2 * JR - 0.2 * JS - 0.2 * KL
  - 0.2 * KM - 0.6 * KN - 0.2 * KO + 0.2 * KP - 0.2 * KQ - 0.2 * KR - 0.2 * KS
  - 0.2 * LM + 0.2 * LN - 0.2 * LO + 0.2 * LP + 0.2 * LQ - 0.2 * LR + 0.2 * LS
  - 0.2 * MN + 0.2 * MO - 0.2 * MP - 0.2 * MQ + 0.2 * MR + 0.2 * MS + 0.2 * NO
  - 0.2 * NP - 0.2 * NQ - 0.2 * NR - 0.2 * NS - 0.2 * OP + 0.2 * OQ + 0.2 * OR
  - 0.6 * OS + 0.2 * PQ + 0.2 * PR - 0.2 * PS + 0.2 * QR + 0.2 * QS - 0.2 * RS
[J] = J - 0.2 * AB - 0.2 * AC - 0.2 * AD + 0.2 * AE - 0.2 * AF + 0.2 * AG
```

```
  + 0.2 * AH + 0.2 * AI - 0.2 * AK - 0.2 * AL + 0.2 * AM + 0.2 * AN - 0.2 * AO
  - 0.6 * AP - 0.2 * AQ + 0.2 * AR - 0.2 * AS - 0.6 * BC - 0.2 * BD + 0.2 * BE
  - 0.2 * BF - 0.2 * BG + 0.2 * BH + 0.2 * BI + 0.2 * BK + 0.2 * BL - 0.2 * BM
  - 0.2 * BN - 0.2 * BO + 0.2 * BP + 0.2 * BQ - 0.2 * BR - 0.2 * BS + 0.2 * CD
  + 0.2 * CE + 0.2 * CF - 0.2 * CG - 0.2 * CH - 0.2 * CI + 0.2 * CK + 0.2 * CL
  - 0.2 * CM + 0.2 * CN - 0.2 * CO - 0.2 * CP - 0.2 * CQ - 0.2 * CR + 0.2 * CS
  - 0.2 * DE - 0.2 * DF - 0.2 * DG + 0.2 * DH + 0.2 * DI - 0.2 * DK + 0.2 * DL
  + 0.2 * DM - 0.6 * DN + 0.2 * DO - 0.2 * DP - 0.2 * DQ - 0.2 * DR + 0.2 * DS
  + 0.2 * EF - 0.6 * EG + 0.2 * EH - 0.2 * EI + 0.2 * EK - 0.2 * EL - 0.2 * EM
  - 0.2 * EN - 0.2 * EO - 0.2 * EP - 0.2 * EQ + 0.2 * ER - 0.2 * ES + 0.2 * FG
  - 0.2 * FH - 0.2 * FI - 0.2 * FK + 0.2 * FL - 0.2 * FM - 0.2 * FN + 0.2 * FO
  + 0.2 * FP - 0.2 * FQ + 0.2 * FR - 0.6 * FS + 0.2 * GH - 0.2 * GI - 0.2 * GK
  + 0.2 * GL - 0.2 * GM + 0.2 * GN + 0.2 * GO - 0.2 * GP + 0.2 * GQ - 0.2 * GR
  - 0.2 * GS - 0.2 * HI - 0.2 * HK - 0.2 * HL - 0.6 * HM - 0.2 * HN + 0.2 * HO
  - 0.2 * HP - 0.2 * HQ - 0.2 * HR + 0.2 * HS - 0.2 * IK + 0.2 * IL + 0.2 * IM
  + 0.2 * IN - 0.2 * IO + 0.2 * IP - 0.6 * IQ - 0.2 * IR - 0.2 * IS - 0.2 * KL
  + 0.2 * KM + 0.2 * KN + 0.2 * KO - 0.2 * KP + 0.2 * KQ - 0.6 * KR - 0.2 * KS
  - 0.2 * LM - 0.2 * LN - 0.6 * LO - 0.2 * LP + 0.2 * LQ - 0.2 * LR - 0.2 * LS
  - 0.2 * MN + 0.2 * MO - 0.2 * MP + 0.2 * MQ + 0.2 * MR - 0.2 * MS - 0.2 * NO
  + 0.2 * NP - 0.2 * NQ - 0.2 * NR + 0.2 * NS + 0.2 * OP - 0.2 * OQ - 0.2 * OR
  - 0.2 * OS - 0.2 * PQ + 0.2 * PR + 0.2 * PS + 0.2 * QR + 0.2 * QS + 0.2 * RS
[K] = K - 0.2 * AB - 0.2 * AC + 0.2 * AD + 0.2 * AE - 0.2 * AF - 0.6 * AG
  - 0.2 * AH + 0.2 * AI - 0.2 * AJ - 0.2 * AL - 0.2 * AM - 0.2 * AN + 0.2 * AO
  - 0.2 * AP + 0.2 * AQ + 0.2 * AR + 0.2 * AS - 0.2 * BC - 0.2 * BD - 0.2 * BE
  + 0.2 * BF - 0.2 * BG + 0.2 * BH + 0.2 * BI + 0.2 * BJ - 0.2 * BL - 0.2 * BM
  + 0.2 * BN + 0.2 * BO - 0.2 * BP - 0.6 * BQ - 0.2 * BR + 0.2 * BS - 0.6 * CD
  - 0.2 * CE + 0.2 * CF - 0.2 * CG - 0.2 * CH + 0.2 * CI + 0.2 * CJ + 0.2 * CL
  + 0.2 * CM - 0.2 * CN - 0.2 * CO - 0.2 * CP + 0.2 * CQ + 0.2 * CR - 0.2 * CS
  + 0.2 * DE + 0.2 * DF + 0.2 * DG - 0.2 * DH - 0.2 * DI - 0.2 * DJ + 0.2 * DL
  + 0.2 * DM - 0.2 * DN + 0.2 * DO - 0.2 * DP - 0.2 * DQ - 0.2 * DR - 0.2 * DS
  - 0.2 * EF - 0.2 * EG - 0.2 * EH + 0.2 * EI + 0.2 * EJ - 0.2 * EL + 0.2 * EM
  + 0.2 * EN - 0.6 * EO + 0.2 * EP - 0.2 * EQ - 0.2 * ER - 0.2 * ES + 0.2 * FG
  - 0.6 * FH + 0.2 * FI - 0.2 * FJ + 0.2 * FL - 0.2 * FM - 0.2 * FN - 0.2 * FO
  - 0.2 * FP - 0.2 * FQ - 0.2 * FR + 0.2 * FS + 0.2 * GH - 0.2 * GI - 0.2 * GJ
  - 0.2 * GL + 0.2 * GM - 0.2 * GN - 0.2 * GO + 0.2 * GP + 0.2 * GQ - 0.2 * GR
  + 0.2 * GS + 0.2 * HI - 0.2 * HJ - 0.2 * HL + 0.2 * HM - 0.2 * HN + 0.2 * HO
  + 0.2 * HP - 0.2 * HQ + 0.2 * HR - 0.2 * HS - 0.2 * IJ - 0.2 * IL - 0.2 * IM
  - 0.6 * IN - 0.2 * IO + 0.2 * IP - 0.2 * IQ - 0.2 * IR - 0.2 * IS - 0.2 * JL
  + 0.2 * JM + 0.2 * JN + 0.2 * JO - 0.2 * JP + 0.2 * JQ - 0.6 * JR - 0.2 * JS
  - 0.2 * LM + 0.2 * LN + 0.2 * LO + 0.2 * LP - 0.2 * LQ + 0.2 * LR - 0.6 * LS
  - 0.2 * MN - 0.2 * MO - 0.6 * MP - 0.2 * MQ + 0.2 * MR - 0.2 * MS - 0.2 * NO
  + 0.2 * NP - 0.2 * NQ + 0.2 * NR + 0.2 * NS - 0.2 * OP + 0.2 * OQ - 0.2 * OR
  - 0.2 * OS + 0.2 * PQ - 0.2 * PR - 0.2 * PS - 0.2 * QR + 0.2 * QS + 0.2 * RS
[L] = L + 0.2 * AB + 0.2 * AC - 0.2 * AD - 0.2 * AE - 0.2 * AF + 0.2 * AG
  + 0.2 * AH - 0.2 * AI - 0.2 * AJ - 0.2 * AK - 0.6 * AM - 0.2 * AN + 0.2 * AO
  - 0.2 * AP - 0.2 * AQ + 0.2 * AR + 0.2 * AS - 0.2 * BC - 0.2 * BD + 0.2 * BE
  + 0.2 * BF - 0.2 * BG - 0.6 * BH - 0.2 * BI + 0.2 * BJ - 0.2 * BK - 0.2 * BM
  - 0.2 * BN - 0.2 * BO + 0.2 * BP - 0.2 * BQ + 0.2 * BR + 0.2 * BS - 0.2 * CD
  - 0.2 * CE - 0.2 * CF + 0.2 * CG - 0.2 * CH + 0.2 * CI + 0.2 * CJ + 0.2 * CK
  - 0.2 * CM - 0.2 * CN + 0.2 * CO + 0.2 * CP - 0.2 * CQ - 0.6 * CR - 0.2 * CS
  - 0.6 * DE - 0.2 * DF + 0.2 * DG - 0.2 * DH - 0.2 * DI + 0.2 * DJ + 0.2 * DK
  + 0.2 * DM + 0.2 * DN - 0.2 * DO - 0.2 * DP - 0.2 * DQ + 0.2 * DR + 0.2 * DS
  + 0.2 * EF + 0.2 * EG + 0.2 * EH - 0.2 * EI - 0.2 * EJ - 0.2 * EK + 0.2 * EM
  + 0.2 * EN - 0.2 * EO + 0.2 * EP - 0.2 * EQ - 0.2 * ER - 0.2 * ES - 0.2 * FG
  - 0.2 * FH - 0.2 * FI + 0.2 * FJ + 0.2 * FK - 0.2 * FM + 0.2 * FN + 0.2 * FO
  - 0.6 * FP + 0.2 * FQ - 0.2 * FR - 0.2 * FS + 0.2 * GH - 0.6 * GI + 0.2 * GJ
  - 0.2 * GK + 0.2 * GM - 0.2 * GN - 0.2 * GO - 0.2 * GP - 0.2 * GQ - 0.2 * GR
  - 0.2 * GS + 0.2 * HI - 0.2 * HJ - 0.2 * HK - 0.2 * HM + 0.2 * HN - 0.2 * HO
  - 0.2 * HP + 0.2 * HQ + 0.2 * HR - 0.2 * HS + 0.2 * IJ - 0.2 * IK - 0.2 * IM
  + 0.2 * IN - 0.2 * IO + 0.2 * IP + 0.2 * IQ - 0.2 * IR + 0.2 * IS - 0.2 * JK
  - 0.2 * JM - 0.2 * JN - 0.6 * JO - 0.2 * JP + 0.2 * JQ - 0.2 * JR - 0.2 * JS
  - 0.2 * KM + 0.2 * KN + 0.2 * KO + 0.2 * KP - 0.2 * KQ + 0.2 * KR - 0.6 * KS
  - 0.2 * MN + 0.2 * MO + 0.2 * MP + 0.2 * MQ - 0.2 * MR + 0.2 * MS - 0.2 * NO
  - 0.2 * NP - 0.6 * NQ - 0.2 * NR + 0.2 * NS - 0.2 * OP + 0.2 * OQ - 0.2 * OR
  + 0.2 * OS - 0.2 * PQ + 0.2 * PR - 0.2 * PS + 0.2 * QR - 0.2 * QS - 0.2 * RS
[M] = M + 0.2 * AB + 0.2 * AC - 0.2 * AD + 0.2 * AE - 0.2 * AF - 0.2 * AG
  - 0.2 * AH - 0.2 * AI + 0.2 * AJ - 0.2 * AK - 0.6 * AL + 0.2 * AN + 0.2 * AO
  + 0.2 * AP - 0.2 * AQ - 0.2 * AR - 0.2 * AS + 0.2 * BC + 0.2 * BD - 0.2 * BE
  - 0.2 * BF - 0.2 * BG + 0.2 * BH + 0.2 * BI - 0.2 * BJ - 0.2 * BK - 0.2 * BL
  - 0.6 * BN - 0.2 * BO + 0.2 * BP - 0.2 * BQ - 0.2 * BR + 0.2 * BS - 0.2 * CD
```

- 0.2 * CE + 0.2 * CF + 0.2 * CG - 0.2 * CH - 0.6 * CI - 0.2 * CJ + 0.2 * CK
- 0.2 * CL - 0.2 * CN - 0.2 * CO - 0.2 * CP + 0.2 * CQ - 0.2 * CR + 0.2 * CS
- 0.2 * DE - 0.2 * DF - 0.2 * DG + 0.2 * DH - 0.2 * DI + 0.2 * DJ + 0.2 * DK
+ 0.2 * DL - 0.2 * DN - 0.2 * DO + 0.2 * DP + 0.2 * DQ - 0.2 * DR - 0.6 * DS
- 0.6 * EF - 0.2 * EG + 0.2 * EH - 0.2 * EI - 0.2 * EJ + 0.2 * EK + 0.2 * EL
+ 0.2 * EN + 0.2 * EO - 0.2 * EP - 0.2 * EQ - 0.2 * ER + 0.2 * ES + 0.2 * FG
+ 0.2 * FH + 0.2 * FI - 0.2 * FJ - 0.2 * FK - 0.2 * FL + 0.2 * FN + 0.2 * FO
- 0.2 * FP + 0.2 * FQ - 0.2 * FR - 0.2 * FS - 0.2 * GH - 0.2 * GI - 0.2 * GJ
+ 0.2 * GK + 0.2 * GL - 0.2 * GN + 0.2 * GO + 0.2 * GP - 0.6 * GQ + 0.2 * GR
- 0.2 * GS + 0.2 * HI - 0.6 * HJ + 0.2 * HK - 0.2 * HL + 0.2 * HN - 0.2 * HO
- 0.2 * HP - 0.2 * HQ - 0.2 * HR - 0.2 * HS + 0.2 * IJ - 0.2 * IK - 0.2 * IL
- 0.2 * IN + 0.2 * IO - 0.2 * IP - 0.2 * IQ + 0.2 * IR + 0.2 * IS + 0.2 * JK
- 0.2 * JL - 0.2 * JN + 0.2 * JO - 0.2 * JP + 0.2 * JQ + 0.2 * JR - 0.2 * JS
- 0.2 * KL - 0.2 * KN - 0.2 * KO - 0.6 * KP - 0.2 * KQ + 0.2 * KR - 0.2 * KS
- 0.2 * LN + 0.2 * LO + 0.2 * LP + 0.2 * LQ - 0.2 * LR + 0.2 * LS - 0.2 * NO
+ 0.2 * NP + 0.2 * NQ + 0.2 * NR - 0.2 * NS - 0.2 * OP - 0.2 * OQ - 0.6 * OR
- 0.2 * OS - 0.2 * PQ + 0.2 * PR - 0.2 * PS - 0.2 * QR + 0.2 * QS + 0.2 * RS

[N] = N - 0.2 * AB + 0.2 * AC + 0.2 * AD - 0.6 * AE + 0.2 * AF - 0.2 * AG
- 0.2 * AH - 0.2 * AI + 0.2 * AJ - 0.2 * AK - 0.2 * AL + 0.2 * AM - 0.2 * AO
- 0.2 * AP - 0.2 * AQ + 0.2 * AR + 0.2 * AS + 0.2 * BC + 0.2 * BD - 0.2 * BE
+ 0.2 * BF - 0.2 * BG - 0.2 * BH - 0.2 * BI - 0.2 * BJ + 0.2 * BK - 0.2 * BL
- 0.6 * BM + 0.2 * BO + 0.2 * BP + 0.2 * BQ - 0.2 * BR - 0.2 * BS + 0.2 * CD
+ 0.2 * CE - 0.2 * CF - 0.2 * CG - 0.2 * CH + 0.2 * CI + 0.2 * CJ - 0.2 * CK
- 0.2 * CL - 0.2 * CM - 0.6 * CO - 0.2 * CP + 0.2 * CQ - 0.2 * CR - 0.2 * CS
- 0.2 * DE - 0.2 * DF + 0.2 * DG + 0.2 * DH - 0.2 * DI - 0.6 * DJ - 0.2 * DK
+ 0.2 * DL - 0.2 * DM - 0.2 * DO - 0.2 * DP - 0.2 * DQ + 0.2 * DR - 0.2 * DS
- 0.2 * EF - 0.2 * EG - 0.2 * EH + 0.2 * EI - 0.2 * EJ + 0.2 * EK + 0.2 * EL
+ 0.2 * EM - 0.2 * EO - 0.2 * EP + 0.2 * EQ + 0.2 * ER - 0.2 * ES - 0.6 * FG
- 0.2 * FH + 0.2 * FI - 0.2 * FJ - 0.2 * FK + 0.2 * FL + 0.2 * FM + 0.2 * FO
+ 0.2 * FP - 0.2 * FQ - 0.2 * FR - 0.2 * FS + 0.2 * GH + 0.2 * GI + 0.2 * GJ
- 0.2 * GK - 0.2 * GL - 0.2 * GM + 0.2 * GO + 0.2 * GP - 0.2 * GQ + 0.2 * GR
- 0.2 * GS - 0.2 * HI - 0.2 * HJ - 0.2 * HK + 0.2 * HL + 0.2 * HM - 0.2 * HO
+ 0.2 * HP + 0.2 * HQ - 0.6 * HR + 0.2 * HS + 0.2 * IJ - 0.6 * IK + 0.2 * IL
- 0.2 * IM + 0.2 * IO - 0.2 * IP - 0.2 * IQ - 0.2 * IR - 0.2 * IS + 0.2 * JK
- 0.2 * JL - 0.2 * JM - 0.2 * JO + 0.2 * JP - 0.2 * JQ - 0.2 * JR + 0.2 * JS
+ 0.2 * KL - 0.2 * KM - 0.2 * KO + 0.2 * KP - 0.2 * KQ + 0.2 * KR + 0.2 * KS
- 0.2 * LM - 0.2 * LO - 0.2 * LP - 0.6 * LQ - 0.2 * LR + 0.2 * LS - 0.2 * MO
+ 0.2 * MP + 0.2 * MQ + 0.2 * MR - 0.2 * MS - 0.2 * OP + 0.2 * OQ + 0.2 * OR
+ 0.2 * OS - 0.2 * PQ - 0.2 * PR - 0.6 * PS - 0.2 * QR + 0.2 * QS - 0.2 * RS

[O] = O + 0.2 * AB - 0.2 * AC - 0.2 * AD - 0.2 * AE - 0.2 * AF - 0.2 * AG
- 0.2 * AH + 0.2 * AI - 0.2 * AJ + 0.2 * AK + 0.2 * AL + 0.2 * AM - 0.2 * AN
+ 0.2 * AP - 0.6 * AQ + 0.2 * AR - 0.2 * AS - 0.2 * BC + 0.2 * BD + 0.2 * BE
- 0.6 * BF + 0.2 * BG - 0.2 * BH - 0.2 * BI - 0.2 * BJ + 0.2 * BK - 0.2 * BL
- 0.2 * BM + 0.2 * BN - 0.2 * BP - 0.2 * BQ - 0.2 * BR + 0.2 * BS + 0.2 * CD
+ 0.2 * CE - 0.2 * CF + 0.2 * CG - 0.2 * CH - 0.2 * CI - 0.2 * CJ - 0.2 * CK
+ 0.2 * CL - 0.2 * CM - 0.6 * CN + 0.2 * CP + 0.2 * CQ + 0.2 * CR - 0.2 * CS
+ 0.2 * DE + 0.2 * DF - 0.2 * DG - 0.2 * DH - 0.2 * DI + 0.2 * DJ + 0.2 * DK
- 0.2 * DL - 0.2 * DM - 0.2 * DN - 0.6 * DP - 0.2 * DQ + 0.2 * DR - 0.2 * DS
- 0.2 * EF - 0.2 * EG + 0.2 * EH + 0.2 * EI - 0.2 * EJ - 0.6 * EK - 0.2 * EL
+ 0.2 * EM - 0.2 * EN - 0.2 * EP - 0.2 * EQ - 0.2 * ER + 0.2 * ES - 0.2 * FG
- 0.2 * FH - 0.2 * FI + 0.2 * FJ - 0.2 * FK + 0.2 * FL + 0.2 * FM + 0.2 * FN
- 0.2 * FP - 0.2 * FQ + 0.2 * FR + 0.2 * FS - 0.6 * GH - 0.2 * GI + 0.2 * GJ
- 0.2 * GK - 0.2 * GL + 0.2 * GM + 0.2 * GN + 0.2 * GP + 0.2 * GQ - 0.2 * GR
- 0.2 * GS + 0.2 * HI + 0.2 * HJ + 0.2 * HK - 0.2 * HL - 0.2 * HM - 0.2 * HN
+ 0.2 * HP + 0.2 * HQ - 0.2 * HR + 0.2 * HS - 0.2 * IJ - 0.2 * IK - 0.2 * IL
+ 0.2 * IM + 0.2 * IN - 0.2 * IP + 0.2 * IQ + 0.2 * IR - 0.6 * IS + 0.2 * JK
- 0.6 * JL + 0.2 * JM - 0.2 * JN + 0.2 * JP - 0.2 * JQ - 0.2 * JR - 0.2 * JS
+ 0.2 * KL - 0.2 * KM - 0.2 * KN + 0.2 * KP + 0.2 * KQ - 0.2 * KR - 0.2 * KS
+ 0.2 * LM - 0.2 * LN - 0.2 * LP + 0.2 * LQ - 0.2 * LR + 0.2 * LS - 0.2 * MN
- 0.2 * MP - 0.2 * MQ - 0.6 * MR - 0.2 * MS - 0.2 * NP + 0.2 * NQ + 0.2 * NR
+ 0.2 * NS - 0.2 * PQ + 0.2 * PR + 0.2 * PS - 0.2 * QR - 0.2 * QS - 0.2 * RS

[P] = P - 0.2 * AB + 0.2 * AC - 0.2 * AD - 0.2 * AE + 0.2 * AF + 0.2 * AG
- 0.2 * AH + 0.2 * AI - 0.6 * AJ - 0.2 * AK - 0.2 * AL + 0.2 * AM - 0.2 * AN
+ 0.2 * AO + 0.2 * AQ - 0.2 * AR - 0.2 * AS + 0.2 * BC - 0.2 * BD - 0.2 * BE
- 0.2 * BF - 0.2 * BG - 0.2 * BH - 0.2 * BI + 0.2 * BJ - 0.2 * BK + 0.2 * BL
+ 0.2 * BM + 0.2 * BN - 0.2 * BO + 0.2 * BQ - 0.6 * BR + 0.2 * BS - 0.2 * CD
+ 0.2 * CE + 0.2 * CF - 0.6 * CG + 0.2 * CH - 0.2 * CI - 0.2 * CJ - 0.2 * CK
+ 0.2 * CL - 0.2 * CM - 0.2 * CN + 0.2 * CO - 0.2 * CQ - 0.2 * CR - 0.2 * CS
+ 0.2 * DE + 0.2 * DF - 0.2 * DG + 0.2 * DH - 0.2 * DI - 0.2 * DJ - 0.2 * DK
- 0.2 * DL + 0.2 * DM - 0.2 * DN - 0.6 * DO + 0.2 * DQ + 0.2 * DR + 0.2 * DS

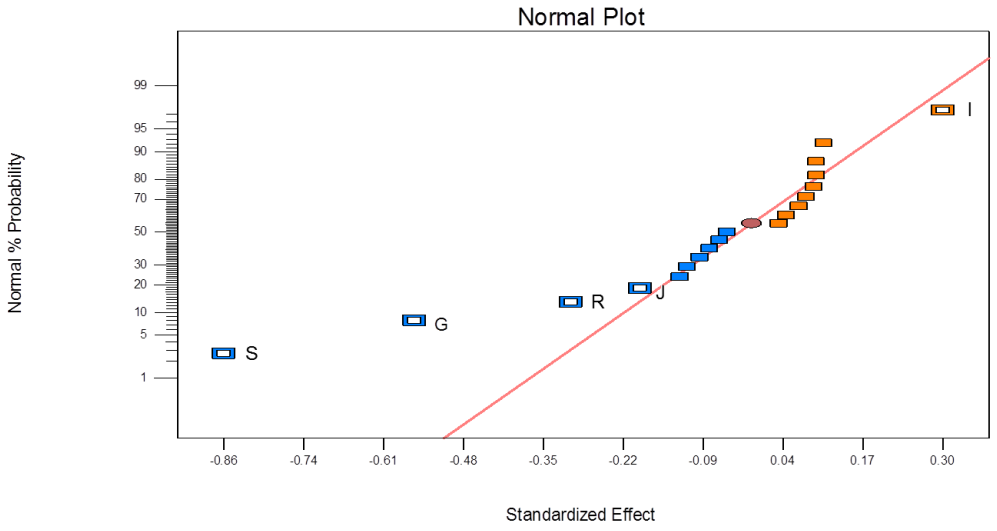
```
  + 0.2 * EF + 0.2 * EG - 0.2 * EH - 0.2 * EI - 0.2 * EJ + 0.2 * EK + 0.2 * EL
  - 0.2 * EM - 0.2 * EN - 0.2 * EO - 0.6 * EQ - 0.2 * ER + 0.2 * ES - 0.2 * FG
  - 0.2 * FH + 0.2 * FI + 0.2 * FJ - 0.2 * FK - 0.6 * FL - 0.2 * FM + 0.2 * FN
  - 0.2 * FO - 0.2 * FQ - 0.2 * FR - 0.2 * FS - 0.2 * GH - 0.2 * GI - 0.2 * GJ
  + 0.2 * GK - 0.2 * GL + 0.2 * GM + 0.2 * GN + 0.2 * GO - 0.2 * GQ - 0.2 * GR
  + 0.2 * GS - 0.6 * HI - 0.2 * HJ + 0.2 * HK - 0.2 * HL - 0.2 * HM + 0.2 * HN
  + 0.2 * HO + 0.2 * HQ + 0.2 * HR - 0.2 * HS + 0.2 * IJ + 0.2 * IK + 0.2 * IL
  - 0.2 * IM - 0.2 * IN - 0.2 * IO + 0.2 * IQ + 0.2 * IR - 0.2 * IS - 0.2 * JK
  - 0.2 * JL - 0.2 * JM + 0.2 * JN + 0.2 * JO - 0.2 * JQ + 0.2 * JR + 0.2 * JS
  + 0.2 * KL - 0.6 * KM + 0.2 * KN - 0.2 * KO + 0.2 * KQ - 0.2 * KR - 0.2 * KS
  + 0.2 * LM - 0.2 * LN - 0.2 * LO - 0.2 * LQ + 0.2 * LR - 0.2 * LS + 0.2 * MN
  - 0.2 * MO - 0.2 * MQ + 0.2 * MR - 0.2 * MS - 0.2 * NO - 0.2 * NQ - 0.2 * NR
  - 0.6 * NS - 0.2 * OQ + 0.2 * OR + 0.2 * OS - 0.2 * QR + 0.2 * QS - 0.2 * RS
[Q] = Q - 0.2 * AB + 0.2 * AC - 0.2 * AD + 0.2 * AE + 0.2 * AF - 0.2 * AG
  + 0.2 * AH - 0.2 * AI - 0.2 * AJ + 0.2 * AK - 0.2 * AL - 0.2 * AM - 0.2 * AN
  - 0.6 * AO + 0.2 * AP + 0.2 * AR - 0.2 * AS - 0.2 * BC + 0.2 * BD - 0.2 * BE
  - 0.2 * BF + 0.2 * BG + 0.2 * BH - 0.2 * BI + 0.2 * BJ - 0.6 * BK - 0.2 * BL
  - 0.2 * BM + 0.2 * BN - 0.2 * BO + 0.2 * BP + 0.2 * BR - 0.2 * BS + 0.2 * CD
  - 0.2 * CE - 0.2 * CF - 0.2 * CG - 0.2 * CH - 0.2 * CI - 0.2 * CJ + 0.2 * CK
  - 0.2 * CL + 0.2 * CM + 0.2 * CN + 0.2 * CO - 0.2 * CP + 0.2 * CR - 0.6 * CS
  - 0.2 * DE + 0.2 * DF + 0.2 * DG - 0.6 * DH + 0.2 * DI - 0.2 * DJ - 0.2 * DK
  - 0.2 * DL + 0.2 * DM - 0.2 * DN - 0.2 * DO + 0.2 * DP - 0.2 * DR - 0.2 * DS
  + 0.2 * EF + 0.2 * EG - 0.2 * EH + 0.2 * EI - 0.2 * EJ - 0.2 * EK - 0.2 * EL
  - 0.2 * EM + 0.2 * EN - 0.2 * EO - 0.6 * EP + 0.2 * ER + 0.2 * ES + 0.2 * FG
  + 0.2 * FH - 0.2 * FI - 0.2 * FJ - 0.2 * FK + 0.2 * FL + 0.2 * FM - 0.2 * FN
  - 0.2 * FO - 0.2 * FP - 0.6 * FR - 0.2 * FS - 0.2 * GH - 0.2 * GI + 0.2 * GJ
  + 0.2 * GK - 0.2 * GL - 0.6 * GM - 0.2 * GN + 0.2 * GO - 0.2 * GP - 0.2 * GR
  - 0.2 * GS - 0.2 * HI - 0.2 * HJ - 0.2 * HK + 0.2 * HL - 0.2 * HM + 0.2 * HN
  + 0.2 * HO + 0.2 * HP - 0.2 * HR - 0.2 * HS - 0.6 * IJ - 0.2 * IK + 0.2 * IL
  - 0.2 * IM - 0.2 * IN + 0.2 * IO + 0.2 * IP + 0.2 * IR + 0.2 * IS + 0.2 * JK
  + 0.2 * JL + 0.2 * JM - 0.2 * JN - 0.2 * JO - 0.2 * JP + 0.2 * JR + 0.2 * JS
  - 0.2 * KL - 0.2 * KM - 0.2 * KN + 0.2 * KO + 0.2 * KP - 0.2 * KR + 0.2 * KS
  + 0.2 * LM - 0.6 * LN + 0.2 * LO - 0.2 * LP + 0.2 * LR - 0.2 * LS + 0.2 * MN
  - 0.2 * MO - 0.2 * MP - 0.2 * MR + 0.2 * MS + 0.2 * NO - 0.2 * NP - 0.2 * NR
  + 0.2 * NS - 0.2 * OP - 0.2 * OR - 0.2 * OS - 0.2 * PR + 0.2 * PS - 0.2 * RS
[R] = R - 0.2 * AB - 0.2 * AC - 0.6 * AD - 0.2 * AE + 0.2 * AF - 0.2 * AG
  - 0.2 * AH - 0.2 * AI + 0.2 * AJ + 0.2 * AK + 0.2 * AL - 0.2 * AM + 0.2 * AN
  + 0.2 * AO - 0.2 * AP + 0.2 * AQ - 0.2 * AS - 0.2 * BC + 0.2 * BD - 0.2 * BE
  + 0.2 * BF + 0.2 * BG - 0.2 * BH + 0.2 * BI - 0.2 * BJ - 0.2 * BK + 0.2 * BL
  - 0.2 * BM - 0.2 * BN - 0.2 * BO - 0.6 * BP + 0.2 * BQ + 0.2 * BS - 0.2 * CD
  + 0.2 * CE - 0.2 * CF - 0.2 * CG + 0.2 * CH + 0.2 * CI - 0.2 * CJ + 0.2 * CK
  - 0.6 * CL - 0.2 * CM - 0.2 * CN + 0.2 * CO - 0.2 * CP + 0.2 * CQ + 0.2 * CS
  + 0.2 * DE - 0.2 * DF - 0.2 * DG - 0.2 * DH - 0.2 * DI - 0.2 * DJ - 0.2 * DK
  + 0.2 * DL - 0.2 * DM + 0.2 * DN + 0.2 * DO + 0.2 * DP - 0.2 * DQ + 0.2 * DS
  - 0.2 * EF + 0.2 * EG + 0.2 * EH - 0.6 * EI + 0.2 * EJ - 0.2 * EK - 0.2 * EL
  - 0.2 * EM + 0.2 * EN - 0.2 * EO - 0.2 * EP + 0.2 * EQ - 0.2 * ES + 0.2 * FG
  + 0.2 * FH - 0.2 * FI + 0.2 * FJ - 0.2 * FK - 0.2 * FL - 0.2 * FM - 0.2 * FN
  + 0.2 * FO - 0.2 * FP - 0.6 * FQ + 0.2 * FS + 0.2 * GH + 0.2 * GI - 0.2 * GJ
  - 0.2 * GK - 0.2 * GL + 0.2 * GM + 0.2 * GN - 0.2 * GO - 0.2 * GP - 0.2 * GQ
  - 0.6 * GS - 0.2 * HI - 0.2 * HJ + 0.2 * HK + 0.2 * HL - 0.2 * HM - 0.6 * HN
  - 0.2 * HO + 0.2 * HP - 0.2 * HQ - 0.2 * HS - 0.2 * IJ - 0.2 * IK - 0.2 * IL
  + 0.2 * IM - 0.2 * IN + 0.2 * IO + 0.2 * IP + 0.2 * IQ - 0.2 * IS - 0.6 * JK
  - 0.2 * JL + 0.2 * JM - 0.2 * JN - 0.2 * JO + 0.2 * JP + 0.2 * JQ + 0.2 * JS
  + 0.2 * KL + 0.2 * KM + 0.2 * KN - 0.2 * KO - 0.2 * KP - 0.2 * KQ + 0.2 * KS
  - 0.2 * LM - 0.2 * LN - 0.2 * LO + 0.2 * LP + 0.2 * LQ - 0.2 * LS + 0.2 * MN
  - 0.6 * MO + 0.2 * MP - 0.2 * MQ + 0.2 * MS + 0.2 * NO - 0.2 * NP - 0.2 * NQ
  - 0.2 * NS + 0.2 * OP - 0.2 * OQ - 0.2 * OS - 0.2 * PQ - 0.2 * PS - 0.2 * QS
[S] = S - 0.2 * AB + 0.2 * AC + 0.2 * AD + 0.2 * AE - 0.2 * AF + 0.2 * AG
  - 0.6 * AH - 0.2 * AI - 0.2 * AJ + 0.2 * AK + 0.2 * AL - 0.2 * AM + 0.2 * AN
  - 0.2 * AO - 0.2 * AP - 0.2 * AQ - 0.2 * AR - 0.2 * BC - 0.2 * BD - 0.6 * BE
  - 0.2 * BF + 0.2 * BG - 0.2 * BH - 0.2 * BI - 0.2 * BJ + 0.2 * BK + 0.2 * BL
  + 0.2 * BM - 0.2 * BN + 0.2 * BO + 0.2 * BP - 0.2 * BQ + 0.2 * BR - 0.2 * CD
  + 0.2 * CE - 0.2 * CF + 0.2 * CG + 0.2 * CH - 0.2 * CI + 0.2 * CJ - 0.2 * CK
  - 0.2 * CL + 0.2 * CM - 0.2 * CN - 0.2 * CO - 0.2 * CP - 0.6 * CQ + 0.2 * CR
  - 0.2 * DE + 0.2 * DF - 0.2 * DG - 0.2 * DH + 0.2 * DI + 0.2 * DJ - 0.2 * DK
  + 0.2 * DL - 0.6 * DM - 0.2 * DN - 0.2 * DO + 0.2 * DP - 0.2 * DQ + 0.2 * DR
  + 0.2 * EF - 0.2 * EG - 0.2 * EH - 0.2 * EI - 0.2 * EJ - 0.2 * EK - 0.2 * EL
  + 0.2 * EM - 0.2 * EN + 0.2 * EO + 0.2 * EP + 0.2 * EQ - 0.2 * ER - 0.2 * FG
  + 0.2 * FH + 0.2 * FI - 0.6 * FJ + 0.2 * FK - 0.2 * FL - 0.2 * FM - 0.2 * FN
  + 0.2 * FO - 0.2 * FP - 0.2 * FQ + 0.2 * FR + 0.2 * GH + 0.2 * GI - 0.2 * GJ
```

```
+ 0.2 * GK - 0.2 * GL - 0.2 * GM - 0.2 * GN - 0.2 * GO + 0.2 * GP - 0.2 * GQ
- 0.6 * GR + 0.2 * HI + 0.2 * HJ - 0.2 * HK - 0.2 * HL - 0.2 * HM + 0.2 * HN
+ 0.2 * HO - 0.2 * HP - 0.2 * HQ - 0.2 * HR - 0.2 * IJ - 0.2 * IK + 0.2 * IL
+ 0.2 * IM - 0.2 * IN - 0.6 * IO - 0.2 * IP + 0.2 * IQ - 0.2 * IR - 0.2 * JK
- 0.2 * JL - 0.2 * JM + 0.2 * JN - 0.2 * JO + 0.2 * JP + 0.2 * JQ + 0.2 * JR
- 0.6 * KL - 0.2 * KM + 0.2 * KN - 0.2 * KO - 0.2 * KP + 0.2 * KQ + 0.2 * KR
+ 0.2 * LM + 0.2 * LN + 0.2 * LO - 0.2 * LP - 0.2 * LQ - 0.2 * LR - 0.2 * MN
- 0.2 * MO - 0.2 * MP + 0.2 * MQ + 0.2 * MR + 0.2 * NO - 0.6 * NP + 0.2 * NQ
- 0.2 * NR + 0.2 * OP - 0.2 * OQ - 0.2 * OR + 0.2 * PQ - 0.2 * PR - 0.2 * QR
```

(b) Show that for the 20-run Plackett-Burman in Table P8.10, the weights (or correlations) that multiply the two-factor interactions in each alias chain are either -0.2, 0.2, or -0.6. Of the 153 interactions that are aliased with each main effect, 144 have weights of -0.2 or 0.2, while nine interactions have weights of -0.6

See the Design Expert output shown above.

(c) Verify that the five largest main effects are *S*, *G*, *R*, *I* and *J*.



Design Expert Output

Response 2 Response rate
ANOVA for selected factorial model
Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	6.36	5	1.27	35.43	< 0.0001	significant
G-Sticker	*1.55*	*1*	*1.55*	*43.03*	*< 0.0001*	
I-Copy Message	*0.44*	*1*	*0.44*	*12.20*	*0.0036*	
J-Letter headline	*0.18*	*1*	*0.18*	*5.13*	*0.0399*	
R-2nd buckslip	*0.46*	*1*	*0.46*	*12.86*	*0.0030*	
S-Interest rate	*3.73*	*1*	*3.73*	*103.91*	*< 0.0001*	
Residual	0.50	14	0.036			
Cor Total	6.87	19				

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	1.30	1	0.042	1.21	1.39	
G-Sticker	-0.28	1	0.042	-0.37	-0.19	1.00
J-Copy Message	0.15	1	0.042	0.057	0.24	1.00
K-Letter headline	-0.096	1	0.042	-0.19	-5.106E-003	1.00
S-2nd buckslip	-0.15	1	0.042	-0.24	-0.061	1.00
T-Interest rate	-0.43	1	0.042	-0.52	-0.34	1.00

(d) Factor *S* (interest rate) and *G* (presence of a sticker) are by far the largest main effects. The correlation between the main effect of *R* (2nd buckslip) and the *SG* interaction is -0.6. This means that a significant *SG* interaction would bias the estimate of the main effect of *R* by -0.6 times the value of the interaction. This suggests that it may not be the main effect of factor *R* that is important, but the two factor interaction between *S* and *G*.

(e) Since this design projects into a full factorial in any three factors, obtain the projection in factors *S*, *G* and *R* and verify that it is a full factorial with some runs replicated. Fit a full factorial model involving all three of these factors and the interactions (you will need to use a regression program to do this). Show that *S*, *G* and the *SG* interaction are significant.

Test Cell	*G*	*R*	*S*	*GR*	*GS*	*RS*	*GRS*	run	Orders	Resp Rate
1	1	1	-1	1	-1	-1	-1	*gr*	52	1.04%
2	1	1	1	1	1	1	1	*grs*	38	0.76%
3	1	-1	1	-1	1	-1	-1	*gs*	42	0.84%
4	-1	-1	-1	1	1	1	-1	(1)	134	2.68%
5	-1	-1	-1	1	1	1	-1	(1)	104	2.08%
6	1	-1	-1	-1	-1	1	1	*g*	60	1.20%
7	1	1	-1	1	-1	-1	-1	*gr*	61	1.22%
8	-1	-1	1	1	-1	-1	1	*s*	68	1.36%
9	1	1	-1	1	-1	-1	-1	*gr*	57	1.14%
10	1	-1	1	-1	1	-1	-1	*gs*	30	0.60%
11	-1	1	-1	-1	1	-1	1	*r*	108	2.16%
12	-1	1	1	-1	-1	1	-1	*rs*	39	0.78%
13	-1	1	1	-1	-1	1	-1	*rs*	40	0.80%
14	-1	1	1	-1	-1	1	-1	*rs*	49	0.98%
15	1	-1	1	-1	1	-1	-1	*gs*	37	0.74%
16	-1	-1	-1	1	1	1	-1	(1)	99	1.98%
17	1	1	-1	1	-1	-1	-1	*gr*	86	1.72%
18	-1	1	1	-1	-1	1	-1	*rs*	43	0.86%
19	1	-1	1	-1	1	-1	-1	*gs*	47	0.94%
20	-1	-1	-1	-1	-1	-1	-1	*gr*	104	2.08%

The projected design is shown in the above table, along with the terms to estimate the interaction. Note that all 8 corners of the 2^3 are represented at least once. The analysis shown below shows that the significant factors are *G* and *S* (*A* and *C*). *R* (*B*) is not significant. The *GS* (*AC*) interaction has a higher *t*-value than the *R* (*C*) factor that it was confounded with in the original Plackett-Burman design.

Design Expert Output

Response 2 Response rate

ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	**Sum of Squares**	**df**	**Mean Square**	**F Value**	**p-value Prob > F**	
Model	6.20	7	0.89	15.87	< 0.0001	significant
A-G - Sticker	*1.31*	*1*	*1.31*	*23.50*	*0.0004*	
B-R - 2nd buckslip	*0.048*	*1*	*0.048*	*0.86*	*0.3718*	
C-S - Interest rate	*1.91*	*1*	*1.91*	*34.23*	*< 0.0001*	
AB	*0.074*	*1*	*0.074*	*1.33*	*0.2706*	
AC	*0.29*	*1*	*0.29*	*5.25*	*0.0408*	
BC	*0.063*	*1*	*0.063*	*1.12*	*0.3099*	
ABC	*0.026*	*1*	*0.026*	*0.46*	*0.5084*	
Pure Error	0.67	12	0.056			
Cor Total	6.87	19				

Factor	**Coefficient Estimate**	**df**	**Standard Error**	**95% CI Low**	**95% CI High**	**VIF**
Intercept	1.33	1	0.066	1.18	1.47	

A-G - Sticker	-0.32	1	0.066	-0.46	-0.18	1.56
B-R - 2nd buckslip	-0.061	1	0.066	-0.21	0.083	1.56
C-S - Interest rate	-0.39	1	0.066	-0.53	-0.24	1.56
AB	0.076	1	0.066	-0.068	0.22	1.56
AC	0.15	1	0.066	7.414E-003	0.30	1.56
BC	-0.070	1	0.066	-0.21	0.074	1.56
ABC	0.045	1	0.066	-0.099	0.19	1.00

8.31. Consider the following experiment:

Run	Treatment Combination	*A*	*B*	*C*	*D=AB*	*E=-AC*
1	*d*	-1	-1	-1	1	-1
2	*ae*	1	-1	-1	-1	1
3	*b*	-1	1	-1	-1	-1
4	*abde*	1	1	-1	1	1
5	*cde*	-1	-1	1	1	1
6	*ac*	1	-1	1	-1	-1
7	*bce*	-1	1	1	-1	1
8	*abcd*	1	1	1	1	-1

Answer the following questions about this experiment:

(a) How many factors did this experiment investigate?

5.

(b) How many factors are in the basic design?

3.

(c) Assume that the factors in the experiment are represented by the initial letters of the alphabet (i.e. A, B, etc.), what are the design generators for the factors beyond the basic design?

$D = AB$ and $E = -AC$, therefore $I = ABD - ACE$.

(d) Is this design a principal fraction?

No, it is not a principal fraction.

(e) What is the complete defining relation?

$I = ABD - ACE - BDCE$.

(f) What is the resolution of this design?

It is a resolution III design.

8.32. Consider the following experiment:

Run	Treatment Combination		*A*	*B*	*C*	*D=ABC*	*y*
1	(1)		-1	-1	-1	-1	8
2	*ad*		1	-1	-1	1	10
3	*bd*		-1	1	-1	1	12
4	*ab*		1	1	-1	-1	7
5	*cd*		-1	-1	1	1	13
6	*ac*		1	-1	1	-1	6
7	*bc*		-1	1	1	-1	5
8	*abcd*		1	1	1	1	11
		Avg (+)	8.5	4.5	8.75	11.5	
		Avg (-)	9.5	9.25	9.25	26	
		Effect	-0.5	-2.375	-0.25	-7.25	

(a) How many factors did this experiment investigate?

4.

(b) What is the resolution of this design?

IV.

(c) Calculate the estimates of the effects.

See table above.

(d) What is the complete defining relation?

I = *ABCD*.

8.33. An unreplicated 2^{5-1} fractional factorial experiment with four center points has been run in a chemical process. The response variable is molecular weight. The experimenter has used the following factors:

Factor	Natural levels	Coded levels (x's)
A – time	20, 40 (minutes)	-1, 1
B – temperature	160, 180 (deg C)	-1, 1
C – concentration	30, 60 (percent)	-1, 1
D – stirring rate	100, 150 (rpm)	-1, 1
E – catalyst type	1, 2 (type)	-1, 1

Suppose that the prediction equation from this experiment is $\hat{y} = 10 + 3x_1 + 2x_2 - 1x_1x_2$. What is the predicted response at *A* = 30, *B* = 165, *C* = 50, *D* = 135, and *E* = 1?

A = 3, is *A* = 0 in coded units, *B* = 165 is *B* = -0.5 in coded units. These are the only two factors in the prediction equation.

$$\hat{y} = 10 + 3(0) + 2(-0.5) - 1(0)(-0.5) = 9$$

8.34. An unreplicated 2^{4-1} fractional factorial experiment has been run. The experimenter has used the following factors:

Factor	Natural levels	Coded levels (x's)
A	20, 50	-1, 1
B	200, 280	-1, 1
C	50, 100	-1, 1
D	150, 200	-1, 1

(a) Suppose that this design has four center points that average 100. The average of the 8 factorial design points is 95. What is the sum of squares for pure quadratic curvature?

$$SS_{\text{Pure quadratic}} = \frac{n_F n_C \left(\bar{y}_F - \bar{y}_C\right)^2}{n_F + n_C} = \frac{8*4\left(95-100\right)^2}{8+4} = 66.67$$

(b) Suppose that the prediction equation that results from this experiment is $\hat{y} = 100 - 2x_1 + 10x_2 - 4x_1x_2$. Find the predicted response at *A* = 41, *B* = 280, *C* = 60 and *D* = 185.

A = 41, is *A* = 0.4 in coded units, *B* = 280 is *B* = 1 in coded units. These are the only two factors in the prediction equation.

$$\hat{y} = 100 - 2\left(0.4\right) + 10\left(1\right) - 4\left(0.4\right)\left(10\right) = 93.2$$

8.35. A 2^{6-2} factorial experiment with three replicates has been run in a pharmaceutical drug manufacturing process. The experimenter has used the following factors:

Factor	Natural levels	Coded levels (x's)
A	50, 100	-1, 1
B	20, 60	-1, 1
C	10, 30	-1, 1
D	12, 18	-1, 1
E	15, 30	-1, 1
F	60, 100	-1, 1

(a) If two main effects and one two-factor interaction are included in the final model, how many degrees of freedom for error will be available?

A 2^{6-2} factorial experiment has 16 runs. We are estimating four model parameters – the intercept, two main effects and one two-factor interactions. This leaves 12 degrees of freedom for error. This assumes a non-hierarchical model.

(b) Suppose that the significant factors are *A*, *C*, *AB* and *AC*. What other effects need to be included to obtain a hierarchical model?

The hierarchical model would include *A*, *B*, *C*, *AB* and *AC*.

8.36. Consider the following design:

Run	*A*	*B*	*C*	*D*	*E*	*y*	Block = *ABC*
1	-1	-1	-1	-1	-1	63	-1

2	1	-1	-1	-1	1	21	1
3	-1	1	-1	-1	1	36	1
4	1	1	-1	-1	-1	99	-1
5	-1	-1	1	-1	1	24	1
6	1	-1	1	-1	-1	66	-1
7	-1	1	1	-1	-1	71	-1
8	1	1	1	-1	1	54	1
9	-1	-1	-1	1	-1	23	-1
10	1	-1	-1	1	1	74	1
11	-1	1	-1	1	1	80	1
12	1	1	-1	1	-1	33	-1
13	-1	-1	1	1	1	63	1
14	1	-1	1	1	-1	21	-1
15	-1	1	1	1	-1	44	-1
16	1	1	1	1	1	96	1
Avg(+)	58	64.125	54.875	54.25	56		
Avg(-)	50.5	44.375	53.625	54.25	52.5		
Effect	3.75	9.875	0.625	0	1.75		

(a) What is the generator for column *E*?

E = ABC.

(b) If *ABC* is confounded with blocks, run 1 above goes in the ___ block. Answer either + or –.

ABC goes in to the – block.

(c) What is the resolution of this design?

Resolution IV design.

(d) Find the estimates of the main effects and their aliases.

See table above.

8.37. Consider the following design:

Run	*A*	*B*	*C*	*D*	*E*	*y*	Block=*ABE*
1	-1	-1	-1	-1	-1	65	-1
2	1	-1	-1	-1	1	25	-1
3	-1	1	-1	-1	1	30	-1
4	1	1	-1	-1	-1	89	-1
5	-1	-1	1	-1	1	25	1
6	1	-1	1	-1	-1	60	1
7	-1	1	1	-1	-1	70	1
8	1	1	1	-1	1	50	1
9	-1	-1	-1	1	1	20	1
10	1	-1	-1	1	-1	70	1
11	-1	1	-1	1	-1	80	1
12	1	1	-1	1	1	30	1
13	-1	-1	1	1	-1	60	-1
14	1	-1	1	1	1	20	-1
15	-1	1	1	1	1	40	-1
16	1	1	1	1	-1	90	-1
Avg(+)	54.25	59.875	51.875	51.25	30		
Avg(–)	48.75	43.125	51.125	51.75	73		
Effect	2.75	8.375	0.375	-0.25	-21.5		

(a) What is the generator for column *E*?

E = –ABCD.

(b) If *ABE* is confounded with blocks, run 16 above goes in the ___ block. Answer either + or –.

Run 16 is in the – block.

(c) What is the resolution of this design?

It is a resolution V design.

(d) Find the estimates of the main effects and their aliases.

See the table above.

8.38. Consider the following design:

Run	*A*	*B*	*C*	*D=–ABC*	*E=–BC*	*y*	Block = *AB*
1	-1	-1	-1	1	-1	50	1
2	1	-1	-1	-1	-1	20	-1
3	-1	1	-1	-1	1	40	-1
4	1	1	-1	1	1	25	1
5	-1	-1	1	-1	1	45	1
6	1	-1	1	1	1	30	-1
7	-1	1	1	1	-1	40	-1
8	1	1	1	-1	-1	30	1

(a) What is the generator for column *D*?

D = –ABC

(b) What is the generator for column *E*?

E = –BC

(c) If this design were run in two blocks with the *AB* interaction confounded with blocks, the run *d* would be in the block where the sign of *AB* is ____? Answer either + or –.

Run *d*, is the combination (-1, -1, -1, 1, -1), it is run 1 and is in the + block.

8.39. Consider the following design:

Run	A	B	C	D	E	y
1	-1	-1	-1	1	1	40
2	1	-1	-1	-1	1	10
3	-1	1	-1	-1	-1	30
4	1	1	-1	1	-1	20
5	-1	-1	1	-1	-1	40
6	1	-1	1	1	-1	30
7	-1	1	1	1	1	20
8	1	1	1	-1	1	30

9	1	1	1	-1	-1
10	-1	1	1	1	-1
11	1	-1	1	1	1
12	-1	-1	1	-1	1
13	1	1	-1	1	1
14	-1	1	-1	-1	1
15	1	-1	-1	-1	-1
16	-1	-1	-1	1	-1

(a) What is the generator for column *D*?

$D = -ABC$

(b) What is the generator for column *E*?

$E = BC$

(c) If this design were folded over, what is the resolution of the combined design?

Resolution IV design.

8.40S. In an article in *Quality Engineering* ("An Application of Fractional Factorial Experimental Designs," 1988, Vol. 1 pp. 19-23) M.B. Kilgo describes an experiment to determine the effect of CO_2 pressure (*A*), CO_2 temperature (*B*), peanut moisture (*C*), CO_2 flow rate (*D*), and peanut particle size (*E*) on the total yield of oil per batch of peanuts (*y*). The levels she used for these factors are shown in Table P8.11. She conducted the 16-run fractional factorial experiment shown in Table P8.12.

Table P8.11

	A	*B*	*C*	*D*	*E*
Coded Level	Pressure (bar)	Temp (C)	Moisture (% by weight)	Flow (liters/min)	Particle Size (mm)
-1	415	25	5	40	1.28
1	550	95	15	60	4.05

Table P8.12

	A	*B*	*C*	*D*	*E*	y
1	415	25	5	40	1.28	63
2	550	25	5	40	4.05	21
3	415	95	5	40	4.05	36
4	550	95	5	40	1.28	99
5	415	25	15	40	4.05	24
6	550	25	15	40	1.28	66
7	415	95	15	40	1.28	71
8	550	95	15	40	4.05	54
9	415	25	5	60	4.05	23
10	550	25	5	60	1.28	74
11	415	95	5	60	1.28	80
12	550	95	5	60	4.05	33
13	415	25	15	60	1.28	63
14	550	25	15	60	4.05	21
15	415	95	15	60	4.05	44
16	550	95	15	60	1.28	96

(a) What type of design has been used? Identify the defining relation and the alias relationships.

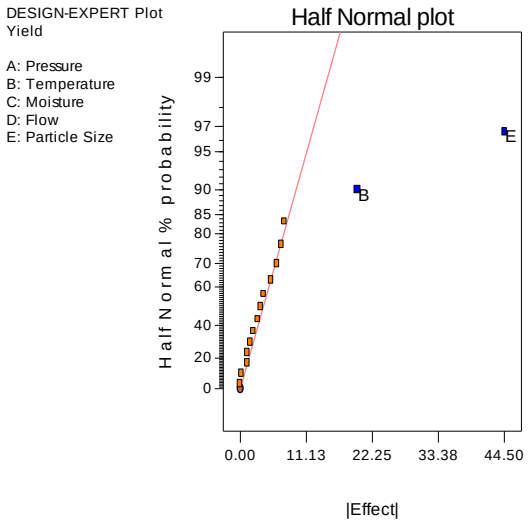
A 2_{V}^{5-1}, 16-run design, with $I=-ABCDE$.

$A(-ABCDE)=$	$-BCDE$	$A=$	$-BCDE$
$B(-ABCDE)=$	$-ACDE$	$B=$	$-ACDE$
$C(-ABCDE)=$	$-ABDE$	$C=$	$-ABDE$
$D(-ABCDE)=$	$-ABCE$	$D=$	$-ABCE$
$E(-ABCDE)=$	$-ABCD$	$E=$	$-ABCD$
$AB(-ABCDE)=$	$-CDE$	$AB=$	$-CDE$
$AC(-ABCDE)=$	$-BDE$	$AC=$	$-BDE$
$AD(-ABCDE)=$	$-BCE$	$AD=$	$-BCE$
$AE(-ABCDE)=$	$-BCD$	$AE=$	$-BCD$
$BC(-ABCDE)=$	$-ADE$	$BC=$	$-ADE$
$BD(-ABCDE)=$	$-ACE$	$BD=$	$-ACE$
$BE(-ABCDE)=$	$-ACD$	$BE=$	$-ACD$
$CD(-ABCDE)=$	$-ABE$	$CD=$	$-ABE$
$CE(-ABCDE)=$	$-ABD$	$CE=$	$-ABD$
$DE(-ABCDE)=$	$-ABC$	$DE=$	$-ABC$

(b) Estimate the factor effects and use a normal probability plot to tentatively identify the important factors.

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Error	A	7.5	225	2.17119
Model	B	19.75	1560.25	15.056
Error	C	1.25	6.25	0.0603107
Error	D	0	0	0
Model	E	44.5	7921	76.4354
Error	AB	5.25	110.25	1.06388
Error	AC	1.25	6.25	0.0603107
Error	AD	-4	64	0.617582
Error	AE	7	196	1.89134
Error	BC	3	36	0.34739
Error	BD	-1.75	12.25	0.118209
Error	BE	0.25	0.25	0.00241243
Error	CD	2.25	20.25	0.195407
Error	CE	-6.25	156.25	1.50777
Error	DE	3.5	49	0.472836
	Lenth's ME	11.5676		
	Lenth's SME	23.4839		



(c) Perform an appropriate statistical analysis to test the hypothesis that the factors identified in part above have a significant effect on the yield of peanut oil.

Design Expert Output

Response: Yield

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	9481.25	2	4740.63	69.89	< 0.0001	significant
B	*1560.25*	*1*	*1560.25*	*23.00*	*0.0003*	
E	*7921.00*	*1*	*7921.00*	*116.78*	*< 0.0001*	
Residual	881.75	13	67.83			
Cor Total	10363.00	15				

The Model F-value of 69.89 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	8.24	R-Squared	0.9149
Mean	54.25	Adj R-Squared	0.9018
C.V.	15.18	Pred R-Squared	0.8711
PRESS	1335.67	Adeq Precision	18.017

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	54.25	1	2.06	49.80	58.70	
B-Temperature	9.88	1	2.06	5.43	14.32	1.00
E-Particle Size	22.25	1	2.06	17.80	26.70	1.00

(d) Fit a model that could be used to predict peanut oil yield in terms of the factors that you have identified as important.

Design Expert Output

Final Equation in Terms of Coded Factors:

Yield =
+54.25
+9.88 * B
+22.25 * E

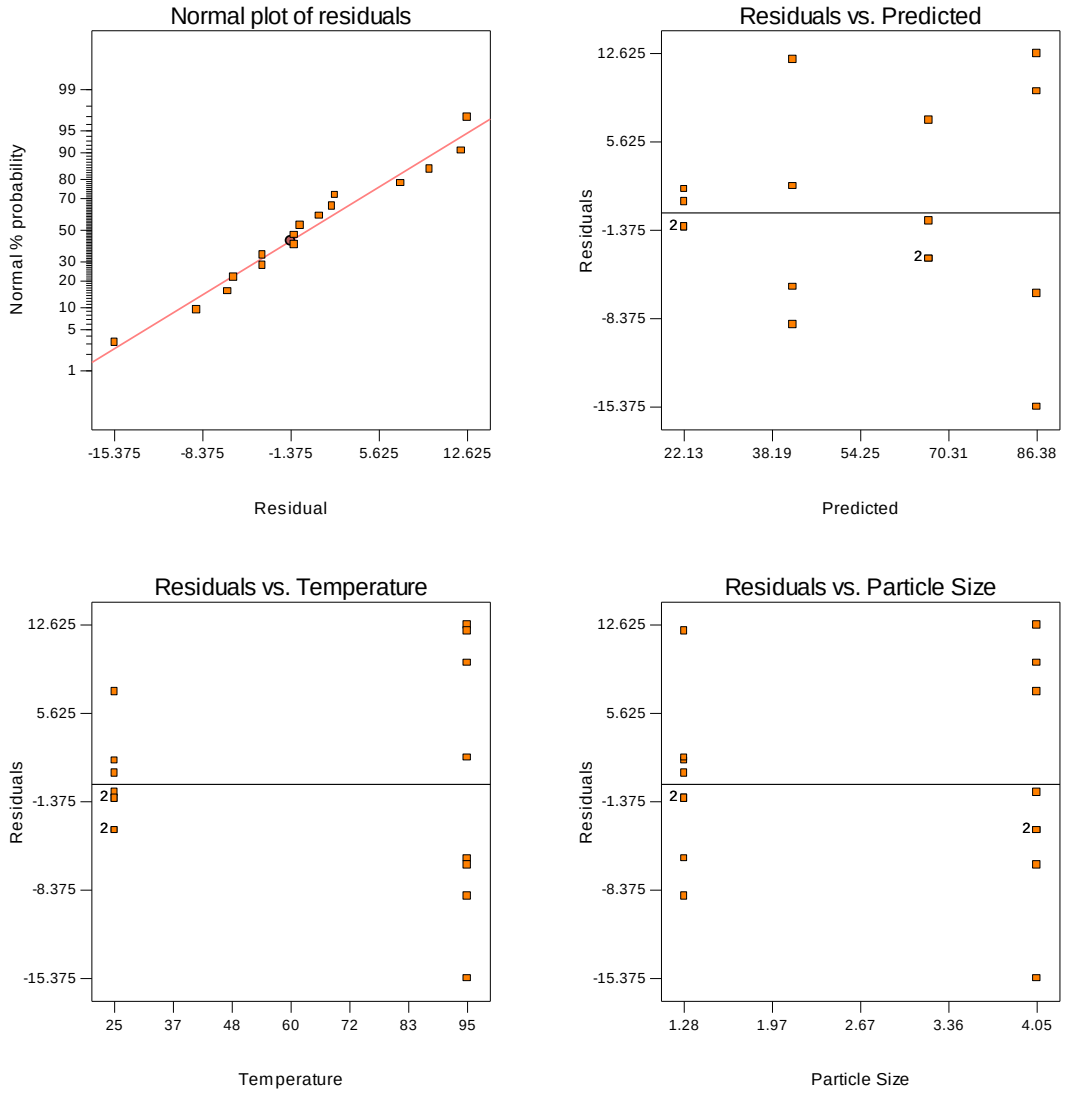
Final Equation in Terms of Actual Factors:

Yield =
-5.49175

+0.28214 * Temperature
+16.06498 * Particle Size

(e) Analyze the residuals from this experiment and comment on model adequacy.

The residual plots are satisfactory. There is a slight tendency for the variability of the residuals to increase with the predicted value of *y*.



8.41S. A 16-run fractional factorial experiment in nine factors was conducted by Chrysler Motors Engineering and described in the article "Sheet Molded Compound Process Improvement," by P.I. Hsieh and D.E. Goodwin (*Fourth Symposium on Taguchi Methods*, American Supplier Institute, Dearborn, MI, 1986, pp. 13-21). The purpose was to reduce the number of defects in the finish of sheet-molded grill opening panels. The design, and the resulting number of defects, *c*, observed on each run, is shown in Table P8.14. This is a resolution III fraction with generators *E=BD*, *F=BCD*, *G=AC*, *H=ACD*, and *J=AB*.

Table P8.14

Run	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*J*	c	$\sqrt{c}$	F&T's Modification

1	–	–	–	–	+	–	+	–	+	56	7.48	7.52
2	+	–	–	–	+	–	–	+	–	17	4.12	4.18
3	–	+	–	–	–	+	+	–	–	2	1.41	1.57
4	+	+	–	–	–	+	–	+	+	4	2.00	2.12
5	–	–	+	–	+	+	–	+	+	3	1.73	1.87
6	+	–	+	–	+	+	+	–	–	4	2.00	2.12
7	–	+	+	–	–	–	–	+	–	50	7.07	7.12
8	+	+	+	–	–	–	+	–	+	2	1.41	1.57
9	–	–	–	+	–	+	+	+	+	1	1.00	1.21
10	+	–	–	+	–	+	–	–	–	0	0.00	0.50
11	–	+	–	+	+	–	+	+	–	3	1.73	1.87
12	+	+	–	+	+	–	–	–	+	12	3.46	3.54
13	–	–	+	+	–	–	–	–	+	3	1.73	1.87
14	+	–	+	+	–	–	+	+	–	4	2.00	2.12
15	–	+	+	+	+	+	–	–	–	0	0.00	0.50
16	+	+	+	+	+	+	+	+	+	0	0.00	0.50

(a) Find the defining relation and the alias relationships in this design.

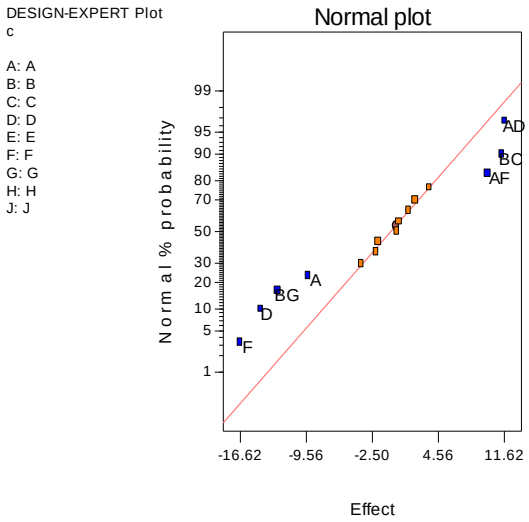
I = ABJ = ACG = BDE = CEF = DGH = FHJ = ABFH = ACDH = ADEJ = AEFG = BCDF = BCGJ = BEGH = CEHJ = DFGJ = ABCEH = ABDFG = ACDFJ = ADEFH = AEGHJ = BCDHJ = BCFGH = BEFGJ = CDEGJ = ABCDEG = ABCEFJ = ABDGHJ = ACFGHJ = BDEFHJ = CDEFGH = ABCDEFGHJ

(b) Estimate the factor effects and use a normal probability plot to tentatively identify the important factors.

The effects are shown below in the *Design Expert* output. The normal probability plot of effects identifies factors *A*, *D*, *F*, and interactions *AD*, *AF*, *BC*, *BG* as important.

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	-9.375	351.562	7.75573
Model	B	-1.875	14.0625	0.310229
Model	C	-3.625	52.5625	1.15957
Model	D	-14.375	826.562	18.2346
Error	E	3.625	52.5625	1.15957
Model	F	-16.625	1105.56	24.3895
Model	G	-2.125	18.0625	0.398472
Error	H	0.375	0.5625	0.0124092
Error	J	0.125	0.0625	0.0013788
Model	AD	11.625	540.563	11.9252
Error	AE	2.125	18.0625	0.398472
Model	AF	9.875	390.063	8.60507
Error	AH	1.375	7.5625	0.166834
Model	BC	11.375	517.563	11.4178
Model	BG	-12.625	637.562	14.0651
	Lenth's ME	13.9775		
	Lenth's SME	28.3764		



(c) Fit an appropriate model using the factors identified in part (b) above.

The analysis of variance and corresponding model is shown below. Factors *B*, *C*, and *G* are included for hierarchal purposes.

Design Expert Output

Response: c

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	4454.13	10	445.41	28.26	0.0009	significant
A	*351.56*	*1*	*351.56*	*22.30*	*0.0052*	
B	*14.06*	*1*	*14.06*	*0.89*	*0.3883*	
C	*52.56*	*1*	*52.56*	*3.33*	*0.1274*	
D	*826.56*	*1*	*826.56*	*52.44*	*0.0008*	
F	*1105.56*	*1*	*1105.56*	*70.14*	*0.0004*	
G	*18.06*	*1*	*18.06*	*1.15*	*0.3333*	
AD	*540.56*	*1*	*540.56*	*34.29*	*0.0021*	
AF	*390.06*	*1*	*390.06*	*24.75*	*0.0042*	
BC	*517.56*	*1*	*517.56*	*32.84*	*0.0023*	
BG	*637.56*	*1*	*637.56*	*40.45*	*0.0014*	
Residual	78.81	5	15.76			
Cor Total	4532.94	15				

The Model F-value of 28.26 implies the model is significant. There is only a 0.09% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	3.97	R-Squared	0.9826
Mean	10.06	Adj R-Squared	0.9478
C.V.	39.46	Pred R-Squared	0.8220
PRESS	807.04	Adeq Precision	17.771

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	10.06	1	0.99	7.51	12.61	
A-A	-4.69	1	0.99	-7.24	-2.14	1.00
B-B	-0.94	1	0.99	-3.49	1.61	1.00
C-C	-1.81	1	0.99	-4.36	0.74	1.00
D-D	-7.19	1	0.99	-9.74	-4.64	1.00
F-F	-8.31	1	0.99	-10.86	-5.76	1.00
G-G	-1.06	1	0.99	-3.61	1.49	1.00
AD	5.81	1	0.99	3.26	8.36	1.00
AF	4.94	1	0.99	2.39	7.49	1.00

BC	5.69	1	0.99	3.14	8.24	1.00
BG	-6.31	1	0.99	-8.86	-3.76	1.00

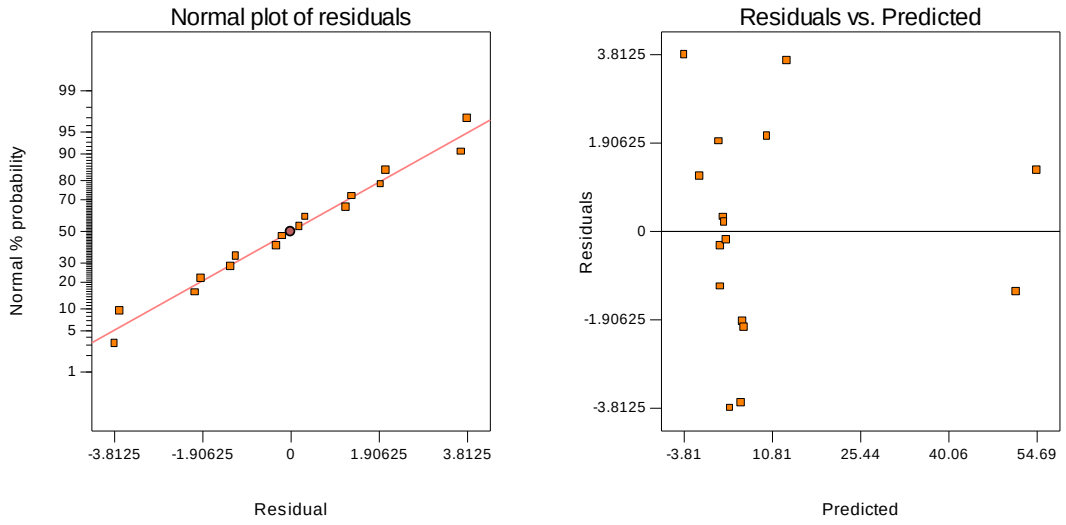
Final Equation in Terms of Coded Factors:

```
c     =
+10.06
-4.69   * A
-0.94   * B
-1.81   * C
-7.19   * D
-8.31   * F
-1.06   * G
+5.81   * A * D
+4.94   * A * F
+5.69   * B * C
-6.31   * B * G
```

Final Equation in Terms of Actual Factors:

```
c     =
+10.06250
-4.68750   * A
-0.93750   * B
-1.81250   * C
-7.18750   * D
-8.31250   * F
-1.06250   * G
+5.81250   * A * D
+4.93750   * A * F
+5.68750   * B * C
-6.31250   * B * G
```

(d) Plot the residuals from this model versus the predicted number of defects. Also, prepare a normal probability plot of the residuals. Comment on the adequacy of these plots.



There is a significant problem with inequality of variance. This is likely caused by the response variable being a count. A transformation may be appropriate.

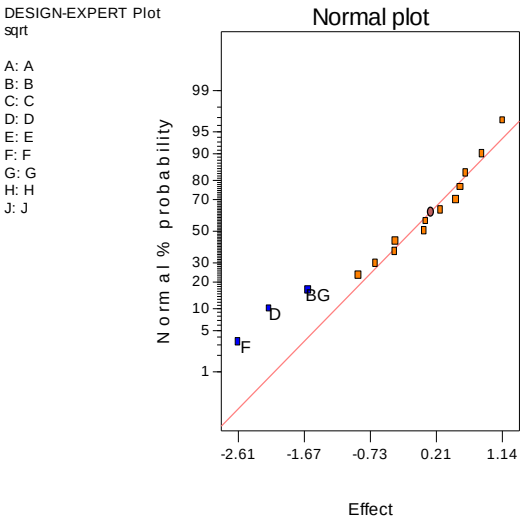
(e) In part (d) you should have noticed an indication that the variance of the response is not constant (considering that the response is a count, you should have expected this). The previous table also shows a transformation on c, the square root, that is a widely used variance stabilizing transformation for count data (refer to the discussion of variance stabilizing transformations in Chapter 3). Repeat

parts (a) through (d) using the transformed response and comment on your results. Specifically, are the residual plots improved?

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Error	A	-0.895	3.2041	4.2936
Model	B	-0.3725	0.555025	0.743752
Error	C	-0.6575	1.72922	2.31722
Model	D	-2.1625	18.7056	25.0662
Error	E	0.4875	0.950625	1.27387
Model	F	-2.6075	27.1962	36.4439
Model	G	-0.385	0.5929	0.794506
Error	H	0.27	0.2916	0.390754
Error	J	0.06	0.0144	0.0192965
Error	AD	1.145	5.2441	7.02727
Error	AE	0.555	1.2321	1.65106
Error	AF	0.86	2.9584	3.96436
Error	AH	0.0425	0.007225	0.00968175
Error	BC	0.6275	1.57502	2.11059
Model	BG	-1.61	10.3684	13.894
	Lenth's ME	2.27978		
	Lenth's SME	4.62829		

The analysis of the data with the square root transformation identifies only *D*, *F*, the *BG* interaction as being significant. The original analysis identified factor *A* and several two factor interactions as being significant.



Design Expert Output

Response: sqrt

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	57.42	5	11.48	6.67	0.0056	significant
B	*0.56*	*1*	*0.56*	*0.32*	*0.5826*	
D	*18.71*	*1*	*18.71*	*10.87*	*0.0081*	
F	*27.20*	*1*	*27.20*	*15.81*	*0.0026*	
G	*0.59*	*1*	*0.59*	*0.34*	*0.5702*	
BG	*10.37*	*1*	*10.37*	*6.03*	*0.0340*	
Residual	17.21	10	1.72			
Cor Total	74.62	15				

The Model F-value of 6.67 implies the model is significant. There is only a 0.56% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	1.31	R-Squared	0.7694
Mean	2.32	Adj R-Squared	0.6541
C.V.	56.51	Pred R-Squared	0.4097
PRESS	44.05	Adeq Precision	8.422

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	2.32	1	0.33	1.59	3.05	
B-B	-0.19	1	0.33	-0.92	0.54	1.00
D-D	-1.08	1	0.33	-1.81	-0.35	1.00
F-F	-1.30	1	0.33	-2.03	-0.57	1.00
G-G	-0.19	1	0.33	-0.92	0.54	1.00
BG	-0.80	1	0.33	-1.54	-0.074	1.00

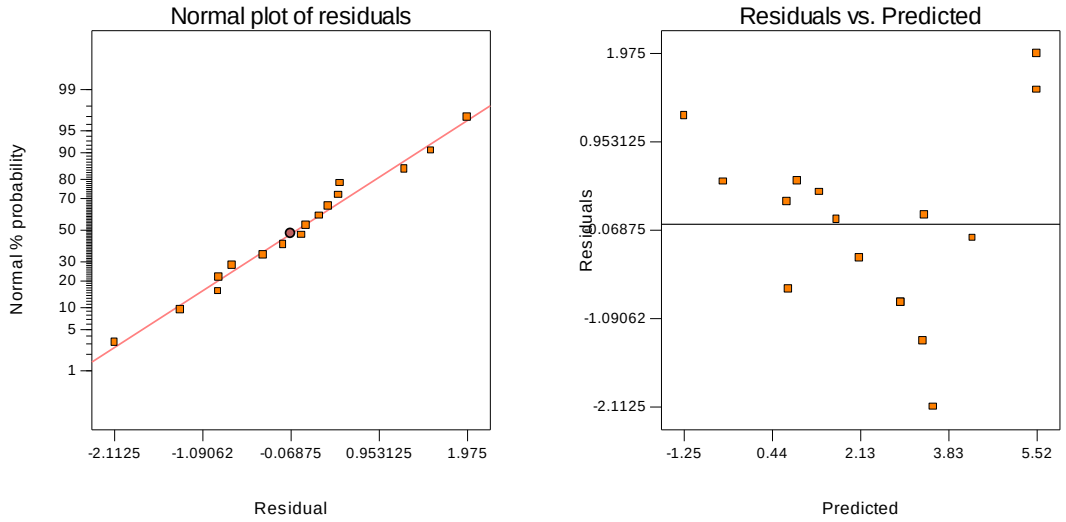
Final Equation in Terms of Coded Factors:

sqrt =
+2.32
-0.19 * B
-1.08 * D
-1.30 * F
-0.19 * G
-0.80 * B * G

Final Equation in Terms of Actual Factors:

sqrt	=
+2.32125	
-0.18625	* B
-1.08125	* D
-1.30375	* F
-0.19250	* G
-0.80500	* B * G

The residual plots are acceptable; although, there appears to be a slight "u" shape to the residuals versus predicted plot.



(f) There is a modification to the square root transformation proposed by Freeman and Tukey ("Transformations Related to the Angular and the Square Root," *Annals of Mathematical Statistics*, Vol. 21, 1950, pp. 607-611) that improves its performance. F&T's modification to the square root transformation is:

$$\frac{1}{2}\left[\sqrt{c}+\sqrt{c+1}\right]$$

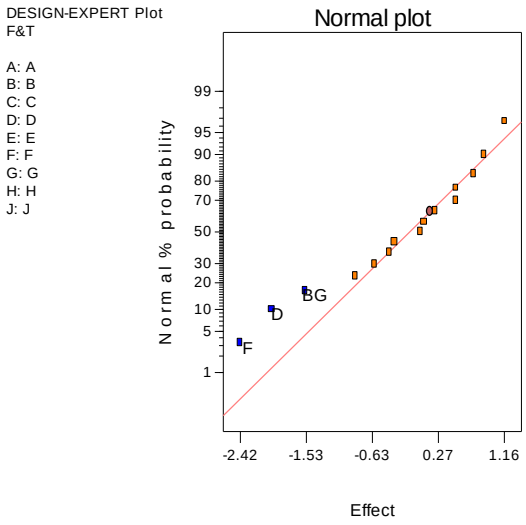
Rework parts (a) through (d) using this transformation and comment on the results. (For an interesting discussion and analysis of this experiment, refer to "Analysis of Factorial Experiments with Defects or Defectives as the Response," by S. Bisgaard and H.T. Fuller, *Quality Engineering*, Vol. 7, 1994-5, pp. 429-443.)

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Error	A	-0.86	2.9584	4.38512
Model	B	-0.325	0.4225	0.626255
Error	C	-0.605	1.4641	2.17018
Model	D	-1.995	15.9201	23.5977
Error	E	0.5025	1.01002	1.49712
Model	F	-2.425	23.5225	34.8664
Model	G	-0.4025	0.648025	0.960541
Error	H	0.225	0.2025	0.300158
Error	J	0.0275	0.003025	0.00448383
Error	AD	1.1625	5.40562	8.01254

Error	AE	0.505	1.0201	1.51205
Error	AF	0.8825	3.11523	4.61757
Error	AH	0.0725	0.021025	0.0311645
Error	BC	0.7525	2.26503	3.35735
Model	BG	-1.54	9.4864	14.0613
	Lenth's ME	2.14001		
	Lenth's SME	4.34453		

As with the square root transformation, factors *D*, *F*, and the *BG* interaction remain significant.



Design Expert Output

Response: F&T

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	50.00	5	10.00	5.73	0.0095	significant
B	*0.42*	*1*	*0.42*	*0.24*	*0.6334*	
D	*15.92*	*1*	*15.92*	*9.12*	*0.0129*	
F	*23.52*	*1*	*23.52*	*13.47*	*0.0043*	
G	*0.65*	*1*	*0.65*	*0.37*	*0.5560*	
BG	*9.49*	*1*	*9.49*	*5.43*	*0.0420*	
Residual	17.47	10	1.75			
Cor Total	67.46	15				

The Model F-value of 5.73 implies the model is significant. There is only a 0.95% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	1.32	R-Squared	0.7411
Mean	2.51	Adj R-Squared	0.6117
C.V.	52.63	Pred R-Squared	0.3373
PRESS	44.71	Adeq Precision	7.862

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	2.51	1	0.33	1.78	3.25	
B-B	-0.16	1	0.33	-0.90	0.57	1.00
D-D	-1.00	1	0.33	-1.73	-0.26	1.00
F-F	-1.21	1	0.33	-1.95	-0.48	1.00
G-G	-0.20	1	0.33	-0.94	0.53	1.00
BG	-0.77	1	0.33	-1.51	-0.034	1.00

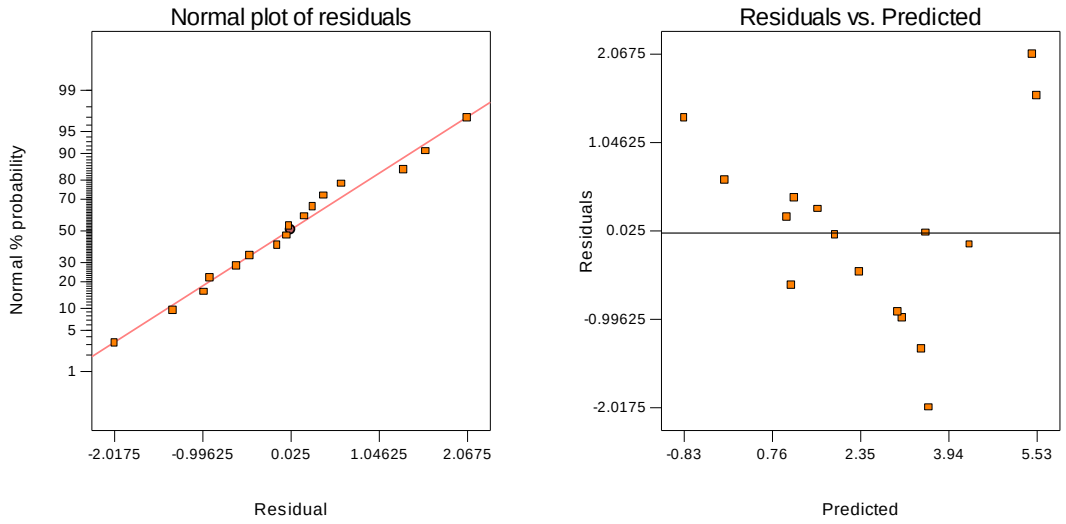
Final Equation in Terms of Coded Factors:

```
F&T     =
+2.51
-0.16     * B
-1.00     * D
-1.21     * F
-0.20     * G
-0.77     * B * G
```

Final Equation in Terms of Actual Factors:

```
F&T     =
+2.51125
-0.16250     * B
-0.99750     * D
-1.21250     * F
-0.20125     * G
-0.77000     * B * G
```

The following interaction plots appear as they did with the square root transformation; a slight "u" shape is observed in the residuals versus predicted plot.



8.42S. An experiment is run in a semiconductor factory to investigate the effect of six factors on transistor gain. The design selected is the 2_{IV}^{6-2} shown in Table P8.15.

Table P8.15

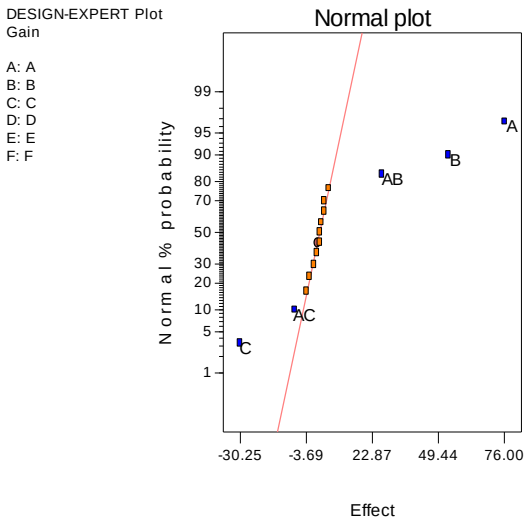
Standard Order	Run Order	*A*	*B*	*C*	*D*	*E*	*F*	Gain
1	2	–	–	–	–	–	–	1455
2	8	+	–	–	–	+	–	1511
3	5	–	+	–	–	+	+	1487
4	9	+	+	–	–	–	+	1596
5	3	–	–	+	–	+	+	1430
6	14	+	–	+	–	–	+	1481
7	11	–	+	+	–	–	–	1458
8	10	+	+	+	–	+	–	1549
9	15	–	–	–	+	–	+	1454
10	13	+	–	–	+	+	+	1517
11	1	–	+	–	+	+	–	1487
12	6	+	+	–	+	–	–	1596

13	12	–	–	+	+	+	–	1446
14	4	+	–	+	+	–	–	1473
15	7	–	+	+	+	–	+	1461
16	16	+	+	+	+	+	+	1563

(a) Use a normal plot of the effects to identify the significant factors.

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	76	23104	55.2714
Model	B	53.75	11556.2	27.6459
Model	C	-30.25	3660.25	8.75637
Error	D	3.75	56.25	0.134566
Error	E	2	16	0.0382766
Error	F	1.75	12.25	0.0293055
Model	AB	26.75	2862.25	6.84732
Model	AC	-8.25	272.25	0.6513
Error	AD	-0.75	2.25	0.00538265
Error	AE	-3.5	49	0.117222
Error	AF	5.25	110.25	0.26375
Error	BD	0.5	1	0.00239229
Error	BF	2.5	25	0.0598072
Error	ABD	3.5	49	0.117222
Error	ABF	-2.5	25	0.0598072
	Lenth's ME	9.63968		
	Lenth's SME	19.57		



(b) Conduct appropriate statistical tests for the model identified in part (a).

Design Expert Output

Response: Gain

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Model	41455.00	5	8291.00	239.62	< 0.0001	significant
A	*23104.00*	*1*	*23104.00*	*667.75*	*< 0.0001*	
B	*11556.25*	*1*	*11556.25*	*334.00*	*< 0.0001*	
C	*3660.25*	*1*	*3660.25*	*105.79*	*< 0.0001*	
AB	*2862.25*	*1*	*2862.25*	*82.72*	*< 0.0001*	
AC	*272.25*	*1*	*272.25*	*7.87*	*0.0186*	

Residual	346.00	10	34.60
Cor Total	41801.00	15	

The Model F-value of 239.62 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	5.88	R-Squared	0.9917
Mean	1497.75	Adj R-Squared	0.9876
C.V.	0.39	Pred R-Squared	0.9788
PRESS	885.76	Adeq Precision	44.419

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	1497.75	1	1.47	1494.47	1501.03	
A-A	38.00	1	1.47	34.72	41.28	1.00
B-B	26.87	1	1.47	23.60	30.15	1.00
C-C	-15.13	1	1.47	-18.40	-11.85	1.00
AB	13.38	1	1.47	10.10	16.65	1.00
AC	-4.12	1	1.47	-7.40	-0.85	1.00

Final Equation in Terms of Coded Factors:

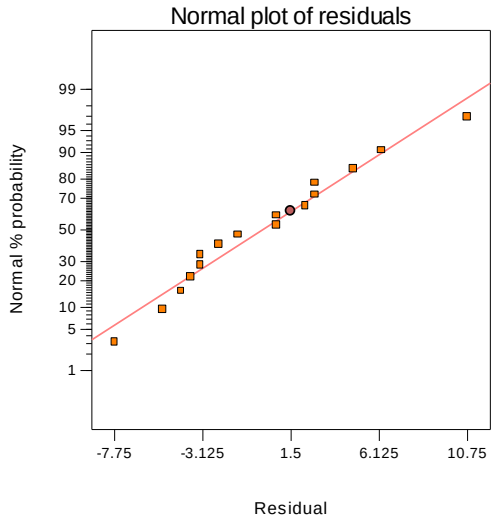
Gain =
+1497.75
+38.00 * A
+26.87 * B
-15.13 * C
+13.38 * A * B
-4.12 * A * C

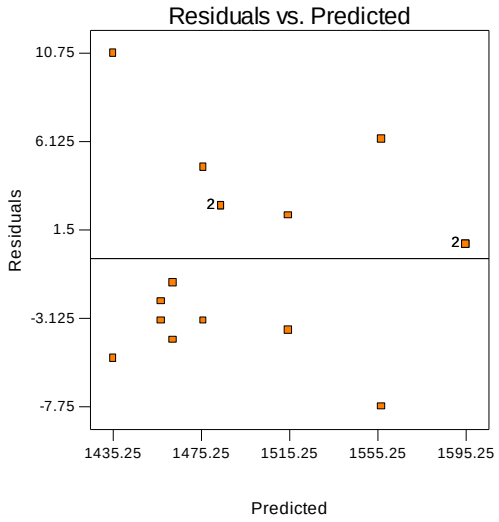
Final Equation in Terms of Actual Factors:

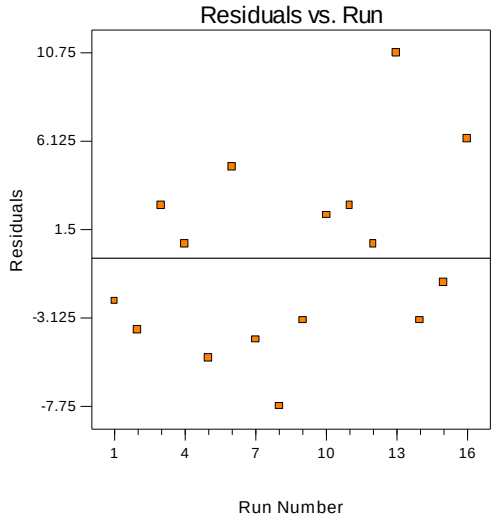
Gain =
+1497.75000
+38.00000 * A
+26.87500 * B
-15.12500 * C
+13.37500 * A * B
-4.12500 * A * C

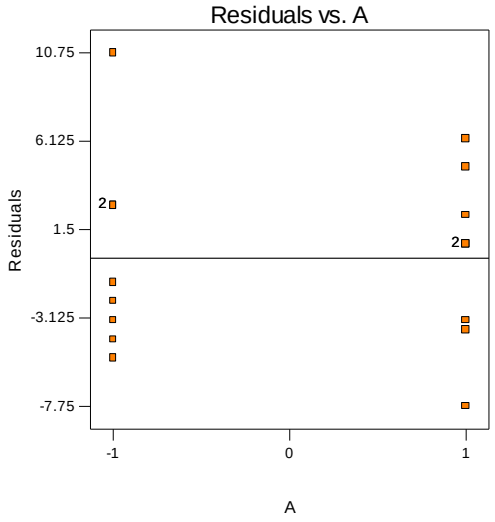
(c) Analyze the residuals and comment on your findings.

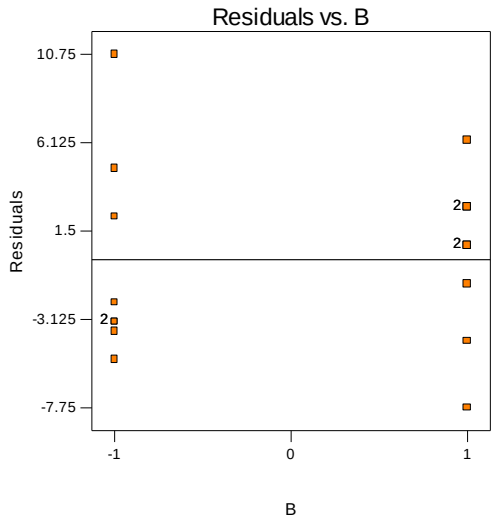
The residual plots are acceptable. The normality and equality of variance assumptions are verified. There does not appear to be any trends or interruptions in the residuals versus run order plot.

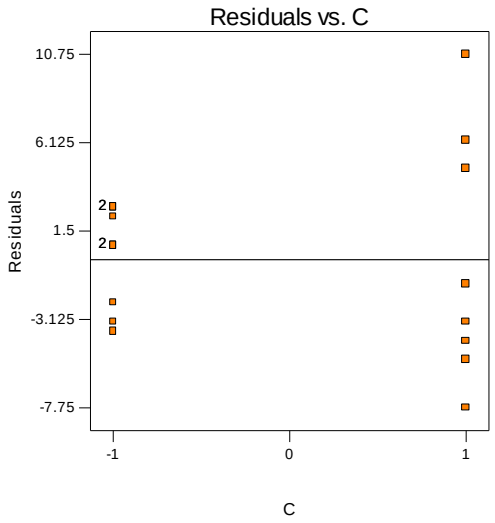


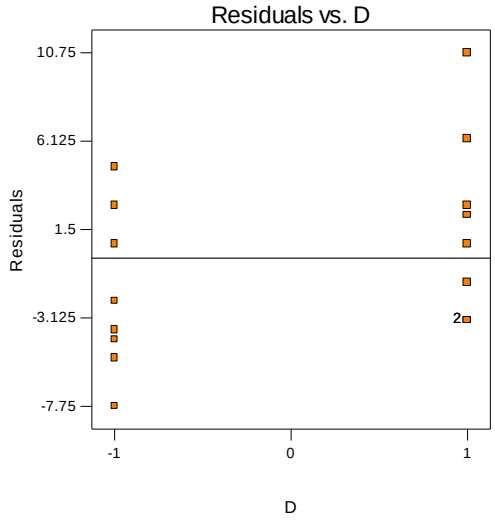


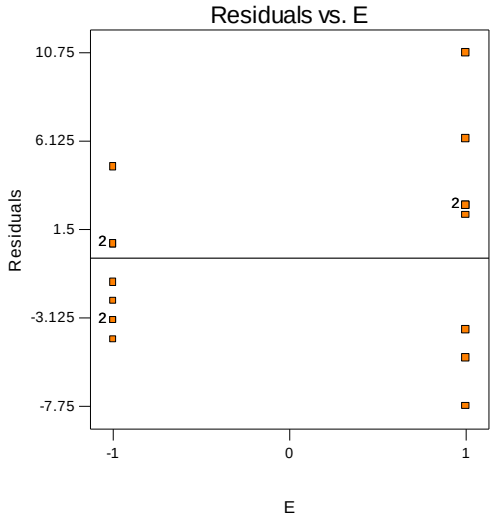

Residuals vs. Run
Residuals
10.75
6.125
1.5
-3.125
-7.75
1
4
7
10
13
16
Run Number

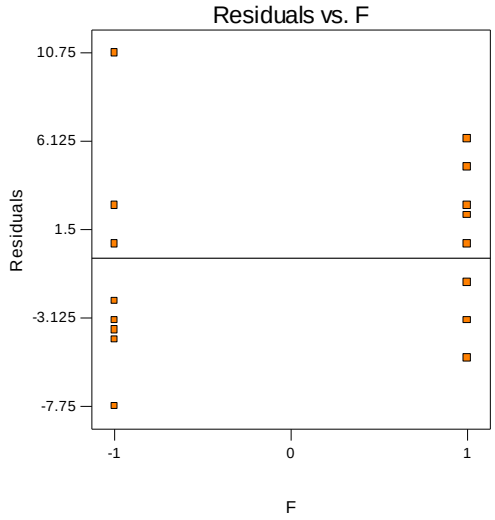

Residuals vs. A
Residuals
10.75
6.125
1.5
-3.125
-7.75
2
2
-1
0
1
A


Residuals vs. B
Residuals
10.75
6.125
1.5
-3.125
-7.75
2
2
2
-1
0
1
B


Residuals vs. C
Residuals
10.75
6.125
1.5
-3.125
-7.75
2
2
-1
0
1
C

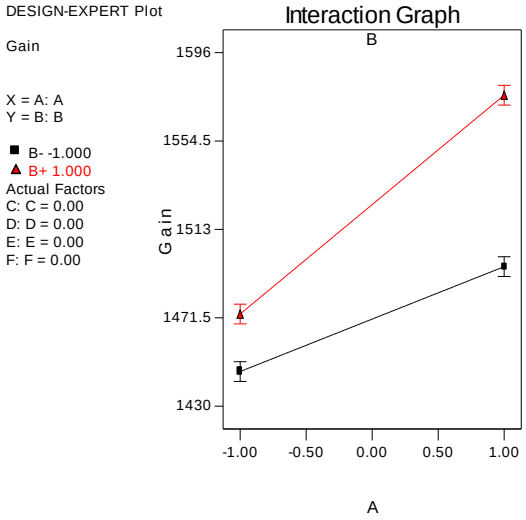

Residuals vs. D
Residuals
10.75
6.125
1.5
-3.125
-7.75
2
-1
0
1
D

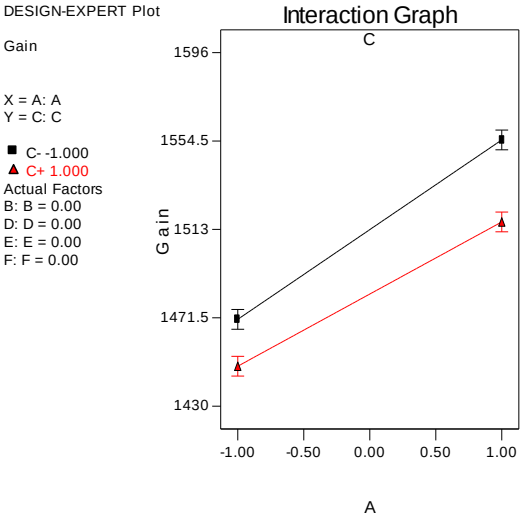

Residuals vs. E
Residuals
10.75
6.125
1.5
-3.125
-7.75
2
2
2
-1
0
1
E

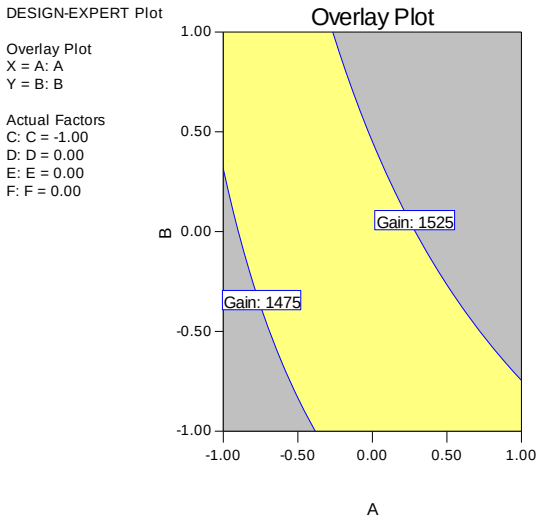


(d) Can you find a set of operating conditions that produce gain of 1500 ± 25?

Yes, see the graphs below.







8.43S. Heat treating is often used to carbonize metal parts, such as gears. The thickness of the carbonized layer is a critical output variable from this process, and it is usually measured by performing a carbon analysis on the gear pitch (top of the gear tooth). Six factors were studied on a 2_{IV}^{6-2} design: *A* = furnace temperature, *B* = cycle time, *C* = carbon concentration, *D* = duration of the carbonizing cycle, *E* = carbon concentration of the diffuse cycle, and *F* = duration of the diffuse cycle. The experiment is shown in Table P8.16.

Table P8.16

Standard Order	Run Order	*A*	*B*	*C*	*D*	*E*	*F*	Pitch
1	5	–	–	–	–	–	–	74
2	7	+	–	–	–	+	–	190
3	8	–	+	–	–	+	+	133
4	2	+	+	–	–	–	+	127
5	10	–	–	+	–	+	+	115
6	12	+	–	+	–	–	+	101
7	16	–	+	+	–	–	–	54
8	1	+	+	+	–	+	–	144
9	6	–	–	–	+	–	+	121
10	9	+	–	–	+	+	+	188
11	14	–	+	–	+	+	–	135
12	13	+	+	–	+	–	–	170
13	11	–	–	+	+	+	–	126
14	3	+	–	+	+	–	–	175
15	15	–	+	+	+	–	+	126
16	4	+	+	+	+	+	+	193

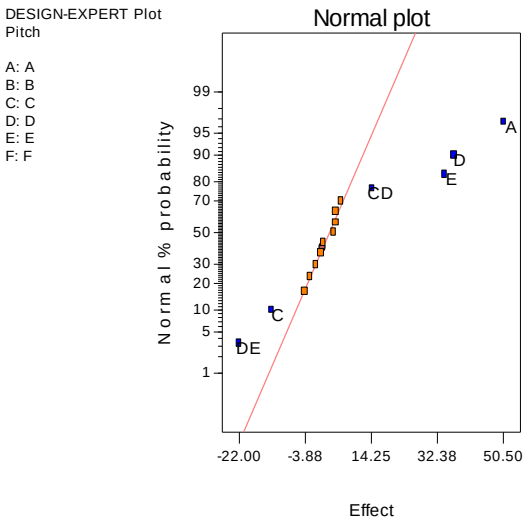
(a) Estimate the factor effects and plot them on a normal probability plot. Select a tentative model.

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Model	Intercept			
Model	A	50.5	10201	41.8777
Error	B	-1	4	0.016421
Model	C	-13	676	2.77515
Model	D	37	5476	22.4804
Model	E	34.5	4761	19.5451

Error	F	4.5	81	0.332526
Error	AB	-4	64	0.262737
Error	AC	-2.5	25	0.102631
Error	AD	4	64	0.262737
Error	AE	1	4	0.016421
Error	BD	4.5	81	0.332526
Model	CD	14.5	841	3.45252
Model	DE	-22	1936	7.94778
Error	ABD	0.5	1	0.00410526
Error	ABF	6	144	0.591157
	Lenth's ME	15.4235		
	Lenth's SME	31.3119		

Factors *A*, *C*, *D*, *E* and the two factor interactions *CD* and *DE* appear to be significant. The *CD* and *DE* interactions are aliased with BF and AF interactions respectively. Because factors B and F are not significant, *CD* and *DE* were included in the model. The model can be found in the Design Expert Output below.



(b) Perform appropriate statistical tests on the model.

Design Expert Output

Response: Pitch

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Model	23891.00	6	3981.83	76.57	< 0.0001	significant
A	*10201.00*	*1*	*10201.00*	*196.17*	*< 0.0001*	
C	*676.00*	*1*	*676.00*	*13.00*	*0.0057*	
D	*5476.00*	*1*	*5476.00*	*105.31*	*< 0.0001*	
E	*4761.00*	*1*	*4761.00*	*91.56*	*< 0.0001*	
CD	*841.00*	*1*	*841.00*	*16.17*	*0.0030*	
DE	*1936.00*	*1*	*1936.00*	*37.23*	*0.0002*	
Residual	468.00	9	52.00			
Cor Total	24359.00	15				

The Model F-value of 76.57 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	7.21	R-Squared	0.9808
Mean	135.75	Adj R-Squared	0.9680
C.V.	5.31	Pred R-Squared	0.9393
PRESS	1479.11	Adeq Precision	28.618

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	135.75	1	1.80	131.67	139.83	
A-A	25.25	1	1.80	21.17	29.33	1.00
C-C	-6.50	1	1.80	-10.58	-2.42	1.00
D-D	18.50	1	1.80	14.42	22.58	1.00
E-E	17.25	1	1.80	13.17	21.33	1.00
CD	7.25	1	1.80	3.17	11.33	1.00
DE	-11.00	1	1.80	-15.08	-6.92	1.00

Final Equation in Terms of Coded Factors:

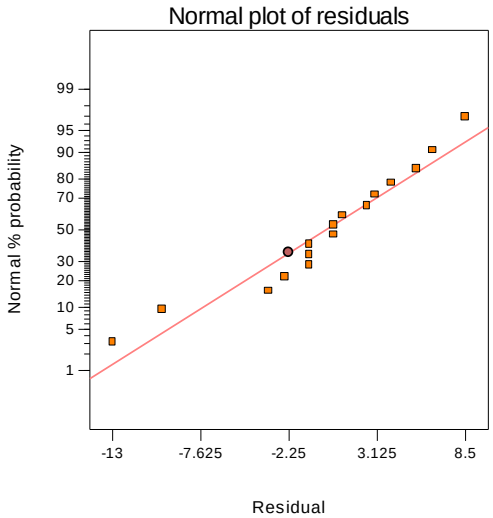
Pitch =
+135.75
+25.25 * A
-6.50 * C
+18.50 * D
+17.25 * E
+7.25 * C * D
-11.00 * D * E

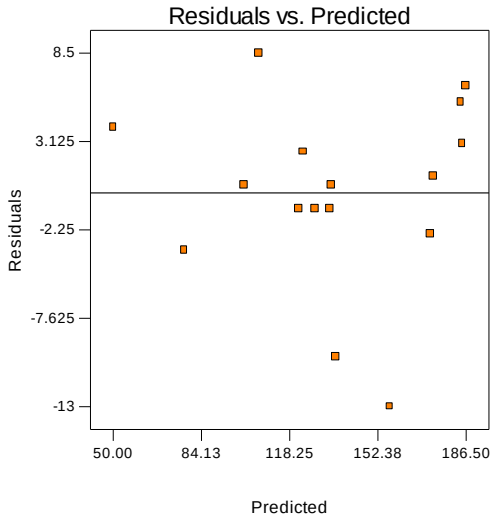
Final Equation in Terms of Actual Factors:

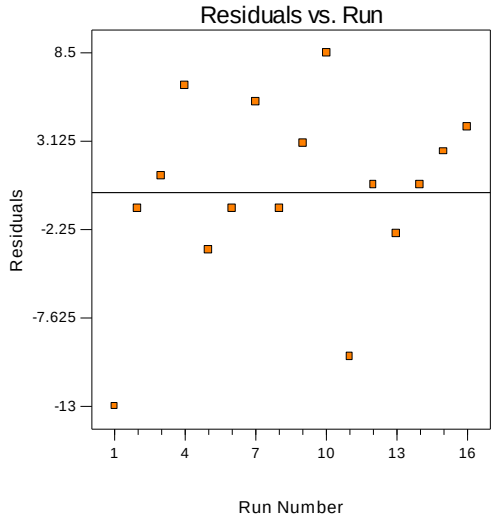
Pitch =
+135.75000
+25.25000 * A
-6.50000 * C
+18.50000 * D
+17.25000 * E
+7.25000 * C * D
-11.00000 * D * E

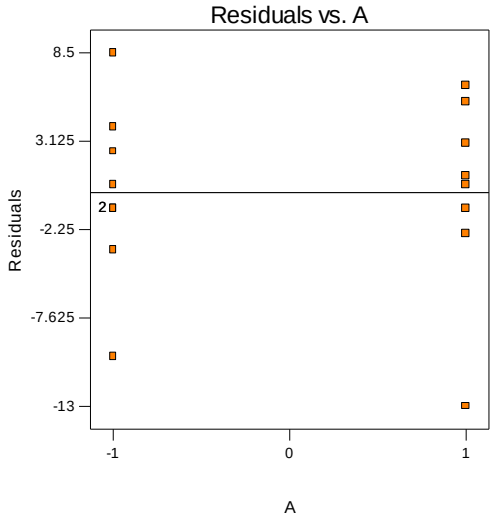
(c) Analyze the residuals and comment on model adequacy.

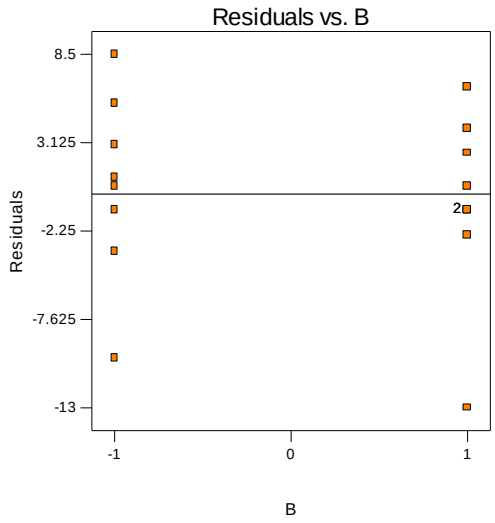
The residual plots are acceptable. The normality and equality of variance assumptions are verified. There does not appear to be any trends or interruptions in the residuals versus run order plot. The plots of the residuals versus factors *C* and *E* identify reduced variation at the lower level of both variables while the plot of residuals versus factor *F* identifies reduced variation at the upper level. Because *C* and *E* are significant factors in the model, this might not affect the decision on the optimum solution for the process. However, factor *F* is not included in the model and may be set at the upper level to reduce variation.

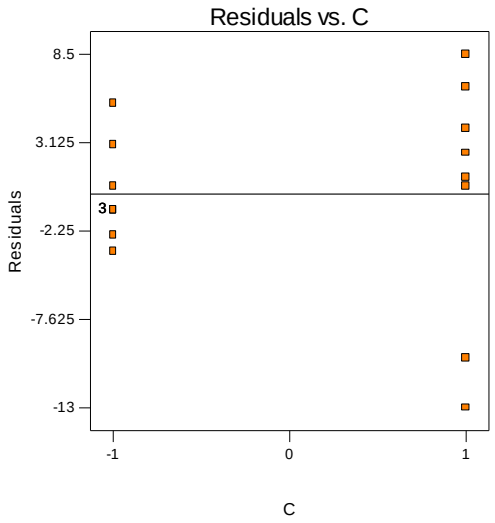


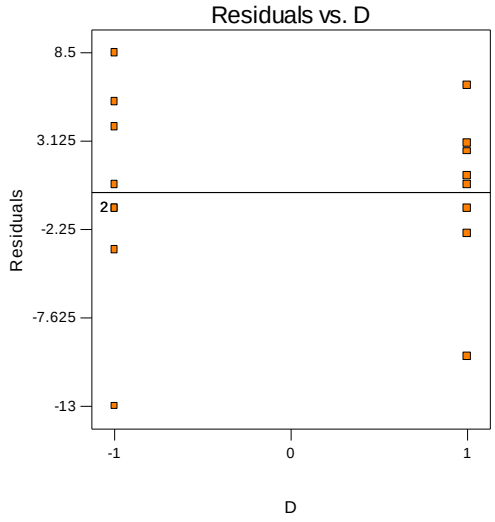


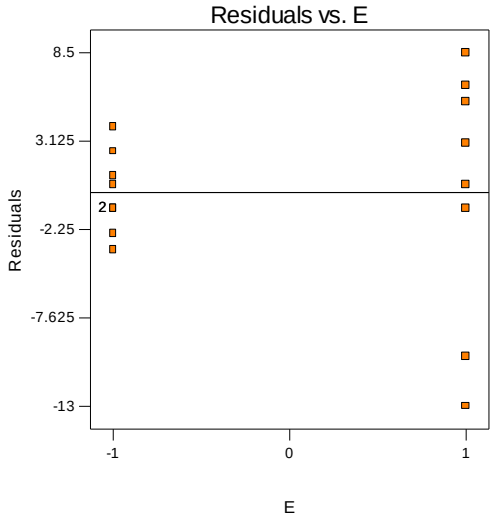

Residuals vs. Run
Residuals
8.5
3.125
-2.25
-7.625
-13
1
4
7
10
13
16
Run Number

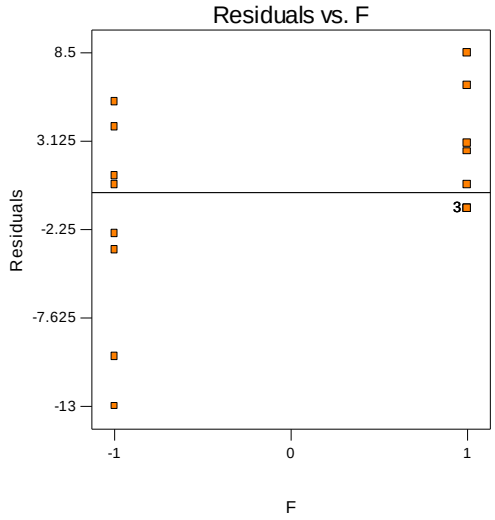

Residuals vs. A
Residuals
8.5
3.125
-2.25
-7.625
-13
2
-1
0
1
A


Residuals vs. B
Residuals
8.5
3.125
-2.25
-7.625
-13
2
-1
0
1
B


Residuals vs. C
Residuals
8.5
3.125
-2.25
-7.625
-13
3
-1
0
1
C

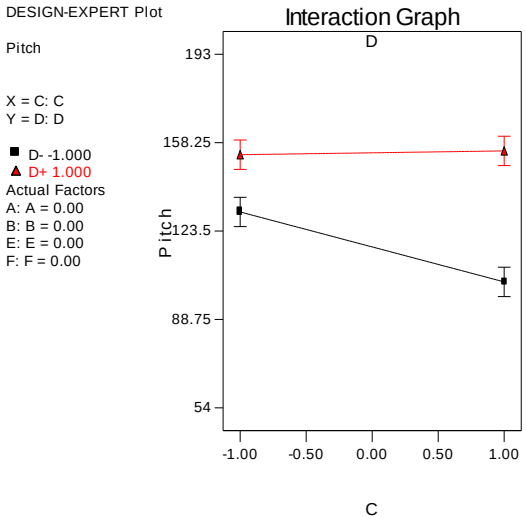

Residuals vs. D
Residuals
8.5
3.125
-2.25
-7.625
-13
2
-1
0
1
D

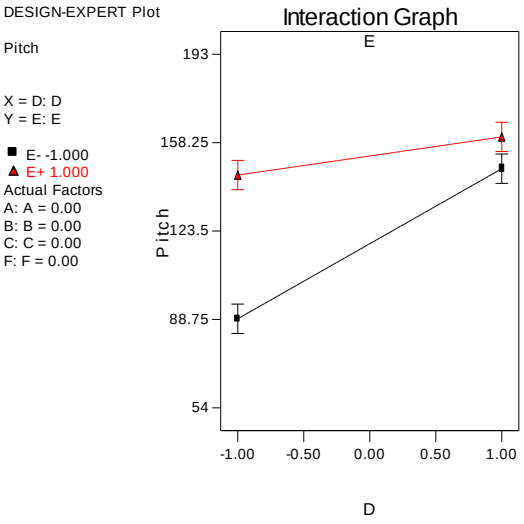

Residuals vs. E
Residuals
8.5
3.125
-2.25
-7.625
-13
2
-1
0
1
E

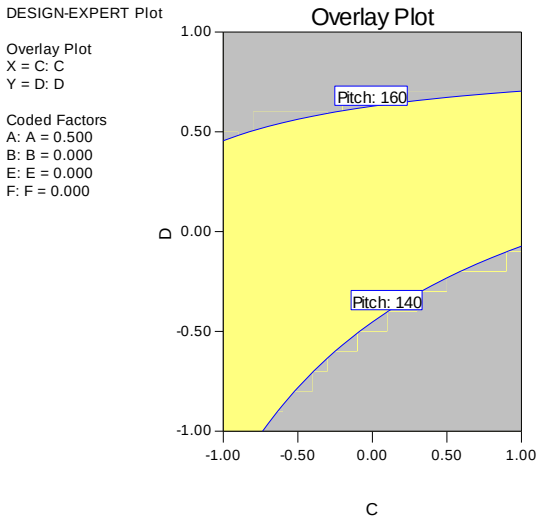


(d) Interpret the results of this experiment. Assume that a layer thickness of between 140 and 160 is desirable.

The graphs below identify a region that is acceptable between 140 and 160.







8.44. An article by L.B. Hare ("In the Soup: A Case Study to Identify Contributors to Filling Variability", Journal of Quality Technology, Vol. 20, pp. 36-43) describes a factorial experiment used to study the filling variability of dry soup mix packages. The factors are *A* = number of mixing ports through which the vegetable oil was added (1, 2), *B* = temperature surrounding the mixer (cooled, ambient), *C* = mixing time (60, 80 sec), *D* = batch weight (1500, 2000 lb), and *E* = number of days between mixing and packaging (1,7). Between 125 and 150 packages of soup were sampled over an eight hour period for each run in the design and the standard deviation of package weight was used as the response variable. The design and resulting data are shown in Table P8.17.

Table P8.17

Std Order	*A* – Mixer Ports	*B* – Temp	*C* – Time	*D* – Batch Weight	*E* – Delay	*y* – Std Dev
1	–1	–1	–1	–1	–1	1.13
2	1	–1	–1	–1	1	1.25
3	–1	1	–1	–1	1	0.97
4	1	1	–1	–1	–1	1.70
5	–1	–1	1	–1	1	1.47
6	1	–1	1	–1	–1	1.28
7	–1	1	1	–1	–1	1.18
8	1	1	1	–1	1	0.98
9	–1	–1	–1	1	1	0.78
10	1	–1	–1	1	–1	1.36
11	–1	1	–1	1	–1	1.85
12	1	1	–1	1	1	0.62
13	–1	–1	1	1	–1	1.09
14	1	–1	1	1	1	1.10
15	–1	1	1	1	1	0.76
16	1	1	1	1	–1	2.10

(a) What is the generator for this design?

The design generator is $I = -ABCDE$.

(b) What is the resolution of this design?

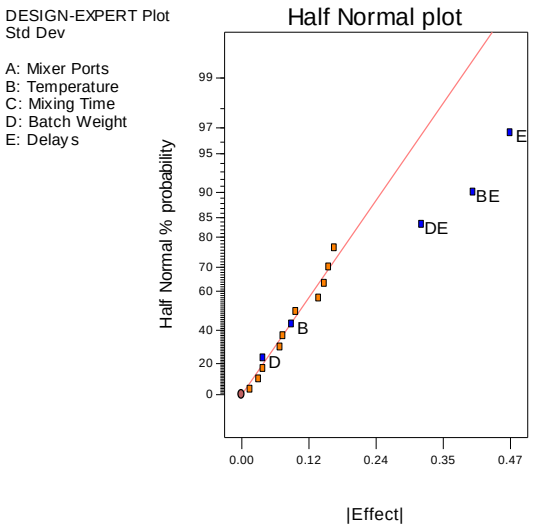
This design is Resolution *V*.

(c) Estimate the factor effects. Which effects are large?

Design Expert Output

	Term	Effect	SumSqr	% Contribtn
Require	Intercept			
Error	A	0.145	0.0841	3.48388
Model	B	0.0875	0.030625	1.26865
Error	C	0.0375	0.005625	0.233018
Model	D	-0.0375	0.005625	0.233018
Model	E	-0.47	0.8836	36.6035
Error	AB	0.015	0.0009	0.0372829
Error	AC	0.095	0.0361	1.49546
Error	AD	0.03	0.0036	0.149132
Error	AE	-0.1525	0.093025	3.8536
Error	BC	-0.0675	0.018225	0.754979
Error	BD	0.1625	0.105625	4.37556
Model	BE	-0.405	0.6561	27.1792
Error	CD	0.0725	0.021025	0.87097
Error	CE	0.135	0.0729	3.01992
Model	DE	-0.315	0.3969	16.4418
	Lenth's ME	0.337389		
	Lenth's SME	0.684948		

Factor *E* and the two factor interactions *BE* and *DE* appear to be significant. Factors *B*, and *D*, are included to satisfy model hierarchy. The analysis of variance and model can be found in the Design Expert Output below.



Design Expert Output

Response: Std Dev

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	1.97	5	0.39	8.94	0.0019	significant
B	*0.031*	*1*	*0.031*	*0.69*	*0.4242*	
D	*5.625E-003*	*1*	*5.625E-003*	*0.13*	*0.7284*	
E	*0.88*	*1*	*0.88*	*20.03*	*0.0012*	
BE	*0.66*	*1*	*0.66*	*14.87*	*0.0032*	
DE	*0.40*	*1*	*0.40*	*9.00*	*0.0134*	
Residual	0.44	10	0.044			
Cor Total	2.41	15				

The Model F-value of 8.94 implies the model is significant. There is only a 0.19% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.21	R-Squared	0.8173
Mean	1.23	Adj R-Squared	0.7259
C.V.	17.13	Pred R-Squared	0.5322
PRESS	1.13	Adeq Precision	9.252

Final Equation in Terms of Coded Factors:

Std Dev =
+1.23
+0.044 * B
-0.019 * D
-0.24 * E
-0.20 * B * E
-0.16 * D * E

Final Equation in Terms of Actual Factors:

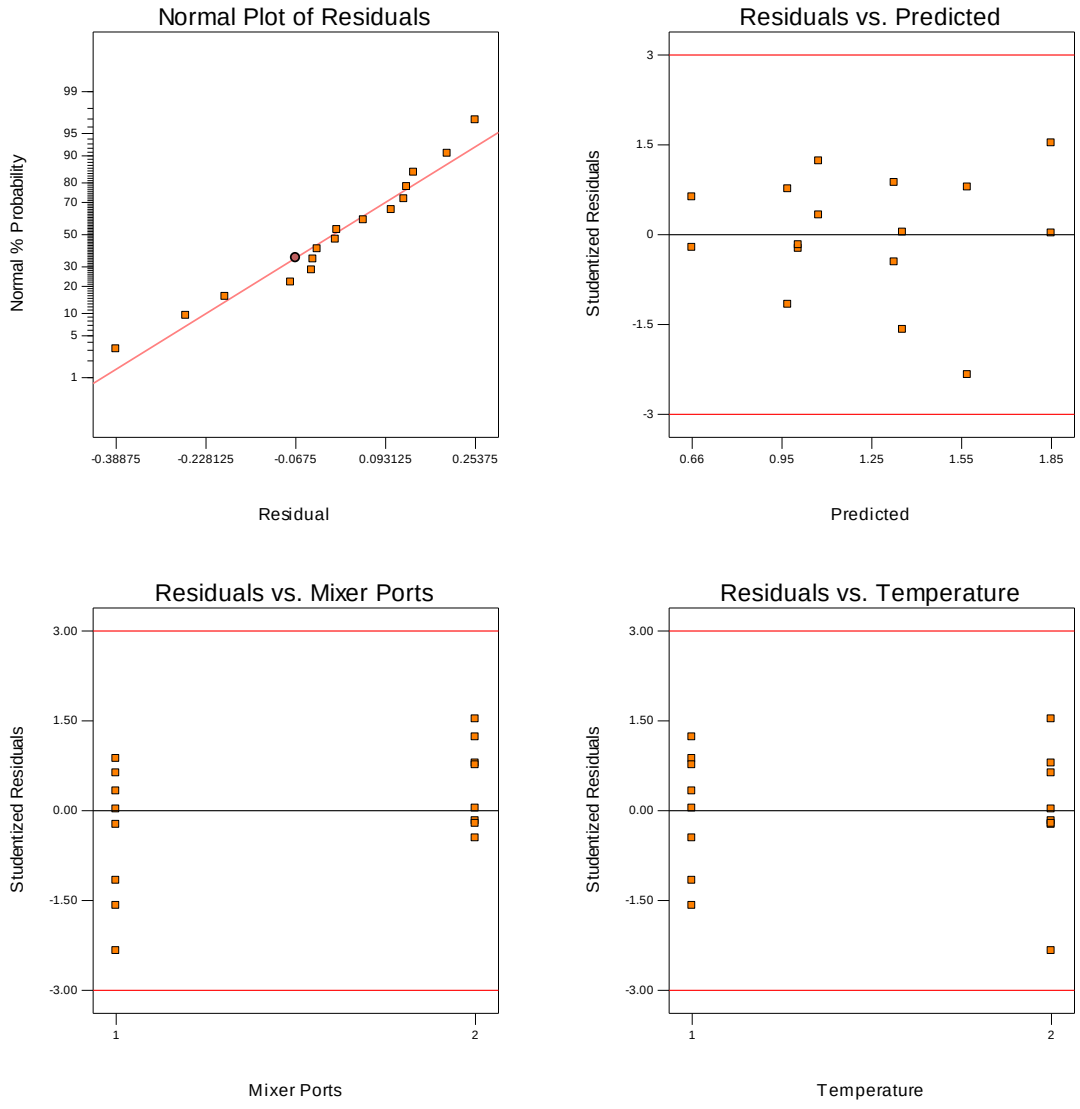
Temperature Cool
Std Dev =
-0.11292
+7.65000E-004 * Batch Weight

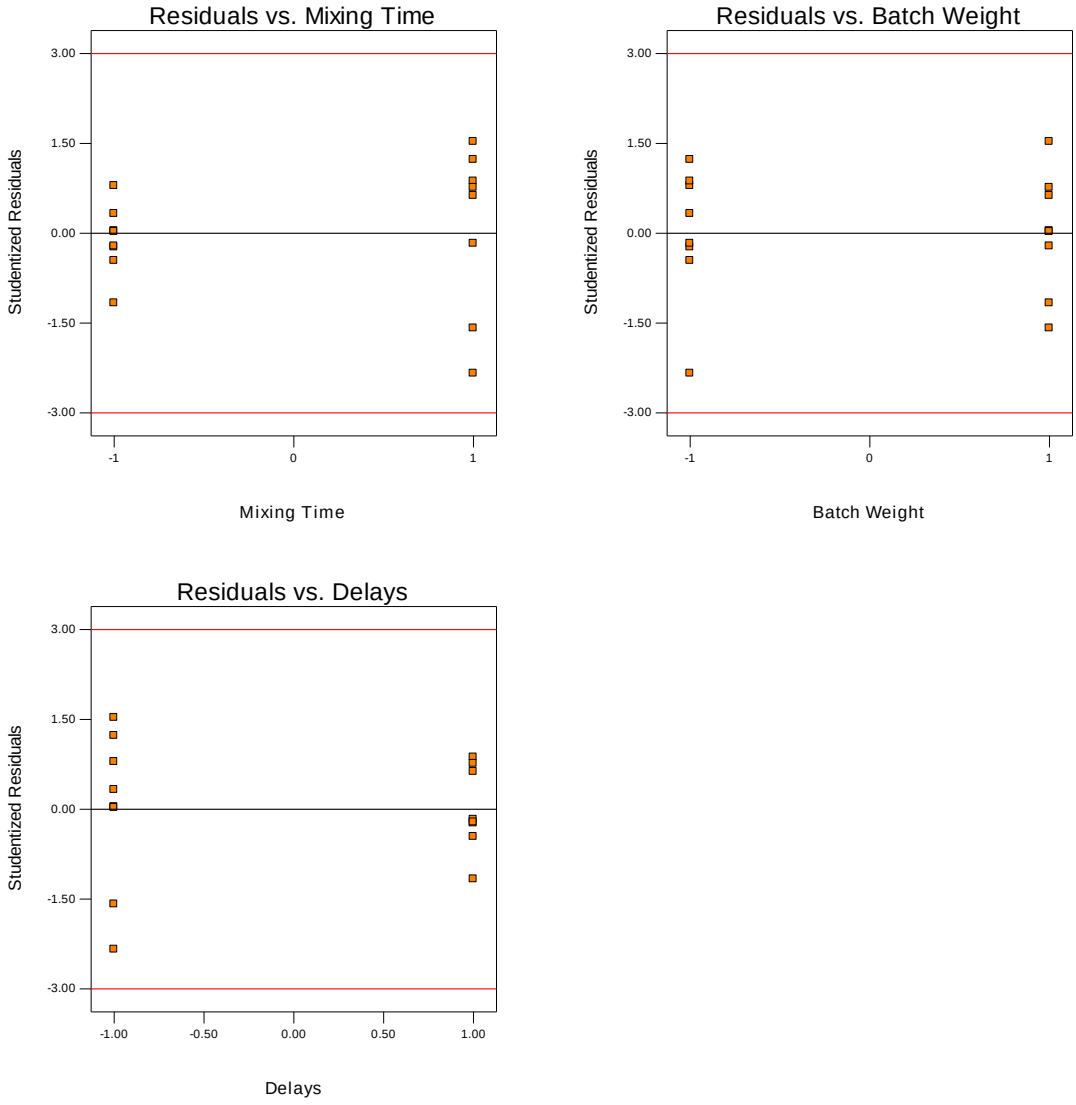
```
          +0.35667    * Delays
     -2.10000E-004    * Batch Weight * Delays

      Temperature     Ambient
          Std Dev     =
          +0.51458
     +7.65000E-004    * Batch Weight
          +0.22167    * Delays
     -2.10000E-004    * Batch Weight * Delays
```

(d) Does a residual analysis indicate any problems with the underlying assumptions?

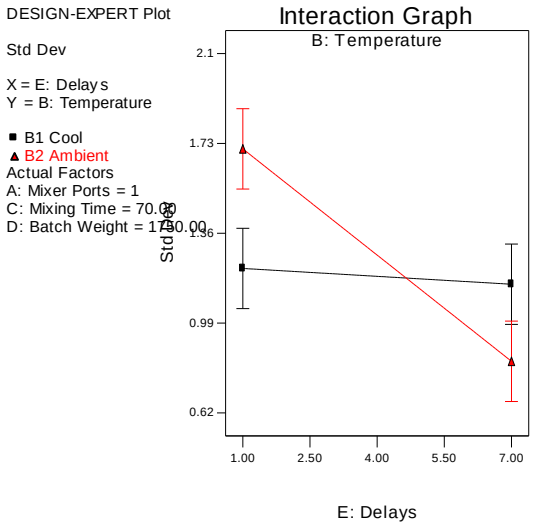
Often a transformation such as the natural log is required for the standard deviation response; however, the following residuals appear to be acceptable without the transformation.

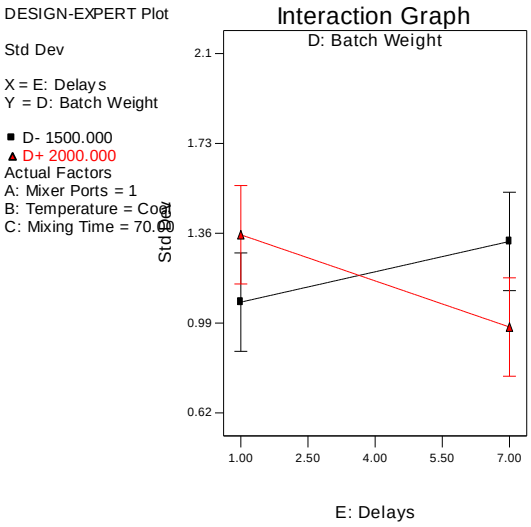




(e) Draw conclusions about this filling process.

From the interaction plots below, the lowest standard deviation can be achieved with the Temperature at ambient, Batch Weight at 2000 lbs, and a Delay of 7 days.





8.45. Consider the 2_{IV}^{6-2} design.

(a) Suppose that the design had been folded over by changing the signs in the column *B* instead of column *A*. What changes would have resulted in the effects that can be estimated from the combined design?

Minitab Output – Fold Over on *A*

Fractional Factorial Design

```
Factors:   6   Base Design:          6, 16   Resolution:   IV
Runs:     32   Replicates:                1   Fraction:    1/2
Blocks:    1   Center pts (total):        0

Design Generators (before folding): E = ABC, F = BCD


Folded on Factors: A


Alias Structure

I + BCDF

A + ABCDF
B + CDF
C + BDF
D + BCF
E + BCDEF
F + BCD
AB + ACDF
AC + ABDF
AD + ABCF
AE + ABCDEF
AF + ABCD
BC + DF
BD + CF
BE + CDEF
BF + CD
CE + BDEF
DE + BCEF
EF + BCDE
ABC + ADF
```

```
ABD + ACF
ABE + ACDEF
ABF + ACD
ACE + ABDEF
ADE + ABCEF
AEF + ABCDE
BCE + DEF
BDE + CEF
BEF + CDE
ABCE + ADEF
ABDE + ACEF
ABEF + ACDE
```

Minitab Output – Fold Over on *B*

Fractional Factorial Design

```
Factors:   6   Base Design:         6, 16   Resolution:   IV
Runs:     32   Replicates:              1   Fraction:    1/2
Blocks:    1   Center pts (total):      0


Design Generators (before folding): E = ABC, F = BCD


Folded on Factors: B


Alias Structure

I + ADEF

A + DEF
B + ABDEF
C + ACDEF
D + AEF
E + ADF
F + ADE
AB + BDEF
AC + CDEF
AD + EF
AE + DF
AF + DE
BC + ABCDEF
BD + ABEF
BE + ABDF
BF + ABDE
CD + ACEF
CE + ACDF
CF + ACDE
ABC + BCDEF
ABD + BEF
ABE + BDF
ABF + BDE
ACD + CEF
ACE + CDF
ACF + CDE
BCD + ABCEF
BCE + ABCDF
BCF + ABCDE
ABCD + BCEF
ABCE + BCDF
ABCF + BCDE
```

Both combined designs are still resolution *IV*, there are some two-factor interactions aliased with other two-factor interactions. In the combined design folded on *A*, all two-factor interactions with *A* are now aliased with four-factor or higher interactions. The sparsity of effects principle would tell us that the higher order interactions are highly unlikely to occur. In the combined design folded on *B*, all two-factor interactions with *B* are now aliased with four-factor or higher interactions.

(b) Suppose that the design had been folded over by changing the sign in the column *E* instead of column *A*. What changes would have resulted in the effects that can be estimated from the combined design?

Minitab Output – Fold Over on *B*

Fractional Factorial Design

```
Factors:   6   Base Design:         6, 16   Resolution:   IV
Runs:     32   Replicates:              1   Fraction:    1/2
Blocks:    1   Center pts (total):      0

Design Generators (before folding): E = ABC, F = BCD

Folded on Factors: E

Alias Structure

I + BCDF

A + ABCDF
B + CDF
C + BDF
D + BCF
E + BCDEF
F + BCD
AB + ACDF
AC + ABDF
AD + ABCF
AE + ABCDEF
AF + ABCD
BC + DF
BD + CF
BE + CDEF
BF + CD
CE + BDEF
DE + BCEF
EF + BCDE
ABC + ADF
ABD + ACF
ABE + ACDEF
ABF + ACD
ACE + ABDEF
ADE + ABCEF
AEF + ABCDE
BCE + DEF
BDE + CEF
BEF + CDE
ABCE + ADEF
ABDE + ACEF
ABEF + ACDE
```

In the combined design folded on *E*, all two-factor interactions with *E* are now aliased with four-factor or higher interactions.

8.46S. Consider the 2_{IV}^{7-3} design. Suppose that fold over of this design is run by changing the signs in column *A*. Determine the alias relationship in the combined design.

Minitab Output

Fractional Factorial Design

```
Factors:   7   Base Design:         7, 16   Resolution:   IV
Runs:     32   Replicates:              1   Fraction:    1/4
Blocks:    1   Center pts (total):      0

Design Generators (before folding): E = ABC, F = BCD, G = ACD
```

```
Folded on Factors: A

Alias Structure

I + BCDF + BDEG + CEFG

A + ABCDF + ABDEG + ACEFG
B + CDF + DEG + BCEFG
C + BDF + EFG + BCDEG
D + BCF + BEG + CDEFG
E + BDG + CFG + BCDEF
F + BCD + CEG + BDEFG
G + BDE + CEF + BCDFG
AB + ACDF + ADEG + ABCEFG
AC + ABDF + AEFG + ABCDEG
AD + ABCF + ABEG + ACDEFG
AE + ABDG + ACFG + ABCDEF
AF + ABCD + ACEG + ABDEFG
AG + ABDE + ACEF + ABCDFG
BC + DF + BEFG + CDEG
BD + CF + EG + BCDEFG
BE + DG + BCFG + CDEF
BF + CD + BCEG + DEFG
BG + DE + BCEF + CDFG
CE + FG + BCDG + BDEF
CG + EF + BCDE + BDFG
ABC + ADF + ABEFG + ACDEG
ABD + ACF + AEG + ABCDEFG
ABE + ADG + ABCFG + ACDEF
ABF + ACD + ABCEG + ADEFG
ABG + ADE + ABCEF + ACDFG
ACE + AFG + ABCDG + ABDEF
ACG + AEF + ABCDE + ABDFG
BCE + BFG + CDG + DEF
BCG + BEF + CDE + DFG
ABCE + ABFG + ACDG + ADEF
ABCG + ABEF + ACDE + ADFG
```

8.47S. Reconsider the 2_{IV}^{7-3} design in Problem 8.46S.

(a) Suppose that a fold over of this design is run by changing the signs in column *B*. Determine the alias relationship in the combined design.

Minitab Output

Fractional Factorial Design

```
Factors:   7   Base Design:         7, 16   Resolution:   IV
Runs:     32   Replicates:              1   Fraction:    1/4
Blocks:    1   Center pts (total):      0

Design Generators (before folding): E = ABC, F = BCD, G = ACD

Folded on Factors: B

Alias Structure

I + ACDG + ADEF + CEFG

A + CDG + DEF + ACEFG
B + ABCDG + ABDEF + BCEFG
C + ADG + EFG + ACDEF
D + ACG + AEF + CDEFG
E + ADF + CFG + ACDEG
F + ADE + CEG + ACDFG
G + ACD + CEF + ADEFG
AB + BCDG + BDEF + ABCEFG
```

```
AC + DG + AEFG + CDEF
AD + CG + EF + ACDEFG
AE + DF + ACFG + CDEG
AF + DE + ACEG + CDFG
AG + CD + ACEF + DEFG
BC + ABDG + BEFG + ABCDEF
BD + ABCG + ABEF + BCDEFG
BE + ABDF + BCFG + ABCDEG
BF + ABDE + BCEG + ABCDFG
BG + ABCD + BCEF + ABDEFG
CE + FG + ACDF + ADEG
CF + EG + ACDE + ADFG
ABC + BDG + ABEFG + BCDEF
ABD + BCG + BEF + ABCDEFG
ABE + BDF + ABCFG + BCDEG
ABF + BDE + ABCEG + BCDFG
ABG + BCD + ABCEF + BDEFG
ACE + AFG + CDF + DEG
ACF + AEG + CDE + DFG
BCE + BFG + ABCDF + ABDEG
BCF + BEG + ABCDE + ABDFG
ABCE + ABFG + BCDF + BDEG
ABCF + ABEG + BCDE + BDFG
```

(b) Compare the aliases from this combined design to those from the combined design from Problem 8.35. What differences resulted by changing the signs in a different column?

Both combined designs are still resolution IV, there are some two-factor interactions aliased with other two-factor interactions. In the combined design folded on *A*, all two-factor interactions with *A* are now aliased with four-factor or higher interactions. The sparsity of effects principle would tell us that the higher order interactions are highly unlikely to occur. In the combined design folded on *B*, all two-factor interactions with *B* are now aliased with four-factor or higher interactions.

8.48S. Consider the 2_{IV}^{7-3} design.

(a) Suppose that a partial fold over of this design is run using column *A* (+ signs only). Determine the alias relationship in the combined design.

By choosing a fold over design in *Design Expert*, sorting on column *A*, and deleting the rows with a minus sign for *A* in the second block, the alias structures are identified below.

Design Expert Output

Factorial Effects Aliases

[Est Terms] Aliased Terms

[Intercept] = Intercept + ABCE + ABFG + ACDG + ADEF + BCDF + BDEG + CEFG
[A] = A - ABCE - ABFG - ACDG - ADEF + ABCDF + ABDEG + ACEFG
[B] = B + ACE + AFG + CDF + DEG + ABCDG + ABDEF + BCEFG
[C] = C + ABE + ADG + BDF + EFG + ABCFG + ACDEF + BCDEG
[D] = D + ACG + AEF + BCF + BEG + ABCDE + ABDFG + CDEFG
[E] = E + ABC + ADF + BDG + CFG + ABEFG + ACDEG + BCDEF
[F] = F + ABG + ADE + BCD + CEG + ABCEF + ACDFG + BDEFG
[G] = G + ABF + ACD + BDE + CEF + ABCEG + ADEFG + BCDFG
[AB] = AB - ACE - AFG + ACDF + ADEG - ABCDG - ABDEF + ABCEFG
[AC] = AC - ABE - ADG + ABDF + AEFG - ABCFG - ACDEF + ABCDEG
[AD] = AD - ACG - AEF + ABCF + ABEG - ABCDE - ABDFG + ACDEFG
[AE] = AE - ABC - ADF + ABDG + ACFG - ABEFG - ACDEG + ABCDEF
[AF] = AF - ABG - ADE + ABCD + ACEG - ABCEF - ACDFG + ABDEFG
[AG] = AG - ABF - ACD + ABDE + ACEF - ABCEG - ADEFG + ABCDFG
[BC] = BC + DF + ABC + ADF + BEFG + CDEG + ABEFG + ACDEG
[BD] = BD + CF + EG + ABCG + ABEF + ACDE + ADFG + BCDEFG
[BE] = BE + DG + ABE + ADG + BCFG + CDEF + ABCFG + ACDEF
[BF] = BF + CD + ABF + ACD + BCEG + DEFG + ABCEG + ADEFG
[BG] = BG + DE + ABG + ADE + BCEF + CDFG + ABCEF + ACDFG
[CE] = CE + FG + ACE + AFG + BCDG + BDEF + ABCDG + ABDEF

[CG] = CG + EF + ACG + AEF + BCDE + BDFG + ABCDE + ABDFG
[ABD] = ABD + ACF + AEG - ABCG - ABEF - ACDE - ADFG + ABCDEFG
[BCE] = BCE + BFG + CDG + DEF + ABCE + ABFG + ACDG + ADEF
[BCG] = BCG + BEF + CDE + DFG + ABCG + ABEF + ACDE + ADFG

Factorial Effects Defining Contrast
I = BCDF = BDEG = CEFG

(b) Rework part (a) using the negative signs to define the partial fold over. Does it make any difference which signs are used to define the partial fold over?

Both partial fold over designs produce the same alias relationships as shown below.

Design Expert Output

Factorial Effects Aliases
[Est Terms] Aliased Terms
[Intercept] = Intercept + ABCE + ABFG + ACDG + ADEF + BCDF + BDEG + CEFG
[A] = A - ABCE - ABFG - ACDG - ADEF + ABCDF + ABDEG + ACEFG
[B] = B + ACE + AFG + CDF + DEG + ABCDG + ABDEF + BCEFG
[C] = C + ABE + ADG + BDF + EFG + ABCFG + ACDEF + BCDEG
[D] = D + ACG + AEF + BCF + BEG + ABCDE + ABDFG + CDEFG
[E] = E + ABC + ADF + BDG + CFG + ABEFG + ACDEG + BCDEF
[F] = F + ABG + ADE + BCD + CEG + ABCEF + ACDFG + BDEFG
[G] = G + ABF + ACD + BDE + CEF + ABCEG + ADEFG + BCDFG
[AB] = AB - ACE - AFG + ACDF + ADEG - ABCDG - ABDEF + ABCEFG
[AC] = AC - ABE - ADG + ABDF + AEFG - ABCFG - ACDEF + ABCDEG
[AD] = AD - ACG - AEF + ABCF + ABEG - ABCDE - ABDFG + ACDEFG
[AE] = AE - ABC - ADF + ABDG + ACFG - ABEFG - ACDEG + ABCDEF
[AF] = AF - ABG - ADE + ABCD + ACEG - ABCEF - ACDFG + ABDEFG
[AG] = AG - ABF - ACD + ABDE + ACEF - ABCEG - ADEFG + ABCDFG
[BC] = BC + DF + ABC + ADF + BEFG + CDEG + ABEFG + ACDEG
[BD] = BD + CF + EG + ABCG + ABEF + ACDE + ADFG + BCDEFG
[BE] = BE + DG + ABE + ADG + BCFG + CDEF + ABCFG + ACDEF
[BF] = BF + CD + ABF + ACD + BCEG + DEFG + ABCEG + ADEFG
[BG] = BG + DE + ABG + ADE + BCEF + CDFG + ABCEF + ACDFG
[CE] = CE + FG + ACE + AFG + BCDG + BDEF + ABCDG + ABDEF
[CG] = CG + EF + ACG + AEF + BCDE + BDFG + ABCDE + ABDFG
[ABD] = ABD + ACF + AEG - ABCG - ABEF - ACDE - ADFG + ABCDEFG
[BCE] = BCE + BFG + CDG + DEF + ABCE + ABFG + ACDG + ADEF
[BCG] = BCG + BEF + CDE + DFG + ABCG + ABEF + ACDE + ADFG

Factorial Effects Defining Contrast
I = BCDF = BDEG = CEFG

8.49. Consider a partial fold over for the 2_{IV}^{6-2} design. Suppose that the signs are reversed in column *A*, but the eight runs are retained are the runs that have positive signs in column *C*. Determine the alias relationship in the combined design.

Design Expert Output

Factorial Effects Aliases
[Est Terms] Aliased Terms
[Intercept] = Intercept - ABE + BCDF - ACDEF
[Block 1] = Block 1 + ABE + ABCE + ADEF + ACDEF
[Block 2] = Block 2 - ABE - ABCE - ADEF - ACDEF
[A] = A + BCE + DEF + ABCDF
[B] = B + ACE + CDF + ABDEF
[C] = C + ABE + BDF + ACDEF
[D] = D + BCF - ABDE - ACEF
[E] = E + ABC + ADF + BCDEF
[F] = F + BCD - ABEF - ACDE
[AB] = AB + ABC + ADF + ACDF
[AC] = AC - BCE - DEF + ABDF
[AD] = AD - BDE - CEF + ABCF
[AE] = AE + ACE + ABDEF + ABCDEF
[AF] = AF - BEF - CDE + ABCD

[BC] = BC + DF - ACE - ABDEF
[BD] = BD + CF + ABEF + ACDE
[BE] = BE + BCE + DEF + CDEF
[BF] = BF + CD + ABDE + ACEF
[CE] = CE - ABC - ADF + BDEF
[DE] = DE + BEF + CDE + BCEF
[EF] = EF + BDE + CEF + BCDE
[ABD] = ABD + ACF + BEF + CDE
[ABF] = ABF + ACD + BDE + CEF
[ADE] = ADE + ABEF + ACDE + ABCEF
[AEF] = AEF + ABDE + ACEF + ABCDE

Factorial Effects Defining Contrast
I = BCDF

8.50. Consider a partial fold over for the 2_{III}^{7-4} design. Suppose that the partial fold over of this design is constructed using column *A* (+ signs only). Determine the alias relationship for the combined design.

Design Expert Output

Factorial Effects Aliases
[Est Terms] Aliased Terms
[Intercept] = Intercept - BD - CE - FG - ABCF + ABCG + ABEF - ABEG + ACDF
- ACDG - ADEF + ADEG + BCDE + BDFG + CEFG - BCDEFG
[Block 1] = Block 1 + BD + CE + FG + ABD + ACE + AFG + BCF + BEG + CDG + DEF
+ ABCF + ABEG + ACDG + ADEF + BCDEFG + ABCDEFG
[Block 2] = Block 2 - BD - CE - FG - ABD - ACE - AFG - BCF - BEG - CDG - DEF
- ABCF - ABEG - ACDG - ADEF - BCDEFG - ABCDEFG
[A] = A + BD + CE + FG + BCG + BEF + CDF + DEG + ABCF + ABEG + ACDG + ADEF
+ ABCDE + ABDFG + ACEFG + BCDEFG
[B] = B + AD + CF + EG + ACG + AEF + CDE + DFG + ABCE + ABFG + BCDG + BDEF
+ ABCDF + ABDEG + BCEFG + ACDEFG
[C] = C + AE + BF + DG + ABG + ADF + BDE + EFG + ABCD + ACFG + BCEG + CDEF
+ ABCEF + ACDEG + BCDFG + ABDEFG
[D] = D + AD + CF + EG + ACF + AEG + BCE + BFG + ABCE + ABFG + BCDG + BDEF
+ ABCDG + ABDEF + CDEFG + ACDEFG
[E] = E + AE + BF + DG + ABF + ADG + BCD + CFG + ABCD + ACFG + BCEG + CDEF
+ ABCEG + ACDEF + BDEFG + ABDEFG
[F] = F + AG + BC + DE + ABE + ACD + BDG + CEG + ABDF + ACEF + BEFG + CDFG
+ ABCFG + ADEFG + BCDEF + ABCDEG
[G] = G + AG + BC + DE + ABC + ADE + BDF + CEF + ABDF + ACEF + BEFG + CDFG
+ ABEFG + ACDFG + BCDEG + ABCDEG
[AB] = AB - AD - CF + CG + EF - EG - ABCE - ABFG + ACDE + ADFG + BCDF - BCDG
- BDEF + BDEG + ABCEFG - ACDEFG
[AC] = AC - AE - BF + BG + DF - DG - ABCD + ABDE - ACFG + AEFG + BCEF - BCEG
- CDEF + CDEG + ABCDFG - ABDEFG
[AF] = AF - AG - BC + BE + CD - DE - ABDF + ABDG - ACEF + ACEG + BCFG - BEFG
- CDFG + DEFG + ABCDEF - ABCDEG

Factorial Effects Defining Contrast
I = ABCG = ABEF = ACDF = ADEG = BCDE = BDFG = CEFG

8.51. Consider a partial fold over for the 2_{III}^{5-2} design. Suppose that the partial fold over of this design is constructed using column *A* (+ signs only). Determine the alias relationship for the combined design.

Design Expert Output

Factorial Effects Aliases
[Est Terms] Aliased Terms
[Intercept] = Intercept + ABCE + ABFG + ACDG + ADEF + BCDF + BDEG + CEFG
[Intercept] = Intercept - BD - CE + BCDE
[Block 1] = Block 1 + BD + CE + ABD + ACE
[Block 2] = Block 2 - BD - CE - ABD - ACE
[A] = A + BD + CE + ABCDE
[B] = B + AD + CDE + ABCE

[C] = C + AE + BDE + ABCD
[D] = D + AD + BCE + ABCE
[E] = E + AE + BCD + ABCD
[AB] = AB - AD - ABCE + ACDE
[AC] = AC - AE - ABCD + ABDE
[BC] = BC + DE + ABE + ACD
[BE] = BE + CD + ABE + ACD
[ABC] = ABC - ABE - ACD + ADE

Factorial Effects Defining Contrast
I = BCDE

8.52. Reconsider the 2^{4-1} design in Example 8.1. The significant factors are *A*, *C*, *D*, *AC* + *BD*, and *AD* + *BC*. Find a partial fold over design that will allow the *AC*, *BD*, *AD*, and *BD* interactions to be estimated.

By constructing a partial fold over reversing the signs in column *A* and using column *A* (+ signs only), the *AC*, *BD*, *AD*, and *BD* interactions can be estimated as shown below. This could also be accomplished by reversing the signs of any one of the factors *A*, *B*, *C*, and *D*.

Design Expert Output

Factorial Effects Aliases
[Est Terms] Aliased Terms
[Intercept] = Intercept - BCD
[Block 1] = Block 1 + BCD + ABCD
[Block 2] = Block 2 - BCD - ABCD
[A] = A + BCD
[B] = B + ACD
[C] = C + ABD
[D] = D + ABC
[AB] = AB - ACD
[AC] = AC - ABD
[AD] = AD - ABC
[BC] = BC + ABC
[BD] = BD + ABD
[CD] = CD + ACD

8.53. Construct a supersaturated design for $k = 8$ factors in $N = 6$ runs.

We have chosen the Hadamard matrix design approach using the following Plackett-Burman design for $N = 12$ runs and $k = 11$ factors. By sorting on the 11th factor, *L*, deleting the rows with the positive levels on *L*, and removing the last 3 factors, the design is reduced to 6 runs with 8 factors. Either the positive or negative levels on *L* could have been chosen. Also, the JMP statistics software package has the capability to generate supersaturated designs.

Run	*I*	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*J*	*K*	*L*
1	+	+	+	–	+	+	+	–	–	–	+	–
2	+	–	+	+	–	+	+	+	–	–	–	+
3	+	+	–	+	+	–	+	+	+	–	–	–
4	+	–	+	–	+	+	–	+	+	+	–	–
5	+	–	–	+	–	+	+	–	+	+	+	–
6	+	–	–	–	+	–	+	+	–	+	+	+
7	+	+	–	–	–	+	–	+	+	–	+	+
8	+	+	+	–	–	–	+	–	+	+	–	+
9	+	+	+	+	–	–	–	+	–	+	+	–
10	+	–	+	+	+	–	–	–	+	–	+	+
11	+	+	–	+	+	+	–	–	–	+	–	+
12	+	–	–	–	–	–	–	–	–	–	–	–

Original Run	Run	*I*	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*

1	1	1	1	1	–1	1	1	1	–1	–1
3	2	1	1	–1	1	1	–1	1	1	1
4	3	1	–1	1	–1	1	1	–1	1	1
5	4	1	–1	–1	1	–1	1	1	–1	1
9	5	1	1	1	1	–1	–1	–1	1	–1
12	6	1	–1	–1	–1	–1	–1	–1	–1	–1

8.54. Consider the 2^{8-3} design in problem 8.29. Suppose that the alias chain involving the *AB* interaction was large. Recommend a partial fold–over design to resolve the ambiguity about this interaction.

The full defining relationship is: $I + ABCF + ABDG + CDFG + ADEFH + ACEGH + BCDEH$; and the alias structure for the *AB* interaction is $AB + CF + DG$. The following 16 runs were created by changing the sign of *A* in the original experiment and then choosing the runs where A is at the low (–) level.

A	*B*	*C*	*D*	*E*	*F*	*G*	*H*
–1	–1	–1	–1	–1	1	1	1
–1	1	–1	–1	–1	–1	–1	–1
–1	–1	1	–1	–1	–1	1	–1
–1	1	1	–1	–1	1	–1	1
–1	–1	–1	1	–1	1	–1	–1
–1	1	–1	1	–1	–1	1	1
–1	–1	1	1	–1	–1	–1	1
–1	1	1	1	–1	1	1	–1
–1	–1	–1	–1	1	1	1	–1
–1	1	–1	–1	1	–1	–1	1
–1	–1	1	–1	1	–1	1	1
–1	1	1	–1	1	1	–1	–1
–1	–1	–1	1	1	1	–1	1
–1	1	–1	1	1	–1	1	–1
–1	–1	1	1	1	–1	–1	–1
–1	1	1	1	1	1	1	1

8.55. Construct a supersaturated design for $k = 12$ factors in $N = 10$ runs.

We have chosen the Hadamard matrix design approach using the following Plackett–Burman design for $N = 20$ runs and $k = 19$ factors. By sorting on the 19th factor, *T*, deleting the rows with the negative levels on *T*, and removing the last 7 factors, the design is reduced to 10 runs with 12 factors. Either the positive or negative levels on *T* could have been chosen. Also, the JMP statistics software package has the capability to generate supersaturated designs.

Run	*I*	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*J*	*K*	*L*	*M*	*N*	*O*	*P*	*Q*	*R*	*S*	*T*
1	+	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–
2	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+
3	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+
4	+	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–
5	+	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–
6	+	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–	–
7	+	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+	–
8	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–	+
9	+	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+	–
10	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–	+
11	+	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+	–
12	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+	+
13	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+	+
14	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+	+
15	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–	+
16	+	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–	–
17	+	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+	–
18	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+	+
19	+	+	–	–	+	+	+	+	–	+	–	+	–	–	–	–	+	+	–	+

20	+	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

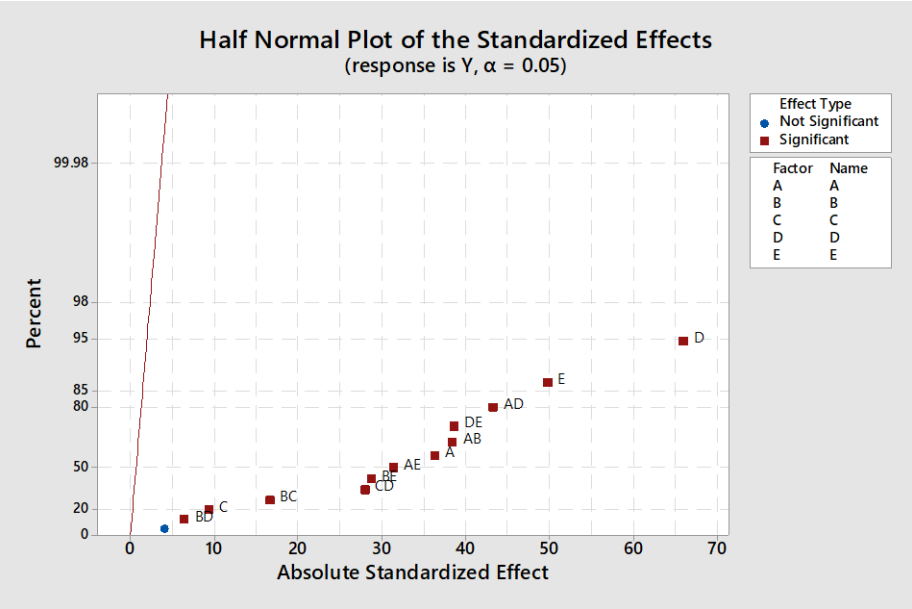
Original Run	Run	*I*	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*J*	*K*	*L*	*M*
2	1	+	–	+	+	–	–	+	+	+	+	–	+	–
3	2	+	+	–	+	+	–	–	+	+	+	+	–	+
8	3	+	–	–	–	–	+	+	–	+	+	–	–	+
10	4	+	–	+	–	–	–	–	+	+	–	+	+	–
12	5	+	–	+	–	+	–	–	–	–	+	+	–	+
13	6	+	+	–	+	–	+	–	–	–	–	+	+	–
14	7	+	+	+	–	+	–	+	–	–	–	–	+	+
15	8	+	+	+	+	–	+	–	+	–	–	–	–	+
18	9	+	–	–	+	+	+	+	–	+	–	+	–	–
19	10	+	+	–	–	+	+	+	+	–	+	–	+	–

8.56. Consider the full 2^5 factorial design in Problem 6.42. Suppose that this experiment had been run as a 2^{5-1} fractional factorial. Set up the fractional design using the principle fraction. Using the 16 runs associated with this design from the original experiment analyze the data and compare your results with those obtained from the analysis of the original full factorial.

The principle fraction for the experiment is:

A	*B*	*C*	*D*	*E*	*y*
–1	–1	–1	–1	1	7.93
1	–1	–1	–1	–1	5.56
–1	1	–1	–1	–1	5.77
1	1	–1	–1	1	12
–1	–1	1	–1	–1	9.17
1	–1	1	–1	1	3.65
–1	1	1	–1	1	6.4
1	1	1	–1	–1	5.69
–1	–1	–1	1	–1	8.82
1	–1	–1	1	1	17.55
–1	1	–1	1	1	8.87
1	1	–1	1	–1	8.94
–1	–1	1	1	1	13.06
1	–1	1	1	–1	11.49
–1	1	1	1	–1	6.25
1	1	1	1	1	26.05

The first analysis that includes the main effects and two factor interactions shows nothing significant. Significant factors are masked because it is a saturated design, i.e. there are not enough degrees of freedom to estimate the full model and 2–factor interactions are aliased with 3–factor interactions. By removing the two factors with the smallest effect, *AC* and *CE*, and re–running the analysis, most of the factors are now significant.



Minitab Output

Factorial Regression: Y versus A, B, C, D, E

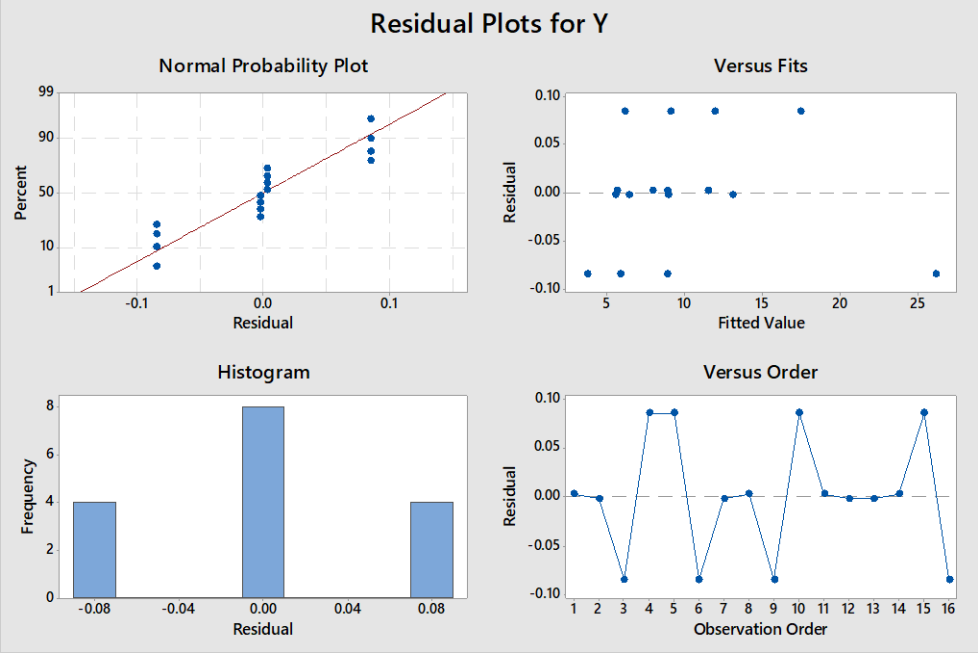
Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	13	461.935	35.533	1228.47	0.001
Linear	5	238.236	47.647	1647.27	0.001
A	1	38.007	38.007	1313.99	0.001
B	1	0.469	0.469	16.22	0.056
C	1	2.496	2.496	86.31	0.011
D	1	125.776	125.776	4348.36	0.000
E	1	71.487	71.487	2471.46	0.000
2-Way Interactions	8	223.699	27.962	966.72	0.001
A*B	1	42.641	42.641	1474.19	0.001
A*D	1	54.023	54.023	1867.68	0.001
A*E	1	28.409	28.409	982.16	0.001
B*C	1	7.981	7.981	275.91	0.004
B*D	1	1.188	1.188	41.08	0.023
B*E	1	23.814	23.814	823.32	0.001
C*D	1	22.610	22.610	781.68	0.001
D*E	1	43.034	43.034	1487.76	0.001
Error	2	0.058	0.029		
Total	15	461.993			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.170074	99.99%	99.91%	99.20%

This model accounts for more of the variation than the full factorial. The R^2 predicted is 99.20%. The issue is that the 2–factor interactions are aliased with the 3–factor interactions and these can only be separated with additional runs. The Residual plots look fine.

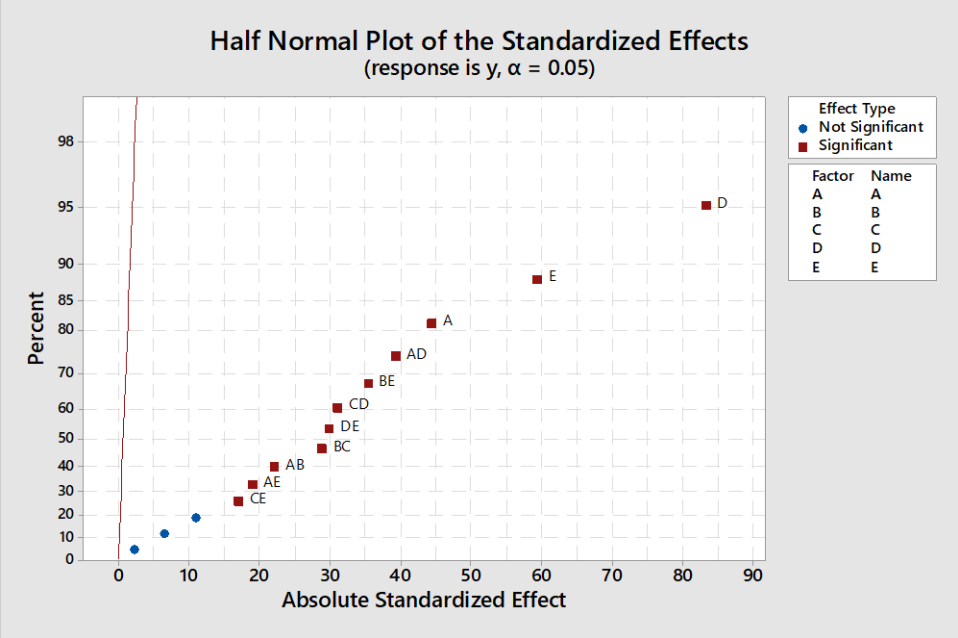


8.57. Consider the full 2^5 factorial design in Problem 6.42. Suppose that this experiment had been run as a 2^{5-1} fractional factorial. Set up the fractional design using the alternate fraction. Using the 16 runs associated with this design from the original experiment analyze the data and compare your results with those obtained from the analysis of the principal fraction in Problem 8.48.

The runs for the alternate fraction are:

A	*B*	*C*	*D*	*E*	*y*
–1	–1	–1	–1	–1	8.11
1	–1	–1	–1	1	5
–1	1	–1	–1	1	7.47
1	1	–1	–1	–1	5.82
–1	–1	1	–1	1	9.86
1	–1	1	–1	–1	7.8
–1	1	1	–1	–1	3.23
1	1	1	–1	1	11.61
–1	–1	–1	1	1	12.43
1	–1	–1	1	–1	14.23
–1	1	–1	1	–1	9.2
1	1	–1	1	1	25.38
–1	–1	1	1	–1	8.68
1	–1	1	1	1	18.85
–1	1	1	1	1	11.78
1	1	1	1	–1	9.12

The first analysis that includes the main effects and two factor interactions shows nothing significant. Significant factors are masked because it is a saturated design, i.e. there are not enough degrees of freedom to estimate the full model and 2–factor interactions are aliased with 3–factor interactions. By removing the factor with the smallest effect, *AC*, and re–running the analysis, most of the factors are now significant.



This model accounts for more of the variation than the full factorial. The R^2 predicted is 98.67%. The issue is that the 2–factor interactions are aliased with the 3–factor interactions and these can only be separated with additional runs. The Residual plots look unusual because of the low number of degrees of freedom for err.

Minitab Output

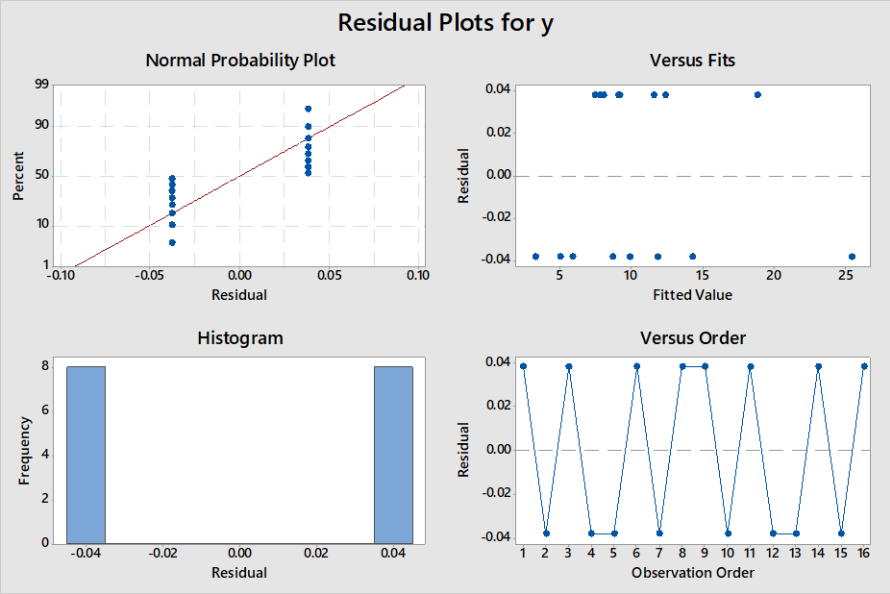
Factorial Regression: y versus A, B, C, D, E

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	14	446.106	31.865	1370.16	0.021
Linear	5	291.616	58.323	2507.85	0.015
A	1	45.731	45.731	1966.41	0.014
B	1	0.114	0.114	4.90	0.270
C	1	2.814	2.814	121.00	0.058
D	1	161.100	161.100	6927.15	0.008
E	1	81.857	81.857	3519.80	0.011
2-Way Interactions	9	154.490	17.166	738.11	0.029
A*B	1	11.306	11.306	486.17	0.029
A*D	1	35.790	35.790	1538.95	0.016
A*E	1	8.338	8.338	358.51	0.034
B*C	1	19.250	19.250	827.74	0.022
B*D	1	0.965	0.965	41.51	0.098
B*E	1	29.025	29.025	1248.06	0.018
C*D	1	22.349	22.349	961.00	0.021
C*E	1	6.695	6.695	287.89	0.037
D*E	1	20.771	20.771	893.13	0.021
Error	1	0.023	0.023		
Total	15	446.130			

Model Summary

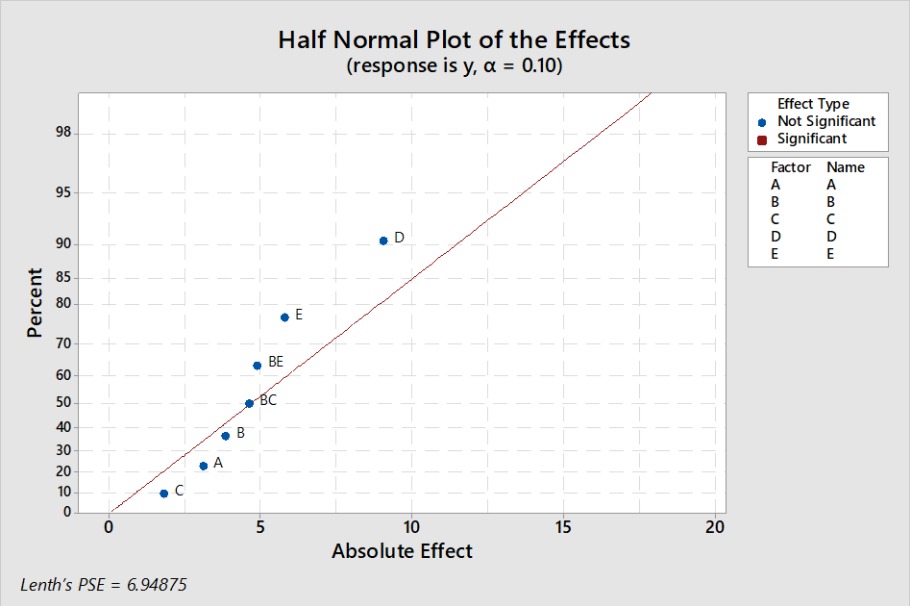
S	R-sq	R-sq(adj)	R-sq(pred)
0.1525	99.99%	99.92%	98.67%



8.58. Consider the full 2^5 factorial design in Problem 6.42. Suppose that this experiment had been run as a 2^{5-2} fractional factorial. Set up the fractional design using the principal fraction. Using the 8 runs associated with this design from the original experiment analyze the data and compare your results with those obtained from the analysis of original design. Then construct the fold–over design and analyze the data from the combined design.

The resulting design is a Resolution III design with main effects aliased with 2–factor interactions.

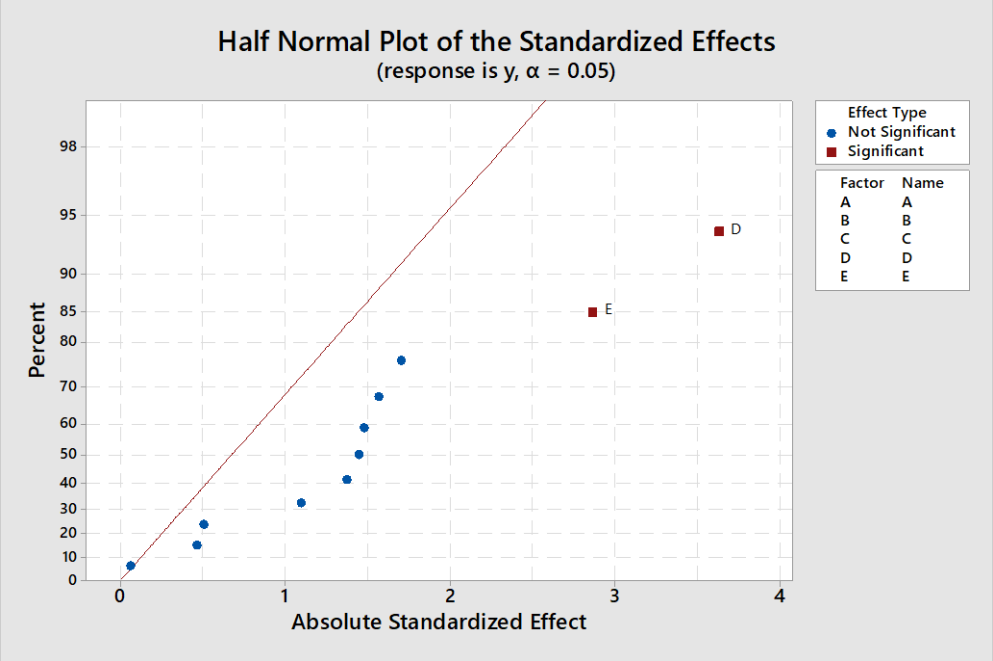
A	*B*	*C*	*D*	*E*	*y*	effect
–1	–1	–1	1	1	12.43	*de*
1	–1	–1	–1	–1	5.56	*a*
–1	1	–1	–1	1	7.47	*be*
1	1	–1	1	–1	8.94	*abd*
–1	–1	1	1	–1	8.68	*cd*
1	–1	1	–1	1	3.65	*ace*
–1	1	1	–1	–1	3.23	*bc*
1	1	1	1	1	26.05	*abcde*



None of the effects are significant in this small design. If the effect of *C* is eliminated, then there is a degree of freedom for error. Even with this, no effect is significant.

By doing the full fold over, i.e. changing the signs on every run, the resulting design is:

A	*B*	*C*	*D*	*E*	*y*	effect
–1	–1	–1	1	1	12.43	*de*
1	–1	–1	–1	–1	5.56	*a*
–1	1	–1	–1	1	7.47	*be*
1	1	–1	1	–1	8.94	*abd*
–1	–1	1	1	–1	8.68	*cd*
1	–1	1	–1	1	3.65	*ace*
–1	1	1	–1	–1	3.23	*bc*
1	1	1	1	1	26.05	*abcde*
1	1	1	–1	–1	5.69	*abc*
–1	1	1	1	1	11.78	*bcde*
1	–1	1	1	–1	11.49	*acd*
–1	–1	1	–1	1	9.86	*ce*
1	1	–1	–1	1	12	*abe*
–1	1	–1	1	–1	9.2	*bd*
1	–1	–1	1	1	17.55	*ade*
–1	–1	–1	–1	–1	8.11	*(1)*



From the half–normal plot, Factors *D* and *E* are significant. The defining relationship is *A*=*BCDE*. This is a Resolution IV design. The model simplifies into the main effects of *D* and *E*. There is one unusual residual.

Minitab Output

Factorial Regression: y versus D, E

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	2	259.16	129.58	8.08	0.005
Linear	2	259.16	129.58	8.08	0.005
D	1	159.71	159.71	9.96	0.008
E	1	99.45	99.45	6.20	0.027
Error	13	208.51	16.04		
Total	15	467.66			

Model Summary

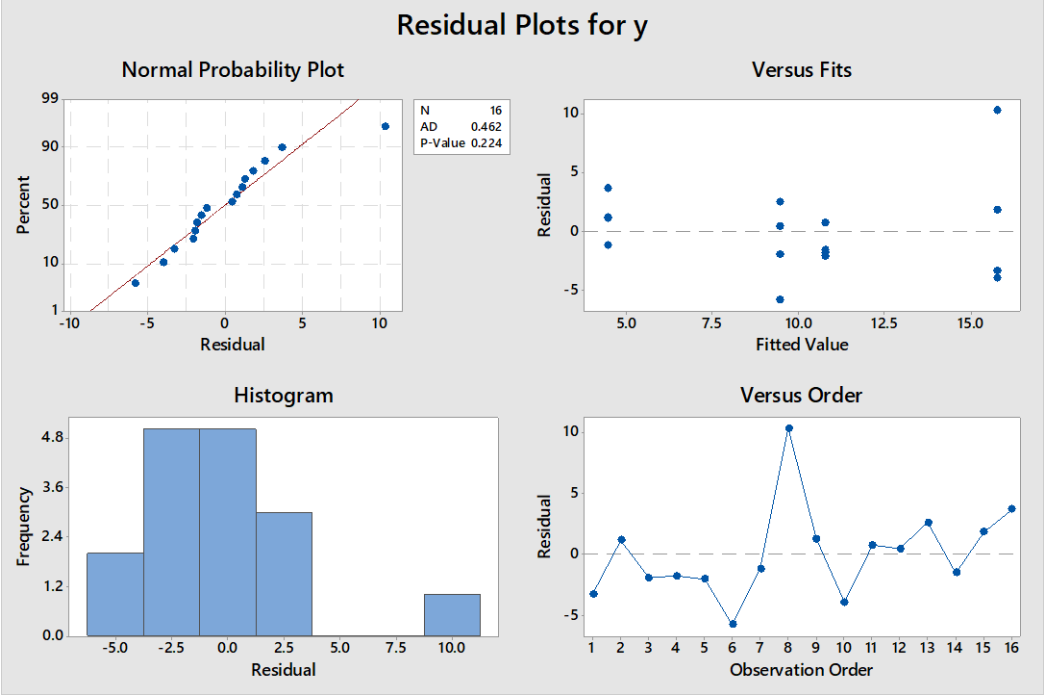
S	R-sq	R-sq(adj)	R-sq(pred)
4.00487	55.42%	48.56%	32.46%

Coded Coefficients

Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant		10.11	1.00	10.09	0.000	
D	6.32	3.16	1.00	3.16	0.008	1.00
E	4.99	2.49	1.00	2.49	0.027	1.00

Regression Equation in Uncoded Units

y = 10.11 + 3.16 D + 2.49 E



8.59. Consider the 2^{5-2} fractional factorial constructed in Problem 8.58. Compare this design with the two one–half fractions of the 2^{5-1}. Is the fold–over design the same as either of the one–half fractions?

The fold–over is not the same as either of the two fractions. The two fractions are resolution V, the fold–over is resolution IV. The half–fraction design has a more complete and better fitting model. But if the experimenter cannot afford to run all of the runs at once, the fold–over is a reasonable alternative.

8.60. The resolution of a two–level fractional factorial design is the number of words in the defining relation.

True **False**

8.61. For a half fraction of a two–level fractional factorial design the maximum possible resolution is equal to the number of factors.

True False

8.62. The 12 run Plackett–Burman for up to 11 factors design is a regular fraction.

True **False**

8.63. The design points of the 2^{k-p} family are the corners of a cube in k–dimensional space and they project into a full factorial in any subset of the original k factors.

True **False**

8.64. It is a good practice to keep the number of factor levels low and the region of interest small in a screening experiment.

True **False**

8.65. Consider a 2^{k-p} fractional factorial design. If the principal fraction is run first – first block – (I = ABCD) and then later augmented with the alternate fraction – second block, the four–factor interaction effect is confounded with blocks.

True False

8.66. In a 2^{k-2} design, every effect has 4 aliases.

True **False**

8.67. In a 2^{k-3} design, the complete defining relation has 15 words.

True False

8.68. The aberration of a fractional factorial design is related to the length of the longest word in the defining relation.

True **False**